Fujisaki-Okamoto Transformation under Average-Case Decryption Error: Tighter and More General Proofs with Applications to PQC

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Abstract. When the Fujisaki-Okamoto (FO) transformation is applied, the decryption error is a crucial parameter that needs to be estimated, as it affects the security bound and may expose vulnerabilities to certain CCA attacks. In general, compared with the worst-case decryption error δ_{wc}, the average-case decryption error δ_{ac} is decoupled from the malicious message, and hence it is simpler to estimate. In light of this, Duman et al. (PKC 2023) proposed FO variants $\mathsf{FOAC}_0 := \mathsf{FO}_m^{\perp} \circ \mathsf{ACWC}_0$ and $\mathsf{FOAC} := \mathsf{FO}_m^{\perp} \circ \mathsf{ACWC}$, and then proved their IND-CCA security in the ROM and QROM based on δ_{ac} and γ-spread. In this paper, we revisit the security proofs of these variants and obtain the following results:

1. We present tighter IND-CCA security proofs of FOAC_0 in the ROM and QROM, while removing the requirement of γ-spread introduced by Duman et al. Moreover, as a direct corollary, we fill the gap in the IND-CCA security proof of BAT (CHES 2022) and give a correct one.
2. We present a tighter IND-CCA security proof of FOAC in the QROM. During this proof, we present a tighter OW-CPA security proof of ACWC in the QROM, which reduces the loss factor of q^2 introduced by Duman et al. to q, where q is the total number of random oracle queries. This actually answers an open question proposed by them.
3. Based on $\mathsf{FO}_m^{\not\perp}$, we define $\mathsf{FOAC}_m^{\not\perp} := \mathsf{FO}_m^{\not\perp} \circ \mathsf{ACWC}$ and present its IND-CCA security proofs in the ROM and QROM. The advantage of $\mathsf{FOAC}_m^{\not\perp}$ is that it neither introduces ciphertext expansion as FOAC_0 does nor requires γ-spread as FOAC does.

In addition, to complete our QROM proofs, we propose the Check Query Replacement technique, which may be of independent interest.

Keywords: quantum random oracle model · security proof · Fujisaki-Okamoto transformation · average-case decryption error.

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1 Introduction

The Fujisaki-Okamoto (FO) [14] is a generic public-key encryption (PKE) transformation that converts a weakly secure PKE scheme into a strongly secure one in the random oracle model (ROM) [3]. Here the weakly secure PKE scheme only has the one-wayness under one-way attacks ($\mathsf{OW\text{-}CPA}$) or the indistinguishability under chosen-plaintext attacks ($\mathsf{IND\text{-}CPA}$), while the strongly secure PKE scheme achieves the indistinguishability under chosen-ciphertext attacks ($\mathsf{IND\text{-}CCA}$). Beyond building PKE schemes, FO also has many key encapsulation mechanism (KEM) variants, e.g., the FO-like transformations $\mathsf{FO}^{\not\perp}$, $\mathsf{FO}^{\not\perp}_m$, $\mathsf{FO}^{\perp}$ and $\mathsf{FO}^{\perp}_m$ [17], which use a $\mathsf{OW\text{-}CPA}$-secure or $\mathsf{IND\text{-}CPA}$-secure PKE scheme to build an $\mathsf{IND\text{-}CCA}$-secure KEM scheme in the ROM. The $\not\perp$ (resp. $\perp$) indicates that the variant is implicit (resp. explicit) rejection type, in which a pseudorandom value (resp. an abort symbol $\perp$) is returned if the ciphertext fails to be decapsulated.

Due to their capacity to elevate security, the FO-like transformations are widely used in the NIST post-quantum cryptography standardisation process [32], and their $\mathsf{IND\text{-}CCA}$ security in the quantum random oracle model (QROM) [6] has been studied by a sequence of works [17,23,5,25,16,15]. Note that many of the underlying PKE schemes designed in the KEM proposals of NIST have a small decryption error, that is, the probability that m is encrypted into c but the decryption of c is not m. This renders the security proof of the original FO no longer applicable to the FO-like transformations. To address this issue, for a PKE scheme P=(Gen,Enc,Dec) with message/randomness spaces $\mathcal{M}/\mathcal{R}$, Hofheinz et al. [17] were the first to formalize the worst-case decryption error δ_{wc} as defined below, which has since been adopted by subsequent works [23,5,25,16,15].

$$\delta_{wc} := \mathop{\mathbb{E}}_{(pk,sk)\leftarrow \mathsf{Gen}}\left[\max_{m\in\mathcal{M}} \Pr_{r\leftarrow_\$ \mathcal{R}}[\mathsf{Dec}(sk,\mathsf{Enc}(pk,m;r))\neq m]\right].$$

Roughly speaking, these subsequent works all prove the $\mathsf{IND\text{-}CCA}$ security of FO-like transformations in the QROM under the assumption that δ_{wc} is negligible. Therefore, to correctly apply FO-like transformations, scheme designers need to ensure that the underlying PKE scheme they designed has a negligible δ_{wc}.

However, this is not always a simple task. Taking the NTRU-based PKE scheme as an example, to make δ_{wc} negligible, scheme designers have to put a limit on the noise and increase the modulus, which expands the ciphertext size and weakens the scheme's security. Hence, scheme designers prefer to adopt the following weaker notion of average-case decryption error:

$$\delta_{ac} := \mathop{\mathbb{E}}_{(pk,sk)\leftarrow \mathsf{Gen}}\left[\Pr_{m\in\mathcal{M},r\leftarrow_\$ \mathcal{R}}[\mathsf{Dec}(sk,\mathsf{Enc}(pk,m;r))\neq m]\right].$$

Compared with δ_{wc}, δ_{ac} no longer focuses on the decryption error of all messages, but rather focuses only on the decryption error of a uniformly random message, making its estimation generally simpler. In fact, a series of works outlined below have investigated how to adjust FO-like transformations so that their $\mathsf{IND\text{-}CCA}$ security can also be proved based on the average-case decryption error δ_{ac}:

- Lyubashevsky and Seiler [30] designed a message-reducing transformation, which we denote as MRT. Then based on δ_{ac} and γ-spread, they proved the $\mathsf{IND\text{-}CCA}$ security of $\mathsf{FO}_m^{\perp} \circ \mathsf{MRT}$ in the ROM. Subsequently, Duman et al. [12] generalized MRT to ACWC_0, and based on δ_{ac} and γ-spread, they proved the $\mathsf{IND\text{-}CCA}$ security of $\mathsf{FOAC}_0 := \mathsf{FO}_m^{\perp} \circ \mathsf{ACWC}_0$ in the ROM and QROM. Notably, FOAC_0 has a drawback that it expands the ciphertext size.
- In [19], Hövelmanns designed the transformation ACWC_1[8], and then proved the $\mathsf{IND\text{-}CCA}$ security of $\mathsf{FOAC}_1 := \mathsf{FO}_m^{\not\perp} \circ \mathsf{ACWC}_1$[9] based on δ_{ac} and γ-spread in the ROM and QROM. Notably, unlike FOAC_0, FOAC_1 avoids the ciphertext expansion by requiring the underlying PKE scheme to be randomness-recoverable. Later, Kim and Park [24] introduced a variant of FOAC_1, which generalized the XOR used in ACWC_1 to a semi-generalized one-time pad SOTP and replaced the $\mathsf{FO}_m^{\not\perp}$ with $\mathsf{FO}_m^{\perp}$. By using this variant, they designed the $\mathsf{NTRU+}$ [33], which has been selected as the KEM winner of the Korean Post-Quantum Cryptography Competition [34].
- Since FOAC_0 and FOAC_1 either expand the ciphertext or require randomness recovery, Duman et al. [12] designed the transformations ACWC and $\mathsf{FOAC} := \mathsf{FO}_m^{\perp} \circ \mathsf{ACWC}$, which avoid these drawbacks. They then proved the $\mathsf{IND\text{-}CCA}$ security of FOAC based on δ_{ac} and γ-spread in the ROM and QROM.

It should be noted that, beyond the basic δ_{ac}, the $\mathsf{IND\text{-}CCA}$ security proofs of FOAC_0 and FOAC given in [12] only require the underlying PKE scheme to satisfy γ-spread; unlike those of FOAC_1, which additionally require the underlying PKE scheme to satisfy randomness recovery. Consequently, FOAC_0 and FOAC are more friendly to scheme designers.

However, unfortunately, the $\mathsf{IND\text{-}CCA}$ security proofs of FOAC_0 and FOAC given in [12] are suboptimal in terms of tightness, and both are much looser than the current widely adopted $\mathsf{IND\text{-}CCA}$ security proofs of $\mathsf{FO}_m^{\perp}$ given in [11,20]. Furthermore, the γ-spread used in [12] remains a constraint for scheme designers. Hence, this leads to a natural question:

Based on the average-case decryption error δ_{ac}, can we give tighter and more general IND-CCA security proofs of FOAC_0 and FOAC?

1.1 Our contributions

In this paper, we answer the above question affirmatively by giving new $\mathsf{IND\text{-}CCA}$ security proofs of FOAC_0 and FOAC based on δ_{ac} in the ROM and QROM. The comparison of security bounds is shown in Table 1.

Tighter and more general security proof of FOAC_0. Our proof of FOAC_0 removes the γ-spread used in [12] and is also valid for dPKE schemes. In the ROM, our proof for dPKE schemes is even tight, and our proof for rPKE schemes has the same tightness as that of FOAC_0 given in [12]. In the QROM, our proofs

[8] It is actually called ACWC in [19]. Here we rename it as ACWC_1 for notational clarity.

[9] [19] also uses the transformation Punc [36] to define FOAC_1. We omit it for simplicity.

for dPKE and rPKE schemes all avoid the quartic security loss incurred in [12].

A fix for BAT's security proof. BAT is a KEM scheme with compact parameters constructed in [13] using FOAC_0, and [13] proved its IND-CCA security by following the IND-CCA security proof of FOAC_0 presented in [12]. However, this is actually incorrect, as [12] requires the underlying PKE scheme to be randomized and γ-spread, while the underlying PKE scheme of BAT is deterministic and [13] did not estimate its γ-spreadness. Fortunately, as shown in Table 1, our proof of FOAC_0 removes the γ-spread and is also valid for dPKE schemes, and hence following our proof can give a correct IND-CCA security proof of BAT. For further details, please refer to Supplementary Material B.

Table 1. IND-CCA security bounds. Here dPKE/rPKE denotes deterministic/randomized public-key encryption, the "Assumption" shows the property needs to be satisfied by the underlying scheme, q denotes the total number of random oracle queries and decapsulation oracle queries, and ϵ denotes the security bound of the underlying scheme.

Transformation	Model	Underlying scheme	Underlying security	Assumption	Security bound
FOAC_0 [12]	ROM	rPKE	OW-CPA	γ-spread	$q^2 \cdot \epsilon + q \cdot \delta_{ac}$
FOAC_0 Thm. 3	ROM	dPKE	OW-CPA	\	$\epsilon + q \cdot \delta_{ac}$
FOAC_0 Thm. 3	ROM	rPKE	OW-CPA	\	$q^2 \cdot \epsilon + q \cdot \delta_{ac}$
FOAC_0 [12]	QROM	rPKE	OW-CPA	γ-spread	$q^2 \cdot \sqrt[4]{\epsilon} + q^2 \cdot \sqrt{\delta_{ac}}$
FOAC_0 Thm. 6	QROM	dPKE	OW-CPA	\	$q \cdot \sqrt{\epsilon} + q \cdot \sqrt{\delta_{ac}}$
FOAC_0 Thm. 6	QROM	rPKE	OW-CPA	\	$q^2 \cdot \sqrt{\epsilon} + q \cdot \sqrt{\delta_{ac}}$
FOAC [12]	QROM	rPKE	OW-CPA	γ-spread	$q^3 \cdot \sqrt{\epsilon} + q^2 \cdot \sqrt{\delta_{ac}}$
FOAC Thm. 8	QROM	rPKE	OW-CPA	γ-spread	$q^2 \cdot \sqrt{\epsilon} + q^2 \cdot \delta_{ac}$
$\mathsf{FOAC}_m^{\not\perp}$ Thm. 21	ROM	dPKE	OW-CPA	\	$\epsilon + q \cdot \delta_{ac}$
$\mathsf{FOAC}_m^{\not\perp}$ Thm. 21	ROM	rPKE	OW-CPA	\	$q^2 \cdot \epsilon + q \cdot \delta_{ac}$
$\mathsf{FOAC}_m^{\not\perp}$ Thm. 23	QROM	dPKE	OW-CPA	\	$q \cdot \sqrt{\epsilon} + q^2 \cdot \delta_{ac}$
$\mathsf{FOAC}_m^{\not\perp}$ Thm. 23	QROM	rPKE	OW-CPA	\	$q^{1.5} \cdot \sqrt{\epsilon} + q^2 \cdot \delta_{ac}$

Tighter security proof of FOAC. In the QROM, our proof of $\mathsf{FOAC}(= \mathsf{FO}_m^{\perp} \circ \mathsf{ACWC})$ reduces the loss factor of q^3 introduced in [12] to q^2, and also achieves a tighter term $q^2 \cdot \delta_{ac}$. During this proof, we present a tighter OW-CPA security proof of ACWC in the QROM, which reduces the loss factor of q^2 introduced in [12] to q, and this actually answers an open question proposed in [12]. For further details, please refer to the technical overview.

Security proof of $\mathsf{FOAC}_m^{\not\perp}$. We define a new KEM transformation $\mathsf{FOAC}_m^{\not\perp} := \mathsf{FO}_m^{\not\perp} \circ \mathsf{ACWC}$ and prove its IND-CCA security in the ROM and QROM, while the tightness of our proof is nearly identical to that of FOAC_0. Here the advantage of $\mathsf{FOAC}_m^{\not\perp}$ is that it neither introduces ciphertext expansion as FOAC_0 does nor requires γ-spread as FOAC does.

A new QROM tool. To complete our QROM proofs, we propose a technique named Check Query Replacement (Theorem 2), which is a new technique related to the compressed oracle [39]. In fact, due to its generality, this technique may have other applications.

$$\begin{array}{ccc}
\begin{array}{c}\mathsf{OW\text{-}CPA}\text{ dPKE/rPKE}\\ \text{Average-case decryption error}\end{array} & \xrightarrow[\text{Thm. 3, Thm. 6}]{\mathsf{FOAC_0=FO_{\mathit{m}}^{\perp}\circ ACWC_0}} & \begin{array}{c}\mathsf{IND\text{-}CCA}\text{ KEM}\\ \text{Ciphertext expansion}\\ \text{Explicit rejection}\end{array}\\
\begin{array}{c}\mathsf{OW\text{-}CPA}\text{ rPKE}\\ \text{Average-case decryption error}\\ \gamma\text{-spread}\end{array} & \xrightarrow[\text{Thm. 8}]{\mathsf{FOAC=FO_{\mathit{m}}^{\perp}\circ ACWC}} & \begin{array}{c}\mathsf{IND\text{-}CCA}\text{ KEM}\\ \text{No ciphertext expansion}\\ \text{Explicit rejection}\end{array}\\
\begin{array}{c}\mathsf{OW\text{-}CPA}\text{ dPKE/rPKE}\\ \text{Average-case decryption error}\end{array} & \xrightarrow[\text{Thm. 21, Thm. 23}]{\mathsf{FOAC_{\mathit{m}}^{\not\perp}=FO_{\mathit{m}}^{\not\perp}\circ ACWC}} & \begin{array}{c}\mathsf{IND\text{-}CCA}\text{ KEM}\\ \text{No ciphertext expansion}\\ \text{Implicit rejection}\end{array}
\end{array}$$

Fig. 1. An overview of the transformations FOAC_0, FOAC and $\mathsf{FOAC}_m^{\not\perp}$.

Insights for scheme designers. To be concise, the security proofs presented in this paper provide the following two insights for scheme designers:

1. In the ROM and QROM, FOAC_0, FOAC and $\mathsf{FOAC}_m^{\not\perp}$ can all transform a $\mathsf{OW\text{-}CPA}$-secure dPKE or rPKE scheme with only average-case decryption error into an $\mathsf{IND\text{-}CCA}$-secure KEM scheme, and there is also no need for excessive concern about the tightness when applying them.
2. NTRU-based PKE schemes are highly suitable for applying FOAC_0, FOAC and $\mathsf{FOAC}_m^{\not\perp}$, e.g. the current BAT [13], NEV [40], MNTRU [2], and DAWN [29]. They can all apply those transformations to obtain an $\mathsf{IND\text{-}CCA}$-secure KEM scheme under the average-case decryption error in the ROM and QROM.

We also provide an overview of FOAC_0, FOAC and $\mathsf{FOAC}_m^{\not\perp}$ in Fig. 1, which clearly shows their corresponding requirements and features.

1.2 Technical overview

Remove the γ-spread for FOAC_0. We first explain how our security proof of $\mathsf{FOAC}_0 = \mathsf{FO}_m^{\perp} \circ \mathsf{ACWC}_0$ removes the γ-spread in the ROM. For clarity, we present the encryption algorithm Enc_0 of ACWC_0 and the decapsulation algorithm Deca_0 of FOAC_0 in Fig. 2. In fact, the key to the $\mathsf{IND\text{-}CCA}$ security proof of FOAC_0 lies in how to simulate Deca_0 using only pk, and our method is similar to that used in [14,17] for handling the decapsulation oracle of $\mathsf{FO}_m^{\perp}$. Specifically, we require that the m obtained in the first step of Deca_0 satisfies the following property:

$$\text{\textit{If } } m \neq \perp \text{\textit{ and }} \mathsf{L_F}(m) = \perp, \text{\textit{ then there is a noticeable probability of}} \\ \mathsf{Enc}_0(pk, m; \mathsf{F}(m)) \neq (c_1, c_2). \tag{1}$$

Here $\mathsf{L_F}$ is the record list used to simulate F on-the-fly, and $\mathsf{L_F}(m) = \bot$ means that F has never been queried on m. Intuitively, this property ensures that the message m passing the re-encryption check has a high probability of satisfying $\mathsf{L_F}(m) \neq \bot$. Consequently, we can search $\mathsf{L_F}$ for the m satisfying $\mathsf{L_F}(m) \neq \bot$ and $\mathsf{Enc_0}(pk, m; \mathsf{L_F}(m)) = (c_1, c_2)$, without needing to compute $\mathsf{Dec_0}(sk, c_1, c_2)$ using the secret key sk. Now, a straightforward way to make (1) hold is to require that the PKE scheme generated by $\mathsf{ACWC_0}$ is γ-spread, i.e. for any m and (c_1, c_2), only a few r satisfy $\mathsf{Enc_0}(pk, m; r) = (c_1, c_2)$. However, we observe that (1) still holds even without introducing the γ-spread. First, it follows from the definition of $\mathsf{Enc_0}$ given in Fig. 2 that $\mathsf{Enc_0}(pk, m; \mathsf{F}(m)) = (c_1, c_2)$ implies $c_2 = m \oplus \mathsf{H}(\mathsf{F}(m))$. Then, since $\mathsf{L_F}(m) = \bot$, $\mathsf{F}(m)$ must be uniformly random by the basic rules of the on-the-fly simulation. This means $\mathsf{F}(m)$ has a negligible probability of being in $\mathsf{L_H}$[10], as the adversary makes at most a polynomial number of queries to H. Hence, based on the basic rules of the on-the-fly simulation again, $\mathsf{H}(\mathsf{F}(m))$ has a noticeable probability of being uniformly random, and this makes the probability of $c_2 \neq m \oplus \mathsf{H}(\mathsf{F}(m))$ noticeable for any m and c_2, which further implies (1) holds. So in summary, based on the specific structure of $\mathsf{ACWC_0}$, we avoid the introduction of γ-spread in the ROM through a flexible control and analysis of the record lists $\mathsf{L_F}$ and $\mathsf{L_H}$.

$\mathsf{Enc_0}(pk, m)$	$\mathsf{Deca_0}(sk, c_1, c_2)$
1 : $r \leftarrow_\$ \mathcal{R}$	1 : $m := \mathsf{Dec_0}(sk, c_1, c_2)$
2 : $c_1 := \mathsf{Enc}(pk, r)$	2 : **if** $m = \bot \vee \mathsf{Enc_0}(pk, m; \mathsf{F}(m)) \neq (c_1, c_2)$
3 : $c_2 := m \oplus \mathsf{H}(r)$	3 : **return** $\bot$
4 : **return** (c_1, c_2)	4 : **else return** $\mathsf{G}(m)$

Fig. 2. $\mathsf{Enc_0}$ and $\mathsf{Deca_0}$. Here F, G and H are random oracles, $\mathsf{Dec_0}$ is the decryption algorithm of $\mathsf{ACWC_0}$, and we assume that Enc does not use randomness for simplicity.

In the QROM, achieving a similar analysis is more complex since currently we can only use the compressed oracle technique [39] to maintain an (imperfect) quantum record list D_F (resp. D_H) for quantum random oracle F (resp. H); these D_F and D_H are also called the databases. Our observation now is that the core operation performed on the classical record lists $\mathsf{L_F}$ and $\mathsf{L_H}$ in the above ROM analysis is to check whether $\mathsf{L_F}(m) = \bot$ and $\mathsf{F}(m) \in \mathsf{L_H}$. So at a high level, if we can establish a conversion framework between quantum random oracle queries and check queries to the database, we should be able to lift the above ROM analysis to the QROM through appropriate rewriting. In light of this, we propose the following Check Query Replacement (CQR) technique.

Theorem 1 (Check Query Replacement, informal). *Let* $\mathsf{H} : \{0,1\}^m \to \{0,1\}^n$ *be a quantum random oracle and* $\mathsf{H_c} : \{0,1\}^m \to \{0,1\}^n$ *be a compressed*

[10] Similar to $\mathsf{L_F}$, $\mathsf{L_H}$ is the record list used to simulate H on-the-fly.

oracle. Let $S \subseteq \{0,1\}^n$ *and define two quantum oracles* $\mathsf{O_0} : |x, b\rangle \mapsto |x, b \oplus [\![\mathsf{H}(x) \in S]\!]\rangle$ *and* $\mathsf{O_1} : |x, D_\mathsf{H}, b\rangle \mapsto |x, D_\mathsf{H}, b \oplus [\![D_\mathsf{H}(x) \in S]\!]\rangle$*. Let* $\mathcal{A}$ *be a quantum algorithm makes at most* q_H *queries to* $\mathsf{H}/\mathsf{H_c}$ *and* q_O *queries to* $\mathsf{O_0}/\mathsf{O_1}$*. Then*

$$|\Pr[1 \leftarrow \mathcal{A}^{\mathsf{H},\mathsf{O_0}}] - \Pr[1 \leftarrow \mathcal{A}^{\mathsf{H_c},\mathsf{O_1}}]| \leq q_\mathsf{O} \cdot \sqrt{|S|/2^n}.$$

The proof idea of Theorem 1 is intuitively explained as follows. First, by [39, Lemma 4], H and $\mathsf{H_c}$ are perfectly indistinguishable. Second, if $\mathcal{A}$ has never queried $\mathsf{H}/\mathsf{H_c}$ with x, we know that $\mathsf{H}(x)$ is uniformly random in $\mathcal{A}$'s view and $D_\mathsf{H}(x) = \perp$, so now the probability that $\mathsf{H}(x) \in S$ is $|S|/2^n$ and $D_\mathsf{H}(x)$ is obviously not in S. Thus, in this case, the difference between $\mathsf{H}(x) \in S$ and $D_\mathsf{H}(x) \in S$ does not exceed $|S|/2^n$, and this bound becomes $\sqrt{|S|/2^n}$ in the specific proof, since both H and $\mathsf{H_c}$ can be quantum accessed. On the other hand, if $\mathcal{A}$ has ever queried $\mathsf{H}/\mathsf{H_c}$ with x, the difference between $\mathsf{H}(x)$ and $D_\mathsf{H}(x)$ is negligible by the basic rules of compressed oracle, and hence the difference between $\mathsf{H}(x) \in S$ and $D_\mathsf{H}(x) \in S$ is also negligible. We point out that the q_O in the upper bound of Theorem 1 stems from the use of a hybrid argument based on the $\mathsf{O_0}/\mathsf{O_1}$ queries.

$\mathsf{Deca_0}(sk, c_1, c_2)$

- $m := \mathsf{Dec_0}(sk, c_1, c_2)$
- **if** $m = \perp$, **return** $\perp$
- **else if** $\mathsf{Enc}(pk, \mathsf{F}(m)) = c_1$
- $\quad \wedge\, m \oplus \mathsf{H}(\mathsf{F}(m)) = c_2$
- **return** $\mathsf{G}(m)$
- **else return** $\perp$

$\mathsf{Deca_0^a}(sk, c_1, c_2)$

- $m := \mathsf{Dec_0}(sk, c_1, c_2)$
- **if** $m = \perp \vee D_\mathsf{H}(\mathsf{F}(m)) = \perp$, **return** $\perp$
- **else if** $\mathsf{Enc}(pk, \mathsf{F}(m)) = c_1$
- $\quad \wedge\, m \oplus D_\mathsf{H}(\mathsf{F}(m)) = c_2$
- **return** $\mathsf{G}(m)$
- **else return** $\perp$

$\mathsf{Deca_0^b}(sk, c_1, c_2)$

- $m := \mathsf{Dec_0}(sk, c_1, c_2)$
- **if** $m = \perp \vee D_\mathsf{F}(m) = \perp \vee D_\mathsf{H}(D_\mathsf{F}(m)) = \perp$, **return** $\perp$
- **else if** $\mathsf{Enc}(pk, D_\mathsf{F}(m)) = c_1$
- $\quad \wedge\, m \oplus D_\mathsf{H}(D_\mathsf{F}(m)) = c_2$
- **return** $\mathsf{G}(m)$
- **else return** $\perp$

$\mathsf{Deca_0^c}(sk, c_1, c_2)$

- $m := \mathsf{Dec_0}(sk, c_1, c_2)$
- **if** $m = \perp \vee D_\mathsf{F}(m) = \perp$, **return** $\perp$
- **else if** $\mathsf{Enc}(pk, D_\mathsf{F}(m)) = c_1$
- $\quad \wedge\, m \oplus \mathsf{H}(D_\mathsf{F}(m)) = c_2$
- **return** $\mathsf{G}(m)$
- **else return** $\perp$

Fig. 3. Decapsulation oracles $\mathsf{Deca_0}$, $\mathsf{Deca_0^a}$, $\mathsf{Deca_0^b}$ and $\mathsf{Deca_0^c}$. Here F, G and H are quantum random oracles, $\mathsf{Dec_0}$ is the decryption algorithm of $\mathsf{ACWC_0}$, and Enc is the underlying encryption algorithm used by $\mathsf{Enc_0}$ as shown in Fig. 2. In this figure, for simplicity, we view D_F and D_H as computational basis states and omit the ket notation.

Next we design three middle oracles $\mathsf{Deca_0^a}$, $\mathsf{Deca_0^b}$ and $\mathsf{Deca_0^c}$ as shown in Fig. 3, which are used to transform the original decapsulation oracle $\mathsf{Deca_0}$ into a new decapsulation oracle without using the secret key sk. Below, we give a high-level explanation of how to analyze the differences among these oracles.

- **$\mathsf{Deca_0}$ to $\mathsf{Deca_0^a}$.** Notice that the H in $\mathsf{Deca_0}$ is a quantum random oracle, while the H in $\mathsf{Deca_0^a}$ is a compressed oracle, and the computation of $\mathsf{Deca}_0^{\mathsf{a}}$ uses the database state D_{H}. In this hop, it can be observed that the check $m \oplus \mathsf{H}(\mathsf{F}(m)) = c_2$ is replaced by the checks $D_{\mathsf{H}}(\mathsf{F}(m)) = \perp$ and $m \oplus D_{\mathsf{H}}(\mathsf{F}(m)) = c_2$. In other words, by defining the set $S := \{m \oplus c_2\}$, this hop replaces the check $\mathsf{H}(\mathsf{F}(m)) \in S$ by the check $D_{\mathsf{H}}(\mathsf{F}(m)) \in S$. Hence by applying Theorem 1, the difference between Deca_0 and $\mathsf{Deca}_0^{\mathsf{a}}$ is $1/\sqrt{2^n}$.

- **$\mathsf{Deca}_0^{\mathsf{a}}$ to $\mathsf{Deca}_0^{\mathsf{b}}$.** Similar to Deca_0 to $\mathsf{Deca}_0^{\mathsf{a}}$, by defining the set $S := \{y : y \neq \perp \wedge D_{\mathsf{H}}(y) \neq \perp \wedge \mathsf{Enc}(pk, y) = c_1 \wedge m \oplus D_{\mathsf{H}}(y) = c_2\}$, we can find that this hop replaces the check $\mathsf{F}(m) \in S$ in $\mathsf{Deca}_0^{\mathsf{a}}$ by the check $D_{\mathsf{F}}(m) \in S$ in $\mathsf{Deca}_0^{\mathsf{b}}$. Since the adversary queries H at most q_{H} times, we have $|S| \leq q_{\mathsf{H}}$, and hence by applying Theorem 1, the difference between $\mathsf{Deca}_0^{\mathsf{a}}$ and $\mathsf{Deca}_0^{\mathsf{b}}$ is $\sqrt{q_{\mathsf{H}}/2^n}$.
- **$\mathsf{Deca}_0^{\mathsf{b}}$ to $\mathsf{Deca}_0^{\mathsf{c}}$.** Similar to Deca_0 to $\mathsf{Deca}_0^{\mathsf{a}}$, by defining the set $S := \{m \oplus c_2\}$, we can find that this hop replaces the check $D_{\mathsf{H}}(D_{\mathsf{F}}(m)) \in S$ in $\mathsf{Deca}_0^{\mathsf{b}}$ by the check $\mathsf{H}(D_{\mathsf{F}}(m)) \in S$ in $\mathsf{Deca}_0^{\mathsf{c}}$. Hence by applying Theorem 1 in reverse, the difference between $\mathsf{Deca}_0^{\mathsf{b}}$ and $\mathsf{Deca}_0^{\mathsf{c}}$ is $1/\sqrt{2^n}$.
- **$\mathsf{Deca}_0^{\mathsf{c}}$ to a new oracle without using the secret key.** By the definition of Enc_0 given in Fig. 2, the check $\mathsf{Enc}(pk, D_{\mathsf{F}}(m)) = c_1 \wedge m \oplus \mathsf{H}(D_{\mathsf{F}}(m)) = c_2$ performed by $\mathsf{Deca}_0^{\mathsf{c}}$ is equivalent to $\mathsf{Enc}_0(pk, m; D_{\mathsf{F}}(m)) = (c_1, c_2)$. Now, it can be noted that all computations of H in $\mathsf{Deca}_0^{\mathsf{c}}$ can be encapsulated within the decryption/encryption algorithm $\mathsf{Dec}_0/\mathsf{Enc}_0$ of ACWC_0. Thus, by using the decryption error of ACWC_0, we can follow the idea of [16] for handling the decapsulation oracle of $\mathsf{FO}_m^{\perp}$ to transform $\mathsf{Deca}_0^{\mathsf{c}}$ into a new decapsulation oracle without using sk, while avoiding the introduction of γ-spread.[11].

Indeed, the above analysis avoids the introduction of γ-spread, and we also stress that the inverse application of Theorem 1 in the hop from $\mathsf{Deca}_0^{\mathsf{b}}$ to $\mathsf{Deca}_0^{\mathsf{c}}$ is crucial, because it is precisely this step that enables us to decouple the computation of H from the decapsulation oracle. In turn, this allows us to reuse the proof from previous works, making our proof more concise and easier to follow.

Tighter security proof of FOAC_0. Our proof outline is shown in Fig. 4, and we emphasize that our Theorem 4 and Theorem 11 both avoid the γ-spread by using the earlier explained technique. In the ROM, our proof has two steps. That is, we first prove that the IND-CCA security of FOAC_0 can be tightly reduced to its IND-CPA security. Then, we prove that the IND-CPA security of FOAC_0 can be reduced to the OW-CPA security of the underlying PKE scheme. Notably, for dPKE schemes, since our Theorem 12 gives a tight IND-CPA security proof of FOAC_0, our IND-CCA security proof of FOAC_0 is likewise tight.

In the QROM, our proof has similar two steps. That is, we first prove that the IND-CCA security of FOAC_0 can be tightly reduced to its IND-CPA security, and then we prove the IND-CPA security of FOAC_0 by using the fact that $\mathsf{FOAC}_0 = \mathsf{FO}_m^{\perp} \circ \mathsf{ACWC}_0 = \mathsf{U}_m^{\perp} \circ \mathsf{T} \circ \mathsf{ACWC}_0$. In more detail, for dPKE schemes, we first prove the security of $\mathsf{U}_m^{\perp}$ and ACWC_0, i.e. Lemma 7 and Lemma 8, respectively, and then we prove the IND-CPA security of FOAC_0 by combining these two lemmas

[11] As a matter of fact, we could also follow the idea of [20] to complete the proof. However, [20] still needs to introduce γ-spread when swapping $\mathsf{eCO.E}$ and the adversary's quantum random oracle queries, i.e. the Eq. (19) of the proof of [20, Theorem 4], and thus we choose to follow the idea of [16] instead.

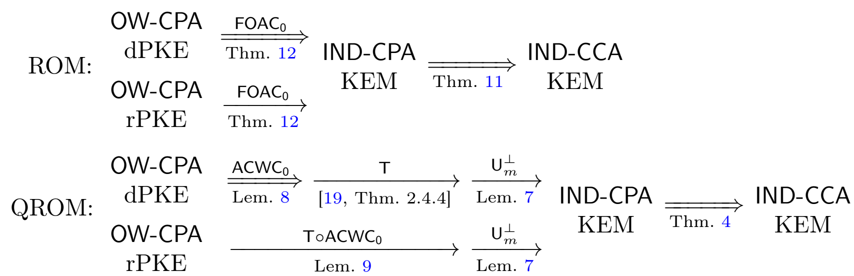

Fig. 4. Proof outline of $\mathsf{FOAC_0}$. The double arrow indicates a tight security proof, while the single arrow indicates a non-tight security proof.

with [19, Theorem 2.4.4]. For rPKE schemes, we first prove the security of $\mathsf{U}_m^\perp$ and $\mathsf{T} \circ \mathsf{ACWC_0}$[12], i.e. Lemma 7 and Lemma 9, respectively, and then we prove the $\mathsf{IND\text{-}CPA}$ security of $\mathsf{FOAC_0}$ by directly combining these two lemmas.

Game $\mathbf{G_0}$	**Game $\mathbf{G_1}$**	**Game $\mathbf{G_2}$**
1. $(pk, sk) \leftarrow \mathsf{Gen}$	1. $(pk, sk) \leftarrow \mathsf{Gen}$	1. $(pk, sk) \leftarrow \mathsf{Gen}$
2. m_1^*, m^* uniformly sample	2. m_1^*, m_2^* uniformly sample	2. m_1^*, m_2^* uniformly sample
$m_2^* := m^* \oplus \mathsf{H}(m_1^*)$	$m^* := m_2^* \oplus \mathsf{H}(m_1^*)$	$c^* := \mathsf{Enc}(pk, m_1^* \vert\vert m_2^*)$
$c^* := \mathsf{Enc}(pk, m_1^* \vert\vert m_2^*)$	$c^* := \mathsf{Enc}(pk, m_1^* \vert\vert m_2^*)$	3. $m' \leftarrow \mathcal{A}^{\mathsf{H}}(pk, c^*)$, measure
3. $m' \leftarrow \mathcal{A}^{\mathsf{H}}(pk, c^*)$	3. $m' \leftarrow \mathcal{A}^{\mathsf{H}}(pk, c^*)$	database register to get D_{H}
4. **return** $[\![m' = m^*]\!]$	4. **return** $[\![m' = m^*]\!]$	4. **return** $[\![m' = m_2^* \oplus D_{\mathsf{H}}(m_1^*)]\!]$

Fig. 5. Games $\mathbf{G_0}$ to $\mathbf{G_2}$. In fact, these games are not used in the proof of Theorem 7, and they are introduced here only for the convenience of explaining our technique.

Tighter security proof of FOAC in the QROM. By using $\mathsf{FOAC} = \mathsf{FO}_m^\perp \circ \mathsf{ACWC}$, our proof has two steps. That is, we first prove Theorem 7, which shows that the $\mathsf{OW\text{-}CPA}$ security of ACWC can be reduced to the $\mathsf{OW\text{-}CPA}$ security of the underlying PKE scheme. Then by combining this proof with the $\mathsf{IND\text{-}CCA}$ security proof of $\mathsf{FO}_m^\perp$ given in [20,22], we prove the $\mathsf{IND\text{-}CCA}$ security of FOAC.

In fact, the key component is Theorem 7, and it is precisely by using this theorem that we achieve a tighter security proof. In Fig. 5, we present three core games for proving Theorem 7, and based on them, our proof outline is as follows:

[12] The proof idea of $\mathsf{T} \circ \mathsf{ACWC_0}$ is similar to that of [19, Theorem 2.4.5], which reprograms two quantum random oracles simultaneously, thereby avoiding the sequential use of One-Way to Hiding theorem [1] twice and in turn yielding a tighter bound.

- $\mathbf{G_0}$ **to** $\mathbf{G_1}$**.** Game $\mathbf{G_0}$ is the $\mathsf{OW\text{-}CPA}$ game of ACWC in the QROM. Obviously, since (m_1^*, m_2^*, m^*) in game $\mathbf{G_0}$ and (m_1^*, m_2^*, m^*) in game $\mathbf{G_1}$ have the same distribution, game $\mathbf{G_1}$ is an equivalent rewrite of game $\mathbf{G_0}$.
- $\mathbf{G_1}$ **to** $\mathbf{G_2}$**.** In game $\mathbf{G_2}$, H is simulated as a compressed oracle, and its database register is measured after $\mathcal{A}$ returns m'. Denote the measurement result as D_{H}, then the final check is replaced with $m' = m_2^* \oplus D_{\mathsf{H}}(m_1^*)$. Now by defining $S := \{m' \oplus m_2^*\}$, it can be observed that game $\mathbf{G_2}$ essentially replaces the check $\mathsf{H}(m_1^*) \in S$ in game $\mathbf{G_1}$ with the check $D_{\mathsf{H}}(m_1^*) \in S$. Therefore, we can use our CQR technique (Theorem 1) to prove that the difference of this game hop is $1/\sqrt{2^n}$. It should be noted that when applying Theorem 1, the quantum algorithm we construct queries only once to $\mathsf{O}_0/\mathsf{O}_1$, and this query is classical since the final check of game $\mathbf{G_1}$ and $\mathbf{G_2}$ is classical.

Based on game $\mathbf{G_2}$, we can construct a $\mathsf{OW\text{-}CPA}$ adversary $\mathcal{B}$ against the underlying PKE scheme. It first runs $\mathcal{A}$ to get m' and D_{H}. Next, it randomly selects an element x from the set $S_D := \{x : D_{\mathsf{H}}(x) \neq \perp\}$ and guesses that x is m_1^*. Finally, it outputs $x||m' \oplus D_{\mathsf{H}}(x)$ as its guess for the challenge plaintext $m_1^*||m_2^*$. Notice that $|S_D| \leq q_{\mathsf{H}}$, where q_{H} is the number of H queries made by $\mathcal{A}$. Hence, the reduction loss incurred by $\mathcal{B}$ is only $O(q_{\mathsf{H}})$, rather than the $O(q_{\mathsf{H}}^2)$ in [12][13].

Tighter security proof of $\mathsf{FOAC}_m^{\not\perp}$**.** In the ROM, based on $\mathsf{FOAC}_m^{\not\perp} = \mathsf{FO}_m^{\not\perp} \circ \mathsf{ACWC}$, our proof is a combination of Lemma 19 and Lemma 20, where Lemma 19 shows that the $\mathsf{IND\text{-}CCA}$ security of $\mathsf{FO}_m^{\not\perp}$ can be tightly reduced to the q-$\mathsf{OW\text{-}CPA}$ security of the underlying PKE scheme, and Lemma 20 proves the q-$\mathsf{OW\text{-}CPA}$ security of ACWC. Notably, for dPKE schemes, since Lemma 20 gives a tight q-$\mathsf{OW\text{-}CPA}$ security proof of ACWC, our $\mathsf{IND\text{-}CCA}$ security proof of $\mathsf{FOAC}_m^{\not\perp}$ is likewise tight. In the QROM, our proof is a combination of [23, Theorem 2] and Theorem 7, where [23, Theorem 2] shows that the $\mathsf{IND\text{-}CCA}$ security of $\mathsf{FO}_m^{\not\perp}$ can be reduced to the $\mathsf{OW\text{-}CPA}$ security of the underlying PKE scheme, and Theorem 7 proves the $\mathsf{OW\text{-}CPA}$ security of ACWC.

2 Preliminaries

2.1 Notation

Let $[n]$ be the set $\{1, 2, \dots, n\}$ for any positive integer n. Let $\Omega_{X \to Y}$ be the set of all functions with domain X and codomain Y. $[\![P]\!]$ denotes a bit which is 1 if the predicate P is true and otherwise 0. $x \leftarrow_\$ S$ denotes the uniformly sampling of an element x from a finite set S. $x \leftarrow \mathfrak{D}$ denotes that x is chosen according to the distribution $\mathfrak{D}$. $\mathcal{T}_{\mathcal{A}}$ denotes the running time of the algorithm $\mathcal{A}$.

[13] [12] proposed an open question, i.e. how to design a variant of the M&R technique [10,9] to get a tighter $\mathsf{OW\text{-}CPA}$ security proof of ACWC. In fact, our Theorem 7 addresses this question, except that we did not design a variant of the M&R technique; instead, we designed a new technique (Theorem 1) based on the compressed oracle.

2.2 Quantum random oracle model

We refer to [31] for detailed quantum background. In the random oracle model (ROM) [3], all parties have classical access to a random function, which is typically implemented by a suitable hash function in practice. In the quantum setting, since such hash functions can be locally computed in superposition by quantum adversaries, the ROM should be lifted to the quantum random oracle model (QROM) [6] where all parties have quantum access to the random oracle. In the QROM, the quantum query to a random oracle H is simulated by the unitary operation $\mathrm{O}_{\mathsf{H}} : |x, y\rangle \mapsto |x, y \oplus \mathsf{H}(x)\rangle$.

2.3 Compressed standard oracle

Roughly speaking, the core idea of the compressed oracle technique [39] is to purify the quantum random oracle via an oracle register, and then encode its quantum state into the (efficiently controlled) database state. In this section, we first introduce the database model, and then introduce one particular version of the compressed oracle named the compressed standard oracle. For simplicity, we assume that the compressed standard oracle is queried at most q times.

Definition 1 (Database). *Define $\mathcal{X} := \{0,1\}^m$ and $\mathcal{Y} := \{0,1\}^n$. Introduce a symbol $\perp$ that $\perp \notin \mathcal{X} \cup \mathcal{Y}$. Then a database D is an ordered list of q-pairs as:*

$$((x_1, y_1), (x_2, y_2), \ldots, (x_l, y_l), (\perp, 0^n), \ldots, (\perp, 0^n)),$$

where $(x_i, y_i) \in \mathcal{X} \times \mathcal{Y}$ for $i \in [l], x_1 < \cdots < x_l$, and all $(\perp, 0^n)$ pairs are at the end of the list. The database set $\mathbf{D}_q$ contains all the above databases.

For an $x \in \mathcal{X}$, we write $D(x) = y$ if there exists a $y \in \mathcal{Y}$ s.t. $(x, y) \in D$, otherwise we write $D(x) = \perp$. Let $n(D)$ denote the number of (x, y) pairs in the database D such that $D(x) \neq \perp$. For any pair $(x, y) \in \mathcal{X} \times \mathcal{Y}$ and any database D which satisfies $D(x) = \perp$ and $n(D) < q$, let $D \cup (x, y)$ be the updated database which first deletes a pair $(\perp, 0^n)$ and then inserts (x, y) appropriately into D to maintain its order. Conversely, for any pair $(x, y) \in \mathcal{X} \times \mathcal{Y}$ and any database D which satisfies $D(x) = y$, let $D \backslash (x, y)$ denote the new database which first removes the pair (x, y) from D and then adds a pair $(\perp, 0^n)$ at the end.

Denote D_q as the quantum register defined over the database set $\mathbf{D}_q$[14], that is, D_q is the Hilbert space with orthonormal basis $\{|D\rangle\}_{D \in \mathbf{D}_q}$, where each basis state is labeled by an element in $\mathbf{D}_q$. In fact, this basis is also called the computational basis by the basic quantum computation. In the following, we refer to D_q as the database register. Define a superposition of the database state as

$$|D \cup (x, \beta_r)\rangle := \frac{1}{\sqrt{2^n}} \sum_{y \in \{0,1\}^n} (-1)^{y \cdot r} |D \cup (x, y)\rangle, \quad (r \in \{0,1\}^n).$$

Then for any $x \in \mathcal{X}$, we define the local decompression procedure $\mathsf{StdDecomp}_x$ to act on D_q as follows:

[14] The quantum implementation details of the datadase set can be found in [7,8].

– For $D \in \mathbf{D}_q$ that $D(x) = \perp$ and $n(D) < q$, $\mathsf{StdDecomp}_x|D\rangle = |D \cup (x, \beta_{0^n})\rangle$.
– For $D \in \mathbf{D}_q$ that $D(x) = \perp$ and $n(D) < q$, $\mathsf{StdDecomp}_x|D \cup (x, \beta_{0^n})\rangle = |D\rangle$ and $\mathsf{StdDecomp}_x|D \cup (x, \beta_r)\rangle = |D \cup (x, \beta_r)\rangle$ for $r \neq 0^n$.
– For $D \in \mathbf{D}_q$ that $D(x) = \perp$ and $n(D) = q$, $\mathsf{StdDecomp}_x|D\rangle = |D\rangle$.

It is obvious that $\mathsf{StdDecomp}_x$ is an involution, since applying it twice results in the identity operation. Meanwhile, it is easy to check that

$$\mathsf{StdDecomp}_x|D \cup (x, y)\rangle = |D \cup (x, y)\rangle + \frac{1}{\sqrt{2^n}}|D\rangle - \frac{1}{\sqrt{2^n}}|D \cup (x, \beta_{0^n})\rangle \quad (2)$$

for any $D \in \mathbf{D}_q$ that $D(x) = \perp$ and $n(D) < q$, which is a very useful property.

Definition 2 (Compressed Standard Oracle). *Let* X *(resp.* Y*) denote the input (resp. output) register defined over* $\mathcal{X}$ *(resp.* $\mathcal{Y}$*). Let* $|D^{\perp}\rangle$ *be the initial state of the database register* D_q*, where* $D^{\perp}$ *is the database containing* q $(\perp, 0^n)$ *pairs. A quantum query to the compressed standard oracle is simulated by performing the following unitary operation on the registers* XYD_q*.*

$$\mathsf{CStO} := \sum_{x \in \mathcal{X}} |x\rangle\langle x|_{\mathsf{X}} \otimes \mathsf{StdDecomp}_x \circ \mathsf{CNOT}^x \circ \mathsf{StdDecomp}_x.$$

Here $\mathsf{CNOT}^x|y, D\rangle_{\mathsf{YD}_q} = |y \oplus D(x), D\rangle_{\mathsf{YD}_q}$ *if* $D(x) \neq \perp$ *and* $|y, D\rangle_{\mathsf{YD}_q}$ *otherwise*[15]*.*

Lemma 1 ([39, Lemma 4]). *For any adversary* $\mathcal{A}$ *making at most* q *oracle queries, the compressed standard oracle as defined in Definition 2 and the quantum random oracle* $\mathsf{H} : \mathcal{X} \to \mathcal{Y}$ *are perfectly indistinguishable.*

Remark 1. As shown in [11, Theorem 4.3], using the compressed standard oracle to perfectly simulate a quantum random oracle with an upper bound of q queries results in a running time increase of $O(q^2)$.

2.4 Check query replacement technique

Here we introduce a theorem named Check Query Replacement (CQR). It shows that under a fixed error, a particular check query to the quantum random oracle H can be replaced with a database check query (i.e., a purified database measurement) when simulating H as a compressed standard oracle. This theorem will be frequently used in our later security proofs.

Theorem 2 (Check Query Replacement). *Let* $\mathsf{H} : \{0,1\}^m \to \{0,1\}^n$ *be a quantum random oracle. Define quantum oralce* $\mathsf{O}_0 : |w, x, b\rangle \mapsto |w, x, b \oplus [\![\mathsf{H}(x) \in S_{w,x}]\!]\rangle$*, which has input/output register* $\mathsf{WX/B}$*, and* W*,* X*,* B *is defined over the set* W*,* $\{0,1\}^m$*,* $\{0,1\}$ *respectively. The set* $S_{w,x} \subseteq \{0,1\}^n$ *used by* O_0 *is determined by* $w \in W$ *and* $x \in \{0,1\}^m$*. Let* $\mathsf{H_c} : \{0,1\}^m \to \{0,1\}^n$ *be*

[15] That is to say, CNOT^x does not change the whole state if $D(x) = \perp$. In fact, this is equivalent to the rule "$y \oplus \perp = y$" defined in [39].

a compressed standard oracle with the database register $\mathsf{D}_{q_\mathsf{H}}$. *Define quantum oralce* $\mathsf{O}_1 : |w, x, D_\mathsf{H}, b\rangle \mapsto |w, x, D_\mathsf{H}, b \oplus [\![D_\mathsf{H}(x) \in S_{w,x}]\!]\rangle$, *which has the same input/output register as* O_0 *and additionally uses* $\mathsf{D}_{q_\mathsf{H}}$ *as the work register.*

Let $\mathcal{A}$ *be a quantum oracle algorithm that makes at most* q_H *queries to* $\mathsf{H}/\mathsf{H}_\mathsf{c}$ *and at most* q_O *queries to* $\mathsf{O}_0/\mathsf{O}_1$. *Define*

$$P_{\text{left}} := \Pr_{z \leftarrow \mathfrak{D}_z}\left[b = 1 : b \leftarrow \mathcal{A}^{\mathsf{H},\mathsf{O}_0}(z)\right] \text{ and } P_{\text{right}} := \Pr_{z \leftarrow \mathfrak{D}_z}\left[b = 1 : b \leftarrow \mathcal{A}^{\mathsf{H}_\mathsf{c},\mathsf{O}_1}(z)\right].$$

Here we denote the classical input of $\mathcal{A}$ *as* z, *and it is sampled according to a distribution* $\mathfrak{D}_z$. *Let* $\Gamma_S := \max_{w \in W, x \in \{0,1\}^m} |S_{w,x}|$, *then we have*

$$|P_{\text{left}} - P_{\text{right}}| \le 6 q_\mathsf{O} \cdot \sqrt{\Gamma_S / 2^n} \text{ and } \left|\sqrt{P_{\text{left}}} - \sqrt{P_{\text{right}}}\right| \le 6 q_\mathsf{O} \cdot \sqrt{\Gamma_S / 2^n}.$$

Proof. The full proof of Theorem 2 is given in Supplementary Material D. □

3 Fujisaki-Okamoto under average-case decryption error

We introduce FO variants FOAC_0, FOAC, and $\mathsf{FOAC}_m^{\not\perp}$ here. Their $\mathsf{IND\text{-}CCA}$ security proofs only require the underlying scheme to have average-case decryption error. Below, we first review the KEM transformations $\mathsf{FO}_m^{\perp}$ and $\mathsf{FO}_m^{\not\perp}$ designed in [17] and the PKE transformations ACWC_0 and ACWC designed in [12].

Transformation $\mathsf{FO}_m^{\perp}$. Let $\mathsf{P} = (\mathsf{Gen}, \mathsf{Enc}, \mathsf{Dec})$ be a rPKE scheme with message space $\mathcal{M}$ and randomness space $\mathcal{R}$. For a given set $\mathcal{K}$, let $\mathsf{F} : \mathcal{M} \to \mathcal{R}$ and $\mathsf{G} : \mathcal{M} \to \mathcal{K}$ be hash functions. We associate a KEM scheme

$$\mathsf{KEM}_m^{\perp} := \mathsf{FO}_m^{\perp}[\mathsf{P}, \mathsf{F}, \mathsf{G}] = (\mathsf{Gen}, \mathsf{Enca}, \mathsf{Deca}_m^{\perp})$$

that has key space $\mathcal{K}$. The constituting algorithms of $\mathsf{KEM}_m^{\perp}$ are given in Fig. 6.

Gen	$\mathsf{Enca}(pk)$	$\mathsf{Deca}_m^{\perp}(sk, c)$
1 : $(pk, sk) \leftarrow \mathsf{Gen}$	1 : $m \leftarrow_\$ \mathcal{M}$	1 : $m := \mathsf{Dec}(sk, c)$
2 : **return** (pk, sk)	2 : $c := \mathsf{Enc}(pk, m; \mathsf{F}(m))$	2 : **if** $m = \perp \lor c \ne \mathsf{Enc}(pk, m; \mathsf{F}(m))$
	3 : $K := \mathsf{G}(m)$	3 : **return** $K := \perp$
	4 : **return** (K, c)	4 : **else return** $K := \mathsf{G}(m)$

Fig. 6. Key encapsulation mechanism $\mathsf{KEM}_m^{\perp}$.

Transformation $\mathsf{FO}_m^{\not\perp}$. Let $\mathsf{P} = (\mathsf{Gen}, \mathsf{Enc}, \mathsf{Dec})$ be a rPKE scheme with message space $\mathcal{M}$, randomness space $\mathcal{R}$ and ciphertext space $\mathcal{C}$. For a given set $\mathcal{K}$, let $\mathsf{F} : \mathcal{M} \to \mathcal{R}$ and $\mathsf{G} : \mathcal{M} \to \mathcal{K}$ be hash functions, let $f : \mathcal{K}^{prf} \times \mathcal{C} \to \mathcal{K}$ be a pseudorandom function (PRF) with key space $\mathcal{K}^{prf}$. We associate a KEM scheme

$$\mathsf{KEM}_m^{\not\perp} := \mathsf{FO}_m^{\not\perp}[f, \mathsf{P}, \mathsf{F}, \mathsf{G}] = (\mathsf{Gen}_m^{\not\perp}, \mathsf{Enca}, \mathsf{Deca}_m^{\not\perp})$$

$\mathsf{Gen}^{\not\perp}_m$	$\mathsf{Enca}(pk)$	$\mathsf{Deca}^{\not\perp}_m(sk' = (sk, s), c)$
$(pk, sk) \leftarrow \mathsf{Gen}$	$m \leftarrow_\$ \leftarrow \mathcal{M}$	$m := \mathsf{Dec}(sk, c)$
$s \leftarrow_\$ \mathcal{K}^{prf}$	$c := \mathsf{Enc}(pk, m; \mathsf{F}(m))$	**if** $m = \perp \vee c \neq \mathsf{Enc}(pk, m; \mathsf{F}(m))$
$sk' := (sk, s)$	$K := \mathsf{G}(m)$	**return** $K := f(s, c)$
return (pk, sk')	**return** (K, c)	**else return** $K := \mathsf{G}(m)$

Fig. 7. Key Encapsulation Mechanism $\mathsf{KEM}^{\not\perp}_m$.

that has key space $\mathcal{K}$. The constituting algorithms of $\mathsf{KEM}^{\not\perp}_m$ are given in Fig. 7.
Transformation ACWC_0. Let $\mathsf{P} = (\mathsf{Gen}, \mathsf{Enc}, \mathsf{Dec})$ be a PKE scheme with message space $\mathcal{M}$, and let $\mathsf{H} : \mathcal{M} \to \{0,1\}^n$ be a hash function. We associate a PKE scheme $\mathsf{ACWC}_0[\mathsf{P}, \mathsf{H}] = (\mathsf{Gen}, \mathsf{Enc}_0, \mathsf{Dec}_0)$ that has message space $\{0,1\}^n$. The constituting algorithms of $\mathsf{ACWC}_0[\mathsf{P}, \mathsf{H}]$ are given in Fig. 8.

Gen	$\mathsf{Enc}_0(pk, m \in \{0,1\}^n)$	$\mathsf{Dec}_0(sk, c_1, c_2)$
$(pk, sk) \leftarrow \mathsf{Gen}$	$r \leftarrow_\$ \mathcal{M}$	$r := \mathsf{Dec}(sk, c_1)$
return (pk, sk)	$c_1 := \mathsf{Enc}(pk, r)$	**if** $r = \perp$
	$c_2 := m \oplus \mathsf{H}(r)$	**return** $m := \perp$
	return (c_1, c_2)	**else return** $m := c_2 \oplus \mathsf{H}(r)$

Fig. 8. Public key encryption $\mathsf{ACWC}_0[\mathsf{P}, \mathsf{H}]$.

Transformation ACWC. Let $\mathsf{P} = (\mathsf{Gen}, \mathsf{Enc}, \mathsf{Dec})$ be a PKE scheme with message space $\{0,1\}^{n_1+n_2}$. Let $\mathsf{H} : \{0,1\}^{n_1} \to \{0,1\}^{n_2}$ be a hash function. We associate a PKE scheme $\mathsf{ACWC}[\mathsf{P}, \mathsf{H}] = (\mathsf{Gen}, \mathsf{Enc}', \mathsf{Dec}')$ that has message space $\{0,1\}^{n_2}$. The constituting algorithms of $\mathsf{ACWC}[\mathsf{P}, \mathsf{H}]$ are given in Fig. 9.

Gen	$\mathsf{Enc}'(pk, m \in \{0,1\}^{n_2})$	$\mathsf{Dec}'(sk, c)$
$(pk, sk) \leftarrow \mathsf{Gen}$	$m_1 \leftarrow_\$ \{0,1\}^{n_1}$	$m_1 \| m_2 := \mathsf{Dec}(sk, c)$
return (pk, sk)	$m_2 := m \oplus \mathsf{H}(m_1)$	$m := m_2 \oplus \mathsf{H}(m_1)$
	$c := \mathsf{Enc}(pk, m_1 \| m_2)$	**return** m
	return c	

Fig. 9. Public key encryption $\mathsf{ACWC}[\mathsf{P}, \mathsf{H}]$.

At this point, based on the above introduced $\mathsf{FO}^{\perp}_m$, $\mathsf{FO}^{\not\perp}_m$, ACWC_0 and ACWC, we define three KEM transformations as follows:

$$\mathsf{FOAC}_0 := \mathsf{FO}^{\perp}_m \circ \mathsf{ACWC}_0, \ \mathsf{FOAC} := \mathsf{FO}^{\perp}_m \circ \mathsf{ACWC}, \ \mathsf{FOAC}^{\not\perp}_m := \mathsf{FO}^{\not\perp}_m \circ \mathsf{ACWC}. \quad (3)$$

These transformations transform a PKE scheme into a KEM scheme, and they all use three hash functions F, G, and H. In the subsequent sections, we will prove their $\mathsf{IND\text{-}CCA}$ security in the ROM and QROM, and the relevant security notions can be found in Supplementary Material E.

Remark 2. The ACWC transformation shown in Fig. 9 is actually a simplified version of the original one defined in [12], because we replace the generalized one-time pad GOTP used in [12] with XOR for the sake of notational simplicity. Nonetheless, we stress that with simple modifications, the security proof of ACWC (Theorem 7) given in this paper is also valid for the original ACWC which uses GOTP. More details on this can be found in Supplementary Material F.1.

4 Security proof of $\mathsf{FOAC_0}$

Here we prove the $\mathsf{IND\text{-}CCA}$ security of $\mathsf{FOAC_0}$ in the ROM and QROM. Our proofs only require that the underlying PKE scheme P have average-case decryption error. Let $\mathsf{F} : \{0,1\}^n \to \mathcal{M}$, $\mathsf{G} : \{0,1\}^n \to \mathcal{K}(= \{0,1\}^k)$ and $\mathsf{H} : \mathcal{M} \to \{0,1\}^n$ be hash functions. Now by Eq. (3), we associate KEM scheme

$$\mathsf{KEMAC_0} := \mathsf{FOAC_0}[\mathsf{P}, \mathsf{F}, \mathsf{G}, \mathsf{H}] = (\mathsf{Gen}, \mathsf{Enca_0}, \mathsf{Deca_0})$$

The constituting algorithms of $\mathsf{KEMAC_0}$ are given in Fig. 10.

Gen	$\mathsf{Enca_0}(pk)$	$\mathsf{Deca_0}(sk, c_1, c_2)$
$(pk, sk) \leftarrow \mathsf{Gen}$	$m \leftarrow_\$ \{0,1\}^n$	$m_1 := \mathsf{Dec}(sk, c_1)$
return (pk, sk)	$m_1 := \mathsf{F}(m)$	**if** $m_1 = \bot$, **return** $\bot$
	$c_1 := \mathsf{Enc}(pk, m_1)$	**else** $m := c_2 \oplus \mathsf{H}(m_1), m_1' := \mathsf{F}(m)$
	$c_2 := m \oplus \mathsf{H}(m_1)$	**if** $c_1 = \mathsf{Enc}(pk, m_1') \wedge c_2 = m \oplus \mathsf{H}(m_1')$
	$K := \mathsf{G}(m)$	**return** $K := \mathsf{G}(m)$
	return (c_1, c_2, K)	**else return** $K := \bot$

Fig. 10. Key encapsulation mechanism $\mathsf{KEMAC_0}$.

4.1 $\mathsf{IND\text{-}CCA}$ security proof of $\mathsf{FOAC_0}$ in the ROM

Theorem 3 ($\mathsf{OW\text{-}CPA}$ **of** $\mathsf{P} \overset{\text{ROM}}{\Rightarrow}$ $\mathsf{IND\text{-}CCA}$ **of** $\mathsf{KEMAC_0}$)**.** *Let* P *be a* δ_{ac}*-average-correct PKE scheme with message space* $\mathcal{M}(= \{0,1\}^m)$*. Let* $\mathcal{A}$ *be an* $\mathsf{IND\text{-}CCA}$ *adversary against* $\mathsf{KEMAC_0}$*, making* q_{D} *classical queries to the decapsulation oracle, and making* q_{F}*,* q_{G} *and* q_{H} *classical queries to the random oracle* F*,* G *and* H*. Then we can construct a* $\mathsf{OW\text{-}CPA}$ *adversary* $\mathcal{B}$ *against* P *such that* $\mathcal{T}_\mathcal{B} \le \mathcal{T}_\mathcal{A} + O(q_{\mathsf{D}} q_{\mathsf{F}} + q_{\mathsf{H}}) \cdot \mathcal{T}_{\mathsf{Enc}}$ *and*

$$\mathrm{Adv}^{\mathsf{IND\text{-}CCA}}_{\mathsf{KEMAC_0},\mathcal{A}} \le \mathrm{Adv}_{\mathsf{P}} + q_{\mathsf{F}} \cdot \delta_{ac} + \frac{q_{\mathsf{D}} q_{\mathsf{H}}}{|\mathcal{M}|} + \frac{2(q_{\mathsf{D}} + q_{\mathsf{F}} + q_{\mathsf{G}}) + 1}{2^n}, \textit{ with}$$

$$\mathrm{Adv}_{\mathsf{P}} = \begin{cases} 3 \cdot \mathrm{Adv}^{\mathsf{OW\text{-}CPA}}_{\mathsf{P},\mathcal{B}} + 6 \cdot \delta_{ac} & \textit{if } \mathsf{P} \textit{ is deterministic,} \\ 3(q_{\mathsf{D}} q_{\mathsf{F}} + q_{\mathsf{H}}) \cdot \mathrm{Adv}^{\mathsf{OW\text{-}CPA}}_{\mathsf{P},\mathcal{B}} & \textit{if } \mathsf{P} \textit{ is randomized.} \end{cases}$$

Here n is the output length of H.

Proof. The full proof of Theorem 3 is given in Supplementary Material G. □

4.2 Quantum accessible decapsulation oracle of $\mathsf{FOAC_0}$

In this section, we analyze how to simulate a quantum-accessible decapsulation oracle for $\mathsf{KEMAC_0}$, denoted qDeca. One may notice that a classical-accessible decapsulation oracle suffices for the IND-CCA security proof. However, to make the proof more consistent[16] and comprehensible, we choose to simulate a quantum-accessible decapsulation oracle for the adversary. We remark that such an enhanced simulation will not cause ambiguity or contradiction, since a classical-accessible oracle can also be simulated by querying the corresponding quantum-accessible oracle, simply by incorporating additional quantum measurements.

As shown in Fig. 10, $\mathsf{KEMAC_0}$ has a key space $\mathcal{K}(=\{0,1\}^k)$ and its ciphertext consists of two parts $c_1 \in \mathcal{C}$ and $c_2 \in \{0,1\}^n$, where $\mathcal{C}$ is the ciphertext space of the underlying PKE scheme P. Based on these properties, we denote $\mathsf{C_1 C_2}$ and K as the input and output register of qDeca, respectively, where $\mathsf{C_1}$ is defined over $\mathcal{C}$, $\mathsf{C_2}$ is defined over $\{0,1\}^n$, and K is defined over $\mathcal{K} \cup \{\perp\}$[17]. Then we define three unitary operations as follows:

$$\mathsf{U_{dec}} : |c_1, y_1, b_1\rangle_{\mathsf{C_1 Y_1 B_1}} \mapsto \begin{cases} |c_1, y_1 \oplus m_1, b_1 \oplus 1\rangle_{\mathsf{C_1 Y_1 B_1}} & \text{if } \mathsf{Dec}(sk, c_1) = m_1 \neq \perp, \\ |c_1, y_1, b_1\rangle_{\mathsf{C_1 Y_1 B_1}} & \text{if } \mathsf{Dec}(sk, c_1) = \perp. \end{cases}$$

$$\begin{aligned} \mathsf{Check_H} :\ & |c_1, c_2, m, m_1', b_2\rangle_{\mathsf{C_1 C_2 M M_1 B_2}} \\ & \mapsto \left|c_1, c_2, m, m_1', b_2 \oplus [\![\mathsf{H}(m_1') \in S_{c_1,c_2,m,m_1'}]\!]\right\rangle_{\mathsf{C_1 C_2 M M_1 B_2}}. \end{aligned} \tag{4}$$

$$\mathsf{Out_G} : |b_1, b_2, m, k\rangle_{\mathsf{B_1 B_2 M K}} \mapsto \begin{cases} |b_1, b_2, m, k \oplus \mathsf{G}(m)\rangle_{\mathsf{B_1 B_2 M K}} & \text{if } b_1 = b_2 = 1, \\ |b_1, b_2, m, k \oplus \perp\rangle_{\mathsf{B_1 B_2 M K}} & \text{if } b_1 = 0 \vee b_2 = 0. \end{cases}$$

In the above definition, the register M is defined over $\{0,1\}^n$, the registers $\mathsf{Y_1}$ and $\mathsf{M_1}$ are both defined over $\mathcal{M}(=\{0,1\}^m)$, and the registers $\mathsf{B_1}$ and $\mathsf{B_2}$ are both defined over $\{0,1\}$. The set $S_{c_1,c_2,m,m_1'}$ is defined as[18]

$$S_{c_1,c_2,m,m_1'} := \begin{cases} \{c_2 \oplus m\} & \text{if } c_1 = \mathsf{Enc}(pk, m_1'), \\ \varnothing & \text{if } c_1 \neq \mathsf{Enc}(pk, m_1'). \end{cases} \tag{5}$$

Essentially, $\mathsf{U_{dec}}$, $\mathsf{Check_H}$ and $\mathsf{Out_G}$ all quantumly compute a fixed function, i.e., their definitions all can be rewritten as $|x, y\rangle \mapsto |x, y \oplus f(x)\rangle$ based on a fixed function f and suitable (x, y). Hence $\mathsf{U_{dec}}$, $\mathsf{Check_H}$ and $\mathsf{Out_G}$ are all involutions.

[16] When the decapsulation oracle is also quantum-accessible, a QROM IND-CCA adversary has a more consistent and concise representation since all its oracles (i.e., its random oracle and decapsulation oracle) are quantum-accessible.

[17] The $\perp$ here is used to denote decapsulation failure, and its quantum representation is explained in Supplementary Material C.

[18] Here for simplicity, we only define $S_{c_1,c_2,m,m_1'}$ for the deterministic Enc; that for the randomized Enc can be defined similarly.

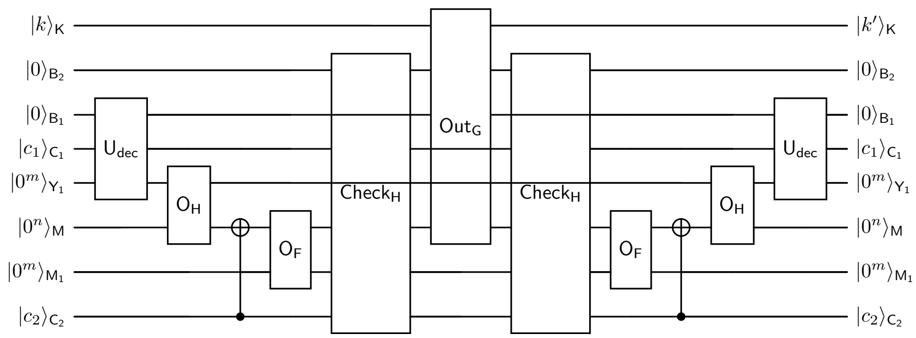

Fig. 11. Quantum circuit of U_{qD}, where $k' := k \oplus \mathsf{Deca}_0(\mathsf{sk}, c_1, c_2)$.

Based on the above defined $\mathsf{U}_{\mathsf{dec}}$, $\mathsf{Check}_{\mathsf{H}}$ and $\mathsf{Out}_{\mathsf{G}}$, we define two composite unitary operations as

$$\begin{aligned} \mathsf{U}_m &:= \mathsf{Check}_{\mathsf{H}} \circ \mathsf{O}_{\mathsf{F}} \circ \mathsf{CNOT} \circ \mathsf{O}_{\mathsf{H}} \circ \mathsf{U}_{\mathsf{dec}}, \\ \mathsf{U}_{\mathsf{qD}} &:= (\mathsf{U}_m)^\dagger \circ \mathsf{Out}_{\mathsf{G}} \circ \mathsf{U}_m. \end{aligned} \tag{6}$$

Here $\mathsf{O}_{\mathsf{H}} : |x, y\rangle \mapsto |x, y \oplus \mathsf{H}(x)\rangle$ has input/output register C_1/M, $\mathsf{O}_{\mathsf{F}} : |x, y\rangle \mapsto |x, y \oplus \mathsf{F}(x)\rangle$ has input/output register M/M_1, and $\mathsf{CNOT} : |x, y\rangle \mapsto |x, y \oplus x\rangle$ has input/output register C_2/M. For clarity, We also provide the quantum circuit of U_{qD} in Fig. 11. Now based on the definition of Deca_0 shown in Fig. 10, it is not hard to verify that

$$\begin{aligned} &\mathsf{U}_{\mathsf{qD}} |c_1, c_2, k\rangle_{\mathsf{C}_1\mathsf{C}_2\mathsf{K}} |0^m, 0^m, 0^n, 0, 0\rangle_{\mathsf{Y}_1\mathsf{M}_1\mathsf{M}\mathsf{B}_1\mathsf{B}_2} \\ &\quad = |c_1, c_2, k \oplus \mathsf{Deca}_0(sk, c_1, c_2)\rangle_{\mathsf{C}_1\mathsf{C}_2\mathsf{K}} |0^m, 0^m, 0^n, 0, 0\rangle_{\mathsf{Y}_1\mathsf{M}_1\mathsf{M}\mathsf{B}_1\mathsf{B}_2}. \end{aligned}$$

By viewing $\mathsf{Y}_1\mathsf{M}_1\mathsf{M}\mathsf{B}_1\mathsf{B}_2$ as the internal registers of the quantum-accessible decapsulation oracle qDeca, and initializing their state as $|0^m, 0^m, 0^n, 0, 0\rangle$, we can use U_{qD} to simulate qDeca, and the state on $\mathsf{Y}_1\mathsf{M}_1\mathsf{M}\mathsf{B}_1\mathsf{B}_2$ remains $|0^m, 0^m, 0^n, 0, 0\rangle$ after such simulation.

4.3 IND-CCA security proof of FOAC_0 in the QROM

Before proving the main theorem of this section, we first prove the following theorem, which shows that the IND-CCA security of FOAC_0 can be tightly reduced to its IND-CPA security in the QROM.

Theorem 4 (IND-CPA of $\mathsf{KEMAC}_0 \overset{\text{QROM}}{\Rightarrow}$ IND-CCA of KEMAC_0)**.** *Let* P *be a δ_{ac}-average-correct PKE scheme with meassage space $\mathcal{M}(= \{0,1\}^m)$. Let $\mathcal{A}$ be an* IND-CCA *adversary against* KEMAC_0*, making q_{D} classical queries to the decapsulation oracle, and making q_{F}, q_{G} and q_{H} quantum queries to the random oracle* F*,* G *and* H*. Then we can construct an* IND-CPA *adversary $\mathcal{B}$ against*

$\mathsf{KEMAC_0}$ *such that* $\mathcal{T}_\mathcal{B} \leq \mathcal{T}_\mathcal{A} + O(q_\mathsf{F}^2) + O(q_\mathsf{D} q_\mathsf{F}) \cdot \mathcal{T}_\mathsf{Enc}$ *and*

$$\mathrm{Adv}^{\mathsf{IND\text{-}CCA}}_{\mathsf{KEMAC_0},\mathcal{A}} \leq \mathrm{Adv}^{\mathsf{IND\text{-}CPA}}_{\mathsf{KEMAC_0},\mathcal{B}} + (80q_\mathsf{F} + 18) \cdot \sqrt{\delta_{ac}} + \frac{14q_\mathsf{D}\sqrt{q_\mathsf{G}} + 1}{\sqrt{|\mathcal{M}|}} + \frac{24q_\mathsf{D}}{\sqrt{2^n}}.$$

Here n *is the output length of* H*. The adversary* $\mathcal{B}$ *queries* F*,* G *and* H *at most* $2q_\mathsf{F}$*,* $q_\mathsf{D} + q_\mathsf{G}$ *and* $2q_\mathsf{D}q_\mathsf{F} + q_\mathsf{H}$ *times, respectively.*

Proof. We first assume that P is deterministic. To prove this theorem, we define a sequence of games $\mathbf{G_0}$ to $\mathbf{G_9}$ as shown in Fig. 12. In these games, the random oracle G is always simulated by the unitary operation $\mathsf{O_G} : |x, y\rangle \mapsto |x, y \oplus \mathsf{G}(x)\rangle$.

GAMES $\mathbf{G_0}$-$\mathbf{G_9}$		$\mathsf{F}(\lvert x, y\rangle)$	
1. $(pk, sk) \leftarrow \mathsf{Gen}$	$//\mathbf{G_0}$-$\mathbf{G_9}$	1. **return** $\lvert x, y \oplus \mathsf{F}(x)\rangle$	$//\mathbf{G_0}$-$\mathbf{G_2}$
$\mathsf{F} \leftarrow_\$ \Omega_{\{0,1\}^n \to \mathcal{M}}$	$//\mathbf{G_0}$-$\mathbf{G_2}$	2. **perform** CStO	$//\mathbf{G_3}$-$\mathbf{G_6}$
$\mathsf{G} \leftarrow_\$ \Omega_{\{0,1\}^n \to \mathcal{K}}$	$//\mathbf{G_0}$-$\mathbf{G_9}$	3. **perform** $\mathsf{CStO}^{m^*}_{m_1^*}$	$//\mathbf{G_7}$-$\mathbf{G_8}$
$\mathsf{H} \leftarrow_\$ \Omega_{\mathcal{M} \to \{0,1\}^n}$	$//\mathbf{G_0}$-$\mathbf{G_9}$	4. **if** $\mathsf{Enc}(pk, \mathsf{F}'(x)) \neq c_1^*$	$//\mathbf{G_9}$
$\mathsf{F}' \leftarrow_\$ \Omega_{\{0,1\}^n \to \mathcal{M}}$	$//\mathbf{G_8}$-$\mathbf{G_9}$	**perform** CStO	
2. $m^* \leftarrow_\$ \{0,1\}^n$	$//\mathbf{G_0}$-$\mathbf{G_9}$	**else return** $\lvert x, y \oplus \mathsf{F}'(x)\rangle$	
$m_1^* := \mathsf{F}(m^*)$	$//\mathbf{G_0}$-$\mathbf{G_6}$		
$m_1^* \leftarrow_\$ \mathcal{M}$	$//\mathbf{G_7}$	$\mathsf{H}(\lvert x, y\rangle)$	
$m_1^* := \mathsf{F}'(m^*)$	$//\mathbf{G_8}$-$\mathbf{G_9}$	1. **return** $\lvert x, y \oplus \mathsf{H}(x)\rangle$	$//\mathbf{G_0}, \mathbf{G_5}$-$\mathbf{G_9}$
$c_1^* := \mathsf{Enc}(pk, m_1^*)$	$//\mathbf{G_0}$-$\mathbf{G_9}$	2. **perform** CStO	$//\mathbf{G_1}$-$\mathbf{G_4}$
$c_2^* := m^* \oplus \mathsf{H}(m_1^*)$	$//\mathbf{G_0}$-$\mathbf{G_9}$		
3. $b \leftarrow_\$ \{0,1\}$	$//\mathbf{G_0}$-$\mathbf{G_9}$	$\mathsf{qDeca}^*(\lvert c_1, c_2, y\rangle)$	
$K_0^* := \mathsf{G}(m^*), K_1^* \leftarrow_\$ \mathcal{K}$	$//\mathbf{G_0}$-$\mathbf{G_9}$	1. **perform** $\mathsf{U}^{\mathbf{i}}_{\mathsf{qD}}$	$//\mathbf{G_i}, \mathbf{i} \in \{0, 1, \ldots, 5\}$
4. $b' \leftarrow \mathcal{A}^{\mathsf{F},\mathsf{G},\mathsf{H},\mathsf{qDeca}^*}(pk, c_1^*, c_2^*, K_b^*)$	$//\mathbf{G_0}$-$\mathbf{G_9}$	2. **perform** $\mathsf{U}^{pk}_{\mathsf{qD}}$	$//\mathbf{G_6}$-$\mathbf{G_9}$
5. **return** $[\![b' = b]\!]$	$//\mathbf{G_0}$-$\mathbf{G_9}$		

Fig. 12. Games $\mathbf{G_0}$ to $\mathbf{G_9}$ in the proof of Theorem 4.

Game $\mathbf{G_0}$: This is the $\mathsf{IND\text{-}CCA}$ game of $\mathsf{KEMAC_0}$ with the adversary $\mathcal{A}$ in the QROM. Its decapsulation oracle qDeca^* is the same as the qDeca simulated by $\mathsf{U_{qD}} = (\mathsf{U}_m)^\dagger \circ \mathsf{Out_G} \circ \mathsf{U}_m$ (Eq. (6)), except that it directly returns $\perp$ to the query (c_1^*, c_2^*). Obviously

$$\mathrm{Adv}^{\mathsf{IND\text{-}CCA}}_{\mathsf{KEMAC_0},\mathcal{A}} = \left|\Pr[1 \leftarrow \mathbf{G_0}] - \frac{1}{2}\right|. \tag{7}$$

Recall that $\mathsf{C_1C_2}$ and K is the input and output register of the decapsulation oracle qDeca, respectively, and it uses internal registers $\mathsf{Y_1M_1MB_1B_2}$. Define a projector on $\mathsf{C_1C_2}$ as $\mathrm{P}_{c_1^*, c_2^*} := |c_1^*, c_2^*\rangle\langle c_1^*, c_2^*|$, and define a unitary operation acting on the register K as $\mathrm{U}_\perp : |k\rangle \mapsto |k \oplus \perp\rangle$. Now one can find that the qDeca^* queried in game $\mathbf{G_0}$ is actually simulated by the following unitary operation:

$$\mathsf{U}^0_{\mathsf{qD}} := \mathrm{U}_\perp \circ \mathrm{P}_{c_1^*, c_2^*} + (\mathsf{U}_m)^\dagger \circ \mathsf{Out}_\mathsf{G} \circ \mathsf{U}_m \circ \left(\mathrm{I} - \mathrm{P}_{c_1^*, c_2^*}\right),$$

where the unitary operation U_m defined in Eq. (6) is

$$\mathsf{U}_m = \mathsf{Check}_\mathsf{H} \circ \mathsf{O}_\mathsf{F} \circ \mathsf{CNOT} \circ \mathsf{O}_\mathsf{H} \circ \mathsf{U}_{\mathsf{dec}}.$$

Game $\mathbf{G_1}$: This game is the same as $\mathbf{G_0}$, except that H is implemented as a compressed standard oracle based on the database register $\mathsf{D}_{q_\mathsf{H}}$, and the $\mathsf{U}^0_{\mathsf{qD}}$ used to simulate qDeca^* is replaced with

$$\mathsf{U}^1_{\mathsf{qD}} := \mathrm{U}_\perp \circ \mathrm{P}_{c_1^*, c_2^*} + (\mathsf{U}^1_m)^\dagger \circ \mathsf{Out}_\mathsf{G} \circ \mathsf{U}^1_m \circ \left(\mathrm{I} - \mathrm{P}_{c_1^*, c_2^*}\right),$$

where

$$\mathsf{U}^1_m := \mathsf{Read}_\mathsf{H} \circ \mathsf{O}_\mathsf{F} \circ \mathsf{CNOT} \circ \mathsf{CStO} \circ \mathsf{U}_{\mathsf{dec}}. \tag{8}$$

The above U^1_m is the same as the U_m, except that O_H is correspondingly replaced by CStO (Definition 2) and that $\mathsf{Check}_\mathsf{H}$ is replaced by a new unitary operation Read_H, which acts on the registers $\mathsf{C}_1\mathsf{C}_2\mathsf{M}\mathsf{M}_1\mathsf{D}_{q_\mathsf{H}}\mathsf{B}_2$ and is defined as

$$\begin{aligned} \mathsf{Read}_\mathsf{H} :\ & |c_1, c_2, m, m_1', D_\mathsf{H}, b_2\rangle \\ & \mapsto \left|c_1, c_2, m, m_1', D_\mathsf{H}, b_2 \oplus [\![D_\mathsf{H}(m_1') \in S_{c_1, c_2, m, m_1'}]\!]\right\rangle. \end{aligned} \tag{9}$$

We use Theorem 2 to compute the difference between the success probabilities of game $\mathbf{G_0}$ and $\mathbf{G_1}$. First we define a quantum oracle algorithm $\mathcal{A}_1$ as follows:

- $\mathcal{A}_1$ has quantum oracle access to H and O_0, where H is a random oracle and O_0 is simulated by the $\mathsf{Check}_\mathsf{H}$ defined in Eq. (4). Here $\mathsf{Check}_\mathsf{H}$ acts on the registers $\mathsf{C}_1\mathsf{C}_2\mathsf{M}\mathsf{M}_1\mathsf{B}_2$, and O_0 has input/output register $\mathsf{C}_1\mathsf{C}_2\mathsf{M}\mathsf{M}_1/\mathsf{B}_2$.
- $\mathcal{A}_1$ runs game $\mathbf{G_0}$ and outputs its final result. In particular, $\mathcal{A}_1$ replies to the H queries by querying its own oracle H, and $\mathcal{A}_1$ replaces performing of $\mathsf{Check}_\mathsf{H}$ with querying to its own oracle O_0.

We denote the classical input of $\mathcal{A}_1$ as z and its distribution as $\mathfrak{D}_z$[19]. Note that game $\mathbf{G_0}$ requires performing $\mathsf{Check}_\mathsf{H}$ twice to simulate one query to qDeca^*; thus $\mathcal{A}_1$ queries H (resp. O_0) at most q_H (resp. $2q_\mathsf{D}$) times. Now we have

$$\Pr\left[1 \leftarrow \mathbf{G_0}\right] = \Pr\left[1 \leftarrow \mathcal{A}_1^{\mathsf{H}, \mathsf{O}_0}(z) : z \leftarrow \mathfrak{D}_z\right].$$

Obviously, if we implement H as a compressed standard oracle based on the database register $\mathsf{D}_{q_\mathsf{H}}$ and replace O_0 by a new oracle O_1 that is simulated by the Read_H defined in Eq. (9), then $\mathcal{A}_1$ will run game $\mathbf{G_1}$. Therefore

$$\Pr\left[1 \leftarrow \mathbf{G_1}\right] = \Pr\left[1 \leftarrow \mathcal{A}_1^{\mathsf{H}_c, \mathsf{O}_1}(z) : z \leftarrow \mathfrak{D}_z\right].$$

[19] The existence of $\mathfrak{D}_z$ is obvious, and we omit its explicit definition here for simplicity.

In the above equation, we write "$\mathsf{H_c}$" to stress that it is a compressed standard oracle. Now by using Theorem 2 with $\mathcal{A}_1$, we can obtain

$$\begin{aligned} |\Pr[1 \leftarrow \mathbf{G_0}] - \Pr[1 \leftarrow \mathbf{G_1}]| &\le 12q_{\mathsf{D}} \cdot \sqrt{\max_{c_1,c_2,m,m_1'} \left|S_{c_1,c_2,m,m_1'}\right| / 2^n} \\ &\overset{(a)}{\le} 12q_{\mathsf{D}} \cdot \sqrt{2^{-n}}. \end{aligned} \tag{10}$$

Here (a) follows from the fact that $|S_{c_1,c_2,m,m_1'}| \le 1$ for any (c_1, c_2, m, m_1'), which is evident from Eq. (5).

Game $\mathbf{G_2}$: This game is the same as $\mathbf{G_1}$, except that the $\mathsf{U}^1_{\mathsf{qD}}$ used to simulate qDeca^* is replaced with

$$\mathsf{U}^2_{\mathsf{qD}} := \mathrm{U}_\perp \circ \mathrm{P}_{c_1^*,c_2^*} + (\mathsf{U}^2_m)^\dagger \circ \mathsf{Out_G} \circ \mathsf{U}^2_m \circ \left(\mathrm{I} - \mathrm{P}_{c_1^*,c_2^*}\right).$$

where

$$\mathsf{U}^2_m := \mathsf{CFRH} \circ \mathsf{CNOT} \circ \mathsf{CStO} \circ \mathsf{U_{dec}}. \tag{11}$$

The above U^2_m is the same as the U^1_m defined in Eq. (8), except that the $\mathsf{Read_H} \circ \mathsf{O_F}$ is replaced with

$$\begin{aligned} \mathsf{CFRH}: \ & |c_1, c_2, m, m_1', D_{\mathsf{H}}, b_2\rangle \\ & \mapsto \left|c_1, c_2, m, m_1', D_{\mathsf{H}}, b_2 \oplus [\![D_{\mathsf{H}}(m_1' \oplus \mathsf{F}(m)) \in S_{c_1,c_2,m,m_1'}]\!]\right\rangle, \end{aligned} \tag{12}$$

which is a new unitary operation acts on the registers $\mathsf{C_1C_2MM_1D_{\mathit{q}_H}B_2}$[20].

Note that game $\mathbf{G_2}$ only replaces the $(\mathsf{U}^1_m)^\dagger \circ \mathsf{Out_G} \circ \mathsf{U}^1_m$ used in game $\mathbf{G_1}$ with $(\mathsf{U}^2_m)^\dagger \circ \mathsf{Out_G} \circ \mathsf{U}^2_m$. In fact, these two unitary operations are identical according to Lemma 2 below; thus game $\mathbf{G_2}$ is merely a restatement of game $\mathbf{G_1}$. Hence

$$\Pr[1 \leftarrow \mathbf{G_1}] = \Pr[1 \leftarrow \mathbf{G_2}]. \tag{13}$$

Lemma 2. *We have* $(\mathsf{U}^1_m)^\dagger \circ \mathsf{Out_G} \circ \mathsf{U}^1_m = (\mathsf{U}^2_m)^\dagger \circ \mathsf{Out_G} \circ \mathsf{U}^2_m$.

Proof. Recall that $\mathsf{O_F} : |x, y\rangle \mapsto |x, y \oplus \mathsf{F}(x)\rangle$, $\mathsf{Read_H}$ defined in Eq. (9) and CFRH defined in Eq. (12) are all involutions. Hence, based on the definition of U^1_m and U^2_m given in Eq. (8) and Eq.(11), respectively, we only need to prove $\mathsf{CFRH} \circ \mathsf{Out_G} \circ \mathsf{CFRH} = \mathsf{O_F} \circ \mathsf{Read_H} \circ \mathsf{Out_G} \circ \mathsf{Read_H} \circ \mathsf{O_F}$ to prove this lemma.

Based on the definitions of CFRH and $\mathsf{Read_H}$, for any computational basis state $|c_1, c_2, m, m_1', D_{\mathsf{H}}, b_2\rangle$ of the registers $\mathsf{C_1C_2MM_1D_{\mathit{q}_H}B_2}$, we can compute

$$\begin{aligned} \mathsf{CFRH}\,&|c_1, c_2, m, m_1', D_{\mathsf{H}}, b_2\rangle \\ &= \left|c_1, c_2, m, m_1', D_{\mathsf{H}}, b_2 \oplus [\![D_{\mathsf{H}}(m_1' \oplus \mathsf{F}(m)) \in S_{c_1,c_2,m,m_1'}]\!]\right\rangle, \\ \mathsf{Read_H} \circ \mathsf{O_F}\,&|c_1, c_2, m, m_1', D_{\mathsf{H}}, b_2\rangle \\ &= \left|c_1, c_2, m, m_1' \oplus \mathsf{F}(m), D_{\mathsf{H}}, b_2 \oplus [\![D_{\mathsf{H}}(m_1' \oplus \mathsf{F}(m)) \in S_{c_1,c_2,m,m_1'}]\!]\right\rangle. \end{aligned}$$

[20] The reason we call the above unitary operation CFRH is that, loosely speaking, it actually reads from D_{H} to check whether $\mathsf{F}(m)$ satisfies $D_{\mathsf{H}}(m_1' \oplus \mathsf{F}(m)) \in S_{c_1,c_2,m,m_1'}$.

This means that CFRH and $\mathsf{O_F} \circ \mathsf{Read_H} \circ \mathsf{O_F}$ actually yield the same result when acting on any computational basis state; thus we have $\mathsf{CFRH} = \mathsf{O_F} \circ \mathsf{Read_H} \circ \mathsf{O_F}$. With this equality, we can then compute

$$\begin{aligned}\mathsf{CFRH} \circ \mathsf{Out_G} \circ \mathsf{CFRH} =& \mathsf{O_F} \circ \mathsf{Read_H} \circ \mathsf{O_F} \circ \mathsf{Out_G} \circ \mathsf{O_F} \circ \mathsf{Read_H} \circ \mathsf{O_F} \\ \overset{(a)}{=}& \mathsf{O_F} \circ \mathsf{Read_H} \circ \mathsf{Out_G} \circ \mathsf{O_F} \circ \mathsf{O_F} \circ \mathsf{Read_H} \circ \mathsf{O_F} \\ \overset{(b)}{=}& \mathsf{O_F} \circ \mathsf{Read_H} \circ \mathsf{Out_G} \circ \mathsf{Read_H} \circ \mathsf{O_F}.\end{aligned}$$

Here (a) follows from the fact that $\mathsf{Out_G}$ commutes with $\mathsf{O_F}$, which is obvious since the register they work together is only M and they both do not change the state of M in the computational basis. (b) is because $\mathsf{O_F}$ is an involution. Now we prove this lemma by the above equation. □

Game $\mathbf{G_3}$: This game is the same as $\mathbf{G_2}$, except that F is implemented as a compressed standard oracle based on the database register $\mathsf{D}_{q_\mathsf{F}}$, and the $\mathsf{U}^2_{\mathsf{qD}}$ used to simulate qDeca^* is replaced with

$$\mathsf{U}^3_{\mathsf{qD}} := \mathrm{U}_\perp \circ \mathrm{P}_{c_1^*,c_2^*} + (\mathsf{U}^3_m)^\dagger \circ \mathsf{Out_G} \circ \mathsf{U}^3_m \circ \left(\mathrm{I} - \mathrm{P}_{c_1^*,c_2^*}\right).$$

where

$$\mathsf{U}^3_m := \mathsf{Read_{F,H}} \circ \mathsf{CNOT} \circ \mathsf{CStO} \circ \mathsf{U_{dec}}. \tag{14}$$

The above U^3_m is the same as the U^2_m defined in Eq. (11), except that the CFRH is replaced with

$$\begin{aligned}\mathsf{Read_{F,H}}:\ & |c_1, c_2, m, m_1', D_\mathsf{F}, D_\mathsf{H}, b_2\rangle \\ & \mapsto \left|c_1, c_2, m, m_1', D_\mathsf{F}, D_\mathsf{H}, b_2 \oplus [\![D_\mathsf{H}(m_1' \oplus D_\mathsf{F}(m)) \in S_{c_1,c_2,m,m_1'}]\!]\right\rangle,\end{aligned} \tag{15}$$

which is a new unitary operation acts on the registers $\mathsf{C_1C_2MM_1D}_{q_\mathsf{F}}\mathsf{D}_{q_\mathsf{H}}\mathsf{B_2}$. Here we stress that we default $D_\mathsf{H}(m_1' \oplus \perp) = \perp$, and thus $D_\mathsf{H}(m_1' \oplus \perp) \notin S_{c_1,c_2,m,m_1'}$.

Next we compute the difference between the success probabilities of game $\mathbf{G_2}$ and $\mathbf{G_3}$. For a fixed $(c_1, c_2, m, m_1', D_\mathsf{H})$ tuple, define the set

$$S_{c_1,c_2,m,m_1',D_\mathsf{H}} := \left\{y : D_\mathsf{H}(m_1' \oplus y) \in S_{c_1,c_2,m,m_1'}\right\}.$$

Since there are at most q_H different x s.t. $D_\mathsf{H}(x) \neq \perp$, it is easy to check that

$$\max_{c_1,c_2,m,m_1',D_\mathsf{H}} |S_{c_1,c_2,m,m_1',D_\mathsf{H}}| \leq q_\mathsf{H}.$$

Then, based on $S_{c_1,c_2,m,m_1',D_\mathsf{H}}$, the CFRH (Eq. (12)) used in U^2_m of game $\mathbf{G_2}$ and the $\mathsf{Read_{F,H}}$ (Eq. (15)) used in U^3_m of game $\mathbf{G_3}$ can both be rewritten as:

$$\begin{aligned}\mathsf{CFRH}:& |c_1, c_2, m, m_1', D_\mathsf{H}, b_2\rangle \\ & \mapsto \left|c_1, c_2, m, m_1', D_\mathsf{H}, b_2 \oplus [\![\mathsf{F}(m) \in S_{c_1,c_2,m,m_1',D_\mathsf{H}}]\!]\right\rangle, \\ \mathsf{Read_{F,H}}:& |c_1, c_2, m, m_1', D_\mathsf{F}, D_\mathsf{H}, b_2\rangle \\ & \mapsto \left|c_1, c_2, m, m_1', D_\mathsf{F}, D_\mathsf{H}, b_2 \oplus [\![D_\mathsf{F}(m) \in S_{c_1,c_2,m,m_1',D_\mathsf{H}}]\!]\right\rangle.\end{aligned}$$

Hence similar to the proof of the game hop from $\mathbf{G_0}$ to $\mathbf{G_1}$, just by simply changing the perspective from H to F, we can design a specific quantum oracle algorithm and then apply Theorem 2 to prove that

$$\begin{aligned} |\Pr[1 \leftarrow \mathbf{G_2}] - \Pr[1 \leftarrow \mathbf{G_3}]| &\le 12 q_{\mathsf{D}} \cdot \sqrt{\max_{c_1,c_2,m,m_1',D_{\mathsf{H}}} \left|S_{c_1,c_2,m,m_1',D_{\mathsf{H}}}\right| / 2^m} \\ &\le 12 q_{\mathsf{D}} \sqrt{q_{\mathsf{H}}} \cdot \sqrt{2^{-m}} = 12 q_{\mathsf{D}} \sqrt{q_{\mathsf{H}}} / \sqrt{|\mathcal{M}|}. \end{aligned} \tag{16}$$

Game $\mathbf{G_4}$: This game is the same as $\mathbf{G_3}$, except that the $\mathsf{U}^3_{\mathsf{qD}}$ used to simulate qDeca^* is replaced with

$$\mathsf{U}^4_{\mathsf{qD}} := \mathrm{U}_\perp \circ \mathrm{P}_{c_1^*,c_2^*} + (\mathsf{U}^4_m)^\dagger \circ \mathsf{Out}_{\mathsf{G}} \circ \mathsf{U}^4_m \circ \left(\mathrm{I} - \mathrm{P}_{c_1^*,c_2^*}\right).$$

where

$$\mathsf{U}^4_m := \mathsf{Read}'_{\mathsf{H}} \circ \mathsf{Read}_{\mathsf{F}} \circ \mathsf{CNOT} \circ \mathsf{CStO} \circ \mathsf{U}_{\mathsf{dec}}. \tag{17}$$

The above U^4_m is the same as the U^3_m defined in Eq. (14), except that the $\mathsf{Read}_{\mathsf{F},\mathsf{H}}$ is replaced with $\mathsf{Read}'_{\mathsf{H}} \circ \mathsf{Read}_{\mathsf{F}}$, where $\mathsf{Read}_{\mathsf{F}}$ acts on the registers $\mathsf{M}\mathsf{M}_1\mathsf{D}_{q_{\mathsf{F}}}$ and

$$\mathsf{Read}_{\mathsf{F}} : |m, m_1', D_{\mathsf{F}}\rangle \mapsto \begin{cases} |m, m_1', D_{\mathsf{F}}\rangle & \text{if } D_{\mathsf{F}}(m) = \perp, \\ |m, m_1' \oplus D_{\mathsf{F}}(m), D_{\mathsf{F}}\rangle & \text{if } D_{\mathsf{F}}(m) \neq \perp. \end{cases} \tag{18}$$

As for the $\mathsf{Read}'_{\mathsf{H}}$, it acts on the registers $\mathsf{C}_1\mathsf{C}_2\mathsf{M}\mathsf{M}_1\mathsf{D}_{q_{\mathsf{F}}}\mathsf{D}_{q_{\mathsf{H}}}\mathsf{B}_2$

$$\begin{aligned} \mathsf{Read}'_{\mathsf{H}} :\ & |c_1, c_2, m, m_1', D_{\mathsf{F}}, D_{\mathsf{H}}, b_2\rangle \\ & \mapsto \left|c_1, c_2, m, m_1', D_{\mathsf{F}}, D_{\mathsf{H}}, b_2 \oplus [\![D_{\mathsf{H}}(m_1') \in S_{c_1,c_2,m,m_1',D_{\mathsf{F}}}]\!]\right\rangle. \end{aligned} \tag{19}$$

In fact, $\mathsf{Read}'_{\mathsf{H}}$ is the same as the $\mathsf{Read}_{\mathsf{H}}$ defined in Eq. (9), except that it additionally acts on $\mathsf{D}_{q_{\mathsf{F}}}$ and replaces $S_{c_1,c_2,m,m_1'}$ with the following new set

$$S_{c_1,c_2,m,m_1',D_{\mathsf{F}}} := \begin{cases} S_{c_1,c_2,m,m_1'} & \text{if } D_{\mathsf{F}}(m) \neq \perp, \\ \varnothing & \text{if } D_{\mathsf{F}}(m) = \perp. \end{cases} \tag{20}$$

Similar to the proof of the game hop from $\mathbf{G_1}$ to $\mathbf{G_2}$, game $\mathbf{G_4}$ is merely a restatement of game $\mathbf{G_3}$ according to the following Lemma 3, and thus

$$\Pr[1 \leftarrow \mathbf{G_3}] = \Pr[1 \leftarrow \mathbf{G_4}]. \tag{21}$$

Lemma 3. *We have* $(\mathsf{U}^3_m)^\dagger \circ \mathsf{Out}_{\mathsf{G}} \circ \mathsf{U}^3_m = (\mathsf{U}^4_m)^\dagger \circ \mathsf{Out}_{\mathsf{G}} \circ \mathsf{U}^4_m$.

Proof. The proof is similar to that of Lemma 2, and we provide the full proof in Supplementary Material H.1. □

Game $\mathbf{G_5}$: There are two differences between this game and $\mathbf{G_4}$. First, the H in game $\mathbf{G_5}$ is no longer simulated as a compressed standard oracle, but instead simulated by $\mathsf{O}_{\mathsf{H}} : |x, y\rangle \mapsto |x, y \oplus \mathsf{H}(x)\rangle$ where $\mathsf{H} \leftarrow_{\$} \Omega_{\mathcal{M} \to \{0,1\}^n}$. Second, in game $\mathbf{G_5}$, the $\mathsf{U}^4_{\mathsf{qD}}$ used to simulate qDeca^* is replaced with

$$\mathsf{U}^5_{\mathsf{qD}} := \mathrm{U}_\perp \circ \mathrm{P}_{c_1^*,c_2^*} + (\mathsf{U}^5_m)^\dagger \circ \mathsf{Out}_{\mathsf{G}} \circ \mathsf{U}^5_m \circ \left(\mathrm{I} - \mathrm{P}_{c_1^*,c_2^*}\right).$$

where

$$\mathsf{U}^5_m := \mathsf{Check}'_{\mathsf{H}} \circ \mathsf{Read}_{\mathsf{F}} \circ \mathsf{CNOT} \circ \mathsf{O}_{\mathsf{H}} \circ \mathsf{U}_{\mathsf{dec}}. \tag{22}$$

The above U^5_m is the same as the U^4_m defined in Eq. (17), except that the CStO is replaced with O_{H} and the $\mathsf{Read}'_{\mathsf{H}}$ is replaced with $\mathsf{Check}'_{\mathsf{H}}$. Here $\mathsf{Check}'_{\mathsf{H}}$ acts on the registers $\mathsf{C}_1\mathsf{C}_2\mathsf{M}\mathsf{M}_1\mathsf{D}_{q_{\mathsf{F}}}\mathsf{B}_2$ and is defined as

$$\begin{aligned}\mathsf{Check}'_{\mathsf{H}}:\ &|c_1, c_2, m, m'_1, D_{\mathsf{F}}, b_2\rangle \\ &\mapsto \left|c_1, c_2, m, m'_1, D_{\mathsf{F}}, b_2 \oplus [\![\mathsf{H}(m'_1) \in S_{c_1,c_2,m,m'_1,D_{\mathsf{F}}}]\!]\right\rangle.\end{aligned}$$

In fact, $\mathsf{Check}'_{\mathsf{H}}$ is the same as the $\mathsf{Check}_{\mathsf{H}}$ defined in Eq. (9), except that it additionally acts on $\mathsf{D}_{q_{\mathsf{F}}}$ and replaces S_{c_1,c_2,m,m'_1} with the set $S_{c_1,c_2,m,m'_1,D_{\mathsf{F}}}$.

By the definition of $S_{c_1,c_2,m,m'_1,D_{\mathsf{F}}}$ given in Eq. (20), it is not hard to check that $\max_{c_1,c_2,m,m'_1,D_{\mathsf{F}}} |S_{c_1,c_2,m,m'_1,D_{\mathsf{F}}}| \le 1$. Hence similar to the proof of the game hop from $\mathbf{G_0}$ to $\mathbf{G_1}$, just by simply changing the perspective from $\mathsf{Check}_{\mathsf{H}}$ and $\mathsf{Read}_{\mathsf{H}}$ to $\mathsf{Check}'_{\mathsf{H}}$ and $\mathsf{Read}'_{\mathsf{H}}$, we can design a specific quantum oracle algorithm and apply Theorem 2 in reverse to prove that

$$\begin{aligned}|\Pr[1 \leftarrow \mathbf{G_4}] - \Pr[1 \leftarrow \mathbf{G_5}]| &\le 12q_{\mathsf{D}} \cdot \sqrt{\max_{c_1,c_2,m,m'_1,D_{\mathsf{F}}} \left|S_{c_1,c_2,m,m'_1,D_{\mathsf{F}}}\right| / 2^n} \\ &\le 12q_{\mathsf{D}} \cdot \sqrt{2^{-n}}.\end{aligned} \tag{23}$$

Game $\mathbf{G_6}$: This game is the same as $\mathbf{G_5}$, except that the $\mathsf{U}^5_{\mathsf{qD}}$ used to simulate qDeca^* is replaced with

$$\mathsf{U}^{pk}_{\mathsf{qD}} := \mathrm{U}_{\perp} \circ \mathrm{P}_{c_1^*,c_2^*} + \mathsf{U}_{\mathsf{search}} \circ \left(\mathrm{I} - \mathrm{P}_{c_1^*,c_2^*}\right), \tag{24}$$

where $\mathsf{U}_{\mathsf{search}}$ acts on the registers $\mathsf{C}_1\mathsf{C}_2\mathsf{K}\mathsf{D}_{q_{\mathsf{F}}}$ and is defined as

$$\mathsf{U}_{\mathsf{search}} : |c_1, c_2, k, D_{\mathsf{F}}\rangle \mapsto |c_1, c_2, k \oplus \mathsf{Search}(pk, c_1, c_2, D_{\mathsf{F}}), D_{\mathsf{F}}\rangle. \tag{25}$$

The function Search used above is defined as follows, which also needs to compute (or query) G and H.

$$\mathsf{Search}(pk, c_1, c_2, D_{\mathsf{F}}) = \begin{cases} \mathsf{G}(m) & m \text{ is the smallest value s.t. } D_{\mathsf{F}}(m) \ne \perp, \\ & c_1 = \mathsf{Enc}(pk, D_{\mathsf{F}}(m)), c_2 = m \oplus \mathsf{H}(D_{\mathsf{F}}(m)), \\ \perp & \text{If such } m \text{ does not exist.} \end{cases} \tag{26}$$

It can be observed that game $\mathbf{G_6}$ only replaces the $(\mathsf{U}^5_m)^\dagger \circ \mathsf{Out}_{\mathsf{G}} \circ \mathsf{U}^5_m$ used in game $\mathbf{G_5}$ with $\mathsf{U}_{\mathsf{search}}$, and the simulation of qDeca^* in game $\mathbf{G_6}$ no longer requires the secret key sk. As for the difference between the success probabilities of game $\mathbf{G_5}$ and $\mathbf{G_6}$, we can prove the following lemma, which demonstrates that the average-case decryption error δ_{ac} of the underlying PKE scheme P can be used to bound this difference.

Lemma 4. $|\Pr[1 \leftarrow \mathbf{G_5}] - \Pr[1 \leftarrow \mathbf{G_6}]| \le 16 \cdot \sqrt{q_{\mathsf{F}}(q_{\mathsf{F}}+1) \cdot \delta_{ac}} + 64q_{\mathsf{F}} \cdot \delta_{ac}$.

Proof (idea). By the definition of the encryption algorithm Enc_0 of ACWC_0 (Fig. 8), the computation of Enc and H in Search, i.e., "$c_1 = \mathsf{Enc}(pk, D_{\mathsf{F}}(m)), c_2 = m \oplus \mathsf{H}(D_{\mathsf{F}}(m))$", can be equally rewritten as "$(c_1, c_2) = \mathsf{Enc}_0(pk, m; D_{\mathsf{F}}(m))$". Now, based on the fact that $\mathsf{FOAC}_0 = \mathsf{FO}_m^{\perp} \circ \mathsf{ACWC}_0$, we can follow the proof idea of [16, Lemma 6] for handling the decapsulation oracle of $\mathsf{FO}_m^{\perp}$ to prove this lemma by means of the worst-case decryption error δ_{wc}^0 of ACWC_0. In fact, the key point is that, $(\mathsf{U}_m^5)^\dagger \circ \mathsf{Out}_{\mathsf{G}} \circ \mathsf{U}_m^5$ and $\mathsf{U}_{\mathsf{search}}$ differ only when acting on certain databases, and such databases are unlikely to be generated if δ_{wc}^0 is small enough. Notice that [12, Lemma 3.1] has proved that the worst-case decryption error δ_{wc}^0 of ACWC_0 is equal to the average-case decryption error δ_{ac} of P; hence we can derive a bound based solely on δ_{ac} by replacing δ_{wc}^0 with δ_{ac}. The full proof of this lemma is given in Supplementary Material H.2. ◻

Game $\mathbf{G_7}$: This game is the same as $\mathbf{G_6}$, except that the $m_1^* = \mathsf{F}(m^*)$ is replaced with $m_1^* \leftarrow \mathcal{M}$ and the CStO used to simulate F is replaced with

$$\mathsf{CStO}_{m_1^*}^{m^*} := \sum_{x \in \{0,1\}^n \setminus \{m^*\}} |x\rangle\langle x| \otimes \mathsf{StdDecomp}_x \circ \mathsf{CNOT}^x \circ \mathsf{StdDecomp}_x + |m^*\rangle\langle m^*| \otimes \mathsf{CNOT}_{m_1^*}. \tag{27}$$

Here $\mathsf{CNOT}_{m_1^*} : |y\rangle \mapsto |y \oplus m_1^*\rangle$ acts on the output register of F. That is to say, for the adversary's m^* query to F, game $\mathbf{G_7}$ no longer simulates the compressed standard oracle but directly writes m_1^* into the output register of F.

As for the difference between the success probabilities of game $\mathbf{G_6}$ and $\mathbf{G_7}$, we can prove the following lemma.

Lemma 5. $|\Pr[1 \leftarrow \mathbf{G_6}] - \Pr[1 \leftarrow \mathbf{G_7}]| \leq 2q_{\mathsf{D}}/\sqrt{|\mathcal{M}|}$.

Proof (idea). In game $\mathbf{G_6}$, the only disturbance to the simulation of $\mathsf{F}(m^*)$ comes from the simulation of qDeca^*, since it performs the $\mathsf{U}_{\mathsf{search}}$ (Eq. (25)) that is not allowed in the implementation of (compressed standard oracle) F. However, such disturbance is limited because of the following observation:

– The computation of $\mathsf{U}_{\mathsf{search}}|c_1, c_2, k, D_{\mathsf{F}}\rangle$ involves $\mathsf{F}(m^*)$ and causes disturbance to its simulation when $(c_1, c_2) = (c_1^*, c_2^*)$. However, the definition of $\mathsf{U}_{\mathsf{qD}}^{pk}$ (Eq. (24)) shows that this case cannot occur. Alternatively, when $\mathsf{U}_{\mathsf{search}}$ acts on $|c_1, c_2, k, D_{\mathsf{F}}\rangle$ with $(c_1, c_2) \neq (c_1^*, c_2^*)$, its disturbance to the simulation of $\mathsf{F}(m^*)$ is limited by the basic rules of the compressed standard oracle.

Translating the above observation into a formal proof can be somewhat cumbersome, and we design a new game sequence to prove this lemma rigorously. The full proof is given in Supplementary Material I. ◻

Game $\mathbf{G_8}$: This game is the same as $\mathbf{G_7}$, except that it introduces a new quantum random oracle $\mathsf{F}'(\leftarrow_\$ \Omega_{\{0,1\}^n \to \mathcal{M}})$ and generates m_1^* as $\mathsf{F}'(m^*)$. Obviously this game is just a restatement of game $\mathbf{G_7}$, and thus

$$\Pr[1 \leftarrow \mathbf{G_7}] = \Pr[1 \leftarrow \mathbf{G_8}]. \tag{28}$$

Game $\mathbf{G_9}$: This game is the same as $\mathbf{G_8}$, except that it uses the following control operation to simulate F for $\mathcal{A}$.

– For the query state $|x, y\rangle$ of the query registers $\mathsf{X_F Y_F}$, perform CStO on the registers $\mathsf{X_F Y_F D}_{q_\mathsf{F}}$ if $\mathsf{Enc}(pk, \mathsf{F}'(x)) \neq c_1^*$, and query F' to return $|x, y \oplus \mathsf{F}'(x)\rangle$ if $\mathsf{Enc}(pk, \mathsf{F}'(x)) = c_1^*$. Here $\mathsf{X_F}/\mathsf{Y_F}$ denotes the input/output register of F and c_1^* is $\mathsf{Enc}(pk, \mathsf{F}'(m_1^*))$.

We would like to point out that the above control operation can be easily implemented by querying F' two times.

By the construction of game $\mathbf{G_8}$, the $\mathsf{CStO}_{m_1^*}^{m^*}$ (Eq. (27)) used in this game to simulate F can be restated as a control operation:

– For the query state $|x, y\rangle$ of the query registers $\mathsf{X_F Y_F}$, perform CStO on the registers $\mathsf{X_F Y_F D}_{q_\mathsf{F}}$ if $x \neq m^*$, and return $|x, y \oplus \mathsf{F}'(x)\rangle$ if $x = m^*$.

Obviously, if the following event $\mathsf{Coll}_{\mathsf{F}', m^*}$ does not occur, we can conclude that $\mathsf{Enc}(pk, \mathsf{F}'(x)) = c_1^*$ if and only if $x = m^*$, and this implies that game $\mathbf{G_9}$ is identical to game $\mathbf{G_8}$.

$$\begin{aligned} \mathsf{Coll}_{\mathsf{F}', m^*}\colon\ & (pk, sk) \leftarrow \mathsf{Gen},\ m^* \leftarrow_\$ \mathcal{M},\ \mathsf{F}' \leftarrow_\$ \Omega_{\{0,1\}^n \to \mathcal{M}} \\ & \exists m \neq m^* \text{ s.t. } \mathsf{Enc}(pk, \mathsf{F}'(m)) = \mathsf{Enc}(pk, \mathsf{F}'(m^*)). \end{aligned} \tag{29}$$

By Lemma 6, we have $\Pr[\mathsf{Coll}_{\mathsf{F}', m^*}] \leq 1/|\mathcal{M}| + 2 \cdot \delta_{ac}$, then by the fundamental lemma [38], we have

$$|\Pr[1 \leftarrow \mathbf{G_8}] - \Pr[1 \leftarrow \mathbf{G_9}]| \leq 1/|\mathcal{M}| + 2 \cdot \delta_{ac}. \tag{30}$$

Next we define an $\mathsf{IND\text{-}CPA}$ adversary $\mathcal{B}$ against $\mathsf{KEMAC_0}$ in the QROM, which runs the adversary $\mathcal{A}$ in game $\mathbf{G_9}$. To avoid confusion with the notation used in the previous proof, we denote the three quantum random oracles used in the $\mathsf{IND\text{-}CPA}$ game of $\mathsf{KEMAC_0}$ as F', G' and H' respectively[21].

1. The input of $\mathcal{B}$ is $(pk, c_1^*, c_2^*, K_b^*)$, where (c_1^*, c_2^*) is the challenge ciphertext of the $\mathsf{IND\text{-}CPA}$ game, $c_1^* := \mathsf{Enc}(pk, \mathsf{F}'(m^*))$, and $c_2^* := m^* \oplus \mathsf{G}'(\mathsf{F}'(m^*))$. Here $m^* \leftarrow_\$ \{0,1\}^n$ is the challenge plaintext.
2. $\mathcal{B}$ prepares the database register $\mathsf{D}_{q_\mathsf{F}}$ and sets its initial state as $|D^\perp\rangle$, where $D^\perp$ is the database containing q_F $(\perp, 0^n)$ pairs. Then $\mathcal{B}$ runs the adversary $\mathcal{A}^{\mathsf{F},\mathsf{G},\mathsf{H},\mathsf{qDeca}^*}(pk, c_1^*, c_2^*, K_b^*)$, outputs its final result, and simulates its oracles as follows:
   (a) When $\mathcal{A}$ queries F with the query state $|x, y\rangle$ of the query registers $\mathsf{X_F Y_F}$, $\mathcal{B}$ performs the following control operation to simulate F: performs CStO on the registers $\mathsf{X_F Y_F D}_{q_\mathsf{F}}$ if $\mathsf{Enc}(pk, \mathsf{F}'(x)) \neq c_1^*$, and queries F' to return $|x, y \oplus \mathsf{F}'(x)\rangle$ if $\mathsf{Enc}(pk, \mathsf{F}'(x)) = c_1^*$.

[21] However in the description of Theorem 4, we still use the notation F, G and H for the adversary $\mathcal{B}$, which is an acceptable abuse of notation.

(b) When $\mathcal{A}$ queries G (resp. H), $\mathcal{B}$ simulates it by directly querying its oracle G' (resp. H').

(c) When $\mathcal{A}$ queries qDeca^* with the query state $|c_1, c_2, k\rangle$ of the query registers $\mathsf{C}_1\mathsf{C}_2\mathsf{K}$, $\mathcal{B}$ simulates it by performing $\mathsf{U}_{\mathsf{qD}}^{pk}$ on the registers $\mathsf{C}_1\mathsf{C}_2\mathsf{KD}_{q_\mathsf{F}}$. Notably, performing $\mathsf{U}_{\mathsf{qD}}^{pk}$ also needs to query G and H. For these queries, $\mathcal{B}$ simulates them by querying its oracle G' and H'.

At this point, it is not hard to check that

$$\mathrm{Adv}_{\mathsf{KEMAC_0},\mathcal{B}}^{\mathsf{IND\text{-}CPA}} = \left|\Pr\left[1 \leftarrow \mathbf{G_9}\right] - \frac{1}{2}\right|. \tag{31}$$

By the construction of $\mathcal{B}$, the oracle F' is queried at most $2q_\mathsf{F}$ times. Notice that a single performance of the $\mathsf{U}_{\mathsf{qD}}^{pk}$ defined in Eq. (24) needs to query G (resp. H) at most 1 (resp. $2q_\mathsf{F}$) times. Hence $\mathcal{B}$ queries G' at most $q_\mathsf{D} + q_\mathsf{G}$ times, and queries H' at most $2q_\mathsf{D}q_\mathsf{F} + q_\mathsf{H}$ times. As for the running time of $\mathcal{B}$, since $\mathcal{B}$ performs CStO with $O(q_\mathsf{F})$ times and computes Enc with $O(q_\mathsf{D}q_\mathsf{F})$ times[22], its running time satisfies $\mathcal{T}_\mathcal{B} \leq \mathcal{T}_\mathcal{A} + O(q_\mathsf{F}^2) + O(q_\mathsf{D}q_\mathsf{F}) \cdot \mathcal{T}_{\mathsf{Enc}}$.

Now by combining Eq. (7), Eq. (10), Eq. (13), Eq. (16), Eq. (21), Eq. (23), Lemma 4, Lemma 5, Eq. (28) to Eq. (31), we finally obtain

$$\mathrm{Adv}_{\mathsf{KEMAC_0},\mathcal{A}}^{\mathsf{IND\text{-}CCA}} \leq \mathrm{Adv}_{\mathsf{KEMAC_0},\mathcal{B}}^{\mathsf{IND\text{-}CPA}} + (80q_\mathsf{F} + 18) \cdot \sqrt{\delta_{ac}} + \frac{14q_\mathsf{D}\sqrt{q_\mathsf{G}} + 1}{\sqrt{|\mathcal{M}|}} + \frac{24q_\mathsf{D}}{\sqrt{2^n}}.$$

This completes the proof of Theorem 4 for deterministic P. For randomized P, the above proof still holds, and a detailed proof is given in Supplementary Material F.2. □

Next we introduce Theorem 5, which shows that the $\mathsf{IND\text{-}CPA}$ security of $\mathsf{FOAC_0}$ can be reduced to the $\mathsf{OW\text{-}CPA}$ security of P in the QROM.

Theorem 5 ($\mathsf{OW\text{-}CPA}$ of $\mathsf{P} \overset{\text{QROM}}{\Rightarrow}$ $\mathsf{IND\text{-}CPA}$ of $\mathsf{KEMAC_0}$). *Let P be a δ_{ac}-average-correct PKE scheme with message space $\mathcal{M}(= \{0,1\}^m)$. Let $\mathcal{A}$ be an $\mathsf{IND\text{-}CPA}$ adversary against $\mathsf{KEMAC_0}$, making q_F, q_G and q_H quantum queries to the random oracle F, G and H. Then we can construct a $\mathsf{OW\text{-}CPA}$ adversary $\mathcal{B}$ against P such that $\mathcal{T}_\mathcal{B} \leq \mathcal{T}_\mathcal{A} + O(q_\mathsf{G} + q_\mathsf{H}) \cdot \mathcal{T}_{\mathsf{Enc}} + O(q_\mathsf{F}^2 + q_\mathsf{G}^2 + q_\mathsf{H}^2)$ and*

$$\mathrm{Adv}_{\mathsf{KEMAC_0},\mathcal{A}}^{\mathsf{IND\text{-}CPA}} \leq \mathrm{Adv}_\mathsf{P} + 6(q_\mathsf{F} + 2q_\mathsf{G} + 3) \cdot \sqrt{\delta_{ac}} + \frac{2(q_\mathsf{F} + 2q_\mathsf{G} + 2)}{\sqrt{2^n}} + \frac{1}{\sqrt{|\mathcal{M}|}}, \textit{with}$$

$$\mathrm{Adv}_\mathsf{P} = \begin{cases} 4(q_\mathsf{F} + 2q_\mathsf{G} + 2) \cdot \sqrt{\mathrm{Adv}_{\mathsf{P},\mathcal{B}}^{\mathsf{OW\text{-}CPA}}} & \textit{if } \mathsf{P} \textit{ is deterministic,} \\ 2(q_\mathsf{F} + q_\mathsf{H} + 2) \cdot \left(\sqrt{\mathrm{Adv}_{\mathsf{P},\mathcal{B}}^{\mathsf{OW\text{-}CPA}}} + \frac{1}{\sqrt{2^n}}\right) & \textit{if } \mathsf{P} \textit{ is randomized.} \end{cases}$$

Here n is the output length of H.

[22] The computation of Enc comes from two parts, i.e. the first part used by $\mathcal{B}$ to simulate F, and the second part used by $\mathcal{B}$ to implement $\mathsf{U}_{\mathsf{qD}}^{pk}$.

Proof. The full proof of Theorem 5 is given in Supplementary Material K. $\square$

Now by directly combining Theorem 4 and Theorem 5, we obtain the main theorem of this section, i.e., the following Theorem 6.

Theorem 6 ($\mathsf{OW\text{-}CPA}$ **of** $\mathsf{P} \overset{\mathrm{QROM}}{\Rightarrow}$ $\mathsf{IND\text{-}CCA}$ **of** $\mathsf{KEMAC_0}$)**.** *Let* P *be a* δ_{ac}*-average-correct PKE scheme with meassage space* $\mathcal{M}(=\{0,1\}^m)$*. Let* $\mathcal{A}$ *be an* $\mathsf{IND\text{-}CCA}$ *adversary against* $\mathsf{KEMAC_0}$*, making* q_{D} *classical queries to the decapsulation oracle, and making* q_{F}*,* q_{G} *and* q_{H} *quantum queries to the random oracle* F*,* G *and* H*. Then we can construct a* $\mathsf{OW\text{-}CPA}$ *adversary* $\mathcal{B}$ *against* P *such that* $\mathcal{T}_{\mathcal{B}} \leq \mathcal{T}_{\mathcal{A}} + O(q_{\mathsf{D}}q_{\mathsf{F}} + q_{\mathsf{G}} + q_{\mathsf{H}}) \cdot \mathcal{T}_{\mathsf{Enc}} + O(q_{\mathsf{F}}^2 + q_{\mathsf{G}}^2 + q_{\mathsf{H}}^2)$ *and*

$$\mathrm{Adv}^{\mathsf{IND\text{-}CCA}}_{\mathsf{KEMAC_0},\mathcal{A}} \leq \mathrm{Adv}_{\mathsf{P}} + (92q+36)\cdot\sqrt{\delta_{ac}} + \frac{14q\sqrt{q}+2}{\sqrt{|\mathcal{M}|}} + \frac{28q+4}{\sqrt{2^n}}, \textit{with}$$

$$\mathrm{Adv}_{\mathsf{P}} = \begin{cases} 8(q+1)\cdot\sqrt{\mathrm{Adv}^{\mathsf{OW\text{-}CPA}}_{\mathsf{P},\mathcal{B}}} & \textit{if } \mathsf{P} \textit{ is deterministic,} \\ (2q_{\mathsf{H}} + 8(q_{\mathsf{D}}+1)q_{\mathsf{F}} + 4)\cdot\left(\sqrt{\mathrm{Adv}^{\mathsf{OW\text{-}CPA}}_{\mathsf{P},\mathcal{B}}} + \frac{1}{\sqrt{2^n}}\right) & \textit{if } \mathsf{P} \textit{ is randomized.} \end{cases}$$

Here n *is the output length of* H *and* $q := q_{\mathsf{D}} + q_{\mathsf{F}} + q_{\mathsf{G}}$*.*

5 Security proofs of FOAC and $\mathsf{FOAC}_m^{\not\perp}$

Since [12] has provided a relatively tight $\mathsf{IND\text{-}CCA}$ security proof of $\mathsf{FOAC}(= \mathsf{FO}_m^{\perp} \circ \mathsf{ACWC})$ in the ROM, here we only prove the $\mathsf{IND\text{-}CCA}$ security of FOAC in the QROM. Then, similar to the proof idea of [12], we choose to first prove the $\mathsf{OW\text{-}CPA}$ security of ACWC in the QROM, i.e, the following theorem.

Theorem 7 ($\mathsf{OW\text{-}CPA}$ **of** $\mathsf{P} \overset{\mathrm{QROM}}{\Rightarrow}$ $\mathsf{OW\text{-}CPA}$ **of** $\mathsf{ACWC}[\mathsf{P},\mathsf{H}]$)**.** *Let* P *be a PKE scheme with average-case decryption error* δ_{ac}*. Let* $\mathcal{A}$ *be a* $\mathsf{OW\text{-}CPA}$ *adversary against* $\mathsf{ACWC}[\mathsf{P},\mathsf{H}]$*, making* q_{H} *quantum queries to* H*. Then we can construct a* $\mathsf{OW\text{-}CPA}$ *adversary* $\mathcal{B}$ *against* P *such that* $\mathcal{T}_{\mathcal{B}} \leq \mathcal{T}_{\mathcal{A}} + O(q_{\mathsf{H}}^2) + O(q_{\mathsf{H}}) \cdot \mathcal{T}_{\mathsf{Enc}}$ *and*

$$\mathrm{Adv}^{\mathsf{OW\text{-}CPA}}_{\mathsf{ACWC}[\mathsf{P},\mathsf{H}],\mathcal{A}} \leq \begin{cases} 2\cdot\mathrm{Adv}^{\mathsf{OW\text{-}CPA}}_{\mathsf{P},\mathcal{B}} + 4\cdot\delta_{ac} + 72/2^{n_2} & \textit{if } \mathsf{P} \textit{ is deterministic,} \\ 2q_{\mathsf{H}}\cdot\mathrm{Adv}^{\mathsf{OW\text{-}CPA}}_{\mathsf{P},\mathcal{B}} + 72/2^{n_2} & \textit{if } \mathsf{P} \textit{ is randomized.} \end{cases}$$

Here n_2 *is the output length of* H*.*

Proof. The core proof idea is to first simulate H as a compressed standard oracle, and then use our CQR technique (Theorem 2). The full proof is given in Supplementary Material L.1. $\square$

By combining Theorem 7 with the $\mathsf{IND\text{-}CCA}$ security proofs of $\mathsf{FO}_m^{\perp}$ presented in [20,22], we can prove the following theorem, which shows that the $\mathsf{IND\text{-}CCA}$ security of FOAC can be reduced to the $\mathsf{OW\text{-}CPA}$ security of P in the QROM.

Theorem 8 (OW-CPA **of** P $\overset{\text{QROM}}{\Rightarrow}$ IND-CCA **of** KEMAC)**.** *Let* P *be a rPKE scheme with δ_{ac}-average-correct and weakly γ-spread. Let $\mathcal{A}$ be an* IND-CCA *adversary against* KEMAC $:=$ FOAC[P, F, G, H]*, making q_{D} classical queries to the decapsulation oracle, and making q_{F}, q_{G} and q_{H} quantum queries to the random oracle* F*,* G *and* H*. Then we can construct a* OW-CPA *adversary $\mathcal{B}$ against* P *such that $\mathcal{T}_{\mathcal{B}} \leq \mathcal{T}_{\mathcal{A}} + O(q_{\mathsf{D}}q + q_{\mathsf{D}}^2 + q_{\mathsf{H}}) \cdot \mathcal{T}_{\mathsf{Enc}} + O(q^2 + q_{\mathsf{D}} + q_{\mathsf{H}}^2)$ and*

$$\begin{aligned}\mathrm{Adv}^{\mathsf{IND\text{-}CCA}}_{\mathsf{KEMAC},\mathcal{A}} \leq & 8(q+q_{\mathsf{D}})^{1.5}\sqrt{q_{\mathsf{D}}} \cdot \sqrt{\mathrm{Adv}^{\mathsf{OW\text{-}CPA}}_{\mathsf{P},\mathcal{B}}} + \frac{72(q+q_{\mathsf{D}})}{\sqrt{2^{n_2}}} + 10(q+1)^2 \cdot \delta' \\ & + 24q_{\mathsf{D}}(q_{\mathsf{F}} + 4q_{\mathsf{D}}) \cdot 2^{-\gamma/2}.\end{aligned}$$

Here $q := q_{\mathsf{F}} + q_{\mathsf{G}}$, $\delta' := \delta_{ac} + (1+\sqrt{n_1+n_2})/\sqrt{2^{n_1}}$, and n_1/n_2 is the input/output length of H*.*

Proof. The full proof of Theorem 8 is given in Supplementary Material L.2. □

As for $\mathsf{FOAC}_m^{\not\perp}$, similar to the above method, we prove the IND-CCA security of $\mathsf{FOAC}_m^{\not\perp}(= \mathsf{FO}_m^{\not\perp} \circ \mathsf{ACWC})$ by combining the security proof of ACWC and that of $\mathsf{FO}_m^{\not\perp}$. The detailed proof is given in Supplementary Material M.

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A Further related works

Besides the average/worst-case decryption error δ_{ac}/δ_{wc}, some other notions of decryption error have also been proposed, such as the key-independent game $\mathsf{FFP\text{-}NK}$ and the find non-generically failing ciphertexts game $\mathsf{FFP\text{-}NG}$ proposed in [20]. However, compared with δ_{ac} or δ_{wc}, computing the upper bound of the success probability for $\mathsf{FFP\text{-}NK}$ and $\mathsf{FFP\text{-}NG}$ is more complicated, since this computation needs to estimate more internal parameters.

In the introduction, we mentioned that Kim and Park [24] designed a variant of FOAC_1 and used this variant to develop $\mathsf{NTRU+}$, the winner of the Korean Post-Quantum Cryptography competition. However, it should be pointed out that our CQR technique (Theorem 1) cannot be directly applied to this variant. This is because it replaces the XOR used by FOAC_1 with a more general semi-generalized one-time pad SOTP, making it challenging to construct a suitable set S to apply Theorem 1. This raises an interesting open question: whether we can modify the CQR technique to give a tighter security proof of such a variant.

In addition, we note that Enc_0 shown in Fig. 2 can be regarded as a combination of Enc and an 1-round Feistel network. So for generic transformations utilizing multi-round Feistel networks, such as OAEP [4], $\mathsf{OAEP+}$ [37], OAEP 3-round [35], among others, our CQR technique might also be used to prove their $\mathsf{IND\text{-}CCA}$ security in the QROM. In fact, at least from a preliminary observation, both Lemma 1 and Lemma 2 of [27], which are used to prove the $\mathsf{IND\text{-}CCA}$ security of OAEP in the QROM, can be given more a concise proof using our CQR technique.

In a concurrent work, Kwag et al. [26] reproved the γ-spreadness of ACWC_0 and ACWC. For ACWC_0, they presented a more precise estimation; for ACWC, they presented a more rigorous computation. Here we would like to point out that there is no longer a need to compute the γ-spreadness of ACWC_0 if we use ACWC_0 and $\mathsf{FO}_m^{\perp}$ in combination, as the security proof of $\mathsf{FOAC}_0(= \mathsf{FO}_m^{\perp} \circ \mathsf{ACWC}_0)$ presented in this paper no longer requires the γ-spreadness. Nonetheless, on the positive side, since [26] provided a more rigorous computation of ACWC's γ-spreadness, by combining this computation with the security proofs of $\mathsf{FOAC}(= \mathsf{FO}_m^{\perp} \circ \mathsf{ACWC})$ and $\mathsf{FOAC}_m^{\not\perp}(= \mathsf{FO}_m^{\not\perp} \circ \mathsf{ACWC})$ presented in this paper, we can provide a more rigorous reference for the scheme designers' parameter selection.

B A fixed security proof of BAT

As mentioned in Section 1.1, the security proof of BAT presented in [13] is incorrect, since the underlying dPKE scheme used by BAT, which we denote as BAT.dPKE, does not meet the requirements specified in the security proof of [12]. In this section, we present a correct IND-CCA security proof of BAT in the ROM and QROM. To make it more accurate and coherent, we adhere to the original symbols used in [13] as much as possible.

In Algorithm 1, we present the encryption algorithm of BAT.dPKE. It should be pointed out that, for the sake of simplicity, the encryption algorithm we present omits the additional check implemented in the original encryption algorithm of BAT.dPKE, namely verifying whether $\|(\gamma s, e)\| > 1.08\sqrt{n(k^2-1)/6}$ holds. Actually, such simplification is acceptable, and we explain the reasoning below. First, this additional check is essentially just a conservative safeguard. It enables the encryption algorithm to reject the encryption of certain bad messages that would significantly increase the decryption error, leading to more optimistic assessments of BAT.dPKE's decryption error. Furthermore, there is in fact no great necessity to adopt such conservatism, as [13] also directly disregards this additional check when estimating the decryption error of BAT.dPKE.

Algorithm 1 BAT.dPKE.Enc

Input: public key $pk := h \in \mathcal{R}$, integers $q, k \in \mathbb{N}$, message $s \in \mathcal{M}' = (\mathcal{R} \bmod 2)$
Output: ciphertext c

1: $c \leftarrow \left\lfloor \frac{(hs \bmod q)}{k} \right\rceil$
2: return c

Note that [13, Theorem 2] has proved that the OW-CPA security of BAT.dPKE can be reduced to the NTRU assumption and the RLWR assumption. In addition, our Theorem 3 and Theorem 6 can both be applied to BAT.dPKE, as these two theorems do not require γ-spread and allow the underlying PKE scheme to be deterministic. Hence, by combining [13, Theorem 2] with Theorem 3 and Theorem 6, we can prove the following two theorems, which ensure the IND-CCA security of BAT$(= \mathsf{FOAC}_0[\mathsf{BAT.dPKE}, \mathsf{F}, \mathsf{G}, \mathsf{H}])$ in the ROM and QROM.

Theorem 9 (NTRU + RLWR $\overset{\text{ROM}}{\Rightarrow}$ IND-CCA **of** BAT)**.** *Let $\mathcal{A}$ be an IND-CCA adversary against BAT, making q_{D} classical queries to the decapsulation oracle, and making q_{F}, q_{G} and q_{H} classical queries to the random oracle F, G and H. Then we can construct a NTRU adversary $\mathcal{B}_1$ and a RLWR adversary $\mathcal{B}_2$ such that $\mathcal{T}_{\mathcal{B}_1} \approx \mathcal{T}_{\mathcal{B}_2} \leq \mathcal{T}_{\mathcal{A}} + O(q_{\mathsf{D}} q_{\mathsf{F}} + q_{\mathsf{H}}) \cdot \mathcal{T}_{\mathsf{BAT.dPKE.Enc}}$ and*

$$\mathrm{Adv}^{\mathsf{IND\text{-}CCA}}_{\mathsf{BAT},\mathcal{A}} \leq 3 \cdot \mathrm{Adv}^{\mathsf{NTRU}}_{\mathcal{R},q,\chi_1,\mathcal{B}_1} + 3 \cdot \mathrm{Adv}^{\mathsf{RLWR}}_{\mathcal{R},q,k,\chi_2,\mathcal{B}_2} + (q_{\mathsf{F}} + 2) \cdot \delta_{ac} + \frac{q_{\mathsf{D}} q_{\mathsf{H}}}{|\mathcal{M}'|} + \frac{2(q_{\mathsf{D}} + q_{\mathsf{F}} + q_{\mathsf{G}}) + 1}{2^n}.$$

Here $\mathcal{M}'$ *is the message space of* BAT.dPKE*,* δ_{ac} *is the average-case decryption error of* BAT.dPKE*, and* n *is the output length of* H*.*

Theorem 10 (NTRU + RLWR $\overset{\text{QROM}}{\Rightarrow}$ IND-CCA **of** BAT)**.** *Let* $\mathcal{A}$ *be an* IND-CCA *adversary against* BAT*, making* q_{D} *classical queries to the decapsulation oracle, and making* q_{F}*,* q_{G} *and* q_{H} *quantum queries to the random oracle* F*,* G *and* H*. Then we can construct a* NTRU *adversary* $\mathcal{B}_1$ *and a* RLWR *adversary* $\mathcal{B}_2$ *such that* $\mathcal{T}_{\mathcal{B}_1} \approx \mathcal{T}_{\mathcal{B}_2} \leq \mathcal{T}_{\mathcal{A}} + O(q_{\mathsf{D}} q_{\mathsf{F}} + q_{\mathsf{G}} + q_{\mathsf{H}}) \cdot \mathcal{T}_{\mathsf{BAT.PKE.dEnc}} + O(q_{\mathsf{F}}^2 + q_{\mathsf{G}}^2 + q_{\mathsf{H}}^2)$ *and*

$$\begin{aligned}\mathrm{Adv}^{\mathsf{IND\text{-}CCA}}_{\mathsf{BAT},\mathcal{A}} \leq & 8(q+1) \cdot \sqrt{\mathrm{Adv}^{\mathsf{NTRU}}_{\mathcal{R},q,\chi_1,\mathcal{B}_1} + \mathrm{Adv}^{\mathsf{RLWR}}_{\mathcal{R},q,k,\chi_2,\mathcal{B}_2}} + (92q+36) \cdot \sqrt{\delta_{ac}} \\ & + \frac{14q\sqrt{q}+2}{\sqrt{|\mathcal{M}'|}} + \frac{28q+4}{\sqrt{2^n}}.\end{aligned}$$

Here $\mathcal{M}'$ *is the message space of* BAT.dPKE*,* δ_{ac} *is the average-case decryption error of* BAT.dPKE*,* $q := q_{\mathsf{D}} + q_{\mathsf{F}} + q_{\mathsf{G}}$*, and* n *is the output length of* H*.*

C Quantum background

A quantum system (register) Q is a complex Hilbert space $\mathcal{H}_Q$ with an inner product $\langle\cdot|\cdot\rangle$, and notation like '$|\cdot\rangle$' or '$\langle\cdot|$' is called Dirac notation. A quantum state on Q is a vector on $\mathcal{H}_Q$. We denote $\mathcal{H}_Q = \mathbb{C}[X]$ if Q is defined over a finite set X. The orthonormal basis of $\mathbb{C}[X]$ is $\{|x\rangle\}_{x\in X}$, where the basis state $|x\rangle$ is labeled by the element x of X. We also refer to $\{|x\rangle\}_{x\in X}$ as the computational basis. The norm of a state $|\psi\rangle$ is defined as $\||\psi\rangle\| := \sqrt{\langle\psi|\psi\rangle}$, and we say $|\psi\rangle$ is a unit state if $\||\psi\rangle\| = 1$. The norm of an operator A is defined as $\|A\| := \max_{|\psi\rangle} \|A|\psi\rangle\|$, where max is over all unit states on Q. Sometimes we also write $|\psi\rangle$ as $|\psi\rangle_Q$ to emphasize that $|\psi\rangle$ is a quantum state on Q.

Composite quantum system. Given quantum systems Q_1 and Q_2, we call the tensor product $Q_1 \otimes Q_2$ the composite quantum system, the state of which is written as $\sum_{x,y} \alpha_{x,y}|x\rangle \otimes |y\rangle \in Q_1 \otimes Q_2$ where $|x\rangle \in Q_1$, $|y\rangle \in Q_2$ are the basis states of Q_1 and Q_2. We sometimes also abbreviate $|x\rangle \otimes |y\rangle$ into $|x, y\rangle$ for simplicity. If the composite quantum system $Q_1 \otimes Q_2$ is in state $|\psi\rangle$, we say that systems Q_0 and Q_1 are entangled if $|\psi\rangle$ can not be written as a product state $|\psi_1\rangle \otimes |\psi_2\rangle$ (here $|\psi_1\rangle \in Q_1$, $|\psi_2\rangle \in Q_2$).

A single qubit is a quantum system defined over $\{0, 1\}$. A single qubit in superposition is a linear combination vector $|b\rangle = \alpha|0\rangle + \beta|1\rangle$ of two computational basis states $|0\rangle$ and $|1\rangle$ with $(\alpha, \beta) \in \mathbb{C}^2$ and $|\alpha|^2 + |\beta|^2 = 1$, α, β are the probability amplitudes of $|b\rangle$. An n-qubit system is $Q^{\otimes n}$ where Q is a single-qubit system.

Evolution of quantum systems. A pure state $|\psi\rangle$ can be manipulated by performing unitary operation U, and the resulting state is $|\psi'\rangle = \mathrm{U}|\psi\rangle$. For any unitary operation U, we have $\mathrm{U}\mathrm{U}^\dagger = \mathrm{I}$, where $\mathrm{U}^\dagger$ is the Hermitian transpose of U and I is the identity operator.

Projector on the quantum system. We introduce a special operation called projector. For the state $|\psi\rangle$ of a quantum system defined over a finite set X, a projector M_S ($S \subseteq X$) applies the operation $\sum_{y\in S} |y\rangle\langle y|$ to $|\psi\rangle$ to get the new state $\sum_{y\in S} |y\rangle\langle y|\psi\rangle$. We stress that any projector M is Hermitian (i.e., we have $\mathrm{M}^\dagger = \mathrm{M}$) and idempotent (i.e., we have $\mathrm{M}^2 = \mathrm{M}$).

Basic measurement. The state $|\psi\rangle$ in a quantum system can be measured in the computational basis. For example, suppose $|\psi\rangle = \sum_{x\in X} \alpha_x|x\rangle$ ($\sum_{x\in X} |\alpha_x|^2 = 1$) under the computational basis $\{|x\rangle\}_{x\in X}$, if we measure $|\psi\rangle$ in basis $\{|x\rangle\}_{x\in X}$, we obtain the measurement result x with probability $|\langle x|\psi\rangle|^2 = |\alpha_x|^2$. Given that result x is obtained, the state $|\psi\rangle$ collapses to state $|x\rangle$. Moreover, we explain an important special class of measurements called projective measurement which is defined by a set of projectors $\mathrm{M}_1, \ldots, \mathrm{M}_n$ where M_i are mutually orthogonal and $\sum_{i=1}^n \mathrm{M}_i = \mathrm{I}$. Upon measuring the state $|\psi\rangle$, the probability of obtaining i ($i \in \{1, \ldots, n\}$) is $\langle\psi|\mathrm{M}_i|\psi\rangle$ ($= \|\mathrm{M}_i|\psi\rangle\|^2$). Given that result i is obtained, the state $|\psi\rangle$ collapses to $\mathrm{M}_i|\psi\rangle/\sqrt{\langle\psi|\mathrm{M}_i|\psi\rangle}$. In our paper, unless otherwise specified, we only use measurements in the computational basis.

Quantum oracle algorithm. A quantum oracle algorithm $\mathcal{A}$ is an algorithm that has quantum access to oracles, and that can perform a mix of classical and quantum unitary algorithms. In this paper, without loss of generality, we default that $\mathcal{A}$'s final output is a single bit $b \in \{0,1\}$. We say that a quantum oracle algorithm $\mathcal{A}^{\mathsf{H}}(z)$ making q oracle queries is a unitary quantum oracle algorithm if its entire execution can be described as

$$\mathbb{M}_{\mathcal{A}} \circ \mathrm{U}_q \circ \mathrm{O}_{\mathsf{H}} \circ \mathrm{U}_{q-1} \circ \mathrm{O}_{\mathsf{H}} \cdots \mathrm{U}_2 \circ \mathrm{O}_{\mathsf{H}} \circ \mathrm{U}_1 \circ \mathrm{O}_{\mathsf{H}} |\psi_z\rangle.$$

Here $|\psi_z\rangle$ is an initial pure state that may depend on z, $\mathrm{U}_1, \ldots, \mathrm{U}_q$ are the fixed unitary operations applied between oracle queries. O_{H} is the unitary operation used to simulate the oracle H. $\mathbb{M}_{\mathcal{A}} := \{\mathrm{M}_0, \mathrm{M}_1\}$ is the final binary projective measurement performed by $\mathcal{A}$ on its quantum register, and the measurement result of $\mathbb{M}_{\mathcal{A}}$ (0 or 1) is the final output of $\mathcal{A}$. For multiple oracles, the unitary quantum oracle algorithm is defined analogously. Next, we introduce the following well-known fact about the quantum oracle algorithm.

Fact 1. *Any quantum oracle algorithm can be transformed into a unitary quantum oracle algorithm with constant factor computational overhead and the same query number.*

Quantum representation of the special symbol $\perp$. In this paper, we sometimes use a special symbol $\perp$ to expand a finite set $\{0,1\}^n$. Roughly speaking, the reason is that when we define a particular unitary operation, we need "$\perp$" to denote "not defined (yet)" or "computational failure".

As for the quantum representation of $\{0,1\}^n \cup \{\perp\}$, we use the extension method given in [7,18]. That is to say, we use a classical encoding function enc that $enc(\perp) := 1||0^n \in \{0,1\}^{n+1}$ and $enc(x) := 0||x \in \{0,1\}^{n+1}$ for any $x \in \{0,1\}^n$. Then the set $\{0,1\}^n \cup \{\perp\}$ can be embedded into the set $\{0,1\}^{n+1}$. Under this representation, the binary operation $x \oplus y$ for $x, y \in \{0,1\}^n \cup \{\perp\}$ actually refers to $enc(x) \oplus enc(y)$, and the quantum register defined over the set $\{0,1\}^n \cup \{\perp\}$ actually refers to the register defined over the set $\{0,1\}^{n+1}$.

Here we also give a more detailed example. Let c_1, c_2 be two ciphertexts with $\mathsf{Dec}(sk, c_1) = m \in \{0,1\}^n$ and $\mathsf{Dec}(sk, c_2) = \perp$. So when making one quantum query to the decryption oracle using the uniform superposition of c_1 and c_2, the adversary can prepare the query state as $1/\sqrt{2}|c_1\rangle_{\mathsf{I}}|0||y\rangle_{\mathsf{O}} + 1/\sqrt{2}|c_2\rangle_{\mathsf{I}}|0||y\rangle_{\mathsf{O}}$, where I/O is the input/output register and y is a fixed value in $\{0,1\}^n$. In fact, the 0 padded in front of y denotes that the current value of the output register is exactly y, not a value like $y \oplus \perp$. After the decryption query, the adversary will get $1/\sqrt{2}|c_1\rangle_{\mathsf{I}}|0||y \oplus m\rangle_{\mathsf{O}} + 1/\sqrt{2}|c_2\rangle_{\mathsf{I}}|1||y\rangle_{\mathsf{O}}$ since we denote $\perp$ by $1||0^n$. Now the adversary "knows"[23] that c_2 failed to decrypt, because it gets the output $1||y$ and finds that this value is actually $y \oplus \perp$.

[23] In fact, an adversary needs to perform a measurement to know something, so here is an intuitive explanation, but this does not lose generality.

D Proof of Theorem 2 (Check Query Replacement)

By the Fact 1 given in Supplementary Material C, we assume that $\mathcal{A}$ is unitary without loss of generality. In the following proof, we first introduce a sequence of games $\mathbf{G}_1^z$ to $\mathbf{G}_3^z$ under a fixed z in the support of the distribution $\mathfrak{D}_z$.

$\mathsf{O}_0(\|w,x,b\rangle)$	$\mathsf{O}_1(\|w,x,D_{\mathsf{H}},b\rangle)$
1. **return** $\|w,x,b \oplus [\![\mathsf{H}(x) \in S_{w,x}]\!]\rangle$	1. **return** $\|w,x,D_{\mathsf{H}},b \oplus [\![D_{\mathsf{H}}(x) \in S_{w,x}]\!]\rangle$

Fig. 13. Quantum oracles O_0 and O_1 used in Theorem 2. Here O_0 and O_1 both have the input/output register WX/B, and O_1 has an additional work register $\mathsf{D}_{q_\mathsf{H}}$.

Game $\mathbf{G}_1^z$: This game samples $\mathsf{H} \leftarrow_\$ \Omega_{\{0,1\}^m \to \{0,1\}^n}$, runs $\mathcal{A}^{\mathsf{H},\mathsf{O}_0}(z)$ and finally outputs $\mathcal{A}$'s result. Here O_0 is defined in Fig. 13. Obviously we have

$$\Pr[1 \leftarrow \mathbf{G}_1^z] = \Pr\left[b = 1 : b \leftarrow \mathcal{A}^{\mathsf{H},\mathsf{O}_0}(z)\right]. \tag{32}$$

Game $\mathbf{G}_2^z$: Based on the database register $\mathsf{D}_{q_\mathsf{H}}$, this game simulates a compressed standard oracle $\mathsf{H}_\mathsf{c} : \{0,1\}^m \to \{0,1\}^n$, then runs $\mathcal{A}^{\mathsf{H}_\mathsf{c},\mathsf{O}_0^*}(z)$ and finally outputs $\mathcal{A}$'s result. The running process of quantum oracle O_0^* is as follows:

1. O_0^* acts on the registers $\mathsf{WXD}_{q_\mathsf{H}}\mathsf{BY}$, where WX/B is the input/output register, $\mathsf{D}_{q_\mathsf{H}}$ and Y (a new register defined over $\{0,1\}^n$) are the work registers. The initial state of register Y is set to $|0^n\rangle$.
2. Implement one query of H_c with the input/output register X/Y, i.e., perform unitary operation CStO (Definition 2) on the registers $\mathsf{XYD}_{q_\mathsf{H}}$.
3. Perform $\mathsf{U}_S : |w,x,b,y\rangle \mapsto |w,x,b \oplus [\![y \in S_{w,x}]\!], y\rangle$ on the registers WXBY.
4. Repeat the above second step, that is, implement one query of H_c with the input/output register X/Y again.

In game $\mathbf{G}_2^z$, O_0^* is obviously implemented using the following unitary operation

$$\mathsf{UO}_0^* := \mathsf{CStO} \circ \mathsf{U}_S \circ \mathsf{CStO}, \tag{33}$$

and the state on register Y will always be $|0^n\rangle$ before and after one implementation of O_0^*. That is to say, Y can be viewed as the internal register of O_0^*. Notice that game $\mathbf{G}_2^z$ is only a rewrite of game $\mathbf{G}_1^z$ by replacing H with a compressed standard oracle and then correspondingly replacing O_0 with O_0^*. Hence by Lemma 1 and Eq. (32), we have

$$\Pr[1 \leftarrow \mathbf{G}_2^z] = \Pr[1 \leftarrow \mathbf{G}_1^z] = \Pr\left[b = 1 : b \leftarrow \mathcal{A}^{\mathsf{H},\mathsf{O}_0}(z)\right]. \tag{34}$$

Game $\mathbf{G}_3^z$: This game is the same as $\mathbf{G}_2^z$, except that O_0^* is replaced with the O_1 defined in Fig. 13. It is obvious that

$$\Pr[1 \leftarrow \mathbf{G}_3^z] = \Pr\left[b = 1 : b \leftarrow \mathcal{A}^{\mathsf{H}_\mathsf{c},\mathsf{O}_1}(z)\right]. \tag{35}$$

Note that the quantum oracle O_1 can be implemented by the following unitary operation UO_1 that acts on the registers $\mathsf{WXD}_{q_\mathsf{H}}\mathsf{B}$.

$$\mathsf{UO}_1 : |w, x, D_\mathsf{H}, b\rangle \mapsto |w, x, D_\mathsf{H}, b \oplus [\![D_\mathsf{H}(x) \in S_{w,x}]\!]\rangle. \tag{36}$$

Next we focus on the final state of games $\mathbf{G_2^{\mathit{z}}}$ and $\mathbf{G_3^{\mathit{z}}}$. Based on the unitary operations UO_0^* and UO_1 defined above, since we have assumed that $\mathcal{A}$ is unitary and z is fixed, the final joint state of games $\mathbf{G_2^{\mathit{z}}}$ and $\mathbf{G_3^{\mathit{z}}}$ just before the final binary projective measurement $\mathbb{M} = \{\mathrm{M}_0, \mathrm{M}_1\}$ can be respectively denoted as $|\Psi_2^z\rangle|0^n\rangle_\mathsf{Y}$ and $|\Psi_3^z\rangle|0^n\rangle_\mathsf{Y}$, where

$$\begin{aligned} |\Psi_2^z\rangle|0^n\rangle_\mathsf{Y} &:= \mathrm{U}_{q_\mathsf{O}} \circ \mathsf{UO}_0^* \cdots \mathrm{U}_2 \circ \mathsf{UO}_0^* \circ \mathrm{U}_1 \circ \mathsf{UO}_0^* |\psi_z\rangle|D_\perp\rangle_{\mathsf{D}_{q_\mathsf{H}}}|0^n\rangle_\mathsf{Y}, \\ |\Psi_3^z\rangle|0^n\rangle_\mathsf{Y} &:= \mathrm{U}_{q_\mathsf{O}} \circ \mathsf{UO}_1 \cdots \mathrm{U}_2 \circ \mathsf{UO}_1 \circ \mathrm{U}_1 \circ \mathsf{UO}_1 |\psi_z\rangle|D_\perp\rangle_{\mathsf{D}_{q_\mathsf{H}}}|0^n\rangle_\mathsf{Y}. \end{aligned} \tag{37}$$

Here $|\psi_z\rangle$ is the initial state related to z on the registers AWXB, and A denotes the register required by $\mathcal{A}$'s internal computation. $|D^\perp\rangle$ is the initial state on the database register $\mathsf{D}_{q_\mathsf{H}}$, where $D^\perp$ is a database containing only $(\perp, 0^n)$ pairs. $\mathrm{U}_1, \ldots, \mathrm{U}_{q_\mathsf{O}}$ are the unitary operations performed between the oracle queries to $\mathsf{O}_0^*/\mathsf{O}_1$, and they can all be viewed as a combination of the unitary operations acting on the registers AWXB and CStO (Definition 2) used to simulate H_c.

Remark on the state $|0^n\rangle_\mathsf{Y}$. In game $\mathbf{G_2^{\mathit{z}}}$, since the state on the register Y is always $|0^n\rangle$ before and after one implementation of O_0^*, the state on Y is also $|0^n\rangle$ before the final measurement $\mathbb{M}$, and hence we denote the final state of game $\mathbf{G_2^{\mathit{z}}}$ as $|\Psi_2^z\rangle|0^n\rangle_\mathsf{Y}$ in Eq. (37). As for the game $\mathbf{G_3^{\mathit{z}}}$, it does not use Y, but in order to make these two games based on the same quantum system, we additionally add the state $|0^n\rangle_\mathsf{Y}$ when we denote the final state of game $\mathbf{G_3^{\mathit{z}}}$ in Eq. (37). Since game $\mathbf{G_3^{\mathit{z}}}$ never uses Y, such addition is an acceptable abuse of notation .

Next we introduce the following Lemma D1, which shows that UO_0^* and UO_1 have a similar effect on a particular type of state. The detailed proof of this lemma is given in Supplementary Material D.1.

Lemma D1 *For any unit joint states on the registers* $\mathsf{AWXD}_{q_\mathsf{H}}\mathsf{B}$ *defined as*

$$|\Phi\rangle := \sum_{\substack{a,w,x \\ D_\mathsf{H}(x)=\perp,b}} \alpha_{x,D_\mathsf{H},b}^{a,w}|a, w, x, D_\mathsf{H}, b\rangle + \sum_{\substack{a,w,x \\ D_\mathsf{H}(x)=\perp,b,r\neq 0^n}} \alpha_{x,D_\mathsf{H},b}^{a,w,r} \sum_y \frac{(-1)^{y\cdot r}}{\sqrt{2^n}}|a, w, x, D_\mathsf{H} \cup (x, y), b\rangle,$$

we have $\|\mathsf{UO}_0^*|\Phi\rangle|0^n\rangle_\mathsf{Y} - \mathsf{UO}_1|\Phi\rangle|0^n\rangle_\mathsf{Y}\| \leq 6 \cdot \sqrt{\varGamma_S/2^n}$.

Remark on the state $|\Phi\rangle$. The state $|\Phi\rangle$ considered in Lemma D1 is actually the same as the state initially defined in the proof of [39, Lemma 5], i.e., a state in which the database cannot be a uniform superposition like $1/\sqrt{2^n}\sum_y |D_\mathsf{H} \cup$

$(x, y)\rangle$. The reason to consider such state is that, when a compressed standard oracle is simulated without any disturbance[24], the specific "$\mathsf{StdDecomp}_x \circ \mathsf{CNOT}^x \circ \mathsf{StdDecomp}_x$" structure of CStO cannot produce a uniform superposition database state.

Now we compute the norm difference between $|\Psi_2^z\rangle|0^n\rangle_{\mathsf{Y}}$ and $|\Psi_3^z\rangle|0^n\rangle_{\mathsf{Y}}$ defined in Eq. (37). For the index $0 \le i \le q_{\mathsf{O}}$, we first define an internal state

$$|\Psi_{2,i}^z\rangle|0^n\rangle_{\mathsf{Y}} := \mathrm{U}_{q_{\mathsf{O}}} \circ \mathsf{UO}_1 \cdots \mathrm{U}_{q_{\mathsf{O}}-i+1} \circ \mathsf{UO}_1 \circ \mathrm{U}_{q_{\mathsf{O}}-i} \circ \mathsf{UO}_0^* \cdots \mathrm{U}_1 \circ \mathsf{UO}_0^* |\psi_z\rangle|D_\perp\rangle_{\mathsf{D}_{q_\mathsf{H}}}|0^n\rangle_{\mathsf{Y}}.$$

In the above state, the first $q_{\mathsf{O}} - i$ oracle queries are simulated by UO_0^*, and the last i oracle queries are simulated by UO_1. At this point we have $|\Psi_{2,0}^z\rangle|0^n\rangle_{\mathsf{Y}} = |\Psi_2^z\rangle|0^n\rangle_{\mathsf{Y}}$ and $|\Psi_{2,q_{\mathsf{O}}}^z\rangle|0^n\rangle_{\mathsf{Y}} = |\Psi_3^z\rangle|0^n\rangle_{\mathsf{Y}}$, and we can compute

$$\begin{aligned} &\left\| |\Psi_{2,i}^z\rangle|0^n\rangle_{\mathsf{Y}} - |\Psi_{2,i+1}^z\rangle|0^n\rangle_{\mathsf{Y}} \right\| \\ =& \left\| (\mathsf{UO}_0^* - \mathsf{UO}_1)\mathrm{U}_{q_{\mathsf{O}}-i-1} \circ \mathsf{UO}_0^* \cdots \mathrm{U}_1 \circ \mathsf{UO}_0^* |\psi_z\rangle|D_\perp\rangle_{\mathsf{D}_{q_\mathsf{H}}}|0^n\rangle_{\mathsf{Y}} \right\| \overset{(a)}{\le} 6 \cdot \sqrt{\Gamma_S/2^n}. \end{aligned}$$

Here (a) uses Lemma D1. In fact, during the operation of $\mathrm{U}_{q_{\mathsf{O}}-i-1} \circ \mathsf{UO}_0^* \cdots \mathrm{U}_1 \circ \mathsf{UO}_0^*$, the compressed standard oracle must be simulated without any disturbance since all of the $\mathrm{U}_1, \ldots, \mathrm{U}_{q_{\mathsf{O}}}$ and UO_0^* can only control the database register by performing CStO. Furthermore, notice that none of $\mathrm{U}_1, \ldots, \mathrm{U}_{q_{\mathsf{O}}}$ act on register Y and the state on Y is always $|0^n\rangle$ before and after one implementation of UO_0^*, so the state $\mathrm{U}_{q_{\mathsf{O}}-i-1} \circ \mathsf{UO}_0^* \cdots \mathrm{U}_1 \circ \mathsf{UO}_0^* |\psi_z\rangle|D_\perp\rangle_{\mathsf{D}_{q_\mathsf{H}}}|0^n\rangle_{\mathsf{Y}}$ can be rewritten as the state $|\Phi\rangle|0^n\rangle_{\mathsf{Y}}$ considered in Lemma D1 under some suitable coefficients $\alpha_{x,D_\mathsf{H},b}^{a,w}$ and $\alpha_{x,D_\mathsf{H},b}^{a,w,r}$, and this is why we can use Lemma D1 in the above inequality (a). Next by the hybrid argument and the fact that $|\Psi_{2,0}^z\rangle|0^n\rangle_{\mathsf{Y}} = |\Psi_2^z\rangle|0^n\rangle_{\mathsf{Y}}$ and $|\Psi_{2,q_{\mathsf{O}}}^z\rangle|0^n\rangle_{\mathsf{Y}} = |\Psi_3^z\rangle|0^n\rangle_{\mathsf{Y}}$, we have

$$\| |\Psi_2^z\rangle|0^n\rangle_{\mathsf{Y}} - |\Psi_3^z\rangle|0^n\rangle_{\mathsf{Y}} \| \le 6q_{\mathsf{O}} \cdot \sqrt{\Gamma_S/2^n}. \tag{38}$$

Based on games $\mathbf{G_2^z}$ and $\mathbf{G_3^z}$, we define the following new games $\mathbf{G_2}$ and $\mathbf{G_3}$.

Game $\mathbf{G_2}$: This game samples $z \leftarrow \mathfrak{D}_z$, then runs game $\mathbf{G_2^z}$ and finally outputs its result. Obviously the joint state of this game just before the final binary projective measurement $\mathbb{M} = \{\mathrm{M}_0, \mathrm{M}_1\}$ can be written as a mixed state $\rho_2 := \mathbb{E}_{z \leftarrow \mathfrak{D}_z} |\Psi_2^z\rangle|0^n\rangle_{\mathsf{Y}}\langle\Psi_2^z|\langle 0^n|_{\mathsf{Y}}$, and by Eq. (34), we have

$$\Pr\left[1 \leftarrow \mathbf{G_2}\right] = \Pr_{z \leftarrow \mathfrak{D}_z}\left[b = 1 : b \leftarrow \mathcal{A}^{\mathsf{H},\mathsf{O}_0}(z)\right] = P_{\text{left}}. \tag{39}$$

Game $\mathbf{G_3}$: This game samples $z \leftarrow \mathfrak{D}_z$, then runs game $\mathbf{G_2^z}$ and finally outputs its result. Obviously the joint state of this game just before the final binary

[24] That is, the database register $\mathsf{D}_{q_\mathsf{H}}$ cannot be controlled by the adversary and is only used to simulate the compressed standard oracle by performing CStO.

projective measurement $\mathbb{M} = \{\mathrm{M}_0, \mathrm{M}_1\}$ can be written as a mixed state $\rho_3 := \mathbb{E}_{z\leftarrow\mathfrak{D}_z} |\Psi_3^z\rangle|0^n\rangle_{\mathsf{Y}}\langle\Psi_3^z|\langle 0^n|_{\mathsf{Y}}$, and by Eq. (35), we have

$$\Pr\left[1 \leftarrow \mathbf{G}_3\right] = \Pr_{z\leftarrow\mathfrak{D}_z}\left[b = 1 : b \leftarrow \mathcal{A}^{\mathsf{H_c},\mathsf{O_1}}(z)\right] = P_{\text{right}}. \tag{40}$$

At this point, we can compute the fidelity [31] between ρ_2 and ρ_3 as follows:

$$\begin{aligned}
F(\rho_2,\rho_3) &= F\left(\mathop{\mathbb{E}}_{z\leftarrow\mathfrak{D}_z} |\Psi_2^z\rangle|0^n\rangle_{\mathsf{Y}}\langle\Psi_2^z|\langle 0^n|_{\mathsf{Y}}, \mathop{\mathbb{E}}_{z\leftarrow\mathfrak{D}_z} |\Psi_3^z\rangle|0^n\rangle_{\mathsf{Y}}\langle\Psi_3^z|\langle 0^n|_{\mathsf{Y}}\right)\\
&\overset{(a)}{\geq} \mathop{\mathbb{E}}_{z\leftarrow\mathfrak{D}_z}\left[F\left(|\Psi_2^z\rangle|0^n\rangle_{\mathsf{Y}}\langle\Psi_2^z|\langle 0^n|_{\mathsf{Y}}, |\Psi_3^z\rangle|0^n\rangle_{\mathsf{Y}}\langle\Psi_3^z|\langle 0^n|_{\mathsf{Y}}\right)\right]\\
&\overset{(b)}{\geq} 1 - \frac{1}{2}\cdot \mathop{\mathbb{E}}_{z\leftarrow\mathfrak{D}_z}\left[\left\| |\Psi_2^z\rangle|0^n\rangle_{\mathsf{Y}} - |\Psi_3^z\rangle|0^n\rangle_{\mathsf{Y}}\right\|^2\right]\\
&\overset{(c)}{\geq} 1 - 36q_{\mathsf{O}}^2 \cdot \frac{\mathit{\Gamma}_S}{2^{n+1}}.
\end{aligned}$$

Here (a) uses the joint concavity of the fidelity [31, (9.95)], (b) follows from the fact that $F(|\Psi\rangle\langle\Psi|, |\Phi\rangle\langle\Phi|) \geq 1 - \||\Psi\rangle - |\Phi\rangle\|^2/2$ for any pure states $|\Psi\rangle$ and $|\Phi\rangle$, and c uses Eq. (38). Based on the above result, we can compute the Bures distance [1] $B(\rho_2,\rho_3)$ between ρ_2 and ρ_3 as follows:

$$\begin{aligned}
B(\rho_2,\rho_3)^2 &= 2(1 - F(\rho_2,\rho_3))\\
&\leq 2\left(1 - \left(1 - 36q_{\mathsf{O}}^2 \cdot \frac{\mathit{\Gamma}_S}{2^{n+1}}\right)\right) = 36q_{\mathsf{O}}^2 \cdot \frac{\mathit{\Gamma}_S}{2^n}.
\end{aligned} \tag{41}$$

Denote $P_{\mathbb{M}}(\rho)$ as the probability that performing measurement $\mathbb{M} = \{\mathrm{M}_0, \mathrm{M}_1\}$ on ρ yields the outcome 1, and then we have

$$\begin{aligned}
P_{\mathbb{M}}(\rho_2) &= \Pr\left[1 \leftarrow \mathbf{G_2}\right] \overset{(a)}{=} P_{\text{left}},\\
P_{\mathbb{M}}(\rho_3) &= \Pr\left[1 \leftarrow \mathbf{G_3}\right] \overset{(b)}{=} P_{\text{right}}.
\end{aligned}$$

Here (a) and (b) use Eq. (39) and Eq. (40), respectively. Now we can compute

$$\begin{aligned}
|P_{\text{left}} - P_{\text{right}}| = |P_{\mathbb{M}}(\rho_2) - P_{\mathbb{M}}(\rho_3)| &\overset{(a)}{\leq} B(\rho_2,\rho_3) \overset{(b)}{\leq} 6q_{\mathsf{O}} \cdot \sqrt{\frac{\mathit{\Gamma}_S}{2^n}},\\
\left|\sqrt{P_{\text{left}}} - \sqrt{P_{\text{right}}}\right| = \left|\sqrt{P_{\mathbb{M}}(\rho_2)} - \sqrt{P_{\mathbb{M}}(\rho_3)}\right| &\overset{(c)}{\leq} B(\rho_2,\rho_3) \overset{(d)}{\leq} 6q_{\mathsf{O}} \cdot \sqrt{\frac{\mathit{\Gamma}_S}{2^n}}.
\end{aligned}$$

Here (a) and (c) follow from [1, Lemma 4], and (b) and (d) use Eq. (41). This completes the proof of Theorem 2. $\square$

D.1 Proof of Lemma D1

We first divide $|\Phi\rangle$ into $|\Phi\rangle = |\Phi_1\rangle + |\Phi_2\rangle$, where

$$\begin{aligned} |\Phi_1\rangle &:= \sum_{\substack{a,w,x\\ D_{\mathsf{H}}(x)=\perp,b}} \alpha^{a,w}_{x,D_{\mathsf{H}},b}|a,w,x,D_{\mathsf{H}},b\rangle, \\ |\Phi_2\rangle &:= \sum_{\substack{a,w,x\\ D_{\mathsf{H}}(x)=\perp,b,r\neq 0^n}} \alpha^{a,w,r}_{x,D_{\mathsf{H}},b}\sum_y \frac{(-1)^{y\cdot r}}{\sqrt{2^n}}|a,w,x,D_{\mathsf{H}}\cup(x,y),b\rangle. \end{aligned} \tag{42}$$

It is obvious that $|\Phi_1\rangle$ and $|\Phi_2\rangle$ are orthogonal. In the following proof, for simplicity, we abbreviate the $\mathsf{StdDecomp}_x$ used by CStO (Definition 2) as S_x.

For the state $|\Phi_1\rangle$, by the definition of UO_0^* given in Eq. (33) and the definition of CStO, we can compute

$$\begin{aligned} \mathsf{UO}_0^*|\Phi_1\rangle|0^n\rangle_{\mathsf{Y}} &= \sum_{\substack{a,w,x\\ D_{\mathsf{H}}(x)=\perp,b}} \alpha^{a,w}_{x,D_{\mathsf{H}},b}\mathsf{CStO}\circ\mathsf{U}_S\circ\mathsf{CStO}|a,w,x,D_{\mathsf{H}},b\rangle|0^n\rangle_{\mathsf{Y}} \\ &= \sum_{\substack{a,w,x\\ D_{\mathsf{H}}(x)=\perp,b}} \frac{\alpha^{a,w}_{x,D_{\mathsf{H}},b}}{\sqrt{2^n}}\mathsf{CStO}\circ\mathsf{U}_S\sum_y \mathsf{S}_x|a,w,x,D_{\mathsf{H}}\cup(x,y),b\rangle|y\rangle_{\mathsf{Y}} \\ &= \sum_{\substack{a,w,x\\ D_{\mathsf{H}}(x)=\perp,b}} \frac{\alpha^{a,w}_{x,D_{\mathsf{H}},b}}{\sqrt{2^n}}\mathsf{CStO}\sum_{y\in S_{w,x}} \mathsf{S}_x|a,w,x,D_{\mathsf{H}}\cup(x,y),b\oplus 1\rangle|y\rangle_{\mathsf{Y}} \\ &\quad + \sum_{\substack{a,w,x\\ D_{\mathsf{H}}(x)=\perp,b}} \frac{\alpha^{a,w}_{x,D_{\mathsf{H}},b}}{\sqrt{2^n}}\mathsf{CStO}\sum_{y\notin S_{w,x}} \mathsf{S}_x|a,w,x,D_{\mathsf{H}}\cup(x,y),b\rangle|y\rangle_{\mathsf{Y}} \\ &= \sum_{\substack{a,w,x\\ D_{\mathsf{H}}(x)=\perp,b}} \frac{\alpha^{a,w}_{x,D_{\mathsf{H}},b}}{\sqrt{2^n}}\sum_{y\in S_{w,x}} \mathsf{S}_x|a,w,x,D_{\mathsf{H}}\cup(x,y),b\oplus 1\rangle|0^n\rangle_{\mathsf{Y}} \\ &\quad + \sum_{\substack{a,w,x\\ D_{\mathsf{H}}(x)=\perp,b}} \frac{\alpha^{a,w}_{x,D_{\mathsf{H}},b}}{\sqrt{2^n}}\sum_{y\notin S_{w,x}} \mathsf{S}_x|a,w,x,D_{\mathsf{H}}\cup(x,y),b\rangle|0^n\rangle_{\mathsf{Y}} \\ &= \sum_{\substack{a,w,x\\ D_{\mathsf{H}}(x)=\perp,b}} \alpha^{a,w}_{x,D_{\mathsf{H}},b}|a,w,x,D_{\mathsf{H}},b\rangle|0^n\rangle_{\mathsf{Y}} \\ &\quad + \sum_{\substack{a,w,x\\ D_{\mathsf{H}}(x)=\perp,b}} \frac{\alpha^{a,w}_{x,D_{\mathsf{H}},b}}{\sqrt{2^n}}\sum_{y\in S_{w,x}} \mathsf{S}_x|a,w,x,D_{\mathsf{H}}\cup(x,y),b\oplus 1\rangle|0^n\rangle_{\mathsf{Y}} \\ &\quad - \sum_{\substack{a,w,x\\ D_{\mathsf{H}}(x)=\perp,b}} \frac{\alpha^{a,w}_{x,D_{\mathsf{H}},b}}{\sqrt{2^n}}\sum_{y\in S_{w,x}} \mathsf{S}_x|a,w,x,D_{\mathsf{H}}\cup(x,y),b\rangle|0^n\rangle_{\mathsf{Y}}. \end{aligned}$$

Then by the definition of UO_1 given in Eq. (36), it is easy to check that

$$\mathsf{UO}_1|\Phi_1\rangle|0^n\rangle_{\mathsf{Y}} = \sum_{\substack{a,w,x \\ D_{\mathsf{H}}(x)=\perp,b}} \alpha^{a,w}_{x,D_{\mathsf{H}},b}|a,w,x,D_{\mathsf{H}},b\rangle|0^n\rangle_{\mathsf{Y}}.$$

Based on the above two equations, we can compute

$$\begin{aligned}
&\|\mathsf{UO}_0^*|\Phi_1\rangle|0^n\rangle_{\mathsf{Y}} - \mathsf{UO}_1|\Phi_1\rangle|0^n\rangle_{\mathsf{Y}}\|^2 \\
=&\left\| \begin{matrix} \displaystyle\sum_{\substack{a,w,x \\ D_{\mathsf{H}}(x)=\perp,b}} \frac{\alpha^{a,w}_{x,D_{\mathsf{H}},b}}{\sqrt{2^n}} \sum_{y\in S_{w,x}} \mathsf{S}_x|a,w,x,D_{\mathsf{H}}\cup(x,y),b\oplus 1\rangle|0^n\rangle_{\mathsf{Y}} \\ \displaystyle - \sum_{\substack{a,w,x \\ D_{\mathsf{H}}(x)=\perp,b}} \frac{\alpha^{a,w}_{x,D_{\mathsf{H}},b}}{\sqrt{2^n}} \sum_{y\in S_{w,x}} \mathsf{S}_x|a,w,x,D_{\mathsf{H}}\cup(x,y),b\rangle|0^n\rangle_{\mathsf{Y}} \end{matrix} \right\|^2 \\
\overset{(a)}{\leq}& 2\cdot \left\| \sum_{\substack{a,w,x \\ D_{\mathsf{H}}(x)=\perp,b}} \frac{\alpha^{a,w}_{x,D_{\mathsf{H}},b}}{\sqrt{2^n}} \sum_{y\in S_{w,x}} \mathsf{S}_x|a,w,x,D_{\mathsf{H}}\cup(x,y),b\oplus 1\rangle|0^n\rangle_{\mathsf{Y}} \right\|^2 \\
&+ 2\cdot \left\| \sum_{\substack{a,w,x \\ D_{\mathsf{H}}(x)=\perp,b}} \frac{\alpha^{a,w}_{x,D_{\mathsf{H}},b}}{\sqrt{2^n}} \sum_{y\in S_{w,x}} \mathsf{S}_x|a,w,x,D_{\mathsf{H}}\cup(x,y),b\rangle|0^n\rangle_{\mathsf{Y}} \right\|^2 \\
=&2\cdot \sum_{\substack{a,w,x \\ D_{\mathsf{H}}(x)=\perp,b}} \left\| \frac{\alpha^{a,w}_{x,D_{\mathsf{H}},b}}{\sqrt{2^n}} \sum_{y\in S_{w,x}} \mathsf{S}_x|a,w,x,D_{\mathsf{H}}\cup(x,y),b\oplus 1\rangle \right\|^2 \\
&+ 2\cdot \sum_{\substack{a,w,x \\ D_{\mathsf{H}}(x)=\perp,b}} \left\| \frac{\alpha^{a,w}_{x,D_{\mathsf{H}},b}}{\sqrt{2^n}} \sum_{y\in S_{w,x}} \mathsf{S}_x|a,w,x,D_{\mathsf{H}}\cup(x,y),b\rangle \right\|^2 \\
\overset{(b)}{=}&2\cdot \sum_{\substack{a,w,x \\ D_{\mathsf{H}}(x)=\perp,b}} \left\| \frac{\alpha^{a,w}_{x,D_{\mathsf{H}},b}}{\sqrt{2^n}} \sum_{y\in S_{w,x}} |a,w,x,D_{\mathsf{H}}\cup(x,y),b\oplus 1\rangle \right\|^2 \\
&+ 2\cdot \sum_{\substack{a,w,x \\ D_{\mathsf{H}}(x)=\perp,b}} \left\| \frac{\alpha^{a,w}_{x,D_{\mathsf{H}},b}}{\sqrt{2^n}} \sum_{y\in S_{w,x}} |a,w,x,D_{\mathsf{H}}\cup(x,y),b\rangle \right\|^2 \\
=&\frac{4}{2^n}\cdot \sum_{\substack{a,w,x \\ D_{\mathsf{H}}(x)=\perp,b}} |S_{w,x}|\cdot \left|\alpha^{a,w}_{x,D_{\mathsf{H}},b}\right|^2 \overset{(c)}{\leq} \frac{4\mathit{\Gamma}_S}{2^n}\cdot \||\Phi_1\rangle\|^2 .
\end{aligned} \tag{43}$$

Here (a) follows from the fact that $\|\sum_{i=1}^q|\psi_i\rangle\|^2 \leq q\cdot\sum_{i=1}^q \||\psi_i\rangle\|^2$ for any states $|\psi_1\rangle,\ldots,|\psi_q\rangle$, (b) follows from the fact that S_x is a unitary operation, and c is due to $\mathit{\Gamma}_S = \max_{w\in W,x\in\{0,1\}^m}|S_{w,x}|$.

For the state $|\Phi_2\rangle$ defined in Eq. (42), we first consider the result of UO_0^* acting on the state $\sum_y (-1)^{y\cdot r}/\sqrt{2^n}|a,w,x,D_{\mathsf{H}}\cup(x,y),b\rangle|0^n\rangle_{\mathsf{Y}}$ ($r\neq 0^n$). In fact, by the definition of UO_0^* given in Eq. (33) and the definition of CStO given in Definition 2, we can compute

$$
\begin{aligned}
&\mathsf{UO}_0^*\sum_y \frac{(-1)^{y\cdot r}}{\sqrt{2^n}}|a,w,x,D_{\mathsf{H}}\cup(x,y),b\rangle|0^n\rangle_{\mathsf{Y}}\\
=&\mathsf{CStO}\circ\mathsf{U}_S\circ\mathsf{CStO}\sum_y \frac{(-1)^{y\cdot r}}{\sqrt{2^n}}|a,w,x,D_{\mathsf{H}}\cup(x,y),b\rangle|0^n\rangle_{\mathsf{Y}}\\
=&\mathsf{CStO}\circ\mathsf{U}_S\sum_y \frac{(-1)^{y\cdot r}}{\sqrt{2^n}}\mathsf{S}_x|a,w,x,D_{\mathsf{H}}\cup(x,y),b\rangle|y\rangle_{\mathsf{Y}}\\
=&\mathsf{CStO}\sum_{y\in S_{w,x}} \frac{(-1)^{y\cdot r}}{\sqrt{2^n}}\mathsf{S}_x|a,w,x,D_{\mathsf{H}}\cup(x,y),b\oplus 1\rangle|y\rangle_{\mathsf{Y}}\\
&+\mathsf{CStO}\sum_{y\notin S_{w,x}} \frac{(-1)^{y\cdot r}}{\sqrt{2^n}}\mathsf{S}_x|a,w,x,D_{\mathsf{H}}\cup(x,y),b\rangle|y\rangle_{\mathsf{Y}}\\
=&\sum_{y\in S_{w,x}} \frac{(-1)^{y\cdot r}}{\sqrt{2^n}}\mathsf{S}_x|a,w,x,D_{\mathsf{H}}\cup(x,y),b\oplus 1\rangle|0^n\rangle_{\mathsf{Y}}\\
&+\sum_{y\notin S_{w,x}} \frac{(-1)^{y\cdot r}}{\sqrt{2^n}}\mathsf{S}_x|a,w,x,D_{\mathsf{H}}\cup(x,y),b\rangle|0^n\rangle_{\mathsf{Y}}.\\
=&\sum_y \frac{(-1)^{y\cdot r}}{\sqrt{2^n}}|a,w,x,D_{\mathsf{H}}\cup(x,y),b\rangle|0^n\rangle_{\mathsf{Y}}\\
&+\sum_{y\in S_{w,x}} \frac{(-1)^{y\cdot r}}{\sqrt{2^n}}\mathsf{S}_x|a,w,x,D_{\mathsf{H}}\cup(x,y),b\oplus 1\rangle|0^n\rangle_{\mathsf{Y}}\\
&-\sum_{y\in S_{w,x}} \frac{(-1)^{y\cdot r}}{\sqrt{2^n}}\mathsf{S}_x|a,w,x,D_{\mathsf{H}}\cup(x,y),b\rangle|0^n\rangle_{\mathsf{Y}}
\end{aligned}
$$

Next we consider the result of UO_1 acting on the state $\sum_y (-1)^{y\cdot r}/\sqrt{2^n}|a,w,x,D_{\mathsf{H}}\cup(x,y),b\rangle|0^n\rangle_{\mathsf{Y}}$ ($r\neq 0^n$). In fact, by the definition of UO_1 given in Eq. (36),

$$
\begin{aligned}
&\mathsf{UO}_1\sum_y \frac{(-1)^{y\cdot r}}{\sqrt{2^n}}|a,w,x,D_{\mathsf{H}}\cup(x,y),b\rangle|0^n\rangle_{\mathsf{Y}}\\
=&\sum_y \frac{(-1)^{y\cdot r}}{\sqrt{2^n}}|a,w,x,D_{\mathsf{H}}\cup(x,y),b\rangle|0^n\rangle_{\mathsf{Y}}\\
&+\sum_{y\in S_{w,x}} \frac{(-1)^{y\cdot r}}{\sqrt{2^n}}|a,w,x,D_{\mathsf{H}}\cup(x,y),b\oplus 1\rangle|0^n\rangle_{\mathsf{Y}}\\
&-\sum_{y\in S_{w,x}} \frac{(-1)^{y\cdot r}}{\sqrt{2^n}}|a,w,x,D_{\mathsf{H}}\cup(x,y),b\rangle|0^n\rangle_{\mathsf{Y}}.
\end{aligned}
$$

Based on the above two equations and the definition of $|\Phi_2\rangle$ given in Eq. (42), we can compute

$$
\begin{aligned}
&\mathsf{UO}_0^*|\Phi_2\rangle|0^n\rangle_{\mathsf{Y}} - \mathsf{UO}_1|\Phi_2\rangle|0^n\rangle_{\mathsf{Y}} \\
=& \sum_{\substack{a,w,x \\ D_{\mathsf{H}}(x)=\perp,b,r\neq 0^n}} \alpha^{a,w,r}_{x,D_{\mathsf{H}},b} \left(\mathsf{UO}_0^* - \mathsf{UO}_1\right) \sum_{y} \frac{(-1)^{y\cdot r}}{\sqrt{2^n}} |a,w,x,D_{\mathsf{H}} \cup (x,y),b\rangle|0^n\rangle_{\mathsf{Y}} \\
\overset{(a)}{=}& \sum_{\substack{a,w,x \\ D_{\mathsf{H}}(x)=\perp,b,r\neq 0^n}} \alpha^{a,w,r}_{x,D_{\mathsf{H}},b} \sum_{y\in S_{w,x}} \frac{(-1)^{y\cdot r}}{2^n} |a,w,x,D_{\mathsf{H}},b\oplus 1\rangle|0^n\rangle_{\mathsf{Y}} \\
&- \sum_{\substack{a,w,x \\ D_{\mathsf{H}}(x)=\perp,b,r\neq 0^n}} \alpha^{a,w,r}_{x,D_{\mathsf{H}},b} \sum_{y\in S_{w,x}} \frac{(-1)^{y\cdot r}}{2^n\sqrt{2^n}} \sum_{y'} |a,w,x,D_{\mathsf{H}} \cup (x,y'),b\oplus 1\rangle|0^n\rangle_{\mathsf{Y}} \\
&- \sum_{\substack{a,w,x \\ D_{\mathsf{H}}(x)=\perp,b,r\neq 0^n}} \alpha^{a,w,r}_{x,D_{\mathsf{H}},b} \sum_{y\in S_{w,x}} \frac{(-1)^{y\cdot r}}{2^n} |a,w,x,D_{\mathsf{H}},b\rangle|0^n\rangle_{\mathsf{Y}} \\
&+ \sum_{\substack{a,w,x \\ D_{\mathsf{H}}(x)=\perp,b,r\neq 0^n}} \alpha^{a,w,r}_{x,D_{\mathsf{H}},b} \sum_{y\in S_{w,x}} \frac{(-1)^{y\cdot r}}{2^n\sqrt{2^n}} \sum_{y'} |a,w,x,D_{\mathsf{H}} \cup (x,y'),b\rangle|0^n\rangle_{\mathsf{Y}} \\
\overset{(b)}{=}& |\phi_1\rangle - |\phi_2\rangle - |\phi_3\rangle + |\phi_4\rangle.
\end{aligned}
$$

The computation of (a) here heavily uses Eq. (2), and (b) here is a notational simplification, that is, we have respectively denoted the previous four superposition states as $|\phi_1\rangle$, $|\phi_2\rangle$, $|\phi_3\rangle$ and $|\phi_4\rangle$. At this point, we have

$$
\begin{aligned}
&\left\|\mathsf{UO}_0^*|\Phi_2\rangle|0^n\rangle_{\mathsf{Y}} - \mathsf{UO}_1|\Phi_2\rangle|0^n\rangle_{\mathsf{Y}}\right\|^2 \\
=& \left\||\phi_1\rangle - |\phi_2\rangle - |\phi_3\rangle + |\phi_4\rangle\right\|^2 \\
\overset{(a)}{\leq}& 4\cdot\left\||\phi_1\rangle\right\|^2 + 4\cdot\left\||\phi_2\rangle\right\|^2 + 4\cdot\left\||\phi_3\rangle\right\|^2 + 4\cdot\left\||\phi_4\rangle\right\|^2 \\
=& \frac{16}{2^n}\cdot \sum_{\substack{a,w,x \\ D_{\mathsf{H}}(x)=\perp,b}} \left| \sum_{r\neq 0^n, y\in S_{w,x}} \frac{(-1)^{y\cdot r}}{\sqrt{2^n}} \alpha^{a,w,r}_{x,D_{\mathsf{H}},b} \right|^2 \\
\overset{(b)}{\leq}& \frac{16}{2^n}\cdot \sum_{\substack{a,w,x \\ D_{\mathsf{H}}(x)=\perp,b}} |S_{w,x}| \sum_{y\in S_{w,x}} \left| \sum_{r\neq 0^n} \frac{(-1)^{y\cdot r}}{\sqrt{2^n}} \alpha^{a,w,r}_{x,D_{\mathsf{H}},b} \right|^2 \\
\overset{(c)}{\leq}& \frac{16\Gamma_S}{2^n}\cdot \sum_{\substack{a,w,x \\ D_{\mathsf{H}}(x)=\perp,b}} \sum_{y\in S_{w,x}} \left| \sum_{r\neq 0^n} \frac{(-1)^{y\cdot r}}{\sqrt{2^n}} \alpha^{a,w,r}_{x,D_{\mathsf{H}},b} \right|^2 \overset{(d)}{\leq} \frac{16\Gamma_S}{2^n}\cdot \left\||\Phi_2\rangle\right\|^2.
\end{aligned}
\tag{44}
$$

Here (a) follows from the fact that $\|\sum_{i=1}^{q}|\psi_i\rangle\|^2 \leq q\cdot\sum_{i=1}^{q}\||\psi_i\rangle\|^2$ for any states $|\psi_1\rangle,\ldots,|\psi_q\rangle$, (b) uses the Cauchy-Schwarz inequality, (c) follows from the fact that $\mathit{\Gamma}_S = \max_{w\in W, x\in\{0,1\}^m} |S_{w,x}|$, and (d) is due to

$$\||\Phi_2\rangle\|^2 = \left\| \sum_{\substack{a,w,x \\ D_{\mathsf{H}}(x)=\perp, b, r\neq 0^n}} \alpha^{a,w,r}_{x,D_{\mathsf{H}},b} \sum_{y\in\{0,1\}^n} \frac{(-1)^{y\cdot r}}{\sqrt{2^n}} |a,w,x,D_{\mathsf{H}}\cup(x,y),b\rangle \right\|^2$$

$$= \sum_{\substack{a,w,x \\ D_{\mathsf{H}}(x)=\perp, b}} \sum_{y\in\{0,1\}^n} \left| \sum_{r\neq 0^n} \frac{(-1)^{y\cdot r}}{\sqrt{2^n}} \alpha^{a,w,r}_{x,D_{\mathsf{H}},b} \right|^2 .$$

Notice that $|\Phi\rangle = |\Phi_1\rangle + |\Phi_2\rangle$, and hence by Eq. (43) and Eq. (44), we can compute

$$\begin{aligned}
&\|\mathsf{UO}_0^*|\Phi\rangle|0^n\rangle_{\mathsf{Y}} - \mathsf{UO}_1|\Phi\rangle|0^n\rangle_{\mathsf{Y}}\| \\
&\overset{(a)}{\leq} \|\mathsf{UO}_0^*|\Phi_1\rangle|0^n\rangle_{\mathsf{Y}} - \mathsf{UO}_1|\Phi_1\rangle|0^n\rangle_{\mathsf{Y}}\| + \|\mathsf{UO}_0^*|\Phi_2\rangle|0^n\rangle_{\mathsf{Y}} - \mathsf{UO}_1|\Phi_2\rangle|0^n\rangle_{\mathsf{Y}}\| \\
&\leq 2\cdot\sqrt{\frac{\mathit{\Gamma}_S}{2^n}}\cdot\||\Phi_1\rangle\| + 4\cdot\sqrt{\frac{\mathit{\Gamma}_S}{2^n}}\cdot\||\Phi_2\rangle\| \overset{(b)}{\leq} 6\cdot\sqrt{\frac{\mathit{\Gamma}_S}{2^n}}.
\end{aligned}$$

Here (a) uses the triangle inequality, and (b) uses the fact that $\||\Phi_1\rangle\|$ and $\||\Phi_2\rangle\|$ are both less than or equal to 1. This completes the proof of Lemma D1. □

E Cryptographic primitives and security definitions

Definition 3 (Public key encryption). *A public key encryption (PKE) scheme* $\mathsf{P} = (\mathsf{Gen}, \mathsf{Enc}, \mathsf{Dec})$ *consists of three polynomial time algorithms (in the security parameter* λ*) with a finite message space* $\mathcal{M}$ *such that*

1. Gen*, the key generation algorithm, is a probabilistic algorithm that on input* 1^λ *outputs a public/secret key pair* (pk, sk).
2. Enc*, the encryption algorithm, on input* pk *and a message* $m \in \mathcal{M}$*, it chooses* $r \leftarrow_\$ \mathcal{R}$ *(*$\mathcal{R}$ *is the randomness space), computes* $\mathsf{Enc}(pk, m; r) \in \mathcal{C}$ *(*$\mathcal{C}$ *is the ciphertext space) and outputs ciphertext* $c := \mathsf{Enc}(pk, m; r)$*. If* Enc *does not use randomness* r *to compute* c*,* Enc *is a deterministic algorithm and outputs* $c := \mathsf{Enc}(pk, m)$.
3. Dec*, the decryption algorithm, is a deterministic algorithm that on input ciphertext* $c \in \mathcal{C}$ *outputs a message* $m := \mathsf{Dec}(sk, c)$*, or a special symbol* $\perp \notin \mathcal{M}$ *to indicate that* c *is not a valid ciphertext.*

We say P *is a randomized PKE (rPKE) scheme if* Enc *uses randomness* r *to compute ciphertext* c*. Otherwise, we say* P *is a deterministic PKE (dPKE) scheme.*

Definition 4 (Worst and average decryption error [12]). *We say an rPKE scheme* $\mathsf{P} = (\mathsf{Gen}, \mathsf{Enc}, \mathsf{Dec})$ *with message/randomness space* $\mathcal{M}/\mathcal{R}$ *is* δ*-worst-correct or has worst-case decryption error* δ *if*

$$\mathbb{E}\left[\max_{m\in\mathcal{M}} \Pr\left[\mathsf{Dec}(sk, c) \neq m : c = \mathsf{Enc}(pk, m; r), r \leftarrow_\$ \mathcal{R}\right]\right] \leq \delta.$$

where the expectation is taken over $(pk, sk) \leftarrow \mathsf{Gen}$ *and the choice of the random oracles involved (if any). Similarly, we say an rPKE or dPKE scheme* $\mathsf{P} = (\mathsf{Gen}, \mathsf{Enc}, \mathsf{Dec})$ *with message space* $\mathcal{M}$ *is* δ*-average-correct or has average-case decryption error* δ *if*

$$\Pr\left[\mathsf{Dec}(sk, c) \neq m : c = \mathsf{Enc}(pk, m), (pk, sk) \leftarrow \mathsf{Gen}, m \leftarrow_\$ \mathcal{M}\right] \leq \delta.$$

Definition 5 (γ**-spread [11]).** *We say an rPKE scheme* $\mathsf{P} = (\mathsf{Gen}, \mathsf{Enc}, \mathsf{Dec})$ *with message space* $\mathcal{M}$*, ciphertext space* $\mathcal{C}$*, and randomness space* $\mathcal{R}$ *is (weakly)* γ*-spread if*

$$\mathbb{E}\left[\max_{m\in\mathcal{M}, c\in\mathcal{C}} \Pr[c = \mathsf{Enc}(pk, m; r) : r \leftarrow_\$ \mathcal{R}]\right] \leq 2^{-\gamma},$$

where the expectation is taken over $(pk, sk) \leftarrow \mathsf{Gen}$ *and the choice of the random oracles involved (if any).*

Definition 6 (Security notions for PKE scheme). *For any polynomial time quantum adversary* $\mathcal{A}$*, we define its* $\mathsf{OW\text{-}CPA}$*,* q-$\mathsf{OW\text{-}CPA}$ *and* $\mathsf{IND\text{-}CPA}$ *advantage against PKE scheme* $\mathsf{P} = (\mathsf{Gen}, \mathsf{Enc}, \mathsf{Dec})$ *as follows:*

$$\begin{aligned}
\mathrm{Adv}^{\mathsf{OW\text{-}CPA}}_{\mathsf{P},\mathcal{A}} &:= \Pr\left[1 \leftarrow \mathbf{G}^{\mathsf{OW\text{-}CPA}}_{\mathsf{P},\mathcal{A}}\right],\\
\mathrm{Adv}^{q\text{-}\mathsf{OW\text{-}CPA}}_{\mathsf{P},\mathcal{A}} &:= \Pr\left[1 \leftarrow \mathbf{G}^{q\text{-}\mathsf{OW\text{-}CPA}}_{\mathsf{P},\mathcal{A}}\right],\\
\mathrm{Adv}^{\mathsf{IND\text{-}CPA}}_{\mathsf{P},\mathcal{A}} &:= \left|\Pr\left[1 \leftarrow \mathbf{G}^{\mathsf{IND\text{-}CPA}}_{\mathsf{P},\mathcal{A}}\right] - \frac{1}{2}\right|.
\end{aligned}$$

Here games $\mathbf{G}^{\mathsf{OW\text{-}CPA}}_{\mathsf{P},\mathcal{A}}$, $\mathbf{G}^{q\text{-}\mathsf{OW\text{-}CPA}}_{\mathsf{P},\mathcal{A}}$ *and* $\mathbf{G}^{\mathsf{IND\text{-}CPA}}_{\mathsf{P},\mathcal{A}}$ *are shown in Fig. 14. We say* P *is* $\mathsf{OW\text{-}CPA}$*-secure,* q-$\mathsf{OW\text{-}CPA}$*-secure and* $\mathsf{IND\text{-}CPA}$*-secure, if for any polynomial time quantum adversary* $\mathcal{A}$*,* $\mathrm{Adv}^{\mathsf{OW\text{-}CPA}}_{\mathsf{P},\mathcal{A}}$*,* $\mathrm{Adv}^{q\text{-}\mathsf{OW\text{-}CPA}}_{\mathsf{P},\mathcal{A}}$ *and* $\mathrm{Adv}^{\mathsf{IND\text{-}CPA}}_{\mathsf{P},\mathcal{A}}$ *is negligible, respectively.*

$\mathbf{G}^{\mathsf{OW\text{-}CPA}}_{\mathsf{P},\mathcal{A}}$	$\mathbf{G}^{q\text{-}\mathsf{OW\text{-}CPA}}_{\mathsf{P},\mathcal{A}}$	$\mathbf{G}^{\mathsf{IND\text{-}CPA}}_{\mathsf{P},\mathcal{A}}$
$(pk, sk) \leftarrow \mathsf{Gen}$	$(pk, sk) \leftarrow \mathsf{Gen}$	$(pk, sk) \leftarrow \mathsf{Gen}$
$m^* \leftarrow_\$ \mathcal{M}$	$m^* \leftarrow_\$ \mathcal{M}$	$b \leftarrow_\$ \{0,1\}$
$c^* := \mathsf{Enc}(pk, m^*)$	$c^* := \mathsf{Enc}(pk, m^*)$	$(m_0^*, m_1^*) \leftarrow \mathcal{A}(pk)$
$m' \leftarrow \mathcal{A}(pk, c^*)$	$\mathcal{Q} \leftarrow \mathcal{A}(pk, c^*)$	$c^* := \mathsf{Enc}(pk, m_b^*)$
$\mathbf{return}\ [\![m' = m^*]\!]$	$\mathbf{return}\ [\![m^* \in \mathcal{Q} \wedge \lvert\mathcal{Q}\rvert \leq q]\!]$	$b' \leftarrow \mathcal{A}(pk, c^*)$
		$\mathbf{return}\ [\![b' = b]\!]$

Fig. 14. Games $\mathbf{G}^{\mathsf{OW\text{-}CPA}}_{\mathsf{P},\mathcal{A}}$, $\mathbf{G}^{q\text{-}\mathsf{OW\text{-}CPA}}_{\mathsf{P},\mathcal{A}}$ and $\mathbf{G}^{\mathsf{IND\text{-}CPA}}_{\mathsf{P},\mathcal{A}}$.

Definition 7 (Key encapsulation mechanism). *A key encapsulation mechanism* $\mathsf{KEM} = (\mathsf{Gen}, \mathsf{Enca}, \mathsf{Deca})$ *consists of three polynomial time algorithms with the key space* $\mathcal{K}$ *and the ciphertext space* $\mathcal{C}$ *such that*

1. Gen*, the key generation algorithm, is a probabilistic algorithm that on input* 1^λ *outputs a public/secret key pair* (pk, sk)*.*
2. Enca*, the encapsulation algorithm, is a probabilistic algorithm that on input* pk *outputs a key* $k \in \mathcal{K}$ *and a ciphertext* $c \in \mathcal{C}$*.*
3. Deca*, the decapsulation algorithm, is a deterministic algorithm that on input ciphertext* c *outputs a key* $k := \mathsf{Deca}(sk, c)$ *or a special symbol* $\perp \notin \mathcal{K}$ *to indicate that* c *is not a valid ciphertext.*

$\mathbf{G}^{\mathsf{IND\text{-}CPA}}_{\mathsf{KEM},\mathcal{A}}$	$\mathbf{G}^{\mathsf{IND\text{-}CCA}}_{\mathsf{KEM},\mathcal{A}}$	$\mathsf{Deca}^*(c)$
$(pk, sk) \leftarrow \mathsf{Gen}$	$(pk, sk) \leftarrow \mathsf{Gen}$	$\mathbf{if}\ c = c^*$
$b \leftarrow_\$ \{0,1\}$	$b \leftarrow_\$ \{0,1\}$	$\quad\mathbf{return}\ \perp$
$(K_0, c^*) \leftarrow \mathsf{Enca}(pk)$	$(K_0, c^*) \leftarrow \mathsf{Enca}(pk)$	$\mathbf{else\ return}\ \mathsf{Deca}(sk, c)$
$K_1 \leftarrow_\$ \mathcal{K}$	$K_1 \leftarrow_\$ \mathcal{K}$	
$b' \leftarrow \mathcal{A}(pk, c^*, K_b)$	$b' \leftarrow \mathcal{A}^{\mathsf{Deca}^*}(pk, c^*, K_b)$	
$\mathbf{return}\ [\![b' = b]\!]$	$\mathbf{return}\ [\![b' = b]\!]$	

Fig. 15. Games $\mathbf{G}^{\mathsf{IND\text{-}CPA}}_{\mathsf{KEM},\mathcal{A}}$ and $\mathbf{G}^{\mathsf{IND\text{-}CCA}}_{\mathsf{KEM},\mathcal{A}}$.

Definition 8 (Security notions for KEM scheme). *For any polynomial time quantum adversary $\mathcal{A}$, we define its* $\mathsf{IND\text{-}CPA}$ *and* $\mathsf{IND\text{-}CCA}$ *advantage against KEM scheme* $\mathsf{KEM} = (\mathsf{Gen}, \mathsf{Enca}, \mathsf{Deca})$ *as follows:*

$$\mathrm{Adv}_{\mathsf{KEM},\mathcal{A}}^{\mathsf{IND\text{-}CPA}} := \left| \Pr\left[1 \leftarrow \mathbf{G}_{\mathsf{KEM},\mathcal{A}}^{\mathsf{IND\text{-}CPA}}\right] - \frac{1}{2} \right|,$$
$$\mathrm{Adv}_{\mathsf{KEM},\mathcal{A}}^{\mathsf{IND\text{-}CCA}} := \left| \Pr\left[1 \leftarrow \mathbf{G}_{\mathsf{KEM},\mathcal{A}}^{\mathsf{IND\text{-}CCA}}\right] - \frac{1}{2} \right|.$$

Here games $\mathbf{G}_{\mathsf{KEM},\mathcal{A}}^{\mathsf{IND\text{-}CPA}}$ *and* $\mathbf{G}_{\mathsf{KEM},\mathcal{A}}^{\mathsf{IND\text{-}CCA}}$ *are shown in Fig. 15. We say* KEM *is* $\mathsf{IND\text{-}CPA}$*-secure/*$\mathsf{IND\text{-}CCA}$*-secure, if for any polynomial time quantum adversary* $\mathcal{A}$*, the advantage* $\mathrm{Adv}_{\mathsf{KEM},\mathcal{A}}^{\mathsf{IND\text{-}CPA}}/\mathrm{Adv}_{\mathsf{KEM},\mathcal{A}}^{\mathsf{IND\text{-}CCA}}$ *is negligible.*

F Some missing discussions

Here we give some detailed discussions that have been omitted in the main text.

F.1 Migrating the proof of ACWC

Here we explain that our security proof of ACWC, i.e., the proof of Theorem 7 given in Supplementary Material L.1, is also valid for the original ACWC defined in [12], which uses a more general XOR named a generalize one-time pad GOTP. In fact, the necessity for [12] to introduce GOTP is that the message space of its underlying PKE schemes may not be $\{0,1\}^n$ (such as $\{-1,0,1\}^n$), so it needs to introduce a more general XOR to compute $m \oplus \mathsf{H}(m_1)$ correspondingly, namely $\mathsf{GOTP}(m, \mathsf{H}(m_1))$. In the following, we first give the definition of the generalize one-time pad, and then introduce the original ACWC defined in [12].

Definition 9 (Adaptive version of [12, Definition 3.5]). *We call function* GOTP*:* $\mathcal{X} \times \mathcal{U} \rightarrow \mathcal{Y}$ *a generalized one-time pad relative to distributions* $\psi_{\mathcal{X}}$*,* $\psi_{\mathcal{U}}$ *and* $\psi_{\mathcal{Y}}$ *if the following properties are all satisfied.*

1. *Decoding: There exists an efficient inversion function* $\mathsf{Inv} : \mathcal{Y} \times \mathcal{U} \rightarrow \mathcal{X}$ *such that for all* $x \in \mathcal{X}$*,* $u \in \mathcal{U}$ *and* $y \in \mathcal{Y}$*,* $\mathsf{Inv}(\mathsf{GOTP}(x, u), u) = x$*.*
2. *Injective-inversion: For all* $y \in \mathcal{Y}$ *and* $x \in \mathcal{X}$*, there exists only one element* $u \in \mathcal{U}$ *such that* $\mathsf{Inv}(y, u) = x$*.*
3. *Randomness-hiding: For all* $u \in \mathcal{U}$*, the random variable* $\mathsf{GOTP}(x, u)$*, for* $x \leftarrow \psi_{\mathcal{X}}$*, has the same distribution as* $\psi_{\mathcal{Y}}$*.*
4. *Message-hiding: For all* $x \in \mathcal{X}$*, the random variable* $\mathsf{GOTP}(x, u)$*, for* $u \leftarrow \psi_{\mathcal{U}}$*, has the same distribution as* $\psi_{\mathcal{Y}}$*.*

Remark 3. We note that the proof of [12, Theorem 3.9] actually uses a property of GOTP that is not explicitly stated in its Definition 3.5, namely the "*Since* $\mathcal{A}$ *did not query* M_1^**, the value* $\mathsf{F}(M_1^*)$ *looks uniformly random to* $\mathcal{A}$*, and therefore* $\mathsf{Inv}(M_2^*, \mathsf{F}(M_1^*))$ *is distributed as* $\psi_{\mathcal{M}'}$" mentioned in the proof of [12, Theorem 3.9]. Here for the convenience of migrating our proof of Theorem 7, we explicitly add and rephrase this property as the above "*Injective-inversion*", which is an injective property of Inv. In fact, this is an acceptable rephrasing, as the inversion function Inv of a typical generalized one-time pad XOR, where $\mathsf{Inv}(y, u) = y \oplus u$, clearly satisfies this property.

Transformation ACWC. Let $\mathsf{P} = (\mathsf{Gen}, \mathsf{Enc}, \mathsf{Dec})$ be a PKE scheme with message space $\mathcal{M}_1 \times \mathcal{M}_2$. Let $\mathsf{H} : \mathcal{M}_1 \rightarrow \mathcal{U}$ be a hash function. Let GOTP: $\mathcal{M}' \times \mathcal{U} \rightarrow \mathcal{M}_2$ be a generalized one-time pad relative to distributions $\psi_{\mathcal{M}'}$, $\psi_{\mathcal{U}}$, and $\psi_{\mathcal{M}_2}$. We associate PKE scheme $\mathsf{ACWC}[\mathsf{P}, \mathsf{GOTP}, \mathsf{H}] = (\mathsf{Gen}, \mathsf{Enc}', \mathsf{Dec}')$ that has message space $\mathcal{M}'$. The constituting algorithms of $\mathsf{ACWC}[\mathsf{P}, \mathsf{GOTP}, \mathsf{H}]$ are given in Fig. 16.

It can be observed that the core properties of XOR that are used by the proof of Theorem 7 are the following three properties:

Gen	$\mathsf{Enc}'(pk, m \in \mathcal{M}')$	$\mathsf{Dec}'(sk, c)$
1 : $(pk, sk) \leftarrow \mathsf{Gen}$	1 : $m_1 \leftarrow \psi_{\mathcal{M}_1}$	1 : $m_1 \| m_2 := \mathsf{Dec}(sk, c)$
2 : **return** (pk, sk)	2 : $m_2 := \mathsf{GOTP}(m, \mathsf{H}(m_1))$	2 : $m := \mathsf{Inv}(m_2, \mathsf{H}(m_1))$
	3 : $c := \mathsf{Enc}(pk, m_1 \| m_2)$	3 : **return** m
	4 : **return** c	

Fig. 16. Public key encryption $\mathsf{ACWC}[\mathsf{P}, \mathsf{GOTP}, \mathsf{H}]$.

1. The game hop from $\mathbf{G_0}$ to $\mathbf{G_1}$ uses the property that (m^*, m_1^*, m_2^*) where $m^* \leftarrow_\$ \mathcal{M}_2$, $m_1^* \leftarrow_\$ \mathcal{M}_1$ and $m_2^* := m^* \oplus \mathsf{H}(m_1^*)$ has the same distribution as (m^*, m_1^*, m_2^*) where $m_2^* \leftarrow_\$ \mathcal{M}_2$, $m_1^* \leftarrow_\$ \mathcal{M}_1$ and $m^* := m_2^* \oplus \mathsf{H}(m_1^*)$.
2. The game hop from $\mathbf{G_2}$ to $\mathbf{G_3}$ uses the property that there exists only one element f such that $m^* := m_2^* \oplus f$ for any m^* and m_2^*.
3. The derivation of Eq. (113) uses the property that $m^* = m_2^* \oplus \mathsf{H}(m_1^*)$ implies $m_2^* = m^* \oplus \mathsf{H}(m_1^*)$.

In the following, by using Definition 9, we show how the above three properties can be satisfied by the GOTP used in $\mathsf{ACWC}[\mathsf{P}, \mathsf{GOTP}, \mathsf{H}]$.

1. In fact, by the randomness-hiding and decoding properties of GOTP, it is easy to verify that (m^*, m_1^*, m_2^*) where $m^* \leftarrow \psi_{\mathcal{M}'}$, $m_1^* \leftarrow \psi_{\mathcal{M}_1}$ and $m_2^* := \mathsf{GOTP}(m^*, \mathsf{H}(m_1^*))$ has the same distribution as (m^*, m_1^*, m_2^*) where $m_1^* \leftarrow \psi_{\mathcal{M}_1}$, $m_2^* \leftarrow \psi_{\mathcal{M}_2}$ and $m^* := \mathsf{Inv}(m_2^*, \mathsf{H}(m_1^*))$. Hence the above first property is satisfied by GOTP.
2. By the injective-inversion property of GOTP, there obviously exists only one element $u \in \mathcal{U}$ such that $m^* = \mathsf{Inv}(m_2^*, u)$ for any m^* and m_2^*. Hence the above second property is satisfied by GOTP.
3. Similar to the game hop from G_6 to G_7 of the proof of [12, Theorem 3.10], by adding a trivial check $m_2^* \in \operatorname{im} \mathsf{GOTP}(-, \Theta)$ to ACWC's $\mathsf{OW\text{-}CPA}$ game, we can deduce that $m^* = \mathsf{Inv}(m_2^*, \mathsf{H}(m_1^*))$ implies $m_2^* = \mathsf{GOTP}(m^*, \mathsf{H}(m_1^*))$. Hence the above third property is satisfied by GOTP.

At this point, we find that all the core properties of XOR that are used by the proof of Theorem 7 are still satisfied by the GOTP, and hence through some trivial restatement and notational replacement, we can migrate the proof of Theorem 7 to that of $\mathsf{ACWC}[\mathsf{P}, \mathsf{GOTP}, \mathsf{H}]$.

We would like to stress that, our migrated proof of $\mathsf{ACWC}[\mathsf{P}, \mathsf{GOTP}, \mathsf{H}]$ has an additional restriction on the set $\mathcal{U}$ and the distribution $\psi_{\mathcal{U}}$ used by GOTP, that is, $\mathcal{U}$ must be $\{0,1\}^n$ and $\psi_{\mathcal{U}}$ must be the uniform distribution. In fact, such restrictions are necessary because we need to simulate H as a compressed standard oracle[25] in our proof of Theorem 7.

[25] Recall that the compressed standard oracle [39] can only be used to perfectly simulate a quantum-accessible uniformly random function with the output space $\{0,1\}^n$.

F.2 Proof of Theorem 4 under randomized PKE schemes

In this section, we present the proof of Theorem 4 when P is an rPKE scheme, and for brevity, we will reuse parts of its proof. One can note that whether P is randomized or deterministic will only affect the output of the random oracle F. Taking the game $\mathbf{G_0}$ shown in Fig. 12 as an example, when P is deterministic, we have $\mathsf{F}(m^*) = m_1^*$ and $c_1^* = \mathsf{Enc}(pk, m_1^*)$, while when P is randomized, we have $\mathsf{F}(m^*) = m_1^*||r^*$ and $c_1^* = \mathsf{Enc}(pk, m_1^*; r^*)$, where F (used to derandomize) needs to output an additional random string r^*. In the following, we explain that the game sequence $\mathbf{G_0}$ to $\mathbf{G_7}$ introduced in the proof of Theorem 4 actually does not focus on the specific output of F.

- **Games $\mathbf{G_0}$ to $\mathbf{G_6}$.** These games are used to change the simulation method for the decapsulation oracle, and finally the simulation in game $\mathbf{G_6}$ no longer uses the secret key sk. That is to say, these games only focus on how much error is generated when using different simulation methods to simulate the decapsulation oracle. Hence these games are independent of the specific output of F[26].
- **Game $\mathbf{G_7}$ to $\mathbf{G_8}$.** The core purpose of this game hop is to illustrate that the computation of $\mathsf{U}_{\mathsf{qD}}^{pk}$ does not cause too much disturbance to the compressed standard oracle simulation of $\mathsf{F}(m^*)$. That is to say, this game hop only focuses on how much disturbance is caused by the computation of $\mathsf{U}_{\mathsf{qD}}^{pk}$ to the simulation of $\mathsf{F}(m^*)$, without regard to what $\mathsf{F}(m^*)$ itself is.

As for the game $\mathbf{G_9}$, since we apply Lemma 6 to compute the probability of the event $\mathsf{Coll}_{\mathsf{F}',m^*}$ (Eq. (29)) and this lemma only holds for dPKE schemes, we need to define a new game to avoid relying on Lemma 6. To address this, we design the following game:

Game $\mathbf{NG_9}$: This game is the same as $\mathbf{G_8}$ (Fig. 12), except that it uses the following control operation to simulate F for $\mathcal{A}$.

- For the query state $|x, y\rangle$ of the query registers $\mathsf{X_F Y_F}$, denote $m_1||r_1 := \mathsf{F}'(x)$, then perform CStO on the registers $\mathsf{X_F Y_F D}_{q_\mathsf{F}}$ if $\mathsf{Enc}(pk, m_1; r_1) \neq c_1^* \vee x \oplus \mathsf{H}(m_1) \neq c_2^*$, and query F' to return $|x, y \oplus \mathsf{F}'(x)\rangle$ if $\mathsf{Enc}(pk, m_1; r_1) = c_1^* \wedge x \oplus \mathsf{H}(m_1) = c_2^*$. Here $\mathsf{X_F}/\mathsf{Y_F}$ denotes the input/output register of F and (c_1^*, c_2^*) is the challenge ciphertext.

In practice, the above control operation can be easily implemented by respectively querying F' and H two times.

[26] One point to note is that the simulation of the decapsulation oracle presented in Section 4.2 only applies to dPKE schemes, as shown by the definition of the set $S_{c_1,c_2,m,m_1'}$ given in Eq. (5). However, this is not an obstacle, as we can make simple adjustments to the definitions and notations to enable the simulation in Section 4.2 to apply to rPKE schemes as well.

Similar with the proof of Eq. (30), we first define the following classical event:

$$\begin{aligned}\mathsf{Coll}_{\mathsf{F}',\mathsf{H},m^*}\colon\ & (pk, sk) \leftarrow \mathsf{Gen},\ m^* \leftarrow_\$ \mathcal{M},\ \mathsf{F}' \leftarrow_\$ \Omega_{\{0,1\}^n \to \mathcal{M}},\ \mathsf{H} \leftarrow_\$ \Omega_{\mathcal{M} \to \{0,1\}^n} \\ & \exists m \neq m^* \text{ s.t. } \mathsf{Enc}(pk, m_1; r_1) = \mathsf{Enc}(pk, m_1^*; r_1^*) \text{ and} \\ & m \oplus \mathsf{H}(m_1) = m^* \oplus \mathsf{H}(m_1^*). \text{ Here } m_1 || r_1 = \mathsf{F}'(m),\ m_1^* || r_1^* = \mathsf{F}'(m^*).\end{aligned}$$

Recall that $(\mathsf{Enc}(pk, m_1^*; r_1^*), m^* \oplus \mathsf{H}(m_1^*))$ is actually the challenge ciphertext (c_1^*, c_2^*) when P is a randomize PKE scheme. Now, if the above event does not occur, we can conclude that $m = m^*$ if and only if $(\mathsf{Enc}(pk, m_1; r_1), m \oplus \mathsf{H}(m_1)) = (\mathsf{Enc}(pk, m_1^*; r_1^*), m^* \oplus \mathsf{H}(m_1^*))$, where $m_1 || r_1 = \mathsf{F}'(m)$, $m_1^* || r_1^* = \mathsf{F}'(m^*)$. Now by the construction of game $\mathbf{G_8}$ in Fig. 12, we have

$$|\Pr[1 \leftarrow \mathbf{G_8}] - \Pr[1 \leftarrow \mathbf{NG_9}]| \leq \Pr[\mathsf{Coll}_{\mathsf{F}',\mathsf{H},m^*}].$$

As for the probability $\Pr[\mathsf{Coll}_{\mathsf{F}',\mathsf{H},m^*}]$, by using [28, Lemma 4] and [12, Lemma 3.1], we can prove $\Pr[\mathsf{Coll}_{\mathsf{F}',\mathsf{H},m^*}] \leq 2 \cdot \delta_{ac}$[27], and thus we obtain

$$|\Pr[1 \leftarrow \mathbf{G_8}] - \Pr[1 \leftarrow \mathbf{NG_9}]| \leq 2 \cdot \delta_{ac} \leq \frac{1}{|\mathcal{M}|} + 2 \cdot \delta_{ac}.$$

The subsequent proof is identical to that of Theorem 4, which involves reducing game $\mathbf{NG_9}$ to an IND-CPA game of $\mathsf{FOAC_0}$ by constructing an adversary $\mathcal{B}$. Note that the $\mathcal{B}$ here requires making additional $2q_\mathsf{F}$ queries to H' compared to the original $\mathcal{B}$ in Theorem 4, as determining the event $\mathsf{Coll}_{\mathsf{F}',\mathsf{H},m^*}$ requires querying H. Since these additional queries only lead to a constant-factor increase in the bound given in Theorem 6, we omit it for simplicity and continue to use the original bound from Theorem 6.

[27] The proof is not complicated and only need the following two steps. First by [28, Lemma 4], the $\Pr[\mathsf{Coll}_{\mathsf{F}',\mathsf{H},m^*}]$ can be upper bounded by using the worst-case decryption error of $\mathsf{ACWC_0}[\mathsf{P},\mathsf{H}]$. Then by [12, Lemma 3.1], the worst-case decryption error of $\mathsf{ACWC_0}[\mathsf{P},\mathsf{H}]$ equals to the average-case decryption error δ_{ac} of P.

G Proof of Theorem 3 (from $\mathsf{OW\text{-}CPA}_\mathsf{P}$ to $\mathsf{IND\text{-}CCA}_{\mathsf{FOAC_0}}$)

For the sake of clarity, we prove Theorem 3 through the following three steps:

1. We prove Theorem 11 in Supplementary Material G.1, which shows that the $\mathsf{IND\text{-}CCA}$ security of $\mathsf{FOAC_0}$ can be tightly reduced to its $\mathsf{IND\text{-}CPA}$ security in the ROM.
2. Then we prove Theorem 12 in Supplementary Material G.2, which shows that the $\mathsf{IND\text{-}CPA}$ security of $\mathsf{FOAC_0}$ can be tightly reduced to the $\mathsf{OW\text{-}CPA}$ security of the underlying PKE scheme P in the ROM.
3. Finally by combining Theorem 11 and Theorem 12 in Supplementary Material G.3, we prove Theorem 3.

To make the proof more concise and readable, we restate the constituting algorithms of $\mathsf{KEMAC_0}$ in Fig. 17, in which we use the $\mathsf{Enc_0}$ and $\mathsf{Dec_0}$ of $\mathsf{ACWC_0}[\mathsf{P},\mathsf{H}]$ (Fig. 8) to represent the encapsulation and decapsulation algorithms, rather than the Enc and Dec of P as shown in Fig. 10.

Gen	$\mathsf{Enca_0}(pk)$	$\mathsf{Deca_0}(sk, c_1, c_2)$
$(pk, sk) \leftarrow \mathsf{Gen}$	$m \leftarrow_\$ \{0,1\}^n$	$m := \mathsf{Dec_0}(sk, c_1, c_2)$
return (pk, sk)	$m_1 := \mathsf{F}(m)$	**if** $m = \perp$, **return** $\perp$
	$(c_1, c_2) := \mathsf{Enc_0}(pk, m; m_1)$	**else** $m'_1 := \mathsf{F}(m)$
	$K := \mathsf{G}(m)$	**if** $(c_1, c_2) := \mathsf{Enc_0}(pk, m; m'_1)$
	return (c_1, c_2, K)	**return** $\mathsf{G}(m)$
		else return $\perp$

Fig. 17. Key encapsulation mechanism $\mathsf{KEMAC_0}$. Here $\mathsf{Enc_0}$ and $\mathsf{Dec_0}$ are defined in Fig. 8, and they both need to compute the hash function H.

G.1 From $\mathsf{IND\text{-}CPA}_{\mathsf{FOAC_0}}$ to $\mathsf{IND\text{-}CCA}_{\mathsf{FOAC_0}}$ in the ROM

Theorem 11 ($\mathsf{IND\text{-}CPA}$ **of** $\mathsf{KEMAC_0} \overset{\text{ROM}}{\Rightarrow}$ $\mathsf{IND\text{-}CCA}$ **of** $\mathsf{KEMAC_0}$)**.** *Let* P *be a* δ_{ac}*-average-correct PKE scheme with meassage space* $\mathcal{M}(=\{0,1\}^m)$*. Let* $\mathcal{A}$ *be an* $\mathsf{IND\text{-}CCA}$ *adversary against* $\mathsf{KEMAC_0}$*, making* q_D *classical queries to the decapsulation oracle, and making* q_F*,* q_G *and* q_H *classical queries to the random oracle* F*,* G *and* H*. Then we can construct an* $\mathsf{IND\text{-}CPA}$ *adversary* $\mathcal{B}$ *against* $\mathsf{KEMAC_0}$ *such that* $\mathcal{T}_\mathcal{B} \le \mathcal{T}_\mathcal{A} + O(q_\mathsf{D} q_\mathsf{F}) \cdot \mathcal{T}_\mathsf{Enc}$ *and*

$$\mathrm{Adv}^{\mathsf{IND\text{-}CCA}}_{\mathsf{KEMAC_0},\mathcal{A}} \le \mathrm{Adv}^{\mathsf{IND\text{-}CPA}}_{\mathsf{KEMAC_0},\mathcal{B}} + q_\mathsf{F} \cdot \delta_{ac} + \frac{q_\mathsf{D} q_\mathsf{H}}{|\mathcal{M}|} + \frac{q_\mathsf{D}}{2^n}.$$

Here n *is the output length of* H*. The adversary* $\mathcal{B}$ *queries* F*,* G *and* H *at most* q_F*,* $q_\mathsf{D} + q_\mathsf{G}$ *and* $q_\mathsf{D} q_\mathsf{F} + q_\mathsf{H}$ *times, respectively.*

Proof. We first assume P is deterministic. To prove this theorem, we introduce a sequence of games $\mathbf{G_0}$ to $\mathbf{G_2}$ as shown in Fig. 18.

GAMES $\mathbf{G_0}$-$\mathbf{G_2}$

1. $(pk, sk) \leftarrow \mathsf{Gen}$	$//\mathbf{G_0}$-$\mathbf{G_2}$
$\mathsf{G} \leftarrow_\$ \Omega_{\{0,1\}^n \to \mathcal{K}}$	$//\mathbf{G_0}$-$\mathbf{G_2}$
2. $m^* \leftarrow_\$ \{0,1\}^n$	$//\mathbf{G_0}$-$\mathbf{G_2}$
$m_1^* := \mathsf{F}(m^*)$	$//\mathbf{G_0}$-$\mathbf{G_2}$
$(c_1^*, c_2^*) := \mathsf{Enc}_0(pk, m^*; m_1^*)$	$//\mathbf{G_0}$-$\mathbf{G_2}$
3. $b \leftarrow_\$ \{0,1\}$	$//\mathbf{G_0}$-$\mathbf{G_2}$
$K_0^* := \mathsf{G}(m^*)$	$//\mathbf{G_0}$-$\mathbf{G_2}$
$K_1^* \leftarrow_\$ \mathcal{K}$	$//\mathbf{G_0}$-$\mathbf{G_2}$
4. $b' \leftarrow \mathcal{A}^{\mathsf{F},\mathsf{G},\mathsf{H},\mathsf{Deca}_0}(pk, c_1^*, c_2^*, K_b^*)$	$//\mathbf{G_0}$
5. $b' \leftarrow \mathcal{A}^{\mathsf{F},\mathsf{G},\mathsf{H},\mathsf{Deca}_1}(pk, c_1^*, c_2^*, K_b^*)$	$//\mathbf{G_1}$
6. $b' \leftarrow \mathcal{B}^{\mathsf{F},\mathsf{G},\mathsf{H}}(pk, c_1^*, c_2^*, K_b^*)$	$//\mathbf{G_2}$
7. **return** $[\![b' = b]\!]$	$//\mathbf{G_0}$-$\mathbf{G_2}$

$\mathsf{F}(x)$ $//\mathbf{G_0}$-$\mathbf{G_2}$

1. **if** $\exists y$ s.t. $(x,y) \in \mathsf{L_F}$, **return** y
2. **else** $y \leftarrow_\$ \mathcal{M}$, $\mathsf{L_F} := \mathsf{L_F} \cup (x,y)$
   return y

$\mathsf{H}(x)$ $//\mathbf{G_0}$-$\mathbf{G_2}$

1. **if** $\exists y$ s.t. $(x,y) \in \mathsf{L_H}$, **return** y
2. **else** $y \leftarrow_\$ \{0,1\}^n$, $\mathsf{L_H} := \mathsf{L_H} \cup (x,y)$
   return y

$\mathsf{Deca}_0(sk, c_1, c_2)$ $//\mathbf{G_0}$

1. **if** $(c_1, c_2) = (c_1^*, c_2^*)$, **return** $\perp$
   else $m := \mathsf{Dec}_0(sk, c_1, c_2)$
2. **if** $m = \perp$, **return** $\perp$
   else $m_1 := \mathsf{F}(m)$
3. **if** $(c_1, c_2) = \mathsf{Enc}_0(pk, m; m_1)$
   return $\mathsf{G}(m)$
   else return $\perp$

$\mathsf{Deca}_1(pk, c_1, c_2)$ $//\mathbf{G_1}$

1. **if** $(c_1, c_2) = (c_1^*, c_2^*)$, **return** $\perp$
   else $m := \mathsf{Search}(pk, c_1, c_2, \mathsf{L_F})$
2. **if** $m \neq \perp$, **return** $\mathsf{G}(m)$
   else return $\perp$

Fig. 18. Games $\mathbf{G_0}$ to $\mathbf{G_2}$ in the proof of Theorem 11. The record lists $\mathsf{L_F}$ and $\mathsf{L_H}$ used here, both of which are empty sets at the beginning of the game.

Game $\mathbf{G_0}$: This is the $\mathsf{IND\text{-}CCA}$ game of KEMAC_0 with the adversary $\mathcal{A}$ in the ROM, and the random oracles F and H are simulated on-the-fly by using the record lists $\mathsf{L_F}$ and $\mathsf{L_H}$, respectively. Obviously

$$\mathrm{Adv}^{\mathsf{IND\text{-}CCA}}_{\mathsf{KEMAC}_0,\mathcal{A}} = \left| \Pr[1 \leftarrow \mathbf{G_0}] - \frac{1}{2} \right|. \tag{45}$$

Game $\mathbf{G_1}$: This game is the same as $\mathbf{G_0}$, except that Deca_0 is replaced with Deca_1, which is a new decapsulation oracle does not use the secret key sk. For a decapsulation query (c_1, c_2), Deca_1 only uses the public key pk to compute $m := \mathsf{Search}(pk, c_1, c_2, \mathsf{L_F})$, and then returns $\mathsf{G}(m)$ if $m \neq \perp$. The function Search used here is defined as

$$\mathsf{Search}\,(pk, c_1, c_2, \mathsf{L_F}) := \begin{cases} m & m \text{ is the minimum value s.t.} \\ & (c_1, c_2) = \mathsf{Enc}_0(pk, m; \mathsf{L_F}(m)), \\ \perp & \text{if such } m \text{ does not exist.} \end{cases} \tag{46}$$

By the definition of the oracles Deca_0 and Deca_1, their returns can only differ in the following three cases:

1. The decapsulation query $(c_1, c_2)(\neq (c_1^*, c_2^*))$ and the record list $\mathsf{L_F}$ just before such query satisfy $\mathsf{Search}(pk, c_1, c_2, \mathsf{L_F}) \neq \bot$, but $\mathsf{Dec_0}(sk, c_1, c_2) = \bot$ or $\mathsf{Dec_0}(sk, c_1, c_2) = m \neq \bot$ and $(c_1, c_2) \neq \mathsf{Enc_0}(pk, m; \mathsf{F}(m))$. In this case, $\mathsf{Deca_0}(sk, c_1, c_2) = \bot$ but $\mathsf{Deca_1}(pk, c_1, c_2) \neq \bot$.
2. The decapsulation query $(c_1, c_2)(\neq (c_1^*, c_2^*))$ and the record list $\mathsf{L_F}$ just before such query satisfy $\mathsf{Search}(pk, c_1, c_2, \mathsf{L_F}) = \bot$, but $\mathsf{Dec_0}(sk, c_1, c_2) = m \neq \bot$ and $(c_1, c_2) = \mathsf{Enc_0}(pk, m; \mathsf{F}(m))$. In this case, $\mathsf{Deca_0}(sk, c_1, c_2) \neq \bot$ but $\mathsf{Deca_1}(pk, c_1, c_2) = \bot$.
3. The decapsulation query $(c_1, c_2)(\neq (c_1^*, c_2^*))$ and the record list $\mathsf{L_F}$ just before such query satisfy $\mathsf{Search}(pk, c_1, c_2, \mathsf{L_F}) \neq \bot$, $\mathsf{Dec_0}(sk, c_1, c_2) = m \neq \bot$ and $(c_1, c_2) = \mathsf{Enc_0}(pk, m; \mathsf{F}(m))$, but $\mathsf{Search}(pk, c_1, c_2, \mathsf{L_F})$ does not equal to m. In this case, neither $\mathsf{Deca_0}(sk, c_1, c_2)$ or $\mathsf{Deca_1}(pk, c_1, c_2)$ equals $\bot$ but these two vlaues are different.

Based on the above three cases, we define the following two classical events Error and Guess that may occur in games $\mathbf{G_0}$ and $\mathbf{G_1}$.

- Error: The adversary $\mathcal{A}$ ever makes a decapsulation query (c_1, c_2) that triggers the above case 1 or case 3.
- Guess: The adversary $\mathcal{A}$ ever makes a decapsulation query (c_1, c_2) that triggers the above case 2.

At this point, it is obvious that games $\mathbf{G_0}$ and $\mathbf{G_1}$ proceed identically if the event $\mathsf{Error} \vee \mathsf{Guess}$ does not occur. Hence by the fundamental lemma [38],

$$|\Pr[1 \leftarrow \mathbf{G_0}] - \Pr[1 \leftarrow \mathbf{G_1}]| \leq \Pr[\mathsf{Error} : \mathbf{G_1}] + \Pr[\mathsf{Guess} : \mathbf{G_1}]. \tag{47}$$

We first consider Guess. By the definition of $\mathsf{Enc_0}$ and $\mathsf{Dec_0}$ given in Fig. 8, if a decapsulation query (c_1, c_2) makes Guess occurs, we have $\mathsf{Search}(pk, c_1, c_2, \mathsf{L_F}) = \bot$, $\mathsf{Dec}(sk, c_1) \neq \bot$, $c_1 = \mathsf{Enc}(pk, \mathsf{F}(m))$ and $c_2 = m \oplus \mathsf{H}(\mathsf{F}(m))$, where $m = \mathsf{H}(\mathsf{Dec}(sk, c_1)) \oplus c_2$ and $\mathsf{L_F}$ is the record list just before the decapsulation query (c_1, c_2). Now we can conclude that $\mathsf{L_F}(m) = \bot$, otherwise by the basic rules of the on-the-fly simulation we have $\mathsf{F}(m) = \mathsf{L_F}(m)$ and then $c_1 = \mathsf{Enc}(pk, \mathsf{L_F}(m)) \wedge c_2 = m \oplus \mathsf{H}(\mathsf{L_F}(m))$, which contradicts with $\mathsf{Search}(pk, c_1, c_2, \mathsf{L_F}) = \bot$ based on the definition of Search given in Eq. (46).

Notice that $\mathsf{L_F}(m) = \bot$ means $\mathsf{F}(m)$ is essentially a uniformly random value over $\mathcal{M}$; hence the probability that $\mathsf{L_H}(\mathsf{F}(m)) \neq \bot$ must be less than or equal to $q_\mathsf{H}/|\mathcal{M}|$, as $\mathcal{A}$ queries H at most q_H times. Here $\mathsf{L_H}$ is the record list of H just before the decapsulation query (c_1, c_2). Now under an error of $q_\mathsf{H}/|\mathcal{M}|$, we can say that $\mathsf{H}(\mathsf{F}(m))$ is a uniformly random value over $\{0,1\}^n$, and hence the probability that $c_1 = \mathsf{Enc}(pk, \mathsf{F}(m)) \wedge c_2 = m \oplus \mathsf{H}(\mathsf{F}(m))$ does not exceed $1/2^n$. Finally, since $\mathcal{A}$ makes at most q_D decapsulation queries, by the union bound,

$$\Pr[\mathsf{Guess} : \mathbf{G_1}] \leq \frac{q_\mathsf{D} q_\mathsf{H}}{|\mathcal{M}|} + \frac{q_\mathsf{D}}{2^n}. \tag{48}$$

As for the event Error, we first consider a fixed (pk, sk) pair. In fact, by the definition of Enc_0 and Dec_0 given in Fig. 8, it is not hard to check that the event Error must imply that $\mathcal{A}$ has queried F by an m such that $\mathsf{Dec}(sk, \mathsf{Enc}(pk, \mathsf{F}(m))) \neq \mathsf{F}(m)$. Recall that in game $\mathbf{G_1}$, F is simulated on-the-fly and queried at most q_F times by $\mathcal{A}$; hence by the union bound,

$$\Pr\left[\mathsf{Error} : \mathbf{G_1}, (pk, sk)\right] \leq q_\mathsf{F} \cdot \Pr\left[\mathsf{Dec}(sk, \mathsf{Enc}(pk, y)) \neq y : y \leftarrow_\$ \mathcal{M}\right].$$

Here $\Pr\left[\mathsf{Error} : \mathbf{G_1}, (pk, sk)\right]$ denotes the probability of Error under a fixed (pk, sk) pair. Then by averaging over $(pk, sk) \leftarrow \mathsf{Gen}$ and utilizing the fact that P has average-case decryption error δ_{ac}, we have

$$\Pr\left[\mathsf{Error} : \mathbf{G_1}\right] \leq q_\mathsf{F} \cdot \delta_{ac}. \tag{49}$$

Game $\mathbf{G_2}$: This game is the same as $\mathbf{G_1}$, except that the adversary $\mathcal{A}$ is replaced with a new adversary $\mathcal{B}$ that does not query the decapsulation oracle. Here $\mathcal{B}$ runs $\mathcal{A}$, outputs its final output, and simulates F, G, H, and Deca_1 for $\mathcal{A}$ as follows:

1. For the F queries, $\mathcal{B}$ replies by querying its own oracle F. Meanwhile, $\mathcal{B}$ sets up a record list $\mathsf{L}_{\mathcal{B},\mathsf{F}}$ to record $\mathcal{A}$'s F queries and the corresponding responses.
2. For the G/H queries, $\mathcal{B}$ replies by querying its own oracle G/H.
3. For the decapsulation query (c_1, c_2), $\mathcal{B}$ returns $\perp$ if $(c_1, c_2) = (c_1^*, c_2^*)$. Otherwise, denote $m := \mathsf{Search}(pk, c_1, c_2, \mathsf{L}_{\mathcal{B},\mathsf{F}})$, $\mathcal{B}$ returns $\mathsf{G}(m)$ (resp. $\perp$) if $m \neq \perp$ (resp. $m = \perp$).

In game $\mathbf{G_1}$, the function Search used by $\mathsf{Deca_1}$ will never return m^* since (c_1^*, c_2^*) is forbidden from being used to query the decapsulation oracle. That is to say, the function Search used by $\mathsf{Deca_1}$ will only search the list that records the F queries made by $\mathcal{A}$, i.e., the list $\mathsf{L}_{\mathcal{B},\mathsf{F}}$ introduced above. This means that game $\mathbf{G_2}$ is actually an equivalent rewrite of game $\mathbf{G_1}$, and thus we have

$$\Pr\left[1 \leftarrow \mathbf{G_1}\right] = \Pr\left[1 \leftarrow \mathbf{G_2}\right]. \tag{50}$$

In addition, it is obvious that game $\mathbf{G_2}$ can be viewed as an $\mathsf{IND\text{-}CPA}$ game of KEMAC_0 with the adversary $\mathcal{B}$ in the ROM, that is,

$$\left|\Pr\left[1 \leftarrow \mathbf{G_2}\right] - \frac{1}{2}\right| = \mathrm{Adv}^{\mathsf{IND\text{-}CPA}}_{\mathsf{KEMAC}_0,\mathcal{B}}. \tag{51}$$

By the definition of the function Search given in Eq.(46), the adversary $\mathcal{B}$ queries F, G and H at most q_F, $q_\mathsf{D} + q_\mathsf{G}$ and $q_\mathsf{D} q_\mathsf{F} + q_\mathsf{H}$ times, respectively. As for the running time, note that $\mathcal{B}$ needs to compute Search at most q_D times, and thus we have $\mathcal{T}_\mathcal{B} \leq \mathcal{T}_\mathcal{A} + O(q_\mathsf{D} q_\mathsf{F}) \cdot \mathcal{T}_\mathsf{Enc}$.

By combining Eq. (45) and Eq. (47) to Eq. (51), we finally get

$$\mathrm{Adv}^{\mathsf{IND\text{-}CCA}}_{\mathsf{KEMAC}_0,\mathcal{A}} \leq \mathrm{Adv}^{\mathsf{IND\text{-}CPA}}_{\mathsf{KEMAC}_0,\mathcal{B}} + q_\mathsf{F} \cdot \delta_{ac} + \frac{q_\mathsf{D} q_\mathsf{H}}{|\mathcal{M}|} + \frac{q_\mathsf{D}}{2^n}.$$

This completes the proof of Theorem 11 for deterministic P.

As for the randomized P, our above proof is still valid, as the game sequence shown in Fig. 18 does not directly use the encryption algorithm Enc of P, but instead relies on the higher-level encryption algorithm $\mathsf{Enc_0}$ of $\mathsf{ACWC_0}[\mathsf{P}, \mathsf{H}]$. Notice that the computation of $\mathsf{Enc_0}$ involves Enc and a random string r as shown in Fig. 8, and thus $\mathsf{Enc_0}$ is a randomized algorithm. That is to say, the proof of Theorem 11 is in fact not sensitive to the randomness of Enc, because regardless of whether Enc is randomized or not, $\mathsf{Enc_0}$ must be randomized and thus the above proof can be directly migrated to randomized P. Therefore, we complete the proof of Theorem 11. □

G.2 From $\mathsf{OW\text{-}CPA_P}$ to $\mathsf{IND\text{-}CPA_{FOAC_0}}$ in the ROM

Before proving the main theorem, we first introduce the following Lemma G1. In fact, Lemma G1 is similar to [28, Lemma 4], and their main difference is that Lemma G1 specifically considers dPKE schemes.

Lemma G1 *Let* $\mathsf{P} = (\mathsf{Gen}, \mathsf{Enc}, \mathsf{Dec})$ *be a dPKE scheme with meassage space* $\mathcal{M}$ *and average-case decryption error* δ_{ac}*. For a fixed* (pk, sk) *pair, define set*

$$S^{coll}_{pk,sk} := \{m \in \mathcal{M} | \exists m' \in \mathcal{M} \text{ s.t. } m' \neq m \wedge \mathsf{Enc}(pk, m') = \mathsf{Enc}(pk, m)\}.$$

Then we have

$$\Pr\left[m \in S^{coll}_{pk,sk} : m \leftarrow_\$ \mathcal{M}, (pk, sk) \leftarrow \mathsf{Gen}\right] \leq 2 \cdot \delta_{ac}.$$

Proof. For a fixed (pk, sk) pair, we first define a subset of $\mathcal{M}$ as:

$$S^{error}_{pk,sk} := \{m \in \mathcal{M} \mid \mathsf{Dec}(sk, \mathsf{Enc}(pk, m)) \neq m\}.$$

Then by the definition of the average-case decryption error, obviously

$$\Pr\left[m \in S^{error}_{pk,sk} : (pk, sk) \leftarrow \mathsf{Gen}, m \leftarrow_\$ \mathcal{M}\right] \leq \delta_{ac}. \tag{52}$$

Denote $\mathcal{C}$ as the ciphertext space of P, and then for a $c \in \mathcal{C}$, define set

$$S^{coll}_{pk,sk,c} := \{m \mid \exists m' \in \mathcal{M} \text{ s.t. } m' \neq m \wedge \mathsf{Enc}(pk, m') = \mathsf{Enc}(pk, m) = c\}.$$

At this point, for two different ciphertexts c_1 and c_2, the set $S^{coll}_{pk,sk,c_1} \cap S^{coll}_{pk,sk,c_2}$ must be empty if S^{coll}_{pk,sk,c_1} and S^{coll}_{pk,sk,c_2} both are non empty. In addition, by the definition of the set $S^{coll}_{pk,sk}$, it is not hard to check that there must exist a subset S of $\mathcal{C}$ such that $S^{coll}_{pk,sk,c}$ is not empty for all $c \in S$ and $S^{coll}_{pk,sk} = \bigcup_{c \in S} S^{coll}_{pk,sk,c}$. That is to say, under some different ciphertexts, the set $S^{coll}_{pk,sk}$ can be decomposed into a union of some disjoint sets.

For the set $S^{coll}_{pk,sk,c}$, there exists at most one element in $S^{coll}_{pk,sk,c}$ such that it does not belong to $S^{error}_{pk,sk}$ because the decryption algorithm Dec is deterministic. Additionally, since $|S^{coll}_{pk,sk,c}| \geq 2$ for any $c \in S$, we have

$$\left|S^{coll}_{pk,sk,c} \backslash S^{error}_{pk,sk}\right| \leq \left|S^{coll}_{pk,sk,c} \cap S^{error}_{pk,sk}\right|.$$

Adding $S^{coll}_{pk,sk} = \bigcup_{c\in S} S^{coll}_{pk,sk,c}$ and the fact that $S^{coll}_{pk,sk,c_1} \cap S^{coll}_{pk,sk,c_2} = \emptyset$ if $c_1 \neq c_2$ and S^{coll}_{pk,sk,c_1} and S^{coll}_{pk,sk,c_2} both are non empty sets, we get

$$\left|S^{coll}_{pk,sk} \backslash S^{error}_{pk,sk}\right| \leq \left|S^{coll}_{pk,sk} \cap S^{error}_{pk,sk}\right|. \tag{53}$$

Now based on $(pk, sk) \leftarrow \mathsf{Gen}$ and $m \leftarrow_\$ \mathcal{M}$, we can compute

$$\begin{aligned}
\Pr\left[m \in S^{coll}_{pk,sk}\right] &= \Pr\left[m \in S^{coll}_{pk,sk} \backslash S^{error}_{pk,sk}\right] + \Pr\left[m \in S^{coll}_{pk,sk} \cap S^{error}_{pk,sk}\right] \\
&\overset{(a)}{\leq} 2 \cdot \Pr\left[m \in S^{coll}_{pk,sk} \cap S^{error}_{pk,sk}\right] \\
&\leq 2 \cdot \Pr\left[m \in S^{error}_{pk,sk}\right] \overset{(b)}{\leq} 2 \cdot \delta_{ac}.
\end{aligned}$$

Here (a) follows from Eq. (53), and (b) follows from Eq. (52). This completes the proof of Lemma G1. □

Theorem 12 **(OW-CPA of** $\mathsf{P} \overset{\text{ROM}}{\Rightarrow}$ **IND-CPA of** $\mathsf{KEMAC_0}$**).** *Let PKE scheme* P *be* δ_{ac}*-average-correct. Let* $\mathcal{A}$ *be an* IND-CPA *adversary against* $\mathsf{KEMAC_0}$*, making* q_F*,* q_G *and* q_H *classical queries to the random oracle* F*,* G *and* H*. Then we can design a* OW-CPA *adversary* $\mathcal{B}$ *against* P *such that* $\mathcal{T}_\mathcal{B} \leq \mathcal{T}_\mathcal{A} + O(q_\mathsf{H}) \cdot \mathcal{T}_\mathsf{Enc}$ *and*

$$\mathrm{Adv}^{\mathsf{IND\text{-}CPA}}_{\mathsf{KEMAC_0},\mathcal{A}} \leq \begin{cases} 3 \cdot \mathrm{Adv}^{\mathsf{OW\text{-}CPA}}_{\mathsf{P},\mathcal{B}} + 6 \cdot \delta_{ac} + \frac{2(q_\mathsf{F}+q_\mathsf{G})+1}{2^n} & \text{if } \mathsf{P} \text{ is deterministic,} \\ 3q_\mathsf{H} \cdot \mathrm{Adv}^{\mathsf{OW\text{-}CPA}}_{\mathsf{P},\mathcal{B}} + \frac{2(q_\mathsf{F}+q_\mathsf{G})+1}{2^n} & \text{if } \mathsf{P} \text{ is randomized.} \end{cases}$$

Here n *is the output length of* H*.*

Proof. Based on $\mathsf{FOAC_0} = \mathsf{FO}^\perp_m \circ \mathsf{ACWC_0}$ (Eq. (3)), we can compute

$$\begin{aligned}
\mathrm{Adv}^{\mathsf{IND\text{-}CPA}}_{\mathsf{KEMAC_0},\mathcal{A}} &\overset{(a)}{\leq} 3 \cdot \mathrm{Adv}^{\mathsf{IND\text{-}CPA}}_{\mathsf{ACWC_0[P,H]},\mathcal{A}_1} + \frac{2(q_\mathsf{F}+q_\mathsf{G})+1}{2^n} \\
&\overset{(b)}{\leq} 3 \cdot \mathrm{Adv}^{q_\mathsf{H}\text{-}\mathsf{OW\text{-}CPA}}_{\mathsf{P},\mathcal{A}_2} + \frac{2(q_\mathsf{F}+q_\mathsf{G})+1}{2^n}.
\end{aligned} \tag{54}$$

Here (a) follows from [20, Theorem 3] and $\mathcal{T}_{\mathcal{A}_1} \approx \mathcal{T}_\mathcal{A}$, and (b) follows from [12, Theorem 3.3] and $\mathcal{T}_{\mathcal{A}_2} \approx \mathcal{T}_{\mathcal{A}_1}$. When P is an rPKE scheme, by using [12, Lemma 2.1], we can further construct a OW-CPA adversary $\mathcal{B}$ such that $\mathcal{T}_\mathcal{B} \approx \mathcal{T}_{\mathcal{A}_2}$ and

$$\mathrm{Adv}^{q_\mathsf{H}\text{-}\mathsf{OW\text{-}CPA}}_{\mathsf{P},\mathcal{A}_2} \leq q_\mathsf{H} \cdot \mathrm{Adv}^{\mathsf{OW\text{-}CPA}}_{\mathsf{P},\mathcal{B}}. \tag{55}$$

When P is a dPKE scheme, based on the adversary $\mathcal{A}_2$, we can construct the following OW-CPA adversary $\mathcal{B}$ against P:

– $\mathcal{B}$ first runs $\mathcal{A}_2$ to get its output set $\mathcal{Q}$ ($|\mathcal{Q}| \leq q_\mathsf{H}$). Then $\mathcal{B}$ searches $\mathcal{Q}$ for the first element x such that $\mathsf{Enc}(pk, x) = c^*$, where c^* is the challenge ciphertext. If this search succeeds, $\mathcal{B}$ outputs x. Otherwise $\mathcal{B}$ outputs $\perp$.

Note that the challenge plaintext m^* must be a uniformly random value over $\mathcal{M}$. Hence by Lemma G1, we can conclude that $\mathsf{Enc}(pk, x) = c^*$ if and only if $x = m^*$ under an error at most $2 \cdot \delta_{ac}$. Hence, when P is a dPKE scheme, we get

$$\mathrm{Adv}^{q_\mathsf{H}\text{-}\mathsf{OW\text{-}CPA}}_{\mathsf{P},\mathcal{A}_2} \leq \mathrm{Adv}^{\mathsf{OW\text{-}CPA}}_{\mathsf{P},\mathcal{B}} + 2 \cdot \delta_{ac}. \tag{56}$$

Here $\mathcal{T}_\mathcal{B} \leq \mathcal{T}_{\mathcal{A}_2} + O(q_\mathsf{H}) \cdot \mathcal{T}_\mathsf{Enc}$. Now by combining Eq. (54) to Eq. (56), we complete the proof of Theorem 12. □

G.3 From $\mathsf{OW\text{-}CPA}_\mathsf{P}$ to $\mathsf{IND\text{-}CCA}_{\mathsf{FOAC}_0}$ in the ROM

By combining Theorem 11 and Theorem 12, we can straightforwardly prove Theorem 3, which we restate below.

Theorem 13 ($\mathsf{OW\text{-}CPA}$ **of** $\mathsf{P} \overset{\text{ROM}}{\Rightarrow}$ $\mathsf{IND\text{-}CCA}$ **of** KEMAC_0)**.** *Let* P *be a* δ_{ac}-*average-correct PKE scheme with message space* $\mathcal{M}(= \{0,1\}^m)$. *Let* $\mathcal{A}$ *be an* $\mathsf{IND\text{-}CCA}$ *adversary against* KEMAC_0, *making* q_D *classical queries to the decapsulation oracle, and making* q_F, q_G *and* q_H *classical queries to the random oracle* F, G *and* H. *Then we can construct a* $\mathsf{OW\text{-}CPA}$ *adversary* $\mathcal{B}$ *against* P *such that* $\mathcal{T}_\mathcal{B} \leq \mathcal{T}_\mathcal{A} + O(q_\mathsf{D} q_\mathsf{F} + q_\mathsf{H}) \cdot \mathcal{T}_\mathsf{Enc}$ *and*

$$\mathrm{Adv}^{\mathsf{IND\text{-}CCA}}_{\mathsf{KEMAC}_0,\mathcal{A}} \leq \mathrm{Adv}_\mathsf{P} + q_\mathsf{F} \cdot \delta_{ac} + \frac{q_\mathsf{D} q_\mathsf{H}}{|\mathcal{M}|} + \frac{2(q_\mathsf{D} + q_\mathsf{F} + q_\mathsf{G}) + 1}{2^n}, \textit{ with}$$

$$\mathrm{Adv}_\mathsf{P} = \begin{cases} 3 \cdot \mathrm{Adv}^{\mathsf{OW\text{-}CPA}}_{\mathsf{P},\mathcal{B}} + 6 \cdot \delta_{ac} & \textit{if } \mathsf{P} \textit{ is deterministic,} \\ 3(q_\mathsf{D} q_\mathsf{F} + q_\mathsf{H}) \cdot \mathrm{Adv}^{\mathsf{OW\text{-}CPA}}_{\mathsf{P},\mathcal{B}} & \textit{if } \mathsf{P} \textit{ is randomized.} \end{cases}$$

Here n *is the output length of* H.

H Some missing proofs of Section 4.3

H.1 Proof of Lemma 3

Similar to the proof of Lemma 2, we only need to show $\mathsf{Read}_{\mathsf{F},\mathsf{H}} \circ \mathsf{Out}_{\mathsf{G}} \circ \mathsf{Read}_{\mathsf{F},\mathsf{H}} = \mathsf{Read}_{\mathsf{F}} \circ \mathsf{Read}'_{\mathsf{H}} \circ \mathsf{Out}_{\mathsf{G}} \circ \mathsf{Read}'_{\mathsf{H}} \circ \mathsf{Read}_{\mathsf{F}}$ to prove this lemma.

By the definitions of $\mathsf{Read}_{\mathsf{F},\mathsf{H}}$, $\mathsf{Read}_{\mathsf{F}}$ and $\mathsf{Read}'_{\mathsf{H}}$ given in Eq. (15), Eq. (18) and Eq. (19), respectively, for any computational basis state $|c_1, c_2, m, m'_1, D_{\mathsf{F}}, D_{\mathsf{H}}, b_2\rangle$ on the registers $\mathsf{C}_1\mathsf{C}_2\mathsf{M}\mathsf{M}_1\mathsf{D}_{q_{\mathsf{F}}}\mathsf{D}_{q_{\mathsf{H}}}\mathsf{B}_2$, we can compute

$$\begin{aligned}
&\mathsf{Read}_{\mathsf{F},\mathsf{H}}|c_1, c_2, m, m'_1, D_{\mathsf{F}}, D_{\mathsf{H}}, b_2\rangle \\
&\quad= \begin{cases} |c_1, c_2, m, m'_1, D_{\mathsf{F}}, D_{\mathsf{H}}, b_2\rangle & \text{if } D_{\mathsf{F}}(m) = \bot, \\ \left| \begin{matrix} c_1, c_2, m, m'_1, D_{\mathsf{F}}, D_{\mathsf{H}}, \\ b_2 \oplus [\![D_{\mathsf{H}}(m'_1 \oplus D_{\mathsf{F}}(m)) \in S_{c_1,c_2,m,m'_1}]\!] \end{matrix} \right\rangle & \text{if } D_{\mathsf{F}}(m) \neq \bot. \end{cases} \\
&\mathsf{Read}'_{\mathsf{H}} \circ \mathsf{Read}_{\mathsf{F}}|c_1, c_2, m, m'_1, D_{\mathsf{F}}, D_{\mathsf{H}}, b_2\rangle \\
&\quad= \begin{cases} |c_1, c_2, m, m'_1, D_{\mathsf{F}}, D_{\mathsf{H}}, b_2\rangle & \text{if } D_{\mathsf{F}}(m) = \bot, \\ \left| \begin{matrix} c_1, c_2, m, m'_1 \oplus D_{\mathsf{F}}(m), D_{\mathsf{F}}, D_{\mathsf{H}}, \\ b_2 \oplus [\![D_{\mathsf{H}}(m'_1 \oplus D_{\mathsf{F}}(m)) \in S_{c_1,c_2,m,m'_1}]\!] \end{matrix} \right\rangle & \text{if } D_{\mathsf{F}}(m) \neq \bot. \end{cases}
\end{aligned}$$

The above equations show that $\mathsf{Read}_{\mathsf{F},\mathsf{H}}$ and $\mathsf{Read}_{\mathsf{F}} \circ \mathsf{Read}'_{\mathsf{H}} \circ \mathsf{Read}_{\mathsf{F}}$ actually have the same result when acting on any computational basis state. Thus we have $\mathsf{Read}_{\mathsf{F},\mathsf{H}} = \mathsf{Read}_{\mathsf{F}} \circ \mathsf{Read}'_{\mathsf{H}} \circ \mathsf{Read}_{\mathsf{F}}$. Then we can compute

$$\begin{aligned}
\mathsf{Read}_{\mathsf{F},\mathsf{H}} \circ \mathsf{Out}_{\mathsf{G}} \circ \mathsf{Read}_{\mathsf{F},\mathsf{H}} &= \mathsf{Read}_{\mathsf{F}} \circ \mathsf{Read}'_{\mathsf{H}} \circ \mathsf{Read}_{\mathsf{F}} \circ \mathsf{Out}_{\mathsf{G}} \circ \mathsf{Read}_{\mathsf{F}} \circ \mathsf{Read}'_{\mathsf{H}} \circ \mathsf{Read}_{\mathsf{F}} \\
&\overset{(a)}{=} \mathsf{Read}_{\mathsf{F}} \circ \mathsf{Read}'_{\mathsf{H}} \circ \mathsf{Out}_{\mathsf{G}} \circ \mathsf{Read}_{\mathsf{F}} \circ \mathsf{Read}_{\mathsf{F}} \circ \mathsf{Read}'_{\mathsf{H}} \circ \mathsf{Read}_{\mathsf{F}} \\
&\overset{(b)}{=} \mathsf{Read}_{\mathsf{F}} \circ \mathsf{Read}'_{\mathsf{H}} \circ \mathsf{Out}_{\mathsf{G}} \circ \mathsf{Read}'_{\mathsf{H}} \circ \mathsf{Read}_{\mathsf{F}}.
\end{aligned}$$

Here (a) uses the fact that $\mathsf{Out}_{\mathsf{G}}$ commutes with $\mathsf{Read}_{\mathsf{F}}$, which is obvious since the register they work together is only M and they both do not change the state of M in the computational basis. (b) is because $\mathsf{Read}_{\mathsf{F}}$ is an involution. Now we prove Lemma 3 by the above equation. □

H.2 Proof of Lemma 4

In order to prove Lemma 4, we first introduce the database read operation [16,39]. To be specific, let f be an arbitrary function with domain $\mathcal{X} \times \mathbf{D}_q$ and codomain $\mathcal{Y}$, where $\mathcal{X}$ and $\mathcal{Y}$ are finite sets and $\mathbf{D}_q$ is the database set defined in Definition 1. Then we call the following unitary operation Read_f a database read.

$$\mathrm{Read}_f : |x, D, y\rangle_{\mathsf{X}\mathsf{D}_q\mathsf{Y}} \mapsto |x, D, y + f(x, D)\rangle_{\mathsf{X}\mathsf{D}_q\mathsf{Y}}. \tag{57}$$

Here X/Y is defined over $\mathcal{X}/\mathcal{Y}$, D_q is the database register, and $+ : \mathcal{Y} \times \mathcal{Y} \to \mathcal{Y}$ is a group operation over $\mathcal{Y}$. In fact, a key property of the database read operation Read_f is that it does not change the database state in the computational basis.

Definition 10 (Compressed Semi-Classical Oracle [16]). *Let $\mathbf{D}_q$ be the database set and $S \subseteq \mathbf{D}_q$. Define function f_S such that $f_S(D) = 1$ if $D \in S$ and $f_S(D) = 0$ otherwise. The compressed semi-classical oracle $\mathcal{O}_S^{CSC}$ performs the following operation on the input state $\sum_{z\in\{0,1\}^*, D\in\mathbf{D}_q} \alpha_{z,D}|z, D\rangle$:*

1. *Initialize a single qubit B with $|0\rangle_\mathsf{B}$, transform $\sum_{z\in\{0,1\}^*, D\in\mathbf{D}_q} \alpha_{z,D}|z, D\rangle|0\rangle_\mathsf{B}$ into $\sum_{z\in\{0,1\}^*, D\in\mathbf{D}_q} \alpha_{z,D}|z, D\rangle|f_S(D)\rangle_\mathsf{B}$.*
2. *Measure B and output the measurement outcome.*

During the execution of a quantum oracle algorithm $\mathcal{A}^{\mathcal{O}_S^{CSC}}$, we denote Find as the classical event that the measurement of $\mathcal{O}_S^{CSC}$ (on B) ever has outcome 1.

Theorem 14 (Compressed Semi-Classical O2H [16, Theorem 1]). *Let D_q be the database register defined over $\mathbf{D}_q$ and $\mathsf{H} : \{0,1\}^m \to \{0,1\}^n$ be the compressed standard oracle implemented on D_q. Let S be a subset of $\mathbf{D}_q$ such that $D^\perp \notin S$ and z be a random string, where $D^\perp$ is the database only contains q pairs $(\perp, 0^n)$. S and z may have arbitrary joint distribution $\mathfrak{D}$. Let $\mathsf{H}\backslash S$ be an oracle that first queries H and then queries $\mathcal{O}_S^{CSC}$ on the database register D_q.*

Let $\mathcal{A}$ be a quantum oracle algorithm (not necessarily unitary) that has oracle access to H and Read_f, and $\mathcal{A}$ queries H (resp. Read_f) at most $q_\mathsf{H} \leq q$[28] (resp. q_R) times. Here Read_f (f may relate to z) is simulated by the database read operation Read_f defined in Eq. (57). Define

$$P_{\text{left}} := \Pr\left[1 \leftarrow \mathcal{A}^{\mathsf{H},\mathsf{Read}_f}(z) : (S, z) \leftarrow \mathfrak{D}\right],$$
$$P_{\text{right}} := \Pr[1 \leftarrow \mathcal{A}^{\mathsf{H}\backslash S,\mathsf{Read}_f}(z) : (S, z) \leftarrow \mathfrak{D}],$$
$$P_{\text{find}} := \Pr[\mathsf{Find} : \mathcal{A}^{\mathsf{H}\backslash S,\mathsf{Read}_f}(z), (S, z) \leftarrow \mathfrak{D}].$$

Then we have

$$|P_{\text{left}} - P_{\text{right}}| \leq \sqrt{(q_\mathsf{H} + 1) \cdot P_{\text{find}}}, \quad \left|\sqrt{P_{\text{left}}} - \sqrt{P_{\text{right}}}\right| \leq \sqrt{(q_\mathsf{H} + 1) \cdot P_{\text{find}}}.$$

Define a projector on the database register D_q as $\mathrm{J}_S := \sum_{D\in S} |D\rangle\langle D|$, then the P_{find} defined above and the CStO defined in Definition 2 satisfy

$$P_{\text{find}} \leq q_\mathsf{H} \cdot \mathop{\mathbb{E}}_{(S,z)\leftarrow\mathfrak{D}} \|[\mathrm{J}_S, \mathsf{CStO}]\|^2.$$

[28] This limitation on q_H is because the database register D_q can only be used to perfectly simulate at most q quantum random oracle queries.

Now we prove Lemma 4. We first analyze the effect of the unitary operation $(\mathsf{U}_m^5)^\dagger \circ \mathsf{Out}_\mathsf{G} \circ \mathsf{U}_m^5$ used in game $\mathbf{G_5}$. Note that $(\mathsf{U}_m^5)^\dagger \circ \mathsf{Out}_\mathsf{G} \circ \mathsf{U}_m^5$ acts on the registers $\mathsf{C}_1\mathsf{C}_2\mathsf{MKY}_1\mathsf{M}_1\mathsf{D}_{q_\mathsf{F}}\mathsf{B}_1\mathsf{B}_2$, where $\mathsf{Y}_1\mathsf{M}_1\mathsf{MB}_1\mathsf{B}_2$ are the internal registers of qDeca^* and their state is always $|0^m, 0^m, 0^n, 0, 0\rangle$ just before and after a single simulation of qDeca^*. Hence for simplicity, we will omit the state of these internal registers when analyzing the effect of $(\mathsf{U}_m^5)^\dagger \circ \mathsf{Out}_\mathsf{G} \circ \mathsf{U}_m^5$.

In fact, based on the definition of U_m^5 given in Eq. (22) and the definition of Out_G given in Eq. (4), $(\mathsf{U}_m^5)^\dagger \circ \mathsf{Out}_\mathsf{G} \circ \mathsf{U}_m^5$ yields the following four results when acting on the computational basis state $|c_1, c_2, k, D_\mathsf{F}\rangle$ on the registers $\mathsf{C}_1\mathsf{C}_2\mathsf{KD}_{q_\mathsf{F}}$.

1. If $\mathsf{Dec}(sk, c_1) = \perp$, then $(\mathsf{U}_m^5)^\dagger \circ \mathsf{Out}_\mathsf{G} \circ \mathsf{U}_m^5 |c_1, c_2, k, D_\mathsf{F}\rangle = |c_1, c_2, k \oplus \perp, D_\mathsf{F}\rangle$.
2. If $\mathsf{Dec}(sk, c_1) = m_1 \neq \perp$, denote $m := c_2 \oplus \mathsf{H}(m_1)$. Then if $D_\mathsf{F}(m) = \perp$, we have $(\mathsf{U}_m^5)^\dagger \circ \mathsf{Out}_\mathsf{G} \circ \mathsf{U}_m^5 |c_1, c_2, k, D_\mathsf{F}\rangle = |c_1, c_2, k \oplus \perp, D_\mathsf{F}\rangle$.
3. If $\mathsf{Dec}(sk, c_1) = m_1 \neq \perp$, denote $m := c_2 \oplus \mathsf{H}(m_1)$. Then if $D_\mathsf{F}(m) \neq \perp$ and $c_1 \neq \mathsf{Enc}(pk, D_\mathsf{F}(m)) \vee c_2 \neq m \oplus \mathsf{H}(D_\mathsf{F}(m))$, we have $(\mathsf{U}_m^5)^\dagger \circ \mathsf{Out}_\mathsf{G} \circ \mathsf{U}_m^5 |c_1, c_2, k, D_\mathsf{F}\rangle = |c_1, c_2, k \oplus \perp, D_\mathsf{F}\rangle$.
4. If $\mathsf{Dec}(sk, c_1) = m_1 \neq \perp$, denote $m := c_2 \oplus \mathsf{H}(m_1)$. Then if $D_\mathsf{F}(m) \neq \perp$, $c_1 = \mathsf{Enc}(pk, D_\mathsf{F}(m))$ and $c_2 = m \oplus \mathsf{H}(D_\mathsf{F}(m))$, we have $(\mathsf{U}_m^5)^\dagger \circ \mathsf{Out}_\mathsf{G} \circ \mathsf{U}_m^5 |c_1, c_2, k, D_\mathsf{F}\rangle = |c_1, c_2, k \oplus \mathsf{G}(m), D_\mathsf{F}\rangle$.

Define a fixed function that needs to compute G and H as

$$\mathsf{Ext}\,(pk, sk, c_1, c_2, D_\mathsf{F}) = \begin{cases} \mathsf{G}(m) & \text{if } \mathsf{Dec}(sk, c_1) = m_1 \neq \perp \\ & \wedge D_\mathsf{F}(m) \neq \perp, \text{where } m := c_2 \oplus \mathsf{H}(m_1) \\ & \wedge c_1 = \mathsf{Enc}(pk, D_\mathsf{F}(m)) \\ & \wedge c_2 = m \oplus \mathsf{H}(D_\mathsf{F}(m)), \\ \perp & \text{otherwise.} \end{cases} \tag{58}$$

At this point, it is easy to check that $(\mathsf{U}_m^5)^\dagger \circ \mathsf{Out}_\mathsf{G} \circ \mathsf{U}_m^5$ can be rewritten as

$$(\mathsf{U}_m^5)^\dagger \circ \mathsf{Out}_\mathsf{G} \circ \mathsf{U}_m^5 : |c_1, c_2, k, D_\mathsf{F}\rangle \mapsto |c_1, c_2, k \oplus \mathsf{Ext}(pk, sk, c_1, c_2, D_\mathsf{F}), D_\mathsf{F}\rangle. \tag{59}$$

Hence similar to the $\mathsf{U}_\mathsf{search}$ defined in Eq. (25), $(\mathsf{U}_m^5)^\dagger \circ \mathsf{Out}_\mathsf{G} \circ \mathsf{U}_m^5$ can also be viewed as a database read operation with respect to a particular function.

Now by the above rewrite of $(\mathrm{U}_m^5)^\dagger \circ \mathsf{Out}_\mathsf{G} \circ \mathrm{U}_m^5$ and the definition of $\mathsf{U}_\mathsf{search}$, we can conclude that

$$(\mathsf{U}_m^5)^\dagger \circ \mathsf{Out}_\mathsf{G} \circ \mathsf{U}_m^5 |c_1, c_2, k, D_\mathsf{F}\rangle \neq \mathsf{U}_\mathsf{Search} |c_1, c_2, k, D_\mathsf{F}\rangle$$

must imply the D_F of $|c_1, c_2, y, D_\mathsf{F}\rangle$ satisfies $D_\mathsf{F} \in R^\mathsf{H}_{pk,sk}$, where

$$R^\mathsf{H}_{pk,sk} := \left\{ D \;\middle|\; D \in \mathbf{D}_{q_\mathsf{F}}, \begin{array}{l} \exists m \; s.t.\; D(m) \neq \perp, \\ c_1 := \mathsf{Enc}(pk, D(m)), c_2 := m \oplus \mathsf{H}(D(m)), \\ \text{but } \mathsf{Dec}(sk, c_1) = \perp, \\ \text{or } \mathsf{Dec}(sk, c_1) = m_1 \neq \perp \wedge c_2 \oplus \mathsf{H}(m_1) \neq m \end{array} \right\}. \tag{60}$$

Then based on the above set $R^\mathsf{H}_{pk,sk}$ and the games $\mathbf{G_5}$ and $\mathbf{G_6}$ shown in Fig. 12, we introduce two new games as follows:

Game $\mathbf{G_5^a}$: This game is the same as $\mathbf{G_5}$, except that it queries the compressed semi-classical oracle $\mathcal{O}^{CSC}_{R^{\mathsf{H}}_{pk,sk}}$ on the database register $\mathsf{D}_{q_\mathsf{F}}$ just after each F queries of the adversary $\mathcal{A}$.

Game $\mathbf{G_6^a}$: This game is the same as $\mathbf{G_6}$, except that it queries the compressed semi-classical oracle $\mathcal{O}^{CSC}_{R^{\mathsf{H}}_{pk,sk}}$ on the database register $\mathsf{D}_{q_\mathsf{F}}$ just after each F queries of the adversary $\mathcal{A}$.

In the following we use Theorem 14 to complete the proof. First we design a new quantum oracle algorithm $\mathcal{A}_1$ as follows to simulate game $\mathbf{G_5}$.

1. The input of $\mathcal{A}_1$ is $(\mathsf{G}, \mathsf{H}, pk, sk)$, where $\mathsf{G} : \{0,1\}^n \to \mathcal{K}$ and $\mathsf{H} : \mathcal{M} \to \{0,1\}^n$ are two fixed functions.
2. $\mathcal{A}_1$ has oracle access to F and Read_E, where F is a compressed standard oracle implemented using $\mathsf{D}_{q_\mathsf{F}}$ and Read_E is an oracle simulated by the $(\mathsf{U}^5_m)^\dagger \circ \mathsf{Out}_\mathsf{G} \circ \mathsf{U}^5_m$ as rewritten in Eq. (59).
3. $\mathcal{A}_1$ runs game $\mathbf{G_5}$ and outputs its final result.
   (a) For the G and H queries of game $\mathbf{G_5}$, $\mathcal{A}_1$ simulates them using its inputs G and H. For the F queries of game $\mathbf{G_5}$, $\mathcal{A}_1$ simulates them by querying its oracle F.
   (b) For the qDeca^* queries of game $\mathbf{G_5}$, $\mathcal{A}_1$ simulates them by implementing $\mathrm{U}_\perp \circ \mathrm{P}_{c_1^*,c_2^*} + \mathsf{Read}_\mathsf{E} \circ \left(\mathrm{I} - \mathrm{P}_{c_1^*,c_2^*}\right)$, which requires two queries to its oracle Read_E[29].

By the above construction, $\mathcal{A}_1$ queries F at most q_F times and Read_E at most $2q_\mathsf{D}$ times. Let $\mathfrak{D}$ be a distribution of the tuple $(R^{\mathsf{H}}_{pk,sk}, \mathsf{G}, \mathsf{H}, pk, sk)$, where $\mathsf{G} \leftarrow_\$ \Omega_{\{0,1\}^n \to \mathcal{K}}$, $\mathsf{H} \leftarrow_\$ \Omega_{\mathcal{M} \to \{0,1\}^n}$, $(pk, sk) \leftarrow \mathsf{Gen}$, and the set $R^{\mathsf{H}}_{pk,sk}$ (Eq. (60)) is completely determined by (pk, sk) and H. Now we have

$$\Pr\left[1 \leftarrow \mathbf{G_5}\right] = \Pr\left[1 \leftarrow \mathcal{A}_1^{\mathsf{F},\mathsf{Read}_\mathsf{E}}(\mathsf{G}, \mathsf{H}, pk, sk) : \left(R^{\mathsf{H}}_{pk,sk}, \mathsf{G}, \mathsf{H}, pk, sk\right) \leftarrow \mathfrak{D}\right],$$

$$\Pr\left[1 \leftarrow \mathbf{G_5^a}\right] = \Pr\left[1 \leftarrow \mathcal{A}_1^{\mathsf{F}\backslash R^{\mathsf{H}}_{pk,sk},\mathsf{Read}_\mathsf{E}}(\mathsf{G}, \mathsf{H}, pk, sk) : \left(R^{\mathsf{H}}_{pk,sk}, \mathsf{G}, \mathsf{H}, pk, sk\right) \leftarrow \mathfrak{D}\right].$$

It is not hard to find that, if we replace Read_E with the new oracle Read_S that is simulated by $\mathsf{U}_{\mathsf{search}}$ defined in Eq. (25), $\mathcal{A}_1$ will run game $\mathbf{G_6}$ and we get

$$\Pr\left[1 \leftarrow \mathbf{G_6}\right] = \Pr\left[1 \leftarrow \mathcal{A}_1^{\mathsf{F},\mathsf{Read}_\mathsf{S}}(\mathsf{G}, \mathsf{H}, pk, sk) : \left(R^{\mathsf{H}}_{pk,sk}, \mathsf{G}, \mathsf{H}, pk, sk\right) \leftarrow \mathfrak{D}\right],$$

$$\Pr\left[1 \leftarrow \mathbf{G_6^a}\right] = \Pr\left[1 \leftarrow \mathcal{A}_1^{\mathsf{F}\backslash R^{\mathsf{H}}_{pk,sk},\mathsf{Read}_\mathsf{S}}(\mathsf{G}, \mathsf{H}, pk, sk) : \left(R^{\mathsf{H}}_{pk,sk}, \mathsf{G}, \mathsf{H}, pk, sk\right) \leftarrow \mathfrak{D}\right].$$

According to the definitions of $(\mathsf{U}^5_m)^\dagger \circ \mathsf{Out}_\mathsf{G} \circ \mathsf{U}^5_m$ and $\mathsf{U}_{\mathsf{search}}$ given respectively in Eq. (59) and Eq. (25), these two unitary operations both can be viewed as

[29] This is obvious since $(\mathsf{U}^5_m)^\dagger \circ \mathsf{Out}_\mathsf{G} \circ \mathsf{U}^5_m$ rewritten in Eq. (59) is an involution.

database read operations with respect to a particular function related to the tuple $(\mathsf{G}, \mathsf{H}, pk, sk)$. Hence using Theorem 14 with $\mathcal{A}_1$, $\mathsf{F}, \mathsf{Read}_\mathsf{E}$ and $\mathfrak{D}$, we get

$$|\Pr[1 \leftarrow \mathbf{G_5}] - \Pr[1 \leftarrow \mathbf{G_5^a}]| \leq \sqrt{q_\mathsf{F}(q_\mathsf{F}+1) \cdot \mathbb{E}\left\|\left[\mathrm{J}_{R^\mathsf{H}_{pk,sk}}, \mathsf{CStO}\right]\right\|^2}. \tag{61}$$

Similarly, by using Theorem 14 with $\mathcal{A}_1$, $\mathsf{F}, \mathsf{Read}_\mathsf{S}$ and $\mathfrak{D}$, we get

$$|\Pr[1 \leftarrow \mathbf{G_6}] - \Pr[1 \leftarrow \mathbf{G_6^a}]| \leq \sqrt{q_\mathsf{F}(q_\mathsf{F}+1) \cdot \mathbb{E}\left\|\left[\mathrm{J}_{R^\mathsf{H}_{pk,sk}}, \mathsf{CStO}\right]\right\|^2}. \tag{62}$$

The expectation in the above equations are take over $(R^\mathsf{H}_{pk,sk}, \mathsf{G}, \mathsf{H}, pk, sk) \leftarrow \mathfrak{D}$.

In addition, by the analysis given in the beginning of this proof, we know that $(\mathsf{U}^5_m)^\dagger \circ \mathsf{Out}_\mathsf{G} \circ \mathsf{U}^5_m$ and $\mathsf{U}_\mathsf{search}$ have the same effect on $|c_1, c_2, k, D_\mathsf{F}\rangle$ if $D_\mathsf{F} \notin R^\mathsf{H}_{pk,sk}$. Hence, $\mathcal{A}_1^{\mathsf{F}\backslash R^\mathsf{H}_{pk,sk}, \mathsf{Read}_\mathsf{E}}(\mathsf{G}, pk, sk)$ and $\mathcal{A}_1^{\mathsf{F}\backslash R^\mathsf{H}_{pk,sk}, \mathsf{Read}_\mathsf{S}}(\mathsf{G}, pk, sk)$ have identical execution process if the compressed semi-classical oracle $\mathcal{O}^{CSC}_{R^\mathsf{H}_{pk,sk}}$ never has a measurement outcome of 1. This means that

$$\begin{aligned}
&\Pr\left[\mathsf{Find} : \mathcal{A}_1^{\mathsf{F}\backslash R^\mathsf{H}_{pk,sk}, \mathsf{Read}_\mathsf{E}}(\mathsf{G}, \mathsf{H}, pk, sk), \left(R^\mathsf{H}_{pk,sk}, \mathsf{G}, \mathsf{H}, pk, sk\right) \leftarrow \mathfrak{D}\right] \\
=&\Pr\left[\mathsf{Find} : \mathcal{A}_1^{\mathsf{F}\backslash R^\mathsf{H}_{pk,sk}, \mathsf{Read}_\mathsf{S}}(\mathsf{G}, \mathsf{H}, pk, sk), \left(R^\mathsf{H}_{pk,sk}, \mathsf{G}, \mathsf{H}, pk, sk\right) \leftarrow \mathfrak{D}\right]
\end{aligned}$$

and

$$\begin{aligned}
&|\Pr[1 \leftarrow \mathbf{G_5^a}] - \Pr[1 \leftarrow \mathbf{G_6^a}]| \\
\leq&\Pr\left[\mathsf{Find} : \mathcal{A}_1^{\mathsf{F}\backslash R^\mathsf{H}_{pk,sk}, \mathsf{Read}_\mathsf{S}}(\mathsf{G}, \mathsf{H}, pk, sk), \left(R^\mathsf{H}_{pk,sk}, \mathsf{G}, \mathsf{H}, pk, sk\right) \leftarrow \mathfrak{D}\right] \\
\overset{(a)}{\leq}& q_\mathsf{F} \cdot \mathbb{E}\left\|\left[\mathrm{J}_{R^\mathsf{H}_{pk,sk}}, \mathsf{CStO}\right]\right\|^2.
\end{aligned} \tag{63}$$

Here (a) uses Theorem 14. Now by combining Eq. (61), Eq. (62), and Eq. (63), we get

$$\begin{aligned}
|\Pr[1 \leftarrow \mathbf{G_5}] - \Pr[1 \leftarrow \mathbf{G_6}]| \leq\ & 2 \cdot \sqrt{q_\mathsf{F}(q_\mathsf{F}+1) \cdot \mathbb{E}\left\|\left[\mathrm{J}_{R^\mathsf{H}_{pk,sk}}, \mathsf{CStO}\right]\right\|^2} \\
&+ q_\mathsf{F} \cdot \mathbb{E}\left\|\left[\mathrm{J}_{R^\mathsf{H}_{pk,sk}}, \mathsf{CStO}\right]\right\|^2.
\end{aligned} \tag{64}$$

The expectation in the above equations is taken over $(R^\mathsf{H}_{pk,sk}, \mathsf{G}, \mathsf{H}, pk, sk) \leftarrow \mathfrak{D}$.

Next we compute the upper bound of $\|[\mathrm{J}_{R^\mathsf{H}_{pk,sk}}, \mathsf{CStO}]\|$. For fixed (pk, sk) and function $\mathsf{G} : \{0,1\}^n \to \mathcal{K}$, we define function $g : \{0,1\}^n \times \mathcal{M} \to \{0,1\}$ as

$$g(x,y) = \begin{cases} 1 & \mathsf{Dec}(sk, c_1) = \bot \\ & \text{or } \mathsf{Dec}(sk, c_1) = y_1 \neq \bot \text{ but } c_2 \oplus \mathsf{H}(y_1) \neq x \\ & \text{where } c_1 := \mathsf{Enc}(pk, y) \wedge c_2 := x \oplus \mathsf{H}(y), \\ 0 & \text{otherwise.} \end{cases} \tag{65}$$

Obviously $g(m,r)=1$ if and only if $\mathsf{Dec}_0(sk,\mathsf{Enc}_0(pk,m;r))\neq m$, where Enc_0 and Dec_0 are respectively the encryption and decryption algorithms of the PKE scheme $\mathsf{ACWC}_0[\mathsf{P},\mathsf{H}]$, which is defined in Fig. 8. Therefore g can be rewritten as

$$g(x,y)=\begin{cases}1 & \text{if } \mathsf{Dec}_0(sk,\mathsf{Enc}_0(pk,x;y))\neq x\\ 0 & \text{otherwise.}\end{cases} \tag{66}$$

Based on function g, we define a relation R_1^g and a corresponding parameter $\Gamma_{R_1^g}$ as follows:

$$R_1^g:=\{(x,y)\in\{0,1\}^n\times\mathcal{M}|g(x,y)=1\},\Gamma_{R_1^g}:=\max_{x\in\{0,1\}^n}|\{y\in\mathcal{M}|g(x,y)=1\}|.$$

Furthermore, we define projectors related to R_1^g on the database register $\mathsf{D}_{q_\mathsf{F}}$:

$$\Sigma^x:=\sum_{\substack{D\ s.t.\ (x,D(x))\in R_1^g\\ x'<x,(x',D(x'))\notin R_1^g}}|D\rangle\langle D|\ (x\in\{0,1\}^n),\ \ \Sigma^\perp:=\mathrm{I}-\sum_{x\in\{0,1\}^n}\Sigma^x.$$

By the definition of $R^{\mathsf{H}}_{pk,sk}$ given in Eq. (60) and the definition of g given in Eq. (65) and Eq. (66), it is not hard to verify that $\mathrm{J}_{R^{\mathsf{H}}_{pk,sk}}=\sum_{x\in\{0,1\}^n}\Sigma^x$, and thus $\Sigma^\perp=\mathrm{I}-\mathrm{J}_{R^{\mathsf{H}}_{pk,sk}}$. Then we have

$$\left\|\left[\mathrm{J}_{R^{\mathsf{H}}_{pk,sk}},\mathsf{CStO}\right]\right\|\overset{(a)}{=}\left\|\left[\mathrm{I}-\mathrm{J}_{R^{\mathsf{H}}_{pk,sk}},\mathsf{CStO}\right]\right\|=\left\|\left[\Sigma^\perp,\mathsf{CStO}\right]\right\|\overset{(b)}{\leq}8\cdot\sqrt{\Gamma_{R_1^g}/|\mathcal{M}|}.$$

Here (a) follows from the basic property of commutor, and (b) uses [16, Lemma 4]. Denote δ^{0}_{wc} as the worst-case decryption error of $\mathsf{ACWC}_0[\mathsf{P},\mathsf{H}]$, we can compute

$$\begin{aligned}\mathop{\mathbb{E}}_{\left(R^{\mathsf{H}}_{pk,sk},\mathsf{G},\mathsf{H},pk,sk\right)\leftarrow\mathfrak{D}}\left\|\left[\mathrm{J}_{R^{\mathsf{H}}_{pk,sk}},\mathsf{CStO}\right]\right\|^2&\leq 64\cdot\mathop{\mathbb{E}}_{\substack{(pk,sk)\leftarrow\mathsf{Gen}\\ \mathsf{H}\leftarrow_{\$}\Omega_{\mathcal{M}\to\{0,1\}^n}}}\Gamma_{R_1^g}/|\mathcal{M}|\\ &\overset{(a)}{\leq}64\cdot\delta^{\mathsf{0}}_{wc}\overset{(b)}{=}64\cdot\delta_{ac}.\end{aligned}\tag{67}$$

Here (a) follows from the definitions of $\Gamma_{R_1^g}$ and δ^{0}_{wc}. (b) follows from [12, Lemma 3.1], which shows that the worst-case decryption error of $\mathsf{ACWC}_0[\mathsf{P},\mathsf{H}]$ and the average-case decryption error of P (i.e., δ_{ac}) are equal. Finally, by combining Eq. (64) and Eq. (67), we get

$$|\Pr[1\leftarrow\mathbf{G_5}]-\Pr[1\leftarrow\mathbf{G_6}]|\leq 16\cdot\sqrt{q_\mathsf{F}(q_\mathsf{F}+1)\cdot\delta_{ac}}+64q_\mathsf{F}\cdot\delta_{ac}.$$

This completes the proof of Lemma 4. □

I Proof of Lemma 5

In order to prove Lemma 5, we introduce a sequence of games $\mathbf{G_6^a}$ to $\mathbf{G_6^d}$ as shown in Fig. 19.

GAMES $\mathbf{G_6^a}$-$\mathbf{G_6^d}$	
1. $(pk, sk) \leftarrow \mathsf{Gen}$	$//\mathbf{G_6^a}$-$\mathbf{G_6^d}$
$\quad \mathsf{G} \leftarrow_\$ \Omega_{\{0,1\}^n \to \mathcal{K}}$	$//\mathbf{G_6^a}$-$\mathbf{G_6^d}$
$\quad \mathsf{H} \leftarrow_\$ \Omega_{\mathcal{M} \to \{0,1\}^n}$	$//\mathbf{G_6^a}$-$\mathbf{G_6^d}$
2. $m^* \leftarrow_\$ \{0,1\}^n$	$//\mathbf{G_6^a}$-$\mathbf{G_6^d}$
$\quad m_1^* \leftarrow_\$ \mathcal{M}$	$//\mathbf{G_6^a}$-$\mathbf{G_6^d}$
$\quad$ **perform** $\mathsf{StdDecomp}_{m^*} \circ \mathsf{U}_{m_1^*}^{m^*}$	$//\mathbf{G_6^a}$-$\mathbf{G_6^c}$
$\quad c_1^* := \mathsf{Enc}(pk, m_1^*)$	$//\mathbf{G_6^a}$-$\mathbf{G_6^d}$
$\quad c_2^* := m^* \oplus \mathsf{H}(m_1^*)$	$//\mathbf{G_6^a}$-$\mathbf{G_6^d}$
3. $b \leftarrow_\$ \{0,1\}$	$//\mathbf{G_6^a}$-$\mathbf{G_6^d}$
$\quad K_0^* := \mathsf{G}(m^*), K_1^* \leftarrow_\$ \mathcal{K}$	$//\mathbf{G_6^a}$-$\mathbf{G_6^d}$
4. $b' \leftarrow \mathcal{A}^{\mathsf{F},\mathsf{G},\mathsf{H},\mathsf{qDeca}^*}(pk, c_1^*, c_2^*, K_b^*)$	$//\mathbf{G_6^a}$-$\mathbf{G_6^d}$
5. **return** $[\![b' = b]\!]$	$//\mathbf{G_6^a}$-$\mathbf{G_6^d}$

$\mathsf{F}(\|x, y\rangle)$	
1. **perform** CStO	$//\mathbf{G_6^a}$-$\mathbf{G_6^b}$
2. **perform** $\mathsf{CStO}_{m_1^*}^{m^*}$	$//\mathbf{G_6^c}$-$\mathbf{G_6^d}$
$\mathsf{G}(\|x, y\rangle)$	
1. **return** $\|x, y \oplus \mathsf{G}(x)\rangle$	$//\mathbf{G_6^a}$-$\mathbf{G_6^d}$
$\mathsf{H}(\|x, y\rangle)$	
1. **return** $\|x, y \oplus \mathsf{H}(x)\rangle$	$//\mathbf{G_6^a}$-$\mathbf{G_6^d}$
$\mathsf{qDeca}^*(\|c_1, c_2, y\rangle)$	
1. **perform** $\mathsf{U}_{\mathsf{qD}}^{pk}$	$//\mathbf{G_6^a}, \mathbf{G_6^d}$
2. **perform** $\mathsf{U}_{pk,\mathsf{qD}}^{\mathsf{SUS}}$	$//\mathbf{G_6^b}$
3. **perform** $\mathsf{U}_{pk,\mathsf{qD}}^{\mathsf{SUUS}}$	$//\mathbf{G_6^c}$

Fig. 19. Games $\mathbf{G_6^a}$ to $\mathbf{G_6^d}$ in the proof of Lemma 5.

Game $\mathbf{G_6^a}$: This game is the same as $\mathbf{G_6}$ (Fig. 12), except that $m_1^* = \mathsf{F}(m^*)$ is replaced with $m_1^* \leftarrow_\$ \mathcal{M}$ and $\mathsf{StdDecomp}_{m^*} \circ \mathsf{U}_{m_1^*}^{m^*}$ is performed on the database register $\mathsf{D}_{q_\mathsf{F}}$ just after the sampling of m_1^*. Here $\mathsf{StdDecomp}_{m^*}$ is the local decompression procedure (Section 2.3) and $\mathsf{U}_{m_1^*}^{m^*}$ is an involution defined as

$$\mathsf{U}_{m_1^*}^{m^*} : |D_\mathsf{F}\rangle \mapsto \begin{cases} |D_\mathsf{F} \cup (m^*, m_1^*)\rangle & \text{if } D_\mathsf{F}(m^*) = \bot, \\ |D_\mathsf{F} \backslash (m^*, m_1^*)\rangle & \text{if } D_\mathsf{F}(m^*) = m_1^*, \\ |D_\mathsf{F}\rangle & \text{otherwise.} \end{cases} \tag{68}$$

In game $\mathbf{G_6}$, the $\mathsf{F}(m^*)$ used to generate m_1^* must be the reply of the first query to F, and this query is also classical. Now, by the definition of the compressed standard oracle given in Definition 2, the simulation process of $\mathsf{F}(m^*)$ can be described as follows:

1. Since this query is the first query to F, the initial state of the database register $\mathsf{D}_{q_\mathsf{F}}$ is $|D_\bot\rangle$, where $D^\bot$ is a database containing q_F $(\bot, 0^n)$ pairs.
2. Initialize the state on $\mathsf{X_F Y_F}$ as $|m^*, 0^m\rangle$, where $\mathsf{X_F}/\mathsf{Y_F}$ is the input/output register of F. Now, the joint state on the registers $\mathsf{X_F Y_F D}_{q_\mathsf{F}}$ is $|m^*, 0^m\rangle|D_\bot\rangle$.

3. Perform $\mathsf{StdDecomp}_{m^*}$ on the database register $\mathsf{D}_{q_\mathsf{F}}$, and then the joint state is transformed into $|m^*, 0^m\rangle \sum_{y\in\{0,1\}^m} \frac{1}{\sqrt{2^m}} |D_\perp \cup (m^*, y)\rangle$.
4. Perform CNOT^{m^*} on the registers $\mathsf{Y}_\mathsf{F}\mathsf{D}_{q_\mathsf{F}}$, and then the joint state is transformed into $\sum_{y\in\{0,1\}^m} \frac{1}{\sqrt{2^m}} |m^*, y\rangle |D_\perp \cup (m^*, y)\rangle$.
5. Perform $\mathsf{StdDecomp}_{m^*}$ on the database register $\mathsf{D}_{q_\mathsf{F}}$, and then the joint state is transformed into

$$|\psi\rangle := \sum_{y\in\{0,1\}^m} \frac{1}{\sqrt{2^m}} |m^*, y\rangle \mathsf{StdDecomp}_{m^*} |D_\perp \cup (m^*, y)\rangle.$$

6. Measure the registers $\mathsf{X}_\mathsf{F}\mathsf{Y}_\mathsf{F}$ and obtain the outcome (m^*, y). Here y is taken as $\mathsf{F}(m^*)$ and the state $|\psi\rangle$ collapses into $\mathsf{StdDecomp}_{m^*} |D_\perp \cup (m^*, m_1^*)\rangle$[30] after obtaining (m^*, y).

Notice that $\mathsf{StdDecomp}_{m^*}$ is a unitary operation, and hence the value $y := \mathsf{F}(m^*)$ (also m_1^* in game $\mathbf{G_6}$) obtained in the above final step must be a uniformly random value over $\mathcal{M}(= \{0,1\}^m)$.

As for the game $\mathbf{G_6^a}$, it is easy to check that the state on the database register $\mathsf{D}_{q_\mathsf{F}}$ is $\mathsf{StdDecomp}_{m^*} |D_\perp \cup (m^*, m_1^*)\rangle$ just after performing $\mathsf{StdDecomp}_{m^*} \circ \mathsf{U}_{m_1^*}^{m^*}$, and the m_1^* here is also a uniformly random value over $\mathcal{M}$, but it is chosen from the beginning rather than obtained through quantum measurement. This means that game $\mathbf{G_6^a}$ is actually an equivalent rewrite of game $\mathbf{G_6}$, and thus

$$\Pr[1 \leftarrow \mathbf{G_6}] = \Pr[1 \leftarrow \mathbf{G_6^a}]. \tag{69}$$

Game $\mathbf{G_6^b}$: This game is the same as $\mathbf{G_6^a}$, except that the $\mathsf{U}_{\mathsf{qD}}^{pk}$ used to simulate qDeca^* is replaced with

$$\mathsf{U}_{pk,\mathsf{qD}}^{\mathsf{SUS}} := \mathrm{U}_\perp \circ \mathrm{P}_{c_1^*,c_2^*} + \mathsf{StdDecomp}_{m^*} \circ \mathsf{U}_{\mathsf{search}} \circ \mathsf{StdDecomp}_{m^*} \circ \left(\mathrm{I} - \mathrm{P}_{c_1^*,c_2^*}\right). \tag{70}$$

Compared with the game $\mathbf{G_6^a}$, game $\mathbf{G_6^b}$ only replaces the $\mathsf{U}_{\mathsf{search}}$ used by $\mathsf{U}_{\mathsf{qD}}^{pk}$ with $\mathsf{StdDecomp}_{m^*} \circ \mathsf{U}_{\mathsf{search}} \circ \mathsf{StdDecomp}_{m^*}$. As for the difference between the success probabilities of these two games, we can prove the following lemma.

Lemma I1 $|\Pr[1 \leftarrow \mathbf{G_6^a}] - \Pr[1 \leftarrow \mathbf{G_6^b}]| \le 2q_\mathsf{D} \cdot \sqrt{2^{-m}}$.

Proof. In this proof, we first consider a fixed (pk, sk) pair and fixed functions $\mathsf{G} : \{0,1\}^n \to \mathcal{K}$, $\mathsf{H} : \mathcal{M} \to \{0,1\}^n$. In addition, let m^*, m_1^*, b, and K_1^* be fixed values in $\{0,1\}^n$, $\mathcal{M}$, $\{0,1\}$ and $\mathcal{K}$, respectively. Then we construct a quantum algorithm $\mathcal{A}_1$ as follows:

1. $\mathcal{A}_1$ has classical input $(\mathsf{G}, \mathsf{H}, pk, sk, m^*, m_1^*, b, K_1^*)$ and runs on the registers $\mathsf{A}\mathsf{D}_{q_\mathsf{F}}$. Here A denotes the register used by $\mathcal{A}_1$ for internal computation and $\mathsf{D}_{q_\mathsf{F}}$ is the database register with initial state $|D_\perp\rangle$, where $D^\perp$ is a database containing q_F $(\perp, 0^n)$ pairs.

[30] We omit the state $|m^*, y\rangle$ of the registers $\mathsf{X}_\mathsf{F}\mathsf{Y}_\mathsf{F}$ since they have been measured.

2. $\mathcal{A}_1$ performs $\mathsf{StdDecomp}_{m^*} \circ \mathsf{U}_{m_1^*}^{m^*}$ on $\mathsf{D}_{q_\mathsf{F}}$ to get the state $\mathsf{StdDecomp}_{m^*}|D_\perp \cup (m^*, m_1^*)\rangle$.
3. Let $c_1^* := \mathsf{Enc}(pk, m_1^*)$, $c_2^* := m^* \oplus \mathsf{H}(m_1^*)$, and $K_0^* := \mathsf{G}(m^*)$. Then $\mathcal{A}_1$ runs $\mathcal{A}^{\mathsf{F},\mathsf{G},\mathsf{H},\mathsf{qDeca}^*}(pk, c_1^*, c_2^*, K_b^*)$ and simulates its oracles as follows:
   (a) $\mathcal{A}_1$ simulates F as a compressed standard oracle based on the database register $\mathsf{D}_{q_\mathsf{F}}$. For the G and H queries of $\mathcal{A}$, $\mathcal{A}_1$ simulates them using its input G and H.
   (b) $\mathcal{A}_1$ simulates qDeca^* by implementing the $\mathsf{U}_{\mathsf{qD}}^{pk}$ defined in Eq. (24).
4. After $\mathcal{A}$ outputs b', $\mathcal{A}_1$ outputs $[\![b = b']\!]$.

It is obvious that $\mathcal{A}_1$ runs game $\mathbf{G_6^a}$ under a fixed $(\mathsf{G}, \mathsf{H}, pk, sk, m^*, m_1^*, b, K_1^*)$, and $\mathcal{A}_1$ will run game $\mathbf{G_6^b}$ if we replace the $\mathsf{U}_{\mathsf{qD}}^{pk}$ used to simulate qDeca^* with $\mathsf{U}_{pk,\mathsf{qD}}^{\mathsf{SUS}}$ (Eq. (70)). Therefore, let $\mathcal{A}_2$ be a quantum algorithm the same as $\mathcal{A}_1$, except that it uses $\mathsf{U}_{pk,\mathsf{qD}}^{\mathsf{SUS}}$ to simulate qDeca^* for $\mathcal{A}$, we have

$$\Pr\left[1 \leftarrow \mathbf{G_6^a} : z\right] = \Pr\left[1 \leftarrow \mathcal{A}_1(z)\right], \ \Pr\left[1 \leftarrow \mathbf{G_6^b} : z\right] = \Pr\left[1 \leftarrow \mathcal{A}_2(z)\right]. \tag{71}$$

Here and in the following, we denote $z := (\mathsf{G}, \mathsf{H}, pk, sk, m^*, m_1^*, b, K_1^*)$ for the sake of simplicity, and we denote $\Pr[1 \leftarrow \mathbf{G} : z]$ as the probability that game $\mathbf{G}$ outputs 1 under fixed $(\mathsf{G}, \mathsf{H}, pk, sk, m^*, m_1^*, b, K_1)$.

Without loss of generality, we assume that $\mathcal{A}^{\mathsf{F},\mathsf{G},\mathsf{H},\mathsf{qDeca}^*}(pk, c_1^*, c_2^*, K_b^*)$ is a unitary quantum oracle algorithm. At this point, the joint state of $\mathcal{A}_1$ just before the final binary measurement $\{\mathrm{M}_0, \mathrm{M}_1\}$, which is used to get the output b' of $\mathcal{A}$, can be written as

$$|\Psi_1\rangle := \mathrm{U}_{q_\mathsf{D}} \circ \mathsf{U}_{\mathsf{qD}}^{pk} \cdots \mathrm{U}_2 \circ \mathsf{U}_{\mathsf{qD}}^{pk} \circ \mathrm{U}_1 \circ \mathsf{U}_{\mathsf{qD}}^{pk} |\psi_z\rangle \otimes \mathsf{StdDecomp}_{m^*}|D_\perp \cup (m^*, m_1^*)\rangle.$$

Similarly, $|\Psi_2\rangle$ denotes the corresponding joint state of $\mathcal{A}_2$, which is

$$\mathrm{U}_{q_\mathsf{D}} \circ \mathsf{U}_{pk,\mathsf{qD}}^{\mathsf{SUS}} \cdots \mathrm{U}_2 \circ \mathsf{U}_{pk,\mathsf{qD}}^{\mathsf{SUS}} \circ \mathrm{U}_1 \circ \mathsf{U}_{pk,\mathsf{qD}}^{\mathsf{SUS}} |\psi_z\rangle \otimes \mathsf{StdDecomp}_{m^*}|D_\perp \cup (m^*, m_1^*)\rangle. \tag{72}$$

Here $|\psi_z\rangle$ denotes the initial state on A, and $\mathsf{StdDecomp}_{m^*}|D_\perp \cup (m^*, m_1^*)\rangle$ is the initial state of $\mathsf{D}_{q_\mathsf{F}}$ since $\mathcal{A}_1$ performs $\mathsf{StdDecomp}_{m^*} \circ \mathsf{U}_{m_1^*}^{m^*}$ in its second step. $\mathrm{U}_1, \ldots, \mathrm{U}_{q_\mathsf{D}}$ here denote the unitary operations performed on $\mathsf{AD}_{q_\mathsf{F}}$ between each queries to qDeca^*. More specifically, $\mathrm{U}_1, \ldots, \mathrm{U}_{q_\mathsf{D}}$ are all composites of the unitary operations acting on A and three particular unitary operations: $\mathsf{O}_\mathsf{G} : |x, y\rangle \mapsto |x, y \oplus \mathsf{G}(x)\rangle$, $\mathsf{O}_\mathsf{H} : |x, y\rangle \mapsto |x, y \oplus \mathsf{G}(x)\rangle$ and CStO (Definition 2). In fact, O_G, O_H, and CStO are used by $\mathcal{A}_1$ and $\mathcal{A}_2$ to simulate the oracles G, H, and F for $\mathcal{A}$, respectively, and O_G and O_H can also be viewed as unitary operations acting on A, as we (implicitly) set the input and output registers of G and H as internal registers of $\mathcal{A}$. Now let $\mathrm{M}_b := |b\rangle\langle b|$, we have

$$\Pr[1 \leftarrow \mathcal{A}_1(z)] = \|\mathrm{M}_b|\Psi_1\rangle\|^2, \ \Pr[1 \leftarrow \mathcal{A}_2(z)] = \|\mathrm{M}_b|\Psi_2\rangle\|^2. \tag{73}$$

Next we define a special class of states that will be used later. For any state $|\psi\rangle$ on the registers $\mathsf{AD}_{q_\mathsf{F}}$, we say it is (m^*, m_1^*)-fixed if it can be written as

$$|\psi\rangle = \sum_{a, D_\mathsf{F}} \alpha_a^{D_\mathsf{F}} |a\rangle \otimes \mathsf{StdDecomp}_{m^*}|D_\mathsf{F} \cup (m^*, m_1^*)\rangle. \tag{74}$$

That is, the superposed database state must be $\mathsf{StdDecomp}_{m^*}|D_\mathsf{F} \cup (m^*, m_1^*)\rangle$ for a fixed (m^*, m_1^*) pair. Then we introduce the following corollary related to the (m^*, m_1^*)-fixed states. Its full proof is given in Supplementary Material I.1.

Corollary 1. *For any (m^*, m_1^*)-fixed state $|\psi\rangle$ of the registers $\mathsf{AD}_{q_\mathsf{F}}$, $\mathsf{CStO}|\psi\rangle$, $\mathrm{U}_1|\psi\rangle, \ldots, \mathrm{U}_{q_\mathsf{D}}|\psi\rangle$, and $\mathsf{U}^{\mathsf{SUS}}_{pk,\mathsf{qD}}|\psi\rangle$ are still (m^*, m_1^*)-fixed states.*

In addition, we can also prove the following corollary, which shows that $\mathsf{U}^{pk}_{\mathsf{qD}}$ and $\mathsf{U}^{\mathsf{SUS}}_{pk,\mathsf{qD}}$ have nearly the same effect when acting on the (m^*, m_1^*)-fixed states. The detailed proof of this corollary is given in Supplementary Material I.2.

Corollary 2. *For any unit (m^*, m_1^*)-fixed state $|\psi\rangle$ of the registers $\mathsf{AD}_{q_\mathsf{F}}$, we have $\|\mathsf{U}^{pk}_{\mathsf{qD}}|\psi\rangle - \mathsf{U}^{\mathsf{SUS}}_{pk,\mathsf{qD}}|\psi\rangle\| \le 2/\sqrt{2^m}$.*

Based on the states $|\Psi_1\rangle$ and $|\Psi_2\rangle$ introduced earlier, we define the following internal states for $0 \le i \le q_\mathsf{D}$.

$$\begin{aligned}|\Psi_{2,i}\rangle := \mathrm{U}_{q_\mathsf{D}} \circ \mathsf{U}^{pk}_{\mathsf{qD}} \cdots \mathrm{U}_{q_\mathsf{D}-i+1} \circ \mathsf{U}^{pk}_{\mathsf{qD}} \\ \circ\, \mathrm{U}_{q_\mathsf{D}-i} \circ \mathsf{U}^{\mathsf{SUS}}_{pk,\mathsf{qD}} \cdots \mathrm{U}_1 \circ \mathsf{U}^{\mathsf{SUS}}_{pk,\mathsf{qD}}|\psi_z\rangle \otimes \mathsf{StdDecomp}_{m^*}|D_\perp \cup (m^*, m_1^*)\rangle.\end{aligned}$$

The special property of $|\Psi_{2,i}\rangle$ is that, the first $q_\mathsf{D} - i$ queries of qDeca^* is simulated by $\mathsf{U}^{\mathsf{SUS}}_{pk,\mathsf{qD}}$, and the last i queries of qDeca^* is simulated by $\mathsf{U}^{pk}_{\mathsf{qD}}$ instead. Obviously we have $|\Psi_{2,0}\rangle = |\Psi_2\rangle$ and $|\Psi_{2,q_\mathsf{D}}\rangle = |\Psi_1\rangle$, and for $0 \le i \le q_\mathsf{D} - 1$ we can compute

$$\begin{aligned}\||\Psi_{2,i}\rangle - |\Psi_{2,i+1}\rangle\| &= \left\| \begin{matrix} (\mathsf{U}^{\mathsf{SUS}}_{pk,\mathsf{qD}} - \mathsf{U}^{pk}_{\mathsf{qD}}) \circ \mathrm{U}_{q_\mathsf{D}-i-1} \circ \mathsf{U}^{\mathsf{SUS}}_{pk,\mathsf{qD}} \cdots \mathrm{U}_1 \circ \mathsf{U}^{\mathsf{SUS}}_{pk,\mathsf{qD}} \\ |\psi_z\rangle \otimes \mathsf{StdDecomp}_{m^*}|D_\perp \cup (m^*, m_1^*)\rangle \end{matrix} \right\| \\ &\overset{(a)}{\le} 2/\sqrt{2^m}.\end{aligned}$$

In fact, based on the definition of the (m^*, m_1^*)-fixed states given in Eq. (74), we know that $|\psi_z\rangle \otimes \mathsf{StdDecomp}_{m^*}|D_\perp \cup (m^*, m_1^*)\rangle$ is a (m^*, m_1^*)-fixed state, and this state is unit. Then by Corollary 1, $\mathrm{U}_{q_\mathsf{D}-i-1} \circ \mathsf{U}^{\mathsf{SUS}}_{pk,\mathsf{qD}} \cdots \mathrm{U}_1 \circ \mathsf{U}^{\mathsf{SUS}}_{pk,\mathsf{qD}}|\psi_z\rangle \otimes \mathsf{StdDecomp}_{m^*}|D_\perp \cup (m^*, m_1^*)\rangle$ is also a unit (m^*, m_1^*)-fixed state. Hence the above inequality (a) is a direct application of Corollary 2.

By Combining the above equation with the fact that $|\Psi_{2,0}\rangle = |\Psi_2\rangle$ and $|\Psi_{2,q_\mathsf{D}}\rangle = |\Psi_1\rangle$, we have

$$\||\Psi_1\rangle - |\Psi_2\rangle\| \le 2q_\mathsf{D}/\sqrt{2^m} \tag{75}$$

by the triangle equality. Then by combining Eq. (71), Eq. (73), Eq. (75), and [1, Lemma 4], we get

$$\left|\Pr\left[1 \leftarrow \mathbf{G^a_6} : z\right] - \Pr\left[1 \leftarrow \mathbf{G^b_6} : z\right]\right| \le 2q_\mathsf{D}/\sqrt{2^m}.$$

Here $z = (\mathsf{G}, \mathsf{H}, pk, sk, m^*, m_1^*, b, K_1^*)$. Finally, by averaging over z, where $\mathsf{G} \leftarrow_\$ \Omega_{\{0,1\}^n \to \mathcal{K}}$, $\mathsf{H} \leftarrow_\$ \Omega_{\mathcal{M} \to \{0,1\}^n}$, $(pk, sk) \leftarrow \mathsf{Gen}$, $m^* \leftarrow_\$ \{0,1\}^n$, $m_1^* \leftarrow_\$ \mathcal{M}$, $b \leftarrow_\$ \{0,1\}$ and $K_1^* \leftarrow_\$ \mathcal{K}$, we obtain

$$\left|\Pr\left[1 \leftarrow \mathbf{G^a_6}\right] - \Pr\left[1 \leftarrow \mathbf{G^b_6}\right]\right| \le 2q_\mathsf{D}/\sqrt{2^m}.$$

This completes the proof of Lemma I1. $\square$

Game $\mathbf{G_6^c}$: This game is the same as $\mathbf{G_6^b}$, except for two differences. First, the CStO used to simulate F in game $\mathbf{G_6^b}$ is replaced with

$$\begin{aligned}\mathsf{CStO}_{m_1^*}^{m^*} := &\sum_{x\in\{0,1\}^n\setminus\{m^*\}} |x\rangle\langle x| \otimes \mathsf{StdDecomp}_x \circ \mathsf{CNOT}^x \circ \mathsf{StdDecomp}_x \\ &+ |m^*\rangle\langle m^*| \otimes \mathsf{CNOT}_{m_1^*}.\end{aligned} \tag{76}$$

Here $\mathsf{CNOT}_{m_1^*} : |y\rangle \mapsto |y \oplus m_1^*\rangle$ acts on the output register of F. That is to say, for the adversary's m^* queries to F, game $\mathbf{G_6^c}$ no longer simulates the compressed standard oracle but directly writes m_1^* on the output register of F instead. Second, the $\mathsf{U}_{pk,\mathsf{qD}}^{\mathsf{SUS}}$ (Eq. (70)) used to simulate qDeca^* in game $\mathbf{G_6^b}$ is replaced with

$$\begin{aligned}\mathsf{U}_{pk,\mathsf{qD}}^{\mathsf{SUUS}} :=& \mathrm{U}_\perp \circ \mathrm{P}_{c_1^*,c_2^*} \\ &+ \mathsf{StdDecomp}_{m^*} \circ \mathsf{U}_{m_1^*}^{m^*} \circ \mathsf{U}_{\mathsf{Search}} \circ \mathsf{U}_{m_1^*}^{m^*} \circ \mathsf{StdDecomp}_{m^*} \circ \left(\mathrm{I} - \mathrm{P}_{c_1^*,c_2^*}\right).\end{aligned} \tag{77}$$

We point out that $\mathsf{U}_{pk,\mathsf{qD}}^{\mathsf{SUUS}}$ defined above can also be rewritten as $\mathsf{StdDecomp}_{m^*} \circ \mathsf{U}_{m_1^*}^{m^*} \circ \mathsf{U}_{\mathsf{qD}}^{pk} \circ \mathsf{U}_{m_1^*}^{m^*} \circ \mathsf{StdDecomp}_{m^*}$ by means of the $\mathsf{U}_{\mathsf{qD}}^{pk}$ defined in Eq. (24).

Next we prove that game $\mathbf{G_6^c}$ is identical with game $\mathbf{G_6^b}$. First, similar to the proof of Lemma I1, we consider a fixed tuple $z := (\mathsf{G}, \mathsf{H}, pk, sk, m^*, m_1^*, b, K_1^*)$ and rewrite game $\mathbf{G_6^b}$ as quantum algorithm $\mathcal{A}_2$ (Eq. (71)) under such z. Then as shown in Eq. (72), the joint state of $\mathcal{A}_2$ on the registers $\mathsf{AD}_{q_\mathsf{F}}$ just before the final measurement can be written as

$$\mathrm{U}_{q_\mathsf{D}} \circ \mathsf{U}_{pk,\mathsf{qD}}^{\mathsf{SUS}} \cdots \mathrm{U}_2 \circ \mathsf{U}_{pk,\mathsf{qD}}^{\mathsf{SUS}} \circ \mathrm{U}_1 \circ \mathsf{U}_{pk,\mathsf{qD}}^{\mathsf{SUS}} |\psi_z\rangle \otimes \mathsf{StdDecomp}_{m^*} |D_\perp \cup (m^*, m_1^*)\rangle,$$

where $\mathrm{U}_1, \ldots, \mathrm{U}_{q_\mathsf{D}}$ are all composites of the unitary operations acting on A and the unitary operation CStO. Here CStO is used to simulate F and $\mathcal{A}$ queries F at most q_F times. Based on this fact, we can rewrite the above state as

$$\mathrm{V}_{q_\mathsf{D}+q_\mathsf{F}} \circ \mathsf{U}_{\mathsf{S/C}} \cdots \mathrm{V}_2 \circ \mathsf{U}_{\mathsf{S/C}} \circ \mathrm{V}_1 \circ \mathsf{U}_{\mathsf{S/C}} |\psi_z\rangle \otimes \mathsf{StdDecomp}_{m^*} |D_\perp \cup (m^*, m_1^*)\rangle. \tag{78}$$

The $\mathsf{U}_{\mathsf{S/C}}$ here can be either CStO or $\mathsf{U}_{pk,\mathsf{qD}}^{\mathsf{SUS}}$[31], and the $\mathrm{V}_1, \ldots, \mathrm{V}_{q_\mathsf{D}+q_\mathsf{F}}$ are all unitary operations acting on the register A. Now we introduce a lemma related to CStO and $\mathsf{U}_{pk,\mathsf{qD}}^{\mathsf{SUS}}$ as follows, which will be used later.

Lemma I2 *For any (m^*, m_1^*)-fixed state $|\psi\rangle$ on the registers $\mathsf{AD}_{q_\mathsf{F}}$, we have $\mathsf{CStO}|\psi\rangle = \mathsf{CStO}_{m_1^*}^{m^*}|\psi\rangle$ and $\mathsf{U}_{pk,\mathsf{qD}}^{\mathsf{SUS}}|\psi\rangle = \mathsf{U}_{pk,\mathsf{qD}}^{\mathsf{SUUS}}|\psi\rangle$.*

Proof. The proof idea of this lemma is similar to Corollary 1, and we give the detailed proof in Supplementary Material I.3.

By the definition of the (m^*, m_1^*)-fixed states given in Eq. (74) and Corollary 1, it is not hard to check that in Eq. (78), the joint state of $\mathsf{AD}_{q_\mathsf{F}}$ just before each application of $\mathsf{U}_{\mathsf{S/C}}$, i.e., just before each application of CStO or $\mathsf{U}_{pk,\mathsf{qD}}^{\mathsf{SUS}}$, must be

[31] For the sake of simplicity, we will not clearly state whether $\mathsf{U}_{\mathsf{S/C}}$ is CStO or $\mathsf{U}_{pk,\mathsf{qD}}^{\mathsf{SUS}}$.

a (m^*, m_1^*)-fixed state. Thus, by Lemma I2, one can observe that the CStO and $\mathsf{U}^{\mathsf{SUS}}_{pk,\mathsf{qD}}$ used (and abstractly represented by $\mathsf{U}_{\mathsf{S/C}}$) in Eq. (78) can be equivalently changed to $\mathsf{CStO}^{m^*}_{m_1^*}$ and $\mathsf{U}^{\mathsf{SUUS}}_{pk,\mathsf{qD}}$, respectively. Note that $\mathcal{A}_2$ is a rewrite of game $\mathbf{G_6^b}$ under a fixed z (Eq. (71)), and $\mathcal{A}_2$ uses CStO and $\mathsf{U}^{\mathsf{SUS}}_{pk,\mathsf{qD}}$ to simulate F and qDeca^* for $\mathcal{A}$, respectively. Hence, by the definition of the game $\mathbf{G_6^c}$, the above observation actually means that game $\mathbf{G_6^c}$ has the same running process as game $\mathbf{G_6^b}$ under a fixed z, that is, $\Pr[1 \leftarrow \mathbf{G_6^b} : z] = \Pr[1 \leftarrow \mathbf{G_6^c} : z]$. Then by averaging over $z = (\mathsf{G}, \mathsf{H}, pk, sk, m^*, m_1^*, b, K_1^*)$, where $\mathsf{G} \leftarrow_\$ \Omega_{\{0,1\}^n \to \mathcal{K}}$, $\mathsf{H} \leftarrow_\$ \Omega_{\mathcal{M} \to \{0,1\}^n}$, $(pk, sk) \leftarrow \mathsf{Gen}$, $m^* \leftarrow_\$ \{0,1\}^n$, $m_1^* \leftarrow_\$ \mathcal{M}$, $b \leftarrow_\$ \{0,1\}$, and $K_1^* \leftarrow_\$ \mathcal{K}$, we have

$$\Pr\left[1 \leftarrow \mathbf{G_6^b}\right] = \Pr\left[1 \leftarrow \mathbf{G_6^c}\right]. \tag{79}$$

Game $\mathbf{G_6^d}$: This game is the same as $\mathbf{G_6^c}$, except that it no longer performs $\mathsf{StdDecomp}_{m^*} \circ \mathsf{U}^{m^*}_{m_1^*}$ after the sampling of m_1^*, and the $\mathsf{U}^{\mathsf{SUUS}}_{pk,\mathsf{qD}}$ used to simulate qDeca^* is replaced with $\mathsf{U}^{pk}_{\mathsf{qD}}$ defined in Eq. (24).

Based on the definition of $\mathsf{CStO}^{m^*}_{m_1^*}$ given in Eq. (76), obviously $\mathsf{CStO}^{m^*}_{m_1^*}$ commutes with $\mathsf{U}^{m^*}_{m_1^*}$ and $\mathsf{StdDecomp}_{m^*}$ since these two operations only affect the value of $D_{\mathsf{F}}(m^*)$. In addition, recall that $\mathsf{U}^{\mathsf{SUUS}}_{pk,\mathsf{qD}}$ can be rewritten as

$$\mathsf{StdDecomp}_{m^*} \circ \mathsf{U}^{m^*}_{m_1^*} \circ \mathsf{U}^{pk}_{\mathsf{qD}} \circ \mathsf{U}^{m^*}_{m_1^*} \circ \mathsf{StdDecomp}_{m^*},$$

and $\mathsf{U}^{m^*}_{m_1^*} \circ \mathsf{StdDecomp}_{m^*}$ is the inverse of $\mathsf{StdDecomp}_{m^*} \circ \mathsf{U}^{m^*}_{m_1^*}$ since $\mathsf{U}^{m^*}_{m_1^*}$ and $\mathsf{StdDecomp}_{m^*}$ both are involutions. Therefore, the $\mathsf{StdDecomp}_{m^*} \circ \mathsf{U}^{m^*}_{m_1^*}$ and $\mathsf{U}^{m^*}_{m_1^*} \circ \mathsf{StdDecomp}_{m^*}$ performed in game $\mathbf{G_6^c}$ can be canceled out as follows:

1. The $\mathsf{StdDecomp}_{m^*} \circ \mathsf{U}^{m^*}_{m_1^*}$ performed after $m_1^* \leftarrow_\$ \mathcal{M}$ can cancel out with the first $\mathsf{U}^{m^*}_{m_1^*} \circ \mathsf{StdDecomp}_{m^*}$ performed in the first implementation of $\mathsf{U}^{\mathsf{SUUS}}_{pk,\mathsf{qD}}$.
2. The second $\mathsf{StdDecomp}_{m^*} \circ \mathsf{U}^{m^*}_{m_1^*}$ performed in the implementation of $\mathsf{U}^{\mathsf{SUUS}}_{pk,\mathsf{qD}}$ can cancel out with the first $\mathsf{U}^{m^*}_{m_1^*} \circ \mathsf{StdDecomp}_{m^*}$ performed in the very next implementation of $\mathsf{U}^{\mathsf{SUUS}}_{pk,\mathsf{qD}}$.
3. As for the second $\mathsf{StdDecomp}_{m^*} \circ \mathsf{U}^{m^*}_{m_1^*}$ performed in the final implementation of $\mathsf{U}^{\mathsf{SUUS}}_{pk,\mathsf{qD}}$, since this operation does not act on the output register of adversary $\mathcal{A}$, we can move it to the end of the game and then delete it without affecting $\mathcal{A}$'s output.

One can observe that game $\mathbf{G_6^c}$ can be transformed into game $\mathbf{G_6^d}$ after the above cancellation of $\mathsf{StdDecomp}_{m^*} \circ \mathsf{U}^{m^*}_{m_1^*}$ and $\mathsf{U}^{m^*}_{m_1^*} \circ \mathsf{StdDecomp}_{m^*}$. Thus we have

$$\Pr\left[1 \leftarrow \mathbf{G_6^c}\right] = \Pr\left[1 \leftarrow \mathbf{G_6^d}\right]. \tag{80}$$

In fact, game $\mathbf{G_6^d}$ defined above is the same as game $\mathbf{G_7}$ defined in Fig. 12. Hence, by combining Eq. (69), Lemma I1, Eq. (79), and Eq. (80), we get

$$|\Pr[1 \leftarrow \mathbf{G_6}] - \Pr[1 \leftarrow \mathbf{G_7}]| \leq 2q_{\mathsf{D}} \cdot \sqrt{2^{-m}} = 2q_{\mathsf{D}}/\sqrt{|\mathcal{M}|}.$$

This completes the proof of Lemma 5. $\square$

I.1 Proof of Corollary 1

We first consider the state $\mathsf{CStO}|\psi\rangle$. In this proof, we denote $\mathsf{StdDecomp}_x$ as S_x for simplicity. In fact, by the definition of the (m^*, m_1^*)-fixed states given in Eq. (74), we can rewrite $|\psi\rangle$ as

$$|\psi\rangle = \sum_{a',x,y,D_{\mathsf{F}}} \alpha_{a',x,y}^{D_{\mathsf{F}}} |a', x, y\rangle \otimes \mathsf{S}_{m^*} |D_{\mathsf{F}} \cup (m^*, m_1^*)\rangle. \tag{81}$$

Here we divide "a" into three parts a', x, and y, where x/y denotes the value of the input/output register of F. We would like to stress that such a division is reasonable, since these two registers are implicitly set as the internal registers of $\mathcal{A}$ and hence are naturally contained in A. Now, by Definition 2, we can compute

$$\begin{aligned}
\mathsf{CStO}|\psi\rangle = & \sum_{a',x\neq m^*,y,D_{\mathsf{F}}} \alpha_{a',x,y}^{D_{\mathsf{F}}} \mathsf{S}_x \circ \mathsf{CNOT}^x \circ \mathsf{S}_x |a', x, y\rangle \otimes \mathsf{S}_{m^*} |D_{\mathsf{F}} \cup (m^*, m_1^*)\rangle \\
& + \sum_{\substack{a',x=m^* \\ y,D_{\mathsf{F}}}} \alpha_{a',x,y}^{D_{\mathsf{F}}} \mathsf{S}_{m^*} \circ \mathsf{CNOT}^{m^*} \circ \mathsf{S}_{m^*} |a', x, y\rangle \otimes \mathsf{S}_{m^*} |D_{\mathsf{F}} \cup (m^*, m_1^*)\rangle \\
= & \sum_{a',x\neq m^*,y,D_{\mathsf{F}}} \alpha_{a',x,y}^{D_{\mathsf{F}}} \mathsf{S}_x \circ \mathsf{CNOT}^x \circ \mathsf{S}_x |a', x, y\rangle \otimes \mathsf{S}_{m^*} |D_{\mathsf{F}} \cup (m^*, m_1^*)\rangle \\
& + \sum_{a',x=m^*,y,D_{\mathsf{F}}} \alpha_{a',x,y}^{D_{\mathsf{F}}} |a', x, y \oplus m_1^*\rangle \otimes \mathsf{S}_{m^*} |D_{\mathsf{F}} \cup (m^*, m_1^*)\rangle.
\end{aligned}$$

Define states

$$\begin{aligned}
|\psi_{\neq m^*}\rangle &:= \sum_{a',x\neq m^*,y,D_{\mathsf{F}}} \alpha_{a',x,y}^{D_{\mathsf{F}}} \mathsf{S}_x \circ \mathsf{CNOT}^x \circ \mathsf{S}_x |a', x, y\rangle \otimes \mathsf{S}_{m^*} |D_{\mathsf{F}} \cup (m^*, m_1^*)\rangle, \\
|\psi_{= m^*}\rangle &:= \sum_{a',x=m^*,y,D_{\mathsf{F}}} \alpha_{a',x,y}^{D_{\mathsf{F}}} |a', x, y \oplus m_1^*\rangle \otimes \mathsf{S}_{m^*} |D_{\mathsf{F}} \cup (m^*, m_1^*)\rangle.
\end{aligned} \tag{82}$$

Obviously $\mathsf{CStO}|\psi\rangle = |\psi_{\neq m^*}\rangle + |\psi_{=m^*}\rangle$ and $|\psi_{=m^*}\rangle$ is a (m^*, m_1^*)-fixed state.

As for $|\psi_{\neq m^*}\rangle$, we first focus on the basis state $\mathsf{S}_x \circ \mathsf{CNOT}^x \circ \mathsf{S}_x |a', x, y\rangle \otimes \mathsf{S}_{m^*} |D_{\mathsf{F}} \cup (m^*, m_1^*)\rangle$ it superposed. In fact, by the definition of the local decompression procedure given in Section 2.3, S_x commutes with S_{m^*} if $x \neq m^*$, and $\mathsf{CNOT}^x : |y, D_{\mathsf{F}}\rangle \mapsto |y \oplus D_{\mathsf{F}}(x), D_{\mathsf{F}}\rangle$ also commutes with S_{m^*} if $x \neq m^*$, since S_{m^*} does not change $D_{\mathsf{F}}(x \neq m^*)$ when acting on the database state $|D_{\mathsf{F}}\rangle$. Hence we have

$$\begin{aligned}
&\mathsf{S}_x \circ \mathsf{CNOT}^x \circ \mathsf{S}_x |a', x, y\rangle \otimes \mathsf{S}_{m^*} |D_{\mathsf{F}} \cup (m^*, m_1^*)\rangle \\
=&\mathsf{S}_{m^*} \circ \mathsf{S}_x \circ \mathsf{CNOT}^x \circ \mathsf{S}_x |a', x, y\rangle \otimes |D_{\mathsf{F}} \cup (m^*, m_1^*)\rangle.
\end{aligned} \tag{83}$$

Notice that the (m^*, m_1^*) pair of the database $D_{\mathsf{F}} \cup (m^*, m_1^*)$ cannot be changed when S_x or CNOT^x with $x \neq m^*$ act on the database state $|D_{\mathsf{F}} \cup (m^*, m_1^*)\rangle$. Hence we can rewrite $\mathsf{S}_x \circ \mathsf{CNOT}^x \circ \mathsf{S}_x |a', x, y\rangle \otimes |D_{\mathsf{F}} \cup (m^*, m_1^*)\rangle$ as

$$\sum_{y',D'_{\mathsf{F}}} \beta_{y,y'}^{D_{\mathsf{F}},D'_{\mathsf{F}}} |a', x, y'\rangle \otimes |D'_{\mathsf{F}} \cup (m^*, m_1^*)\rangle$$

based on a tuple of coefficients $\{\beta_{y,y'}^{D_\mathsf{F},D'_\mathsf{F}}: y' \in \mathcal{M}, D'_\mathsf{F} \in \mathbf{D}_{q_\mathsf{F}}\}$. Substituting this rewrite into Eq. (83) yields

$$\begin{aligned}&\mathsf{S}_x \circ \mathsf{CNOT}^x \circ \mathsf{S}_x|a', x, y\rangle \otimes \mathsf{S}_{m^*}|D_\mathsf{F} \cup (m^*, m_1^*)\rangle \\ =&\sum_{y', D'_\mathsf{F}} \beta_{y,y'}^{D_\mathsf{F},D'_\mathsf{F}}|a', x, y'\rangle \otimes \mathsf{S}_{m^*}|D'_\mathsf{F} \cup (m^*, m_1^*)\rangle.\end{aligned}$$

Then by substituting the above equation into the definition of $|\psi_{\neq m^*}\rangle$ given in Eq. (82), it is easy to see that $|\psi_{\neq m^*}\rangle$ is also a (m^*, m_1^*)-fixed state. Now $\mathsf{CStO}|\psi\rangle = |\psi_{\neq m^*}\rangle + |\psi_{=m^*}\rangle$ must be a (m^*, m_1^*)-fixed state, since the sum of two (m^*, m_1^*)-fixed states obviously is still a (m^*, m_1^*)-fixed state.

As for the states $\mathrm{U}_1|\psi\rangle, \ldots, \mathrm{U}_{q_\mathsf{D}}|\psi\rangle$, recall that $\mathrm{U}_1, \ldots, \mathrm{U}_{q_\mathsf{D}}$ are all composites of the unitary operations acting on A and three particular unitary operations O_G, O_H and CStO, and only CStO can act on the database register $\mathsf{D}_{q_\mathsf{F}}$. Thus by the above proof, for any (m^*, m_1^*)-fixed state $|\psi\rangle$, it is clear that $\mathrm{U}_1|\psi\rangle, \ldots, \mathrm{U}_{q_\mathsf{D}}|\psi\rangle$ are all still (m^*, m_1^*)-fixed states.

Next we consider $\mathsf{U}_{pk,\mathsf{qD}}^{\mathsf{SUS}}|\psi\rangle$. Similar to Eq. (81), we can rewrite $|\psi\rangle$ into the following state.

$$|\psi\rangle = \sum_{a', c_1, c_2, k, D_\mathsf{F}} \alpha_{a', c_1, c_2, k}^{D_\mathsf{F}}|a', c_1, c_2, k\rangle \otimes \mathsf{S}_{m^*}|D_\mathsf{F} \cup (m^*, m_1^*)\rangle. \tag{84}$$

Here we divide "a" into four parts a', c_1, c_2, and k, where $(c_1, c_2)/k$ denotes the value of the input/output register of qDeca^*. We would like to stress again that such a division is reasonable, since these two registers are implicitly set as the internal registers of $\mathcal{A}$ and hence are naturally contained in A. Now, by the definition of $\mathsf{U}_{pk,\mathsf{qD}}^{\mathsf{SUS}}$ given in Eq. (70), we can compute

$$\begin{aligned}\mathsf{U}_{pk,\mathsf{qD}}^{\mathsf{SUS}}|\psi\rangle =&\mathrm{U}_\perp \circ \mathrm{P}_{c_1^*, c_2^*}|\psi\rangle + \mathsf{S}_{m^*} \circ \mathsf{U}_\mathsf{search} \circ \mathsf{S}_{m^*} \circ \left(\mathrm{I} - \mathrm{P}_{c_1^*, c_2^*}\right)|\psi\rangle \\ \overset{(a)}{=}&\mathrm{U}_\perp \circ \mathrm{P}_{c_1^*, c_2^*}|\psi\rangle \\ &+ \sum_{\substack{a', k, D_\mathsf{F} \\ (c_1, c_2) \neq (c_1^*, c_2^*)}} \alpha_{a', c_1, c_2, k}^{D_\mathsf{F}} \mathsf{S}_{m^*} \circ \mathsf{U}_\mathsf{search}|a', c_1, c_2, k\rangle \otimes |D_\mathsf{F} \cup (m^*, m_1^*)\rangle \\ \overset{(b)}{=}&\mathrm{U}_\perp \circ \mathrm{P}_{c_1^*, c_2^*}|\psi\rangle \\ &+ \sum_{\substack{a', k, D_\mathsf{F} \\ (c_1, c_2) \neq (c_1^*, c_2^*)}} \alpha_{a', c_1, c_2, k}^{D_\mathsf{F}}|a', c_1, c_2, k \oplus \mathsf{Search}(pk, c_1, c_2, D_\mathsf{F})\rangle \\ &\qquad\qquad\qquad\qquad \otimes \mathsf{S}_{m^*}|D_\mathsf{F} \cup (m^*, m_1^*)\rangle.\end{aligned} \tag{85}$$

Here (a) follows from the fact that S_{m^*} is an involution, and (b) is because that when $(c_1, c_2) \neq (c_1^*, c_2^*)$[32] we have

$$\begin{aligned}&\mathsf{U}_\mathsf{search}|c_1, c_2, k, D_\mathsf{F} \cup (m^*, m_1^*)\rangle \\ &=|c_1, c_2, k \oplus \mathsf{Search}(pk, c_1, c_2, D_\mathsf{F}), D_\mathsf{F} \cup (m^*, m_1^*)\rangle,\end{aligned} \tag{86}$$

[32] Recall that $c_1^* := \mathsf{Enc}(pk, m_1^*)$ and $c_2^* := m^* \oplus \mathsf{H}(m_1^*)$.

which is easy to verify by the definitions of $\mathsf{U}_{\mathsf{search}}$ and Search given in Eq. (25) and Eq. (26), respectively.

Since the operation $\mathrm{U}_{\perp} \circ \mathrm{P}_{c_1^*,c_2^*}$ does not act on the database register $\mathsf{D}_{q_{\mathsf{F}}}$, $\mathrm{U}_{\perp} \circ \mathrm{P}_{c_1^*,c_2^*}|\psi\rangle$ is still a (m^*, m_1^*)-fixed state. Furthermore, it is obvious that the second state of the final equality of Eq. (85) is also a (m^*, m_1^*)-fixed state, and thus $\mathsf{U}^{\mathsf{SUS}}_{pk,\mathsf{qD}}|\psi\rangle$ is a (m^*, m_1^*)-fixed state since it is a sum of two such states. This completes the proof of Corollary 1. □

I.2 Proof of Corollary 2

In this proof, we denote $\mathsf{StdDecomp}_x$ as S_x for simplicity. Similar to Eq. (84), we rewrite the unit (m^*, m_1^*)-fixed state $|\psi\rangle$ as the following state.

$$|\psi\rangle = \sum_{a',c_1,c_2,k,D_{\mathsf{F}}} \alpha^{D_{\mathsf{F}}}_{a',c_1,c_2,k}|a', c_1, c_2, k\rangle \otimes \mathsf{S}_{m^*}|D_{\mathsf{F}} \cup (m^*, m_1^*)\rangle. \tag{87}$$

Here we divide "a" into four parts a', c_1, c_2 and k, where $(c_1, c_2)/k$ denotes the value of the input/output register of qDeca^*.

For the basis state $|a', c_1, c_2, k\rangle \otimes \mathsf{S}_{m^*}|D_{\mathsf{F}} \cup (m^*, m_1^*)\rangle$ such that $(c_1, c_2) \neq (c_1^*, c_2^*)$, we can compute

$$\begin{aligned}
&\mathsf{U}_{\mathsf{search}}|a', c_1, c_2, k\rangle \otimes \mathsf{S}_{m^*}|D_{\mathsf{F}} \cup (m^*, m_1^*)\rangle\\
\overset{(a)}{=}&\mathsf{U}_{\mathsf{search}}|a', c_1, c_2, k\rangle \otimes \left(|D_{\mathsf{F}} \cup (m^*, m_1^*)\rangle + \frac{1}{\sqrt{2^m}}|D_{\mathsf{F}}\rangle - \frac{1}{\sqrt{2^m}}|D_{\mathsf{F}} \cup (m^*, \beta_{0^m})\rangle\right)\\
\overset{(b)}{=}&|a', c_1, c_2, k \oplus \mathsf{Search}(pk, c_1, c_2, D_{\mathsf{F}})\rangle \otimes |D_{\mathsf{F}} \cup (m^*, m_1^*)\rangle\\
&+ \frac{1}{\sqrt{2^m}}|a', c_1, c_2, k \oplus \mathsf{Search}(pk, c_1, c_2, D_{\mathsf{F}})\rangle \otimes |D_{\mathsf{F}}\rangle\\
&- \frac{1}{\sqrt{2^m}}\mathsf{U}_{\mathsf{Search}}|a', c_1, c_2, k\rangle \otimes |D_{\mathsf{F}} \cup (m^*, \beta_{0^m})\rangle.
\end{aligned}$$

Here (a) uses Eq. (2), and (b) uses Eq. (86). As for $\mathsf{S}_{m^*} \circ \mathsf{U}_{\mathsf{search}} \circ \mathsf{S}_{m^*}$, we can similarly compute

$$\begin{aligned}
&\mathsf{S}_{m^*} \circ \mathsf{U}_{\mathsf{search}} \circ \mathsf{S}_{m^*}|a', c_1, c_2, k\rangle \otimes \mathsf{S}_{m^*}|D_{\mathsf{F}} \cup (m^*, m_1^*)\rangle\\
\overset{(a)}{=}&\mathsf{S}_{m^*} \circ \mathsf{U}_{\mathsf{search}}|a', c_1, c_2, k\rangle \otimes |D_{\mathsf{F}} \cup (m^*, m_1^*)\rangle\\
\overset{(b)}{=}&|a', c_1, c_2, k \oplus \mathsf{Search}(pk, c_1, c_2, D_{\mathsf{F}})\rangle \otimes \mathsf{S}_{m^*}|D_{\mathsf{F}} \cup (m^*, m_1^*)\rangle\\
\overset{(c)}{=}&|a', c_1, c_2, k \oplus \mathsf{Search}(pk, c_1, c_2, D_{\mathsf{F}})\rangle \otimes |D_{\mathsf{F}} \cup (m^*, m_1^*)\rangle\\
&+ \frac{1}{\sqrt{2^m}}|a', c_1, c_2, k \oplus \mathsf{Search}(pk, c_1, c_2, D_{\mathsf{F}})\rangle \otimes |D_{\mathsf{F}}\rangle\\
&- \frac{1}{\sqrt{2^m}}|a', c_1, c_2, k \oplus \mathsf{Search}(pk, c_1, c_2, D_{\mathsf{F}})\rangle \otimes |D_{\mathsf{F}} \cup (m^*, \beta_{0^m})\rangle.
\end{aligned}$$

Here (a) follows from the fact that S_{m^*} is an involution. (b) uses Eq. (86) and (c) uses Eq. (2). Now, for the basis state $|a', c_1, c_2, k\rangle \otimes \mathsf{S}_{m^*}|D_{\mathsf{F}} \cup (m^*, m_1^*)\rangle$ such that $(c_1, c_2) \neq (c_1^*, c_2^*)$, based on the above two equations we get

$$
\begin{aligned}
&(\mathsf{U}_{\mathsf{search}} - \mathsf{S}_{m^*} \circ \mathsf{U}_{\mathsf{search}} \circ \mathsf{S}_{m^*})|a', c_1, c_2, k\rangle \otimes \mathsf{S}_{m^*}|D_{\mathsf{F}} \cup (m^*, m_1^*)\rangle \\
=& \frac{1}{\sqrt{2^m}}|a', c_1, c_2, k \oplus \mathsf{Search}(pk, c_1, c_2, D_{\mathsf{F}})\rangle \otimes |D_{\mathsf{F}} \cup (m^*, \beta_{0^m})\rangle \\
&- \frac{1}{\sqrt{2^m}}\mathsf{U}_{\mathsf{search}}|a', c_1, c_2, k\rangle \otimes |D_{\mathsf{F}} \cup (m^*, \beta_{0^m})\rangle.
\end{aligned}
\tag{88}
$$

Next we define the following two states related to $|\psi\rangle$ rewritten in Eq. (87):

$$
\begin{aligned}
|\psi_1\rangle :=& \sum_{\substack{a', k, D_{\mathsf{F}} \\ (c_1, c_2) \neq (c_1^*, c_2^*)}} \alpha^{D_{\mathsf{F}}}_{a', c_1, c_2, k}|a', c_1, c_2, k \oplus \mathsf{Search}(pk, c_1, c_2, D_{\mathsf{F}})\rangle \\
&\qquad\qquad \otimes|D_{\mathsf{F}} \cup (m^*, \beta_{0^m})\rangle, \\
|\psi_2\rangle :=& \sum_{\substack{a', k, D_{\mathsf{F}} \\ (c_1, c_2) \neq (c_1^*, c_2^*)}} \alpha^{D_{\mathsf{F}}}_{a', c_1, c_2, k}\mathsf{U}_{\mathsf{search}}|a', c_1, c_2, k\rangle \otimes |D_{\mathsf{F}} \cup (m^*, \beta_{0^m})\rangle.
\end{aligned}
\tag{89}
$$

For $|\psi_1\rangle$, we have

$$
\begin{aligned}
\||\psi_1\rangle\|^2 =& \left\| \sum_{\substack{a', k, D_{\mathsf{F}} \\ (c_1, c_2) \neq (c_1^*, c_2^*)}} \alpha^{D_{\mathsf{F}}}_{a', c_1, c_2, k}|a', c_1, c_2, k \oplus \mathsf{Search}(pk, c_1, c_2, D_{\mathsf{F}})\rangle \otimes|D_{\mathsf{F}} \cup (m^*, \beta_{0^m})\rangle \right\|^2 \\
\overset{(a)}{=}& \sum_{\substack{a', k, D_{\mathsf{F}} \\ (c_1, c_2) \neq (c_1^*, c_2^*)}} \left\| \alpha^{D_{\mathsf{F}}}_{a', c_1, c_2, k}|a', c_1, c_2, k \oplus \mathsf{Search}(pk, c_1, c_2, D_{\mathsf{F}})\rangle \otimes|D_{\mathsf{F}} \cup (m^*, \beta_{0^m})\rangle \right\|^2 \\
=& \sum_{\substack{a', k, D_{\mathsf{F}} \\ (c_1, c_2) \neq (c_1^*, c_2^*)}} \left| \alpha^{D_{\mathsf{F}}}_{a', c_1, c_2, k} \right|^2 \overset{(b)}{\leq} \||\psi\rangle\|^2 = 1.
\end{aligned}
\tag{90}
$$

Here (a) follows from the fact that $\||\psi\rangle + |\phi\rangle\|^2 = \||\psi\rangle\|^2 + \||\phi\rangle\|^2$ for any states $|\psi\rangle$ and $|\phi\rangle$[33]. (b) uses Eq. (87). Similarly, for the state $|\psi_2\rangle$, we have

$$
\||\psi_2\rangle\|^2 = \left\| \sum_{\substack{a', k, D_{\mathsf{F}} \\ (c_1, c_2) \neq (c_1^*, c_2^*)}} \alpha^{D_{\mathsf{F}}}_{a', c_1, c_2, k}\mathsf{U}_{\mathsf{Search}}|a', c_1, c_2, k\rangle \otimes |D_{\mathsf{F}} \cup (m^*, \beta_{0^m})\rangle \right\|^2
$$

[33] We point out that we have applied the property that $y_1 \oplus \mathsf{Search}(pk, c_1, c_2, D_{\mathsf{F}}) \neq y_2 \oplus \mathsf{Search}(pk, c_1, c_2, D_{\mathsf{F}})$ for any $y_1 \neq y_2$ such that $y_1, y_2 \in \mathcal{K} \cup \{\perp\}$ during the derivation of that fact. Indeed, this property is obvious because by our quantum representation of the output register of qDeca^* given in Section 4.2, the "$\oplus$" used here is actually the XOR operation on the set $\{0,1\}^{k+1}$.

$$
\begin{aligned}
&\overset{(a)}{=} \left\| \sum_{\substack{a',k,D_{\mathsf{F}} \\ (c_1,c_2)\neq(c_1^*,c_2^*)}} \alpha^{D_{\mathsf{F}}}_{a',c_1,c_2,k} |a',c_1,c_2,k\rangle \otimes |D_{\mathsf{F}} \cup (m^*, \beta_{0^m})\rangle \right\|^2 \\
&\overset{(b)}{=} \sum_{\substack{a',k,D_{\mathsf{F}} \\ (c_1,c_2)\neq(c_1^*,c_2^*)}} \left\| \alpha^{D_{\mathsf{F}}}_{a',c_1,c_2,k} |a',c_1,c_2,k\rangle \otimes |D_{\mathsf{F}} \cup (m^*, \beta_{0^m})\rangle \right\|^2 \qquad (91) \\
&= \sum_{\substack{a',k,D_{\mathsf{F}} \\ (c_1,c_2)\neq(c_1^*,c_2^*)}} \left| \alpha^{D_{\mathsf{F}}}_{a',c_1,c_2,k} \right|^2 \overset{(c)}{\leq} \| |\psi\rangle \|^2 = 1.
\end{aligned}
$$

Here (a) is because $\mathsf{U}_{\mathsf{search}}$ is a unitary operation, and (b) follows from the fact that $\| |\psi\rangle + |\phi\rangle \|^2 = \| |\psi\rangle \|^2 + \| |\phi\rangle \|^2$ for any states $|\psi\rangle$ and $|\phi\rangle$. (c) uses Eq. (87).

Now by the definitions of $\mathsf{U}^{pk}_{\mathsf{qD}}$ and $\mathsf{U}^{\mathsf{SUS}}_{pk,\mathsf{qD}}$ given in Eq. (24) and Eq. (70), respectively, we can compute

$$
\begin{aligned}
&\left\| \mathsf{U}^{pk}_{\mathsf{qD}} |\psi\rangle - \mathsf{U}^{\mathsf{SUS}}_{pk,\mathsf{qD}} |\psi\rangle \right\|^2 \\
&= \left\| \mathsf{U}_{\mathsf{search}} \circ \left(\mathrm{I} - \mathrm{P}_{c_1^*,c_2^*} \right) |\psi\rangle - \mathsf{S}_{m^*} \circ \mathsf{U}_{\mathsf{search}} \circ \mathsf{S}_{m^*} \circ \left(\mathrm{I} - \mathrm{P}_{c_1^*,c_2^*} \right) |\psi\rangle \right\|^2 \\
&\overset{(a)}{=} \left\| \left(\mathsf{U}_{\mathsf{search}} - \mathsf{S}_{m^*} \circ \mathsf{U}_{\mathsf{search}} \circ \mathsf{S}_{m^*} \right) \sum_{\substack{a',k,D_{\mathsf{F}} \\ (c_1,c_2)\neq(c_1^*,c_2^*)}} \begin{array}{r} \alpha^{D_{\mathsf{F}}}_{a',c_1,c_2,k} |a',c_1,c_2,k\rangle \\ \otimes \mathsf{S}_{m^*} |D_{\mathsf{F}} \cup (m^*, m_1^*)\rangle \end{array} \right\|^2 \\
&\overset{(b)}{=} \frac{1}{2^m} \cdot \left\| \begin{array}{l} \displaystyle\sum_{\substack{a',k,D_{\mathsf{F}} \\ (c_1,c_2)\neq(c_1^*,c_2^*)}} \begin{array}{r} \alpha^{D_{\mathsf{F}}}_{a',c_1,c_2,k} |a',c_1,c_2,k \oplus \mathsf{Search}(pk,c_1,c_2,D_{\mathsf{F}})\rangle \\ \otimes |D_{\mathsf{F}} \cup (m^*, \beta_{0^m})\rangle \end{array} \\ \displaystyle - \sum_{\substack{a',k,D_{\mathsf{F}} \\ (c_1,c_2)\neq(c_1^*,c_2^*)}} \alpha^{D_{\mathsf{F}}}_{a',c_1,c_2,k} \mathsf{U}_{\mathsf{search}} |a',c_1,c_2,k\rangle \otimes |D_{\mathsf{F}} \cup (m^*, \beta_{0^m})\rangle \end{array} \right\|^2 \\
&\overset{(c)}{=} \frac{1}{2^m} \cdot \| |\psi_1\rangle - |\psi_2\rangle \|^2 \overset{(d)}{\leq} \frac{2}{2^m} \cdot \left(\| |\psi_1\rangle \|^2 + \| |\psi_2\rangle \|^2 \right) \overset{(e)}{\leq} \frac{4}{2^m}.
\end{aligned}
$$

Here (a) follows from the rewrite of $|\psi\rangle$ given in Eq. (87). (b) uses Eq.(88) and (c) uses Eq. (89). (d) follows from the fact that $\| |\psi\rangle - |\phi\rangle \|^2 \leq 2 \cdot \| |\psi\rangle \|^2 + 2 \cdot \| |\phi\rangle \|^2$ for any states $|\psi\rangle$ and $|\phi\rangle$. (e) follows from Eq. (90) and Eq. (91). This completes the proof of Corollary 2. □

I.3 Proof of Lemma I2

In this proof, we denote $\mathsf{StdDecomp}_x$ as S_x for simplicity. Similar to Eq. (81), we rewrite the (m^*, m_1^*)-fixed state $|\psi\rangle$ as the following state.

$$
|\psi\rangle = \sum_{a',x,y,D_{\mathsf{F}}} \alpha^{D_{\mathsf{F}}}_{a',x,y} |a',x,y\rangle \otimes \mathsf{S}_{m^*} |D_{\mathsf{F}} \cup (m^*, m_1^*)\rangle.
$$

Here we divides "a" into three parts a', x and y, where x/y denotes the value of the input/output register of F.

For the basis state $|a', x, y\rangle \otimes \mathsf{S}_{m^*}|D_{\mathsf{F}} \cup (m^*, m_1^*)\rangle$ of $|\psi\rangle$, we consider two cases where $x = m^*$ and $x \neq m^*$. For the first case, by the definition of CStO given in Definition 2, the computation process of $\mathsf{CStO}|a', m^*, y\rangle \mathsf{S}_{m^*}|D_{\mathsf{F}} \cup (m^*, m_1^*)\rangle$ can be described as

$$
\begin{aligned}
|a', m^*, y\rangle \mathsf{S}_{m^*}|D_{\mathsf{F}} \cup (m^*, m_1^*)\rangle &\xrightarrow{\mathsf{S}_{m^*}} |a', m^*, y\rangle |D_{\mathsf{F}} \cup (m^*, m_1^*)\rangle \\
&\xrightarrow{\mathsf{CNOT}^{m^*}} |a', m^*, y \oplus m_1^*\rangle |D_{\mathsf{F}} \cup (m^*, m_1^*)\rangle \\
&\xrightarrow{\mathsf{S}_{m^*}} |a', m^*, y \oplus m_1^*\rangle \mathsf{S}_{m^*}|D_{\mathsf{F}} \cup (m^*, m_1^*)\rangle.
\end{aligned}
$$

Then by the definition of $\mathsf{CStO}_{m_1^*}^{m^*}$ given in Eq. (76), we have

$$
\mathsf{CStO}_{m_1^*}^{m^*}|a', m^*, y\rangle \mathsf{S}_{m^*}|D_{\mathsf{F}} \cup (m^*, m_1^*)\rangle = |a', m^*, y \oplus m_1^*\rangle \mathsf{S}_{m^*}|D_{\mathsf{F}} \cup (m^*, m_1^*)\rangle.
$$

This means that CStO and $\mathsf{CStO}_{m_1^*}^{m^*}$ have the same effect on the first case. For the second case where $x \neq m^*$, it is easy to see that CStO and $\mathsf{CStO}_{m_1^*}^{m^*}$ both act by applying $\mathsf{StdDecomp}_x \circ \mathsf{CNOT}^x \circ \mathsf{StdDecomp}_x$ on the state $|a', x, y\rangle \otimes \mathsf{S}_{m^*}|D_{\mathsf{F}} \cup (m^*, m_1^*)\rangle$, and hence they also have the same effect on the second case. Now we have proved that $\mathsf{CStO}|\psi\rangle = \mathsf{CStO}_{m_1^*}^{m^*}|\psi\rangle$.

Next we prove $\mathsf{U}_{pk,\mathsf{qD}}^{\mathsf{SUS}}|\psi\rangle = \mathsf{U}_{pk,\mathsf{qD}}^{\mathsf{SUUS}}|\psi\rangle$. In fact, similar to Eq. (84), we can rewrite the (m^*, m_1^*)-fixed state $|\psi\rangle$ into the following state.

$$
|\psi\rangle = \sum_{a', c_1, c_2, k, D_{\mathsf{F}}} \alpha_{a', c_1, c_2, k}^{D_{\mathsf{F}}} |a', c_1, c_2, k\rangle \otimes \mathsf{S}_{m^*}|D_{\mathsf{F}} \cup (m^*, m_1^*)\rangle.
$$

Here we divide "a" into four parts a', c_1, c_2 and k, where $(c_1, c_2)/k$ denotes the value of the input/output register of qDeca^*.

For the basis state $|a', c_1, c_2, k\rangle \otimes \mathsf{S}_{m^*}|D_{\mathsf{F}} \cup (m^*, m_1^*)\rangle$, we consider two cases where $(c_1, c_2) = (c_1^*, c_2^*)$ and $(c_1, c_2) \neq (c_1^*, c_2^*)$. Here $c_1^* := \mathsf{Enc}(pk, m_1^*)$ and $c_2^* := m^* \oplus \mathsf{H}(m_1^*)$. For the first case, by the definitions of $\mathsf{U}_{pk,\mathsf{qD}}^{\mathsf{SUS}}$ and $\mathsf{U}_{pk,\mathsf{qD}}^{\mathsf{SUUS}}$ respectively given in Eq. (70) and Eq. (77), it is obvious that they both transform $|a', c_1^*, c_2^*, k\rangle \otimes \mathsf{S}_{m^*}|D_{\mathsf{F}} \cup (m^*, m_1^*)\rangle$ into $|a', c_1^*, c_2^*, k \oplus \perp\rangle \otimes \mathsf{S}_{m^*}|D_{\mathsf{F}} \cup (m^*, m_1^*)\rangle$. For the second case where $(c_1, c_2) \neq (c_1^*, c_2^*)$, we can compute

$$
\begin{aligned}
&\mathsf{U}_{pk,\mathsf{qD}}^{\mathsf{SUS}}|a', c_1, c_2, k\rangle \otimes \mathsf{S}_{m^*}|D_{\mathsf{F}} \cup (m^*, m_1^*)\rangle \\
=&\mathsf{S}_{m^*} \circ \mathsf{U}_{\mathsf{search}} \circ \mathsf{S}_{m^*}|a', c_1, c_2, k\rangle \otimes \mathsf{S}_{m^*}|D_{\mathsf{F}} \cup (m^*, m_1^*)\rangle \\
\overset{(a)}{=}&\mathsf{S}_{m^*} \circ \mathsf{U}_{\mathsf{search}}|a', c_1, c_2, k\rangle \otimes |D_{\mathsf{F}} \cup (m^*, m_1^*)\rangle \\
\overset{(b)}{=}&\mathsf{S}_{m^*}|a', c_1, c_2, k \oplus \mathsf{Search}(pk, c_1, c_2, D_{\mathsf{F}})\rangle \otimes |D_{\mathsf{F}} \cup (m^*, m_1^*)\rangle \\
=&|a', c_1, c_2, k \oplus \mathsf{Search}(pk, c_1, c_2, D_{\mathsf{F}})\rangle \otimes \mathsf{S}_{m^*}|D_{\mathsf{F}} \cup (m^*, m_1^*)\rangle.
\end{aligned}
$$

Here (a) is because S_{m^*} is an involution, and (b) uses Eq. (86). As for the $\mathsf{U}_{pk,\mathsf{qD}}^{\mathsf{SUUS}}$, recall that it can be rewritten as $\mathsf{S}_{m^*} \circ \mathsf{U}_{m_1^*}^{m^*} \circ \mathsf{U}_{\mathsf{qD}}^{pk} \circ \mathsf{U}_{m_1^*}^{m^*} \circ \mathsf{S}_{m^*}$, and hence we can similarly compute

$$
\begin{aligned}
&\mathsf{U}^{\mathsf{SUUS}}_{pk,\mathsf{qD}}|a',c_1,c_2,k\rangle\otimes\mathsf{S}_{m^*}|D_{\mathsf{F}}\cup(m^*,m_1^*)\rangle\\
=&\mathsf{S}_{m^*}\circ\mathsf{U}^{m^*}_{m_1^*}\circ\mathsf{U}^{pk}_{\mathsf{qD}}|a',c_1,c_2,k\rangle\otimes|D_{\mathsf{F}}\rangle\\
=&\mathsf{S}_{m^*}\circ\mathsf{U}^{m^*}_{m_1^*}\circ\mathsf{U}_{\mathsf{search}}|a',c_1,c_2,k\rangle\otimes|D_{\mathsf{F}}\rangle\\
=&\mathsf{S}_{m^*}\circ\mathsf{U}^{m^*}_{m_1^*}|a',c_1,c_2,k\oplus\mathsf{Search}(pk,c_1,c_2,D_{\mathsf{F}})\rangle\otimes|D_{\mathsf{F}}\rangle\\
=&|a',c_1,c_2,k\oplus\mathsf{Search}(pk,c_1,c_2,D_{\mathsf{F}})\rangle\otimes\mathsf{S}_{m^*}|D_{\mathsf{F}}\cup(m^*,m_1^*)\rangle.
\end{aligned}
$$

Based on the above computation, we can find that $\mathsf{U}^{\mathsf{SUS}}_{pk,\mathsf{qD}}$ and $\mathsf{U}^{\mathsf{SUUS}}_{pk,\mathsf{qD}}$ have the same effect when acting on the basis state $|a',c_1,c_2,k\rangle\otimes\mathsf{S}_{m^*}|D_{\mathsf{F}}\cup(m^*,m_1^*)\rangle$, regardless of whether $(c_1,c_2)=(c_1^*,c_2^*)$ or otherwise. That is to say, we have $\mathsf{U}^{\mathsf{SUS}}_{pk,\mathsf{qD}}|\psi\rangle=\mathsf{U}^{\mathsf{SUUS}}_{pk,\mathsf{qD}}|\psi\rangle$. This completes the proof of Lemma I2. □

J Compute the probability $\Pr[\mathsf{Coll}_{\mathsf{F}',m^*}]$

Lemma 6. *Let $\mathsf{P}=(\mathsf{Gen},\mathsf{Enc},\mathsf{Dec})$ be a dPKE scheme with message space $\mathcal{M}$ and average-case decryption error δ_{ac}, then we have*

$$\Pr\left[\mathsf{Coll}_{\mathsf{F}',m^*}\right]\le\frac{1}{|\mathcal{M}|}+2\cdot\delta_{ac},$$

where $\mathsf{Coll}_{\mathsf{F}',m^*}$ is the probability that $\exists m\neq m^*$ such that $\mathsf{Enc}(pk,\mathsf{F}'(m))=\mathsf{Enc}(pk,\mathsf{F}'(m^*))$ for $(pk,sk)\leftarrow\mathsf{Gen}$, $m^*\leftarrow_\$\mathcal{M}$, and $\mathsf{F}'\leftarrow_\$\Omega_{\{0,1\}^n\to\mathcal{M}}$.

Proof. Since $m\neq m^*$ implies that $\mathsf{F}'(m^*)$ and $\mathsf{F}'(m)$ are two independent and uniform random values over $\mathcal{M}$, the event $\mathsf{Coll}_{\mathsf{F}',m^*}$ implies the following event

$\mathsf{Coll}_{\mathsf{Enc}}$: $(pk,sk)\leftarrow\mathsf{Gen}$, $x^*\leftarrow_\$\mathcal{M}$, $x\leftarrow_\$\mathcal{M}$, $\mathsf{Enc}(pk,x)=\mathsf{Enc}(pk,x^*)$.

Then, we can compute

$$
\begin{aligned}
\Pr[\mathsf{Coll}_{\mathsf{F}',m^*}]&\le\Pr[\mathsf{Coll}_{\mathsf{Enc}}]\\
&=\sum_{x\in\mathcal{M}}\frac{1}{|\mathcal{M}|}\Pr\left[\begin{array}{l}\mathsf{Enc}(pk,x)=\mathsf{Enc}(pk,x^*)\\ :x^*\leftarrow_\$\mathcal{M},(pk,sk)\leftarrow\mathsf{Gen}\end{array}\right]\\
&\overset{(a)}{\le}\sum_{x\in\mathcal{M}}\left(\frac{1}{|\mathcal{M}|^2}+\frac{1}{|\mathcal{M}|}\Pr\left[\begin{array}{l}x\neq x^*\wedge\mathsf{Enc}(pk,x)=\mathsf{Enc}(pk,x^*)\\ :x^*\leftarrow_\$\mathcal{M},(pk,sk)\leftarrow\mathsf{Gen}\end{array}\right]\right)\\
&\overset{(b)}{\le}\frac{1}{|\mathcal{M}|}+\Pr\left[\begin{array}{l}\exists\bar{x}\in\mathcal{M}\text{ s.t. }\bar{x}\neq x^*\wedge\mathsf{Enc}(pk,\bar{x})=\mathsf{Enc}(pk,x^*)\\ :x^*\leftarrow_\$\mathcal{M},(pk,sk)\leftarrow\mathsf{Gen}\end{array}\right]\\
&\overset{(c)}{\le}\frac{1}{|\mathcal{M}|}+2\cdot\delta_{ac}.
\end{aligned}
\tag{92}
$$

Here (a) follows from the fact that, the probability of $x^*=x$ must be $1/|\mathcal{M}|$ for any fixed $x\in\mathcal{M}$ and uniformly chosen x^* from $\mathcal{M}$. (b) uses the fact that $x\neq x^*\wedge\mathsf{Enc}(pk,x)=\mathsf{Enc}(pk,x^*)$ implies that there exists a $\bar{x}\in\mathcal{M}$ such that $\bar{x}\neq x^*\wedge\mathsf{Enc}(pk,\bar{x})=\mathsf{Enc}(pk,x^*)$. (c) follows from Lemma G1. This completes the proof of Lemma 6. □

K Proof of Theorem 5 (from OW-CPA$_{\mathsf{P}}$ to IND-CPA$_{\mathsf{FOAC_0}}$)

Before proving this theorem, we first introduce the following proven theorems.

Definition 11 (Semi-Classical Oracle [1]). *For a subset $S \subseteq \mathcal{X}$, the indicator function f_S from $\mathcal{X}$ to $\{0,1\}$ equals 1 iff its input $x \in S$. The semi-classical oracle, $\mathcal{O}_S^{SC}$, is a quantum accessible oracle that performs the following two operations on its query state $\sum_{x\in\mathcal{X}} \alpha_x |x\rangle$:*

1. *Initialize a single qubit register B with the state $|0\rangle_{\mathsf{B}}$.*
2. *Transform $\sum_{x\in\mathcal{X}} \alpha_x |x\rangle|0\rangle_{\mathsf{B}}$ into $\sum_{x\in\mathcal{X}} \alpha_x |x\rangle|f_S(x)\rangle_{\mathsf{B}}$ and then measure B.*

During the execution of quantum oracle algorithm $\mathcal{A}^{\mathcal{O}_S^{SC}}$, let Find be the classical event that the measurement of $\mathcal{O}_S^{SC}$ (on B) ever has outcome 1.

Theorem 15 (Semi-Classical O2H [1, Theorem 1]). *Let $S \subseteq \mathcal{X}$ be random and $\mathsf{G}, \mathsf{H} : \mathcal{X} \to \mathcal{Y}$ be random functions satisfying $\forall x \notin S$, $\mathsf{G}(x) = \mathsf{H}(x)$. Let z be a random bitstring. ($\mathsf{G}, \mathsf{H}, S, z$ may have arbitrary joint distribution $\mathfrak{D}$.) Let $\mathsf{H}\backslash S$ be an oracle that first queries $\mathcal{O}_S^{SC}$ on the input register of H and then queries H. Let $\mathcal{A}$ be a quantum oracle algorithm with query times q. Define*

$$P_{\text{left}} := \Pr\left[b = 1 : b \leftarrow \mathcal{A}^{\mathsf{H}}(z), (\mathsf{G}, \mathsf{H}, S, z) \leftarrow \mathfrak{D}\right],$$
$$P_{\text{right}} := \Pr\left[b = 1 \wedge \neg\mathsf{Find} : b \leftarrow \mathcal{A}^{\mathsf{H}\backslash S}(z), (\mathsf{G}, \mathsf{H}, S, z) \leftarrow \mathfrak{D}\right].$$

Let $P_{\text{find}} := \Pr[\mathsf{Find} : \mathcal{A}^{\mathsf{H}\backslash S}(z), (\mathsf{G}, \mathsf{H}, S, z) \leftarrow \mathfrak{D}]$. Then

$$|P_{\text{left}} - P_{\text{right}}| \le \sqrt{(q+1)\cdot P_{\text{find}}} \text{ and } \left|\sqrt{P_{\text{left}}} - \sqrt{P_{\text{right}}}\right| \le \sqrt{(q+1)\cdot P_{\text{find}}}.$$

Remark 4. There are several other definitions of P_{right} in [1, Theorem 1]. In our security proofs, we only need the above definition of P_{right}.

Theorem 16 (Search in Semi-Classical Oracle [1, Theorem 2]). *Let $S \subseteq \mathcal{X}$ be random and z be a random bitstring. (S, z) may have an arbitrary joint distribution $\mathfrak{D}$. Let $\mathcal{A}$ be any quantum oracle algorithm making at most q queries to a semi-classical oracle with domain $\mathcal{X}$.*

Let $\mathcal{B}$ be an algorithm that on input z, chooses $i \leftarrow_{\$} [q]$; runs $\mathcal{A}^{\mathcal{O}_{\varnothing}^{SC}}(z)$ until (just before) the i-th query, measures the input register and outputs the measurement outcome x. Then

$$\Pr\left[\mathsf{Find} : \mathcal{A}^{\mathcal{O}_S^{SC}}(z), (S, z) \leftarrow \mathfrak{D}\right] \le 4q \cdot \Pr[x \in S : x \leftarrow \mathcal{B}(z), (S, z) \leftarrow \mathfrak{D}].$$

Theorem 17 (Double-Sided O2H [5, Lemma 5]). *Let $\mathsf{G}, \mathsf{H} : \mathcal{X} \to \mathcal{Y}$ be random functions, $S := \{w^*\}$ ($w^* \in \mathcal{X}$) be a random set and z be a random bitstring. The tuple $(\mathsf{G}, \mathsf{H}, S, z)$ may have arbitrary joint distribution $\mathfrak{D}$ and satisfies that $\forall x \notin S$, $\mathsf{G}(x) = \mathsf{H}(x)$. Let $\mathcal{A}$ be a quantum oracle algorithm with query times q. Define*

$$P_{\text{left}} := \Pr\left[b = 1 : b \leftarrow \mathcal{A}^{\mathsf{H}}(z), (\mathsf{G}, \mathsf{H}, S, z) \leftarrow \mathfrak{D}\right],$$
$$P_{\text{right}} := \Pr\left[b = 1 : b \leftarrow \mathcal{A}^{\mathsf{G}}(z), (\mathsf{G}, \mathsf{H}, S, z) \leftarrow \mathfrak{D}\right].$$

Then, we can construct a new quantum oracle algorithm $\mathcal{B}^{\mathsf{G},\mathsf{H}}(z)$*, which makes at most* q *quieries to* G *and* H *respectively and*

$$|P_{\text{left}} - P_{\text{right}}| \leq 2\sqrt{\text{Adv}(\mathcal{B})} \text{ and } \left|\sqrt{P_{\text{left}}} - \sqrt{P_{\text{right}}}\right| \leq 2\sqrt{\text{Adv}(\mathcal{B})}.$$

Here $\text{Adv}(\mathcal{B}) := \Pr[w \in S : w \leftarrow \mathcal{B}^{\mathsf{G},\mathsf{H}}(z), (\mathsf{G}, \mathsf{H}, S, z) \leftarrow \mathfrak{D}]$*. The running time of* $\mathcal{B}^{\mathsf{G},\mathsf{H}}(z)$ *satisfies* $\mathcal{T}_{\mathcal{B}} \approx \mathcal{T}_{\mathcal{A}}$*.*

Recall that FOAC_0 designed in Eq. (3) is a composite of $\mathsf{FO}_m^{\perp}$ and ACWC_0, where $\mathsf{FO}_m^{\perp}$ designed in [17] is also a composite of two modular transformations $\mathsf{U}_m^{\perp}$ and T. Thus, we have

$$\mathsf{FOAC}_0 = \mathsf{U}_m^{\perp} \circ \mathsf{T} \circ \mathsf{ACWC}_0. \tag{93}$$

The definitions of $\mathsf{U}_m^{\perp}$ and T are given as follows.

Transformation T: Let $\mathsf{P} = (\mathsf{Gen}, \mathsf{Enc}, \mathsf{Dec})$ be an rPKE scheme with message space $\{0,1\}^n$ and randomness space $\mathcal{M}$. Let $\mathsf{F} : \{0,1\}^n \to \mathcal{M}$ be a hash function. We associate dPKE scheme $\mathsf{T}[\mathsf{P}, \mathsf{F}] := (\mathsf{Gen}, \mathsf{Enc}', \mathsf{Dec}')$. The constituting algorithms of $\mathsf{T}[\mathsf{P}, \mathsf{F}]$ are shown in Fig. 20.

Gen	$\mathsf{Enc}'(pk, m \in \{0,1\}^n)$	$\mathsf{Dec}'(sk, c)$
$(pk, sk) \leftarrow \mathsf{Gen}$	$c := \mathsf{Enc}(pk, m; \mathsf{F}(m))$	$m := \mathsf{Dec}(sk, c)$
return (pk, sk)	**return** c	**if** $m = \perp \vee c \neq \mathsf{Enc}(pk, m; \mathsf{F}(m))$
		return $\perp$
		else return m

Fig. 20. Deterministic Public Key Encryption $\mathsf{T}[\mathsf{P},\mathsf{F}]$.

Transformation $\mathsf{U}_m^{\perp}$: Let $\mathsf{dPKE} = (\mathsf{Gen}, \mathsf{dEnc}, \mathsf{dDec})$ be a dPKE scheme with message space $\{0,1\}^n$. For a given set $\mathcal{K}$, let $\mathsf{G} : \{0,1\}^n \to \mathcal{K}$ be a hash function. We associate KEM scheme $\mathsf{U}_m^{\perp}[\mathsf{dPKE}, \mathsf{G}] := (\mathsf{Gen}, \mathsf{Enca}, \mathsf{Deca})$ that has key space $\mathcal{K}$. The constituting algorithms of $\mathsf{U}_m^{\perp}[\mathsf{dPKE},\mathsf{G}]$ are given in Fig. 21.

Gen	$\mathsf{Enca}(pk)$	$\mathsf{Deca}(sk, c)$
$(pk, sk) \leftarrow \mathsf{Gen}$	$m \leftarrow_{\$} \{0,1\}^n$	$m := \mathsf{dDec}(sk, c)$
return (pk, sk)	$c := \mathsf{dEnc}(pk, m)$	**if** $m = \perp$
	$K := \mathsf{G}(m)$	**return** $\perp$
	return (K, c)	**else return** $K := \mathsf{G}(m)$

Fig. 21. Key Encapsulation Mechanism $\mathsf{U}_m^{\perp}[\mathsf{dPKE}, \mathsf{G}]$.

Next, we introduce three lemmas related to the transformations $\mathsf{U}_m^{\perp}$, T and ACWC_0. For $\mathsf{U}_m^{\perp}$, we introduce the following lemma, which shows that the IND-CPA security of $\mathsf{U}_m^{\perp}[\mathsf{dPKE}, \mathsf{G}]$ can be reduced to the OW-CPA security of dPKE in the QROM.

Lemma 7 **(OW-CPA of dPKE $\overset{\text{QROM}}{\Rightarrow}$ IND-CPA of $\mathsf{U}_m^{\perp}$).** *Let $\mathcal{A}$ be an IND-CPA adversary against $\mathsf{U}_m^{\perp}[\mathsf{dPKE}, \mathsf{G}]$, making q_{G} quantum queries to the random oracle G. Then we can construct a OW-CPA adversary $\mathcal{B}$ against dPKE such that $\mathcal{T}_{\mathcal{B}} \leq \mathcal{T}_{\mathcal{A}} + O(q_{\mathsf{G}}) \cdot \mathcal{T}_{\mathsf{dEnc}} + O(q_{\mathsf{G}}^2)$ and*

$$\mathrm{Adv}^{\mathsf{IND\text{-}CPA}}_{\mathsf{U}_m^{\perp}[\mathsf{dPKE},\mathsf{G}],\mathcal{A}} \leq \sqrt{\mathrm{Adv}^{\mathsf{OW\text{-}CPA}}_{\mathsf{dPKE},\mathcal{B}}} + \sqrt{\Pr\left[\mathsf{Coll}_{\mathsf{dPKE}}\right]}.$$

Here $\mathsf{Coll}_{\mathsf{dPKE}}$ is the following event:

$$(pk, sk) \leftarrow \mathsf{Gen},\ m^* \leftarrow_{\$} \{0,1\}^n,\ \exists m \neq m^* \ s.t.\ \mathsf{dEnc}(pk, m) = \mathsf{dEnc}(pk, m^*).$$

Proof. The proof of this lemma is identical to that of [15, Theorem 9], with the only modification being that the MRE-O2H theorem (employed in the proof of [15, Theorem 9]) is replaced by Theorem 17. □

For the transformation T, we introduce the following proven lemma, which shows that the OW-CPA security of $\mathsf{T}[\mathsf{P}, \mathsf{F}]$ can be reduced to the OW-CPA security of P in the QROM without the square-root advantage loss.

Lemma K1 ([19, Theorem 2.4.4]) *Let $\mathcal{A}$ be a OW-CPA adversary against $\mathsf{T}[\mathsf{P}, \mathsf{F}]$, making q_{F} quantum queries to the random oracle F. Then we can construct a OW-CPA adversary $\mathcal{B}$ against P such that $\mathcal{B}$ is $\mathcal{T}_{\mathcal{B}} \leq \mathcal{T}_{\mathcal{A}} + O(q_{\mathsf{F}}^2)$ and*

$$\mathrm{Adv}^{\mathsf{OW\text{-}CPA}}_{\mathsf{T}[\mathsf{P},\mathsf{F}],\mathcal{A}} \leq 4(q_{\mathsf{F}} + 2)^2 \cdot \mathrm{Adv}^{\mathsf{OW\text{-}CPA}}_{\mathsf{P},\mathcal{B}}.$$

As for the transformation ACWC_0, we can prove the following lemma, which gives a OW-CPA proof of $\mathsf{ACWC}_0[\mathsf{P}, \mathsf{H}]$ in the QROM. In fact, the proof of this lemma is a directly apply of Theorem 17, and we give the detailed proof in Supplementary Material K.3.

Lemma 8 **(OW-CPA of P $\overset{\text{QROM}}{\Rightarrow}$ OW-CPA of ACWC_0).** *Let dPKE scheme P be δ_{ac}-average-correct. Let $\mathcal{A}$ be a OW-CPA adversary against the $\mathsf{ACWC}_0[\mathsf{P}, \mathsf{H}]$, making q_{H} quantum queries to the random oracle H. Then we can construct a OW-CPA adversary $\mathcal{B}$ against P such that $\mathcal{T}_{\mathcal{B}} \leq \mathcal{T}_{\mathcal{A}} + O(q_{\mathsf{H}}) \cdot \mathcal{T}_{\mathsf{Enc}} + O(q_{\mathsf{H}}^2)$ and*

$$\mathrm{Adv}^{\mathsf{OW\text{-}CPA}}_{\mathsf{ACWC}_0[\mathsf{P},\mathsf{H}],\mathcal{A}} \leq \left(2 \cdot \sqrt{\mathrm{Adv}^{\mathsf{OW\text{-}CPA}}_{\mathsf{P},\mathcal{B}}} + \frac{1}{\sqrt{2^n}} + 3 \cdot \sqrt{\delta_{ac}}\right)^2.$$

Here n is the output length of H.

K.1 Proof of Theorem 5 under dPKE Schemes

Here we assume that P is a dPKE scheme and prove Theorem 5. In fact, according to Eq. (93), we can compute

$$\begin{aligned}\mathrm{Adv}^{\mathsf{IND\text{-}CPA}}_{\mathsf{KEMAC}_0,\mathcal{A}} &= \mathrm{Adv}^{\mathsf{IND\text{-}CPA}}_{\mathsf{U}^{\perp}_m[\mathsf{T}[\mathsf{ACWC}_0[\mathsf{P},\mathsf{H}],\mathsf{F}],\mathsf{G}],\mathcal{A}} \\ &\overset{(a)}{\leq} \sqrt{\mathrm{Adv}^{\mathsf{OW\text{-}CPA}}_{\mathsf{T}[\mathsf{ACWC}_0[\mathsf{P},\mathsf{H}],\mathsf{F}],\mathcal{B}}} + \sqrt{\Pr\left[\mathsf{Coll}_{\mathsf{T}[\mathsf{ACWC}_0[\mathsf{P},\mathsf{H}],\mathsf{F}]}\right]}.\end{aligned} \tag{94}$$

Here (a) follows from Lemma 7, and the running time of $\mathcal{B}$ is $\mathcal{T}_{\mathcal{B}} \leq \mathcal{T}_{\mathcal{A}} + O(q_{\mathsf{G}}) \cdot \mathcal{T}_{\mathsf{Enc}_{\mathsf{TA}}} + O(q_{\mathsf{G}}^2)$. The event $\mathsf{Coll}_{\mathsf{T}[\mathsf{ACWC}_0[\mathsf{P},\mathsf{H}],\mathsf{F}]}$ denotes $\exists m \neq m^*$ such that $\mathsf{Enc}_{\mathsf{TA}}(pk, m) = \mathsf{Enc}_{\mathsf{TA}}(pk, m^*)$, under the condition that $m^* \leftarrow_{\$} \mathcal{M}_{\mathsf{TA}}$ and $(pk, sk) \leftarrow \mathsf{Gen}_{\mathsf{TA}}$. The $\mathsf{Enc}_{\mathsf{TA}}$, $\mathcal{M}_{\mathsf{TA}}$, and $\mathsf{Gen}_{\mathsf{TA}}$ respectively denote the encryption algorithm, message space and key generation algorithm of $\mathsf{T}[\mathsf{ACWC}_0[\mathsf{P},\mathsf{H}],\mathsf{F}]$.

Based on the construction of ACWC_0 and T given in Fig. 8 and Fig. 20, respectively, it is obvious that the encryption algorithm of $\mathsf{T}[\mathsf{ACWC}_0[\mathsf{P},\mathsf{H}],\mathsf{F}]$ has a similar running times to the encryption algorithm of P. Hence, $\mathcal{B}$ also satisfies $\mathcal{T}_{\mathcal{B}} \leq \mathcal{T}_{\mathcal{A}} + O(q_{\mathsf{G}}) \cdot \mathcal{T}_{\mathsf{Enc}} + O(q_{\mathsf{G}}^2)$, and we can rewrite $\mathsf{Coll}_{\mathsf{T}[\mathsf{ACWC}_0[\mathsf{P},\mathsf{H}],\mathsf{F}]}$ as

$$\begin{gathered}m^* \leftarrow_{\$} \{0,1\}^n,\ (pk, sk) \leftarrow \mathsf{Gen},\ \mathsf{F} \leftarrow_{\$} \Omega_{\{0,1\}^n \to \mathcal{M}},\ \mathsf{H} \leftarrow_{\$} \Omega_{\mathcal{M} \to \{0,1\}^n},\ \exists m \neq m^* \\ \text{s.t. } \mathsf{Enc}(pk, \mathsf{F}(m)) = \mathsf{Enc}(pk, \mathsf{F}(m^*)) \wedge m \oplus \mathsf{H}(\mathsf{F}(m)) = m^* \oplus \mathsf{H}(\mathsf{F}(m^*)).\end{gathered}$$

Note that the above event implies the following classical event

$$\begin{gathered}\mathsf{Coll}_{\mathsf{F},m^*}\colon (pk, sk) \leftarrow \mathsf{Gen},\ m^* \leftarrow_{\$} \{0,1\}^n,\ \mathsf{F} \leftarrow_{\$} \Omega_{\{0,1\}^n \to \mathcal{M}},\ \exists m \neq m^* \text{ s.t.} \\ \mathsf{Enc}(pk, \mathsf{F}(m)) = \mathsf{Enc}(pk, \mathsf{F}(m^*)).\end{gathered}$$

Thus, by Lemma 6, we have

$$\Pr\left[\mathsf{Coll}_{\mathsf{T}[\mathsf{ACWC}_0[\mathsf{P},\mathsf{H}],\mathsf{F}]}\right] \leq \Pr\left[\mathsf{Coll}_{\mathsf{F},m^*}\right] \leq \frac{1}{|\mathcal{M}|} + 2 \cdot \delta_{ac}. \tag{95}$$

Before giving the upper bound of $\mathrm{Adv}^{\mathsf{OW\text{-}CPA}}_{\mathsf{T}[\mathsf{ACWC}_0[\mathsf{P},\mathsf{H}],\mathsf{F}],\mathcal{B}}$, we first count the number of queries of $\mathcal{B}$ to the random oracles F and H. Notice that Lemma 7 only considers the G queries, and hence $\mathcal{B}$ will inherit $\mathcal{A}$'s at most q_{F} (resp. q_{H}) queries to F (resp. H). In addition, by the proof of Lemma 7, $\mathcal{B}$ also needs to run the encryption algorithm of $\mathsf{T}[\mathsf{ACWC}_0[\mathsf{P},\mathsf{H}],\mathsf{F}]$ at most $2q_{\mathsf{G}}$ times. So by the construction of ACWC_0 and T given in Fig. 8 and Fig. 20, respectively, $\mathcal{B}$ needs to query F and H both at most $2q_{\mathsf{G}}$ times to complete such computation. In summary, $\mathcal{B}$ needs to query F (resp. H) at most $q_{\mathsf{F}} + 2q_{\mathsf{G}}$ (resp. $2q_{\mathsf{G}} + q_{\mathsf{H}}$) times.

Based on the query times counted above, we can construct a $\mathsf{OW\text{-}CPA}$ adversary $\mathcal{C}$ against $\mathsf{ACWC}_0[\mathsf{P},\mathsf{H}]$ by using Lemma K1 such that

$$\sqrt{\mathrm{Adv}^{\mathsf{OW\text{-}CPA}}_{\mathsf{T}[\mathsf{ACWC}_0[\mathsf{P},\mathsf{H}],\mathsf{F}],\mathcal{B}}} \leq 2(q_{\mathsf{F}} + 2q_{\mathsf{G}} + 2) \cdot \sqrt{\mathrm{Adv}^{\mathsf{OW\text{-}CPA}}_{\mathsf{ACWC}_0[\mathsf{P},\mathsf{H}],\mathcal{C}}}. \tag{96}$$

The running time of $\mathcal{C}$ satisfies $\mathcal{T}_{\mathcal{C}} \leq \mathcal{T}_{\mathcal{B}} + O(q_{\mathsf{F}}^2 + q_{\mathsf{G}}^2)$ and $\mathcal{C}$ will inherit $\mathcal{B}$'s H queries. Thus, $\mathcal{C}$ queries H at most $2q_{\mathsf{G}} + q_{\mathsf{H}}$ times. Next, using Lemma 8, we can construct a OW-CPA adversary $\mathcal{D}$ against P with running time $\mathcal{T}_{\mathcal{D}} \leq \mathcal{T}_{\mathcal{C}} + O(q_{\mathsf{G}} + q_{\mathsf{H}}) \cdot \mathcal{T}_{\mathsf{Enc}} + O(q_{\mathsf{G}}^2 + q_{\mathsf{H}}^2)$ and

$$\sqrt{\mathrm{Adv}^{\mathsf{OW\text{-}CPA}}_{\mathsf{ACWC_0[P,H]},\mathcal{C}}} \leq 2 \cdot \sqrt{\mathrm{Adv}^{\mathsf{OW\text{-}CPA}}_{\mathsf{P},\mathcal{D}}} + \frac{1}{\sqrt{2^n}} + 3 \cdot \sqrt{\delta_{ac}}. \tag{97}$$

Finally, by Eq. (94) to (97) and renaming $\mathcal{D}$ into $\mathcal{B}$, we obtain

$$\begin{aligned}\mathrm{Adv}^{\mathsf{IND\text{-}CPA}}_{\mathsf{KEMAC_0},\mathcal{A}} \leq & 4(q_{\mathsf{F}} + 2q_{\mathsf{G}} + 2) \cdot \sqrt{\mathrm{Adv}^{\mathsf{OW\text{-}CPA}}_{\mathsf{P},\mathcal{B}}} + 6(q_{\mathsf{F}} + 2q_{\mathsf{G}} + 3) \cdot \sqrt{\delta_{ac}} \\ & + \frac{2(q_{\mathsf{F}} + 2q_{\mathsf{G}} + 2)}{\sqrt{2^n}} + \frac{1}{\sqrt{|\mathcal{M}|}}.\end{aligned}$$

As for the running time of $\mathcal{B}$, by the previous proof, we have $\mathcal{T}_{\mathcal{B}} \leq \mathcal{T}_{\mathcal{A}} + O(q_{\mathsf{G}} + q_{\mathsf{H}}) \cdot \mathcal{T}_{\mathsf{Enc}} + O(q_{\mathsf{F}}^2 + q_{\mathsf{G}}^2 + q_{\mathsf{H}}^2)$. This completes the proof of Theorem 5 under dPKE schemes. ☐

K.2 Proof of Theorem 5 under rPKE Schemes

We first prove the following lemma about $\mathsf{T} \circ \mathsf{ACWC_0}$, which shows that its OW-CPA security can be reduced to the OW-CPA security of the underlying rPKE scheme P in the QROM. The core proof idea of this lemma is similar to that of [19, Theorem 2.4.5], that is, reprogramming two quantum random oracles simultaneously to avoid the sequential use of One-Way to Hiding twice.

Lemma 9 **(OW-CPA of $\mathsf{P} \overset{\mathrm{QROM}}{\Rightarrow}$ OW-CPA of $\mathsf{T} + \mathsf{ACWC_0}$).** *Let $\mathcal{A}$ be a OW-CPA adversary against $\mathsf{T[ACWC_0[P,H],F]}$, making q_{F} and q_{H} quantum queries to the random oracle F and H. Then we can construct a OW-CPA adversary $\mathcal{B}$ against P such that $\mathcal{T}_{\mathcal{B}} \leq \mathcal{T}_{\mathcal{A}} + O(q_{\mathsf{F}}^2 + q_{\mathsf{H}}^2)$ and*

$$\mathrm{Adv}^{\mathsf{OW\text{-}CPA}}_{\mathsf{T[ACWC_0[P,H],F]},\mathcal{A}} \leq 4(q_{\mathsf{F}} + q_{\mathsf{H}} + 2)^2 \cdot \mathrm{Adv}^{\mathsf{OW\text{-}CPA}}_{\mathsf{P},\mathcal{B}} + \frac{4(q_{\mathsf{F}} + q_{\mathsf{H}} + 2)^2}{2^n}.$$

Proof. To prove this lemma, we introduce four games $\mathbf{G_0}$ to $\mathbf{G_3}$ as shown in Fig. 22. The game $\mathbf{G_0}$ here is the OW-CPA game of $\mathsf{T[ACWC_0[P,H],F]}$ with the adversary $\mathcal{A}$ in the QROM. Hence we have

$$\mathrm{Adv}^{\mathsf{OW\text{-}CPA}}_{\mathsf{T[ACWC_0[P,H],F]},\mathcal{A}} = \Pr\left[1 \leftarrow \mathbf{G_0}\right]. \tag{98}$$

The distribution $\mathfrak{D}$ used in games $\mathbf{G_1}$ to $\mathbf{G_3}$ is a joint distribution of the tuple $(\mathsf{G}, \mathsf{G}', S, \mathsf{F}, \mathsf{H}, m^*, pk, sk, y_1)$, where $\mathsf{F} \leftarrow_\$ \Omega_{\{0,1\}^n \to \mathcal{M}}$, $\mathsf{H} \leftarrow_\$ \Omega_{\mathcal{M} \to \{0,1\}^n}$, $m^* \leftarrow_\$ \{0,1\}^n$, $(pk, sk) \leftarrow_\$ \mathsf{Gen}$, $m_1^* \| r^* := \mathsf{F}(m^*)$, $y_1 := \mathsf{H}(m_1^*)$, $S := \{(0, m^*), (1, m_1^*)\}$, and function $\mathsf{G} : (\{0\} \times \{0,1\}^n) \cup (\{1\} \times \mathcal{M}) \to \mathcal{M} \cup \{0,1\}^n$ is defined as $\mathsf{G}(0, x) = \mathsf{F}(x)$ and $\mathsf{G}(1, x) = \mathsf{H}(x)$. The G' is the same as G, except that $\mathsf{G}'(0, m^*)$ and $\mathsf{G}'(1, m_1^*)$ are fresh random values over $\mathcal{M}$ and $\{0,1\}^n$, respectively.

Game $\mathbf{G_1}$: This game is the same as $\mathbf{G_0}$, except that the adversary $\mathcal{A}$ is replaced with $\mathcal{A}_1$, which runs $\mathcal{A}$ and simulates $\mathsf{F}(\cdot)$ and $\mathsf{H}(\cdot)$ for $\mathcal{A}$ by querying $\mathsf{G}(0, \cdot)$

and $\mathsf{G}(1,\cdot)$, respectively[34]. In addition, after getting the output m' from $\mathcal{A}$, $\mathcal{A}_1$ performs a query to G with $(0, m')$. By the definition of the distribution $\mathfrak{D}$, it is not hard to check that

$$\Pr[1 \leftarrow \mathbf{G_1}] = \Pr[1 \leftarrow \mathbf{G_0}]. \quad (99)$$

Obviously, the number of queries of $\mathcal{A}_1$ to G is $q := q_{\mathsf{F}} + q_{\mathsf{H}} + 1$ by its construction.

Game $\mathbf{G_0}$	**GAMES $\mathbf{G_1}$-$\mathbf{G_3}$**	
1. $(pk, sk) \leftarrow \mathsf{Gen}$	1. $(\mathsf{G}, \mathsf{G}', S, \mathsf{F}, \mathsf{H}, m^*, (pk, sk), y_1) \leftarrow_\$ \mathfrak{D}$	$//\mathbf{G_1}$-$\mathbf{G_3}$
$\mathsf{F} \leftarrow_\$ \Omega_{\{0,1\}^n \to \mathcal{M}}$	2. $c_1^* := \mathsf{Enc}(pk, m_1^* \| r^*)$	$//\mathbf{G_1}$-$\mathbf{G_3}$
$\mathsf{H} \leftarrow_\$ \Omega_{\mathcal{M} \to \{0,1\}^n}$	$c_2^* := m^* \oplus y_1$	$//\mathbf{G_1}$-$\mathbf{G_3}$
2. $m^* \leftarrow_\$ \{0,1\}^n$	3. $m' \leftarrow \mathcal{A}_1^{\mathsf{G}}(pk, c_1^* \| c_2^*)$	$//\mathbf{G_1}$
$m_1^* \| r^* := \mathsf{F}(m^*)$	$m' \leftarrow \mathcal{A}_1^{\mathsf{G}' \setminus S}(pk, c_1^* \| c_2^*)$	$//\mathbf{G_2}$
$c_1^* := \mathsf{Enc}(pk, m_1^*; r^*)$	$(b, x) \leftarrow \mathcal{A}_2^{\mathsf{G}'}(pk, c_1^* \| c_2^*)$	$//\mathbf{G_3}$
$c_2^* := m^* \oplus \mathsf{H}(m_1^*)$	4. **return** $[\![m' = m^*]\!]$	$//\mathbf{G_0}$-$\mathbf{G_2}$
3. $m' \leftarrow \mathcal{A}^{\mathsf{F},\mathsf{H}}(pk, c_1^* \| c_2^*)$	**return** $[\![(b, x) \in S]\!]$	$//\mathbf{G_3}$
4. **return** $[\![m' = m^*]\!]$		

Fig. 22. Games $\mathbf{G_0}$ to $\mathbf{G_3}$ in the proof of Lemma 9.

Game $\mathbf{G_2}$: This game is the same as $\mathbf{G_1}$, except that the G queried by $\mathcal{A}_1$ is replaced with the punctured oracle $\mathsf{G}' \setminus S$.

Denote Find_S as the event that the semi-classical oracle $\mathcal{O}_S^{SC}$ ever outputs 1. Now by directly applying Theorem 15, we can obtain

$$\left| \sqrt{\Pr[1 \leftarrow \mathbf{G_1}]} - \sqrt{\Pr[b = 1 \wedge \neg \mathsf{Find}_S : b \leftarrow \mathbf{G_2}]} \right| \leq \sqrt{(q+1) \cdot \Pr[\mathsf{Find}_S : \mathbf{G_2}]}.$$

Since $b = 1$ and Find_S cannot occur simultaneously in game $\mathbf{G_2}$, the probability $\Pr[b = 1 \wedge \neg \mathsf{Find}_S : b \leftarrow \mathbf{G_2}]$ is 0, and thus we have

$$\Pr[1 \leftarrow \mathbf{G_1}] \leq (q+1) \cdot \Pr[\mathsf{Find}_S : \mathbf{G_2}]. \quad (100)$$

Game $\mathbf{G_3}$: This game is the same as $\mathbf{G_2}$, except that the adversary $\mathcal{A}_1$ is replaced with $\mathcal{A}_2$ and the final check is replaced with $(b, x) \in S$, where (b, x) is the output of $\mathcal{A}_2$. Here $\mathcal{A}_2$ chooses $i \leftarrow_\$ [q]$; runs $\mathcal{A}_1^{\mathsf{G}'}(pk, c_1^* \| c_2^*)$ until (just before) the i-th query; then measures the query input register in the computational basis and outputs the measurement outcomes (b, x).

[34] The quantum queries of F and H can be simulated correspondingly.

By using Theorem 16, we can compute

$$\begin{aligned}\Pr\left[\mathsf{Find}_S : \mathbf{G_2}\right] \leq& 4q \cdot \Pr\left[1 \leftarrow \mathbf{G_3}\right] \\ =& 4q \cdot \Pr\left[(b,x) \in S : (b,x) \in \mathcal{A}_2^{\mathsf{G}'}(pk, c_1^*, c_2^*)\right] \\ \leq& \Pr\left[x = m^* : (b,x) \in \mathcal{A}_2^{\mathsf{G}'}(pk, c_1^*, c_2^*)\right] \\ &+ \Pr\left[x = m_1^* : (b,x) \in \mathcal{A}_2^{\mathsf{G}'}(pk, c_1^*, c_2^*)\right].\end{aligned} \tag{101}$$

In the above equation we omit the sampling of $\mathfrak{D}$ for simplicity. At this point, since $\mathsf{G}'(0, m^*)$ and $\mathsf{G}'(1, m_1^*)$ have been reprogrammed into fresh random values, the $m_1^*||r^*$ and y_1 used to generate (c_1^*, c_2^*) must be uniformly random in $\mathcal{A}_2$'s view. Therefore, we have

$$\Pr\left[x = m^* : (b,x) \in \mathcal{A}_2^{\mathsf{G}'}(pk, c_1^*, c_2^*)\right] = \frac{1}{2^n} \tag{102}$$

and one can easily construct a $\mathsf{OW\text{-}CPA}$ adversary $\mathcal{B}$ against P such that

$$\Pr\left[x = m_1^* : \mathcal{A}_2^{\mathsf{G}'}(pk, c_1^*, c_2^*)\right] = \mathrm{Adv}_{\mathsf{P},\mathcal{B}}^{\mathsf{OW\text{-}CPA}}. \tag{103}$$

Briefly speaking, $\mathcal{B}$ first runs $\mathcal{A}_2^{\mathsf{G}'}(pk, c_1^*, c_2^*)$ to get its output (b, x) and then guesses the challenge plaintext m_1^* as x. Here (c_1^*, c_2^*) is the challenge ciphertext of the $\mathsf{OW\text{-}CPA}$ game of P, and $\mathcal{B}$ uses a $2q$-wise independent function to simulate G' for $\mathcal{A}_2$. Obviously we have $\mathcal{T}_{\mathcal{B}} \leq \mathcal{T}_{\mathcal{A}} + O(q_{\mathsf{F}}^2 + q_{\mathsf{H}}^2)$.

Finally, by combining Eq. (98) to Eq. (103), we obtain

$$\mathrm{Adv}_{\mathsf{T}[\mathsf{ACWC_0}[\mathsf{P},\mathsf{H}],\mathsf{F}],\mathcal{A}}^{\mathsf{OW\text{-}CPA}} \leq 4(q_{\mathsf{F}} + q_{\mathsf{H}} + 2)^2 \cdot \mathrm{Adv}_{\mathsf{P},\mathcal{B}}^{\mathsf{OW\text{-}CPA}} + \frac{4(q_{\mathsf{F}} + q_{\mathsf{H}} + 2)^2}{2^n}.$$

This completes the proof of Lemma 9. □

At this point, we can compute

$$\begin{aligned}\mathrm{Adv}_{\mathsf{KEMAC_0},\mathcal{A}}^{\mathsf{IND\text{-}CPA}} &= \mathrm{Adv}_{\mathsf{U}_m^{\perp}[\mathsf{T}[\mathsf{ACWC_0}[\mathsf{P},\mathsf{H}],\mathsf{F}],\mathsf{G}],\mathcal{A}}^{\mathsf{IND\text{-}CPA}} \\ &\overset{(a)}{\leq} \sqrt{\mathrm{Adv}_{\mathsf{T}[\mathsf{ACWC_0}[\mathsf{P},\mathsf{H}],\mathsf{F}],\mathcal{A}_1}^{\mathsf{OW\text{-}CPA}}} + \sqrt{\Pr\left[\mathsf{Coll}_{\mathsf{T}[\mathsf{ACWC_0}[\mathsf{P},\mathsf{H}],\mathsf{F}]}\right]} \\ &\overset{(b)}{\leq} 2(q_{\mathsf{F}} + q_{\mathsf{H}} + 2) \cdot \left(\sqrt{\mathrm{Adv}_{\mathsf{P},\mathcal{B}}^{\mathsf{OW\text{-}CPA}}} + \frac{1}{\sqrt{2^n}}\right) + \frac{1}{\sqrt{|\mathcal{M}|}} + 2 \cdot \sqrt{\delta_{ac}}.\end{aligned}$$

Here (a) follows from Lemma 7, (b) follows from Lemma 9 and Eq. (95), and the running time of $\mathcal{B}$ satisfies $\mathcal{T}_{\mathcal{B}} \leq \mathcal{T}_{\mathcal{A}} + O(q_{\mathsf{G}}) \cdot \mathcal{T}_{\mathsf{Enc}} + O(q_{\mathsf{F}}^2 + q_{\mathsf{G}}^2 + q_{\mathsf{H}}^2)$. This completes the proof of Theorem 5 under rPKE schemes. □

K.3 Proof of Lemma 8

To prove this lemma, we introduce two games $\mathbf{G_0}$ and $\mathbf{G_1}$ as shown in Fig. 23.

Game $\mathbf{G_0}$	**Game $\mathbf{G_1}$**
1. $(pk, sk) \leftarrow \mathsf{Gen}$	1. $(pk, sk) \leftarrow \mathsf{Gen}$
$\mathsf{H} \leftarrow_\$ \Omega_{\mathcal{M}\to\{0,1\}^n}$	$\mathsf{H} \leftarrow_\$ \Omega_{\mathcal{M}\to\{0,1\}^n}$
2. $m^* \leftarrow_\$ \{0,1\}^n$	2. $m^* \leftarrow_\$ \{0,1\}^n$
$r^* \leftarrow_\$ \mathcal{M}$	$m \leftarrow_\$ \{0,1\}^n$
$c_1^* := \mathsf{Enc}(pk, r^*)$	$r^* \leftarrow_\$ \mathcal{M}$
$c_2^* := m^* \oplus \mathsf{H}(r^*)$	$c_1^* := \mathsf{Enc}(pk, r^*)$
3. $m' \leftarrow \mathcal{A}^{\mathsf{H}}(pk, c_1^*, c_2^*)$	$c_2^* := m^* \oplus m$
4. **return** $[\![m' = m^*]\!]$	3. $m' \leftarrow \mathcal{A}^{\mathsf{H}}(pk, c_1^*, c_2^*)$
	4. **return** $[\![m' = m^*]\!]$

Fig. 23. Games $\mathbf{G_0}$ and $\mathbf{G_1}$ in the proof of Lemma 8.

Game $\mathbf{G_0}$: This is the $\mathsf{OW\text{-}CPA}$ game of $\mathsf{ACWC_0[P,H]}$ with the adversary $\mathcal{A}$ in the QROM. Hence we have

$$\mathrm{Adv}^{\mathsf{OW\text{-}CPA}}_{\mathsf{ACWC_0[P,H]},\mathcal{A}} = \Pr\left[1 \leftarrow \mathbf{G_0}\right]. \tag{104}$$

Game $\mathbf{G_1}$: This game is the same as $\mathbf{G_0}$, except that the $\mathsf{H}(r^*)$ used to generate c_2^* is replaced by m, a uniformly random value over $\{0,1\}^n$. Obviously the adversary $\mathcal{A}$ in this game is independent with the m^*, thus we have

$$\Pr\left[1 \leftarrow \mathbf{G_1}\right] = \frac{1}{2^n}. \tag{105}$$

Next we compute the difference between the success probability of games $\mathbf{G_0}$ and $\mathbf{G_1}$. Let $\mathfrak{D}$ be a joint distribution of $(\mathsf{G}, \mathsf{H}, \{r^*\}, z)$, where $\mathsf{H} \leftarrow_\$ \Omega_{\mathcal{M}\to\{0,1\}^n}$, $r^* \leftarrow_\$ \mathcal{M}$, and G is the same as H except that $\mathsf{G}(r^*)$ is a fresh uniform random value over $\{0,1\}^n$. As for the z, it is (pk, c_1^*, c_2^*), where $(pk, sk) \leftarrow \mathsf{Gen}$, $m^* \leftarrow_\$ \{0,1\}^n$, $c_1^* := \mathsf{Enc}(pk, r^*)$, and $c_2^* := m^* \oplus \mathsf{H}(r^*)$. Now, it is easy to check that games $\mathbf{G_0}$ and $\mathbf{G_1}$ satisfy

$$\Pr\left[1 \leftarrow \mathbf{G_0}\right] = \Pr\left[b = 1 : b = [\![m' = m^*]\!], m' \leftarrow \mathcal{A}^{\mathsf{H}}(z), (\mathsf{G}, \mathsf{H}, \{r^*\}, z) \leftarrow \mathfrak{D}\right],$$
$$\Pr\left[1 \leftarrow \mathbf{G_1}\right] = \Pr\left[b = 1 : b = [\![m' = m^*]\!], m' \leftarrow \mathcal{A}^{\mathsf{G}}(z), (\mathsf{G}, \mathsf{H}, \{r^*\}, z) \leftarrow \mathfrak{D}\right].$$

Then, by using Theorem 17 under the distribution $\mathfrak{D}$, we can obtain a new adversary $\mathcal{A}_1$ such that $\mathcal{T}_{\mathcal{A}_1} \approx \mathcal{T}_{\mathcal{A}}$ and

$$\begin{aligned}&\left|\sqrt{\Pr\left[1 \leftarrow \mathbf{G_0}\right]} - \sqrt{\Pr\left[1 \leftarrow \mathbf{G_1}\right]}\right| \\ \le\, &2 \cdot \sqrt{\Pr\left[w = r^* : w \leftarrow \mathcal{A}_1^{\mathsf{G},\mathsf{H}}(z), (\mathsf{G}, \mathsf{H}, \{r^*\}, z) \leftarrow \mathfrak{D}\right]}.\end{aligned} \tag{106}$$

Here $\mathcal{A}_1$ queries G and H both at most q_{H} times.

Based on the above adversary $\mathcal{A}_1$, we construct a $\mathsf{OW\text{-}CPA}$ adversary $\mathcal{B}$ against P as follows:

1. The input of $\mathcal{B}$ is (pk, c_1^*), where $c_1^* := \mathsf{Enc}(pk, r^*)$ is the challenge ciphertext and $r^*(\leftarrow_\$ \mathcal{M})$ is the challenge plaintext of the OW-CPA game.
2. $\mathcal{B}$ samples $m^* \leftarrow_\$ \{0,1\}^n$, $m \leftarrow_\$ \{0,1\}^n$, and chooses a $2q_{\mathsf{H}}$-wise function $g : \mathcal{M} \to \{0,1\}^n$. Then $\mathcal{B}$ runs $\mathcal{A}_1^{\mathsf{G},\mathsf{H}}(pk, c_1^*, m^* \oplus m)$, outputs its final result and simulates its oracles as follows:
   (a) When $\mathcal{A}_1$ queries G with the query state $|x, y\rangle$, $\mathcal{B}$ replies it by performing $\mathsf{O}_g : |x, y\rangle \mapsto |x, y \oplus g(x)\rangle$.
   (b) When $\mathcal{A}_1$ queries H with the query state $|x, y\rangle$, $\mathcal{B}$ replies it by performing the following control operation: Perform O_g if $\mathsf{Enc}(pk, x) \neq c_1^*$, and perform $\mathsf{U}_m : |x, y\rangle \mapsto |x, y \oplus m\rangle$ if $\mathsf{Enc}(pk, x) = c_1^*$.

Note that $\mathcal{B}$ needs to compute Enc at most $2q_{\mathsf{H}}$ times during the simulation of H, and hence the running time of $\mathcal{B}$ satisfies $\mathcal{T}_{\mathcal{B}} \leq \mathcal{T}_{\mathcal{A}} + O(q_{\mathsf{H}}) \cdot \mathcal{T}_{\mathsf{Enc}} + O(q_{\mathsf{H}}^2)$.

In the above construction, since $\mathcal{B}$ cannot get the challenge plaintext r^* directly, it checks if x equals r^* by testing if $\mathsf{Enc}(pk, x)$ equals c_1^* during the simulation of H. Therefore, if the event $\mathsf{Coll}_{pk}^{r^*}$ defined as follows does not occur, $\mathcal{B}$ simulates G and H for $\mathcal{A}_1$ perfectly[35]

$$\mathsf{Coll}_{pk}^{r^*}\text{: } r^* \leftarrow_\$ \mathcal{M},\ (pk, sk) \leftarrow \mathsf{Gen},\ \exists r \neq r^* \text{ s.t. } \mathsf{Enc}(pk, r) = \mathsf{Enc}(pk, r^*)$$

At this point, we can compute

$$\begin{aligned} &\Pr\left[w = r^* : w \leftarrow \mathcal{A}_1^{\mathsf{G},\mathsf{H}}(z), (\mathsf{G}, \mathsf{H}, \{r^*\}, z) \leftarrow \mathfrak{D}\right] \\ \leq &\mathrm{Adv}_{\mathsf{P},\mathcal{B}}^{\mathsf{OW\text{-}CPA}} + \Pr\left[\mathsf{Coll}_{pk}^{r^*}\right] \overset{(a)}{\leq} \mathrm{Adv}_{\mathsf{P},\mathcal{B}}^{\mathsf{OW\text{-}CPA}} + 2 \cdot \delta_{ac}. \end{aligned} \tag{107}$$

Here (a) uses Lemma G1. Finally, by combining Eq. (104) to (107), we get

$$\mathrm{Adv}_{\mathsf{ACWC_0[P,H]},\mathcal{A}}^{\mathsf{OW\text{-}CPA}} \leq \left(2 \cdot \sqrt{\mathrm{Adv}_{\mathsf{P},\mathcal{B}}^{\mathsf{OW\text{-}CPA}}} + \sqrt{\frac{1}{2^n}} + 3 \cdot \sqrt{\delta_{ac}}\right)^2.$$

This completes the proof of Lemma 8. □

[35] Here the "perfectly" means that, in $\mathcal{A}_1$'s view, the oracles G and H simulated by $\mathcal{B}$ is perfectly indistinguishable with the oracles G and H obtained by sampling $(\mathsf{G}, \mathsf{H}, \{r^*\}, z)$ according to the distribution $\mathfrak{D}$.

L Some missing proofs of Section 5

L.1 Proof of Theorem 7 (from $\mathsf{OW\text{-}CPA_P}$ to $\mathsf{OW\text{-}CPA_{ACWC}}$)

In order to prove this theorem, we introduce a sequence of games $\mathbf{G_0}$ to $\mathbf{G_3}$ as shown in Fig. 24.

GAMES $\mathbf{G_0}$-$\mathbf{G_3}$

1. $(pk, sk) \leftarrow \mathsf{Gen}$	$//\mathbf{G_0}$-$\mathbf{G_3}$
$\mathsf{H} \leftarrow_\$ \Omega_{\{0,1\}^{n_1} \to \{0,1\}^{n_2}}$	$//\mathbf{G_0}$-$\mathbf{G_3}$
2. $m^* \leftarrow_\$ \{0,1\}^{n_2}$	$//\mathbf{G_0}$
$m_1^* \leftarrow_\$ \{0,1\}^{n_1}$	$//\mathbf{G_0}$-$\mathbf{G_3}$
$m_2^* := m^* \oplus \mathsf{H}(m_1^*)$	$//\mathbf{G_0}$
$m_2^* \leftarrow_\$ \{0,1\}^{n_2}$	$//\mathbf{G_1}$-$\mathbf{G_3}$
$m^* := m_2^* \oplus \mathsf{H}(m_1^*)$	$//\mathbf{G_1}$
$c^* := \mathsf{Enc}(pk, m_1^* \vert\vert m_2^*)$	$//\mathbf{G_0}$-$\mathbf{G_3}$
3. $m' \leftarrow \mathcal{A}^{\mathsf{F}}(pk, c^*)$	$//\mathbf{G_0}$-$\mathbf{G_3}$
Measure register $\mathsf{D}_{q_\mathsf{F}}$ to get D	$//\mathbf{G_3}$
4. **return** $[\![m' = m^*]\!]$	$//\mathbf{G_0}$-$\mathbf{G_1}$
return $[\![m' = m_2^* \oplus \mathsf{H}(m_1^*)]\!]$	$//\mathbf{G_2}$
return $[\![m' = m_2^* \oplus D(m_1^*)]\!]$	$//\mathbf{G_3}$

$\mathsf{H}(|x, y\rangle)$

1. **return** $|x, y \oplus \mathsf{H}(x)\rangle$	$//\mathbf{G_0}$-$\mathbf{G_2}$
2. **perform** CStO	$//\mathbf{G_3}$

Fig. 24. Games $\mathbf{G_0}$ to $\mathbf{G_3}$ in the proof of Theorem 7.

Game $\mathbf{G_0}$: This is the $\mathsf{OW\text{-}CPA}$ game of $\mathsf{ACWC}[\mathsf{P}, \mathsf{H}]$ with the adversary $\mathcal{A}$ in the QROM. Thus we have

$$\mathrm{Adv}^{\mathsf{OW\text{-}CPA}}_{\mathsf{ACWC}[\mathsf{P},\mathsf{H}],\mathcal{A}} = \Pr[1 \leftarrow \mathbf{G_0}]. \tag{108}$$

Game $\mathbf{G_1}$: In this game, instead of sampling $m^* \leftarrow_\$ \{0,1\}^{n_2}$, $m_1^* \leftarrow_\$ \{0,1\}^{n_1}$, and let $m_2^* := m^* \oplus \mathsf{H}(m_1)$, we instead sample $m_2^* \leftarrow_\$ \{0,1\}^{n_2}$, $m_1^* \leftarrow_\$ \{0,1\}^{n_1}$, and let $m^* := m_2^* \oplus \mathsf{H}(m_1^*)$. Obviously this is a conceptual change, and thus

$$\Pr[1 \leftarrow \mathbf{G_0}] = \Pr[1 \leftarrow \mathbf{G_1}]. \tag{109}$$

Game $\mathbf{G_2}$: This game no longer computes $m_2^* \oplus \mathsf{H}(m_1^*)$ before the m' is generated. Instead, after m' is generated, it checks whether m' equals $m_2^* \oplus \mathsf{H}(m_1^*)$. Obviously game $\mathbf{G_2}$ is a rewrite of game $\mathbf{G_1}$, hence we have

$$\Pr[1 \leftarrow \mathbf{G_1}] = \Pr[1 \leftarrow \mathbf{G_2}]. \tag{110}$$

Game $\mathbf{G_3}$: Compared to game $\mathbf{G_2}$, this new game has three differences. First, H is simulated as a compressed standard oracle based on the database register $\mathsf{D}_{q_\mathsf{H}}$ (Section 2.3). Second, after the generation of m', the $\mathsf{D}_{q_\mathsf{H}}$ is measured to get the outcome D. Third, the final check is replaced with $[\![m' = m_2^* \oplus \mathsf{H}(m_1^*)]\!]$.

In the following, we use our CQR technique (Theorem 2) to compute the difference between games $\mathbf{G_2}$ and $\mathbf{G_3}$. We first construct a quantum oracle algorithm $\mathcal{A}_1$ as shown in Fig. 25 to play game $\mathbf{G_2}$. This algorithm has oracle access to H and O_0: H is a quantum random oracle with input/output space $\{0,1\}^{n_1}/\{0,1\}^{n_2}$, while O_0 is a checking oracle with the input/output register I/B as defined in Fig. 25. The set $S_{x_1,x_2,x}$ used by O_0 is

$$S_{x_1,x_2,x} := \{y : x_2 = x_1 \oplus y\}. \tag{111}$$

At this point, it is obvious that $\mathcal{A}_1^{\mathsf{H},\mathsf{O}_0}$ perfectly runs game $\mathbf{G_2}$. Additionally, if we replace H by a corresponding compressed standard oracle $\mathsf{H_c}$ that acts on the database register $\mathsf{D}_{q_\mathsf{H}}$, and replace O_0 with the O_1 defined in Fig. 25, then $A^{\mathsf{H_c},\mathsf{O}_1}$ perfectly runs game $\mathbf{G_3}$. The reason is that, by the basic rules of quantum computing, the final query of O_1 and the subsequent measurement on B performed by $\mathcal{A}_1^{\mathsf{H_c},\mathsf{O}_1}$ are equivalent to the last two steps performed by game $\mathbf{G_3}$.

$\mathcal{A}_1^{\mathsf{H},\mathsf{O}_0}$:

1. $(pk, sk) \leftarrow \mathsf{Gen}$
2. $m_1^* \leftarrow_\$ \{0,1\}^{n_1}, m_2^* \leftarrow_\$ \{0,1\}^{n_2}, c^* := \mathsf{Enc}(pk, m_1^*||m_2^*)$
3. $m' \leftarrow \mathcal{A}^{\mathsf{H}}(pk, c^*)$
4. Set the state on IB as $|m_2^*, m', m_1^*\rangle_\mathsf{I}|0\rangle_\mathsf{B}$ and then query O_0
5. Measure B and output the measurement outcome

$\mathsf{O}_0(|x_1, x_2, x\rangle_\mathsf{I}|b\rangle_\mathsf{B})$

1. **return** $|x_1, x_2, x\rangle_\mathsf{I}|b \oplus [\![\mathsf{H}(x) \in S_{x_1,x_2,x}]\!]\rangle_\mathsf{B}$

$\mathsf{O}_1(|x_1, x_2, x, D_\mathsf{H}\rangle_{\mathsf{ID}_{q_\mathsf{H}}}|b\rangle_\mathsf{B})$

1. **return** $|x_1, x_2, x, D_\mathsf{H}\rangle_{\mathsf{ID}_{q_\mathsf{H}}}|b \oplus [\![D_\mathsf{H}(x) \in S_{x_1,x_2,x}]\!]\rangle_\mathsf{B}$

Fig. 25. Quantum oracle algorithm $\mathcal{A}_1$ and quantum oracles O_0 and O_1.

That is to say, we have $\Pr[1 \leftarrow \mathbf{G}_2] = \Pr[1 \leftarrow \mathcal{A}_1^{\mathsf{H},\mathsf{O}_0}]$ and $\Pr[1 \leftarrow \mathbf{G}_3] = \Pr[1 \leftarrow \mathcal{A}_1^{\mathsf{H_c},\mathsf{O}_1}]$. Notice that $\mathcal{A}_1$ queries $\mathsf{O}_0/\mathsf{O}_1$ one times, and the set $S_{x_1,x_2,x}$ defined in Eq. (111) satisfies $|S_{x_1,x_2,x}| \le 1$ for any $x_1 \in \{0,1\}^{n_1}$, $x_2 \in \{0,1\}^{n_2}$, and $x \in \{0,1\}^{n_1}$. Hence by using Theorem 2, we have

$$\left|\sqrt{\Pr[1 \leftarrow \mathbf{G}_2]} - \sqrt{\Pr[1 \leftarrow \mathbf{G}_3]}\right| \le 6\sqrt{1/2^{n_2}}. \tag{112}$$

As for the success probability of game $\mathbf{G_3}$, we first assume that P is deterministic and construct the following adversary $\mathcal{B}$ against the $\mathsf{OW\text{-}CPA}$ security of P:

1. The input of $\mathcal{B}$ is (pk, c^*), where c^* is the challenge ciphertext of the q_{H}-OW-CPA game.
2. $\mathcal{B}$ initializes the database register $\mathsf{D}_{q_{\mathsf{H}}}$ to simulate H, and then runs the adversary $\mathcal{A}$ in game $\mathbf{G_3}$ to get its output m'.
3. After getting m', $\mathcal{B}$ measures $\mathsf{D}_{q_{\mathsf{H}}}$ to get the measurement outcome D, and then searches the set $\{x||(m' \oplus D(x)) : D(x) \neq \perp\}$ for the first element $x_1||x_2$ that satifies $\mathsf{Enc}(pk, x_1||x_2) = c^*$. If this search succeeds, $\mathcal{B}$ outputs $x_1||x_2$. Otherwise $\mathcal{B}$ outputs $\perp$.

Since the adversary $\mathcal{A}$ makes at most q_{H} queries to H, $\mathcal{B}$ computes Enc at most q_{H} times. Moreover, if game $\mathbf{G_3}$ succeeds, i.e., the m' returned by $\mathcal{A}$ satisfies $m' = m_2^* \oplus D(m_1^*)$, we know that $D(m_1^*) \neq \perp$ and $m_2^* = m' \oplus D(m_1^*)$. At this point, we can find that the $m_1^*||m_2^*$ must belong to the set $\{x||(m' \oplus D(x)) : D(x) \neq \perp\}$. In addition, according to Lemma G1, we know that $\mathsf{Enc}(pk, x_1||x_2) = c^*$ if and only if $x_1||x_2 = m_1^*||m_2^*$ under an error at most $2 \cdot \delta_{ac}$. Namely, we have

$$\Pr[1 \leftarrow \mathbf{G_3}] \leq \mathrm{Adv}^{\mathsf{OW\text{-}CPA}}_{\mathsf{P},\mathcal{B}} + 2 \cdot \delta_{ac}. \tag{113}$$

For the rPKE scheme P, the above proof no longer holds because Lemma G1 is only valid for dPKE schemes. Based on this, we design a new adversary $\mathcal{B}$ against the OW-CPA security of P, which is the same as the above $\mathcal{B}$ except that it finally outputs a uniformly random value of the set $\{x||(m' \oplus D(x)) : D(x) \neq \perp\}$. Obviously, this new adversary $\mathcal{B}$ satisfies that

$$\Pr[1 \leftarrow \mathbf{G_3}] \leq q_{\mathsf{H}} \cdot \mathrm{Adv}^{\mathsf{OW\text{-}CPA}}_{\mathsf{P},\mathcal{B}}. \tag{114}$$

Finally, by combining Eq. (108) to Eq. (110) and Eq. (112) to Eq. (114), we have

$$\mathrm{Adv}^{\mathsf{OW\text{-}CPA}}_{\mathsf{ACWC[P,H]},\mathcal{A}} \leq \begin{cases} 2 \cdot \mathrm{Adv}^{\mathsf{OW\text{-}CPA}}_{\mathsf{P},\mathcal{B}} + 4 \cdot \delta_{ac} + 72/2^{n_2} & \text{if } \mathsf{P} \text{ is deterministic,} \\ 2q_{\mathsf{H}} \cdot \mathrm{Adv}^{\mathsf{OW\text{-}CPA}}_{\mathsf{P},\mathcal{B}} + 72/2^{n_2} & \text{if } \mathsf{P} \text{ is randomized.} \end{cases}$$

By the construction of $\mathcal{B}$, we also have $\mathcal{T}_{\mathcal{B}} \leq \mathcal{T}_{\mathcal{A}} + O(q_{\mathsf{H}}^2) + O(q_{\mathsf{H}}) \cdot \mathcal{T}_{\mathsf{Enc}}$. This completes the proof of Theorem 7. ◻

L.2 Proof of Theorem 8 (from $\mathsf{OW\text{-}CPA_P}$ to $\mathsf{IND\text{-}CCA_{KEMAC}}$)

Before proving this theorem, we first introduce the following proven theorem.

Theorem 18 (Adaptive version of [22, Theorem 1]). *Let rPKE scheme* $\mathsf{P} = (\mathsf{Gen}, \mathsf{Enc}, \mathsf{Dec})$ *be* δ_{wc}*-worst-correct and weakly* γ*-spread. Let* $\mathcal{A}$ *be an* IND-CCA *adversary against* $\mathsf{KEM}^{\perp}_{m} := \mathsf{FO}^{\perp}_{m}[\mathsf{P}, \mathsf{F}, \mathsf{G}]$*, making* q_{D} *classical queries to the decapsulation oracle, and making* q_{F} *and* q_{G} *quantum queries to the random oracle* F *and* G*. Then we can construct a* OW-CPA *adversary* $\mathcal{B}$ *against* P *such that* $\mathcal{T}_{\mathcal{B}} \leq \mathcal{T}_{\mathcal{A}} + O(q_{\mathsf{D}} q + q_{\mathsf{D}}^2) \cdot \mathcal{T}_{\mathsf{Enc}} + O(q^2 + q_{\mathsf{D}})$ *and*

$$\mathrm{Adv}^{\mathsf{IND\text{-}CCA}}_{\mathsf{KEM}^{\perp}_{m},\mathcal{A}} \leq 8(q + q_{\mathsf{D}}) \cdot \sqrt{\mathrm{Adv}^{\mathsf{OW\text{-}CPA}}_{\mathsf{P},\mathcal{B}}} + 10(q+1)^2 \cdot \delta_{wc} + 24 q_{\mathsf{D}}(q_{\mathsf{F}} + 4q_{\mathsf{D}}) \cdot 2^{-\gamma/2}.$$

Here $q := q_{\mathsf{F}} + q_{\mathsf{G}}$*.*

Remark 5. In fact, the original [22, Theorem 1] requires P to be $\mathsf{IND\text{-}CPA}$-secure; but based on the results presented in [20], this theorem still holds even when P is only $\mathsf{OW\text{-}CPA}$-secure. Furthermore, we emphasize that the adversary $\mathcal{B}$ in the above theorem needs to run the encryption algorithm of P at most $q_{\mathsf{D}}q + q_{\mathsf{D}}^2$ times to simulate the decapsulation oracle for $\mathcal{A}$.

Notice that [12, Lemma 3.6] has shown that $\mathsf{ACWC}[\mathsf{P},\mathsf{H}]$ is $(\delta_{ac} + (1 + \sqrt{n_1 + n_2})/\sqrt{2^{n_1}})$-worst-correct if P is δ_{ac}-average-correct, and [12, Lemma 3.8] has shown that $\mathsf{ACWC}[\mathsf{P},\mathsf{H}]$ is weakly γ-spread if P is weakly γ-spread. Here n_1/n_2 is the input/output length of H. Based on such facts, for an $\mathsf{IND\text{-}CCA}$ adversary $\mathcal{A}$ against $\mathsf{KEMAC} = \mathsf{FOAC}[\mathsf{P},\mathsf{F},\mathsf{G},\mathsf{H}]$, making q_{D} classical queries to the decapsulation oracle, and making q_{F}, q_{G} and q_{H} quantum queries to the random oracle F, G and H, respectively, we can construct a $\mathsf{OW\text{-}CPA}$ adversary $\mathcal{B}$ against P by combining Theorem 7 and Theorem 18 such that

$$\begin{aligned}\mathrm{Adv}^{\mathsf{IND\text{-}CCA}}_{\mathsf{KEMAC},\mathcal{A}} \leq\ & 8(q+q_{\mathsf{D}})^{1.5}\sqrt{q_{\mathsf{D}}} \cdot \sqrt{\mathrm{Adv}^{\mathsf{OW\text{-}CPA}}_{\mathsf{P},\mathcal{B}}} + \frac{72(q+q_{\mathsf{D}})}{\sqrt{2^{n_2}}} + 10(q+1)^2 \cdot \delta' \\ & + 24q_{\mathsf{D}}(q_{\mathsf{F}} + 4q_{\mathsf{D}}) \cdot 2^{-\gamma/2}.\end{aligned}$$

Here $\delta' := \delta_{ac} + (1 + \sqrt{n_1 + n_2})/\sqrt{2^{n_1}}$ and the running time of $\mathcal{B}$ satisfies $\mathcal{T}_{\mathcal{B}} \leq \mathcal{T}_{\mathcal{A}} + O(q_{\mathsf{D}}q + q_{\mathsf{D}}^2 + q_{\mathsf{H}}) \cdot \mathcal{T}_{\mathsf{Enc}} + O(q^2 + q_{\mathsf{D}} + q_{\mathsf{H}}^2)$. This completes the proof of Theorem 8. $\square$

M Security proof of $\mathsf{FOAC}_m^{\not\perp}$

M.1 IND-CCA security proof of $\mathsf{FOAC}_m^{\not\perp}$ in the ROM

We first introduce the following theorem about $\mathsf{FO}_m^{\not\perp}$, which shows that its IND-CCA security can be tightly reduced to the q-OW-CPA security of the underlying PKE scheme in the ROM. Since this theorem is a simple corollary of combining [17, Theorem 3.1] and [17, Theorem 3.4], we omit its detailed proof for simplicity.

Theorem 19 **(q-OW-CPA of P $\overset{\text{ROM}}{\Rightarrow}$ IND-CCA of $\mathsf{KEM}_m^{\not\perp}$).** *Let* P *be a PKE scheme with worst-case decryption error δ_{wc}. Let $\mathcal{A}$ be an* IND-CCA *adversary against $\mathsf{KEM}_m^{\not\perp} := \mathsf{FO}_m^{\not\perp}[f, \mathsf{P}, \mathsf{F}, \mathsf{G}]$, making q_D classical queries to the decapsulation oracle, and making q_F and q_G classical queries to the random oracle* F *and* G*. Then we can construct a* PRF*-adversary $\mathcal{B}_1$ against f and a q-*OW-CPA *adversary $\mathcal{B}_2$ against* P *such that*

$$\mathrm{Adv}^{\mathsf{IND\text{-}CCA}}_{\mathsf{KEM}_m^{\not\perp},\mathcal{A}} \leq \mathrm{Adv}^{\mathsf{PRF}}_{f,\mathcal{B}_1} + \mathrm{Adv}^{q\text{-}\mathsf{OW\text{-}CPA}}_{\mathsf{P},\mathcal{B}_2} + (q_\mathsf{F} + q_\mathsf{D}) \cdot \delta_{wc}.$$

Here $q := q_\mathsf{F} + q_\mathsf{D} + 1$ and $\mathcal{B}_1$ makes at most q_D classical queries. The running time of $\mathcal{B}_1$ and $\mathcal{B}_2$ satisfy $\mathcal{T}_{\mathcal{B}_1} \approx \mathcal{T}_{\mathcal{B}_2} \approx \mathcal{T}_\mathcal{A}$.

Then we prove the following theorem, which shows that the q-OW-CPA security of ACWC can be reduced to the OW-CPA security of the underlying PKE scheme in the ROM.

Theorem 20 **(q-OW-CPA of P $\overset{\text{ROM}}{\Rightarrow}$ q-OW-CPA of ACWC).** *Let* P *be a PKE scheme with average-case decryption error δ_{ac}. Let $\mathcal{A}$ be a q-*OW-CPA *adversary against* ACWC[P, H]*, making q_H classical queries to* H*. Then we can construct a* OW-CPA *adversary $\mathcal{B}$ against* P *such that $\mathcal{T}_\mathcal{B} \leq \mathcal{T}_\mathcal{A} + O(qq_\mathsf{H}) + O(qq_\mathsf{H}) \cdot \mathcal{T}_{\mathsf{Enc}}$ and*

$$\mathrm{Adv}^{q\text{-}\mathsf{OW\text{-}CPA}}_{\mathsf{ACWC}[\mathsf{P},\mathsf{H}],\mathcal{A}} \leq \begin{cases} \mathrm{Adv}^{\mathsf{OW\text{-}CPA}}_{\mathsf{P},\mathcal{B}} + 2 \cdot \delta_{ac} & \text{if } \mathsf{P} \text{ is deterministic,} \\ qq_\mathsf{H} \cdot \mathrm{Adv}^{\mathsf{OW\text{-}CPA}}_{\mathsf{P},\mathcal{B}} & \text{if } \mathsf{P} \text{ is randomized.} \end{cases}$$

Here n_2 is the output length of H*.*

Proof. By using [12, Theorem 3.9], we can obtain

$$\mathrm{Adv}^{q\text{-}\mathsf{OW\text{-}CPA}}_{\mathsf{ACWC}[\mathsf{P},\mathsf{H}],\mathcal{A}} \leq \mathrm{Adv}^{qq_\mathsf{H}\text{-}\mathsf{OW\text{-}CPA}}_{\mathsf{P},\mathcal{A}_1}. \tag{115}$$

The running time of $\mathcal{A}_1$ here satisfies $\mathcal{T}_{\mathcal{A}_1} \leq \mathcal{T}_\mathcal{A} + O(qq_\mathsf{H})$. For a dPKE scheme P, according to Lemma G1, we can conclude that $\mathsf{Enc}(pk, x_1||x_2) = c^*$ if and only if $x_1||x_2 = m_1^*||m_2^*$ under an error at most $2 \cdot \delta_{ac}$, where $m_1^*||m_2^*$ is the challenge plaintext and c^* is the challenge ciphertext of the OW-CPA game of P. Based on this fact, we construct an adversary $\mathcal{B}$ against the OW-CPA security of P as:

– $\mathcal{B}$ first runs $\mathcal{A}_1$ to get its final output $\mathcal{Q}(|\mathcal{Q}| \leq qq_\mathsf{H})$, and then searches $\mathcal{Q}$ for the first element $x_1||x_2$ that satisfies $\mathsf{Enc}(pk, x_1||x_2) = c^*$. If such search succeeds, $\mathcal{B}$ outputs $x_1||x_2$. Otherwise $\mathcal{B}$ outputs $\perp$.

Obviously, the running time of $\mathcal{B}$ satisfies $\mathcal{T}_{\mathcal{B}} \leq \mathcal{T}_{\mathcal{A}_1} + O(qq_{\mathsf{H}}) \cdot \mathcal{T}_{\mathsf{Enc}}$ and

$$\mathrm{Adv}^{qq_{\mathsf{H}}\text{-}\mathsf{OW\text{-}CPA}}_{\mathsf{P},\mathcal{A}_1} \leq \mathrm{Adv}^{\mathsf{OW\text{-}CPA}}_{\mathsf{P},\mathcal{B}} + 2 \cdot \delta_{ac}. \tag{116}$$

For a rPKE scheme P, the above proof no longer holds because Lemma G1 is only valid for dPKE schemes. Based on this, we design a new adversary $\mathcal{B}$ against the OW-CPA security of P, which is the same as the above $\mathcal{B}$ except that it finally outputs a uniformly random value of the set $\mathcal{Q}$. Obviously, this new adversary $\mathcal{B}$ satisfies that $\mathcal{T}_{\mathcal{B}} \approx \mathcal{T}_{\mathcal{A}_1}$ and

$$\mathrm{Adv}^{qq_{\mathsf{H}}\text{-}\mathsf{OW\text{-}CPA}}_{\mathsf{P},\mathcal{A}_1} \leq qq_{\mathsf{H}} \cdot \mathrm{Adv}^{\mathsf{OW\text{-}CPA}}_{\mathsf{P},\mathcal{B}}. \tag{117}$$

Now, by combining Eq. (115) to (117), we complete the proof of Theorem 20. $\square$

By combining the above Theorem 19 and Theorem 20, we can prove the main theorem of this subsection.

Theorem 21 (OW-CPA of $\mathsf{P} \overset{\mathrm{ROM}}{\Rightarrow}$ IND-CCA of $\mathsf{KEM}^{\not\perp}_m$). *Let P be a PKE scheme with average-case decryption error δ_{ac}. Let $\mathcal{A}$ be an IND-CCA adversary against $\mathsf{KEMAC}^{\not\perp}_m := \mathsf{FOAC}^{\not\perp}_m[f, \mathsf{P}, \mathsf{F}, \mathsf{G}, \mathsf{H}]$, making q_{D} classical queries to the decapsulation oracle, and making q_{F}, q_{G} and q_{H} classical queries to the random oracle F, G and H. Then we can construct a PRF-adversary $\mathcal{B}_1$ against f and a OW-CPA adversary $\mathcal{B}_2$ against P such that*

$$\mathrm{Adv}^{\mathsf{IND\text{-}CCA}}_{\mathsf{KEMAC}^{\not\perp}_m,\mathcal{A}} \leq \mathrm{Adv}^{\mathsf{PRF}}_{f,\mathcal{B}_1} + \mathrm{Adv}_{\mathsf{P}} + \frac{q}{2^{n_2}} + (q_{\mathsf{F}} + q_{\mathsf{D}}) \cdot \delta', \text{ with}$$

$$\mathrm{Adv}_{\mathsf{P}} = \begin{cases} \mathrm{Adv}^{\mathsf{OW\text{-}CPA}}_{\mathsf{P},\mathcal{B}_2} + 2 \cdot \delta_{ac} & \text{if } \mathsf{P} \text{ is deterministic,} \\ qq_{\mathsf{H}} \cdot \mathrm{Adv}^{\mathsf{OW\text{-}CPA}}_{\mathsf{P},\mathcal{B}_2} & \text{if } \mathsf{P} \text{ is randomized.} \end{cases}$$

Here $q := q_{\mathsf{F}} + q_{\mathsf{D}} + 1$, $\delta' := \delta_{ac} + (1 + \sqrt{n_1 + n_2})/\sqrt{2^{n_1}}$, n_1/n_2 is the input/output length of H, and $\mathcal{B}_1$ makes at most q_{D} classical queries. The running time of $\mathcal{B}_1$ and $\mathcal{B}_2$ satisfy $\max\{\mathcal{T}_{\mathcal{B}_1}, \mathcal{T}_{\mathcal{B}_2}\} \leq \mathcal{T}_{\mathcal{A}} + O(qq_{\mathsf{H}}) + O(qq_{\mathsf{H}}) \cdot \mathcal{T}_{\mathsf{Enc}}$.

M.2 IND-CCA security proof of $\mathsf{FOAC}^{\not\perp}_m$ in the QROM

We first introduce the following theorem about $\mathsf{FO}^{\not\perp}_m$, which shows that the IND-CCA security of $\mathsf{FO}^{\not\perp}_m$ can be reduced to the OW-CPA security of the underlying PKE scheme in the QROM.

Theorem 22 (Adaptive version of [23, Theorem 2]). *Let P be a PKE scheme with worst-case decryption error δ_{wc}. Let $\mathcal{A}$ be an IND-CCA adversary against $\mathsf{KEM}^{\not\perp}_m := \mathsf{FO}^{\not\perp}_m[f, \mathsf{P}, \mathsf{F}, \mathsf{G}]$, making q_{D} classical queries to the decapsulation oracle, and making q_{F} and q_{G} quantum queries to the random oracle F and G. Then we can construct a PRF-adversary $\mathcal{B}_1$ against f and a OW-CPA adversary $\mathcal{B}_2$ against P such that*

$$\mathrm{Adv}^{\mathsf{IND\text{-}CCA}}_{\mathsf{KEM}^{\not\perp}_m,\mathcal{A}} \leq \mathrm{Adv}^{\mathsf{PRF}}_{f,\mathcal{B}_1} + 2(q_{\mathsf{F}} + q_{\mathsf{G}}) \cdot \sqrt{\mathrm{Adv}^{\mathsf{OW\text{-}CPA}}_{\mathsf{P},\mathcal{B}_2}} + 16(q_{\mathsf{F}} + 1)^2 \cdot \delta_{wc}.$$

Here $\mathcal{B}_1$ makes at most q_{D} classical queries, and the running time of $\mathcal{B}_1$ and $\mathcal{B}_2$ satisfy $\mathcal{T}_{\mathcal{B}_1} \approx \mathcal{T}_{\mathcal{B}_2} \approx \mathcal{T}_{\mathcal{A}}$.

Remark 6. The above theorem optimizes the term "$4q_{\mathsf{F}} \cdot \sqrt{\delta_{wc}}$" provided by [23, Theorem 2] to "$16(q_{\mathsf{F}}+1)^2 \cdot \delta_{wc}$", which is feasible because directly replacing [23, Lemma 2] used in the proof of [23, Theorem 2] with [21, Lemma 2.9] yields the term "$16(q_{\mathsf{F}}+1)^2 \cdot \delta_{wc}$".

Notice that [12, Lemma 3.6] has shown that $\mathsf{ACWC}[\mathsf{P},\mathsf{H}]$ is $(\delta_{ac} + (1 + \sqrt{n_1+n_2})/\sqrt{2^{n_1}})$-worst-correct if P is δ_{ac}-average-correct, where n_1/n_2 is the input/output length of H. Therefore by combining Theorem 22 and Theorem 7, we can prove the main theorem of this subsection.

Theorem 23 (OW-CPA **of** $\mathsf{P} \overset{\text{QROM}}{\Rightarrow}$ IND-CCA **of** $\mathsf{KEM}_m^{\not\perp}$)**.** *Let* P *be a PKE scheme with average-case decryption error* δ_{ac}*. Let* $\mathcal{A}$ *be an* IND-CCA *adversary against* $\mathsf{KEMAC}_m^{\not\perp} := \mathsf{FOAC}_m^{\not\perp}[f,\mathsf{P},\mathsf{F},\mathsf{G},\mathsf{H}]$*, making* q_{D} *classical queries to the decapsulation oracle, and making* q_{F}*,* q_{G} *and* q_{H} *quantum queries to the random oracle* F*,* G *and* H*. Then we can construct a* PRF*-adversary* $\mathcal{B}_1$ *against* f *and a* OW-CPA *adversary* $\mathcal{B}_2$ *against* P *such that*

$$\mathrm{Adv}^{\mathsf{IND\text{-}CCA}}_{\mathsf{KEM}_m^{\not\perp},\mathcal{A}} \leq \mathrm{Adv}^{\mathsf{PRF}}_{f,\mathcal{B}_1} + \mathrm{Adv}_{\mathsf{P}} + 16(q_{\mathsf{F}}+1)^2 \cdot \delta', \textit{ with}$$

$$\mathrm{Adv}_{\mathsf{P}} = \begin{cases} 2(q_{\mathsf{F}}+q_{\mathsf{G}}) \cdot \sqrt{\mathrm{Adv}^{\mathsf{OW\text{-}CPA}}_{\mathsf{P},\mathcal{B}_2} + 4 \cdot \delta_{ac} + 72/2^{n_2}} & \textit{if } \mathsf{P} \textit{ is deterministic,} \\ 2\sqrt{2}(q_{\mathsf{F}}+q_{\mathsf{G}})\sqrt{q_{\mathsf{H}}}\sqrt{\cdot \mathrm{Adv}^{\mathsf{OW\text{-}CPA}}_{\mathsf{P},\mathcal{B}_2} + 72/2^{n_2}} & \textit{if } \mathsf{P} \textit{ is randomized.} \end{cases}$$

Here $\delta' := \delta_{ac} + (1+\sqrt{n_1+n_2})/\sqrt{2^{n_1}}$*,* n_1/n_2 *is the input/output length of* H*, and* $\mathcal{B}_1$ *makes at most* q_{D} *classical queries. The running time of* $\mathcal{B}_1$ *and* $\mathcal{B}_2$ *satisfy* $\max\{\mathcal{T}_{\mathcal{B}_1}, \mathcal{T}_{\mathcal{B}_2}\} \leq \mathcal{T}_{\mathcal{A}} + O(q_{\mathsf{H}}^2) + O(q_{\mathsf{H}}) \cdot \mathcal{T}_{\mathsf{Enc}}$.